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(54) **INSECTICIDAL PEPTIDE PRODUCTION
AND COMBINATION OF CYSTEINE RICH
PEPTIDES**

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(57) **ABSTRACT**

New insecticidal proteins, nucleotides, peptides, their
expression in plants, methods of producing the peptides,
new processes, production techniques, new peptides, new
formulations, and new organisms, a process which increases
the insecticidal peptide production yield from yeast expres-
sion systems. The present invention is also related and
discloses selected endotoxins we call cysteine rich insecti-
cidal peptides (CRIPS) which are peptides derived from
Bacillus thuringiensis (Bt) and their genes and endotoxins in
combination with toxic peptides known as Inhibitor Cystine
Knot (ICK) genes and peptides as well as with other types
of insecticidal peptides such as trypsin modulating oostatic
factor (TMOF) peptide sequences used in various formula-
tions and combinations; of both genes and peptides, useful
for the control of insects.

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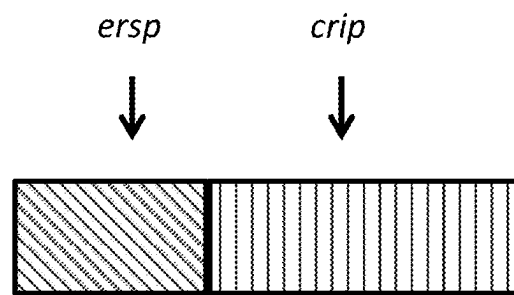


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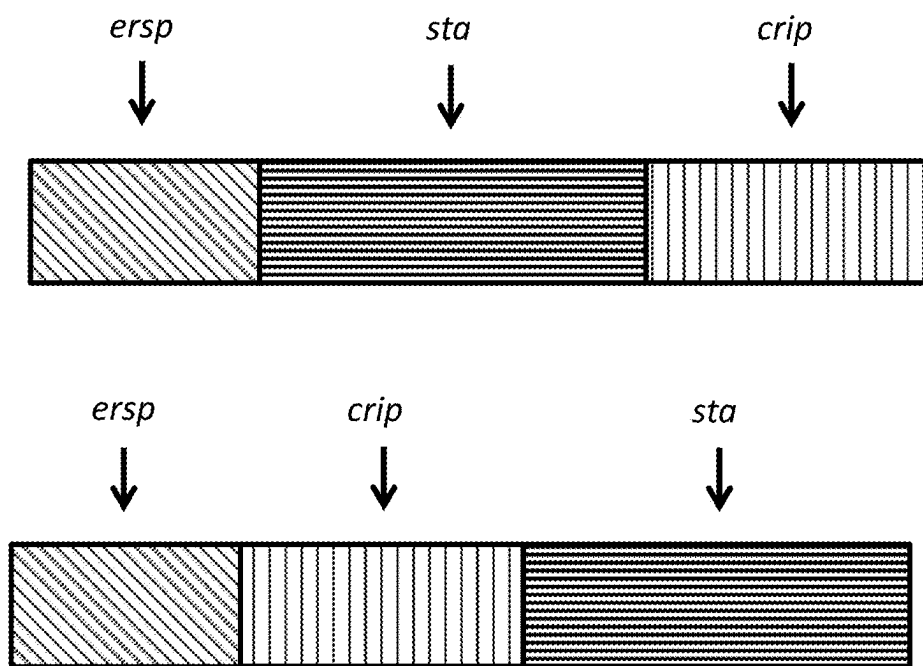


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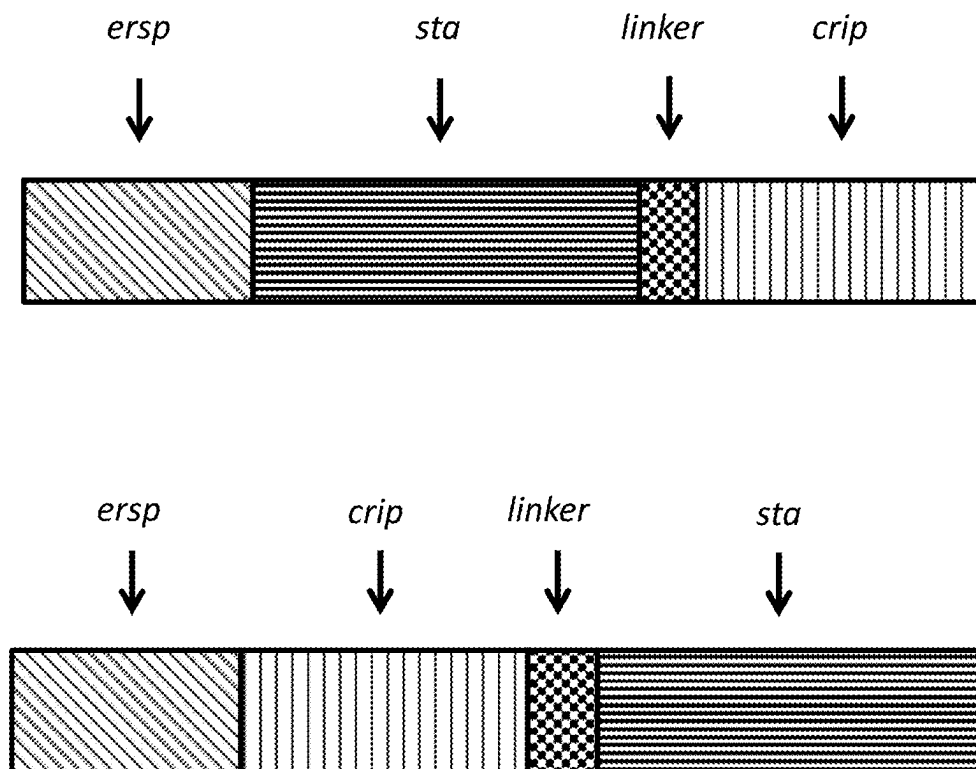


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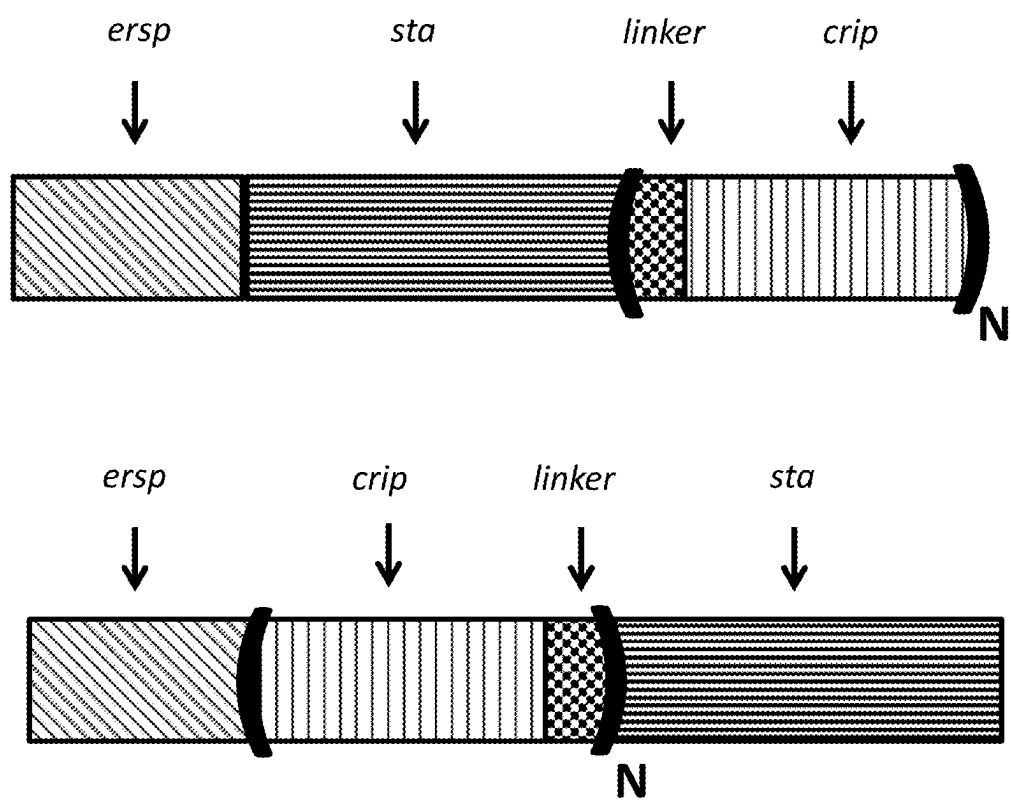


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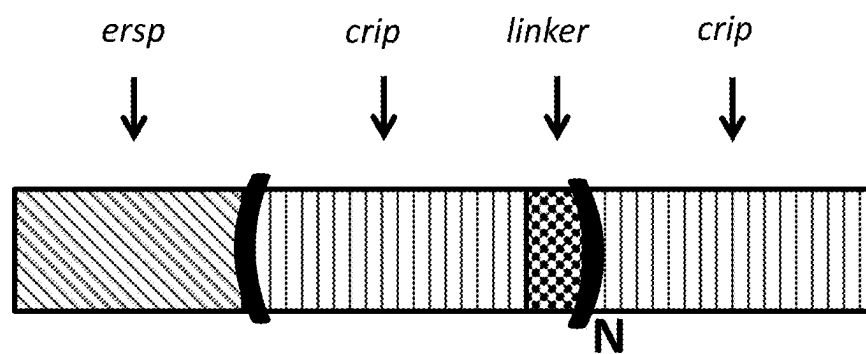


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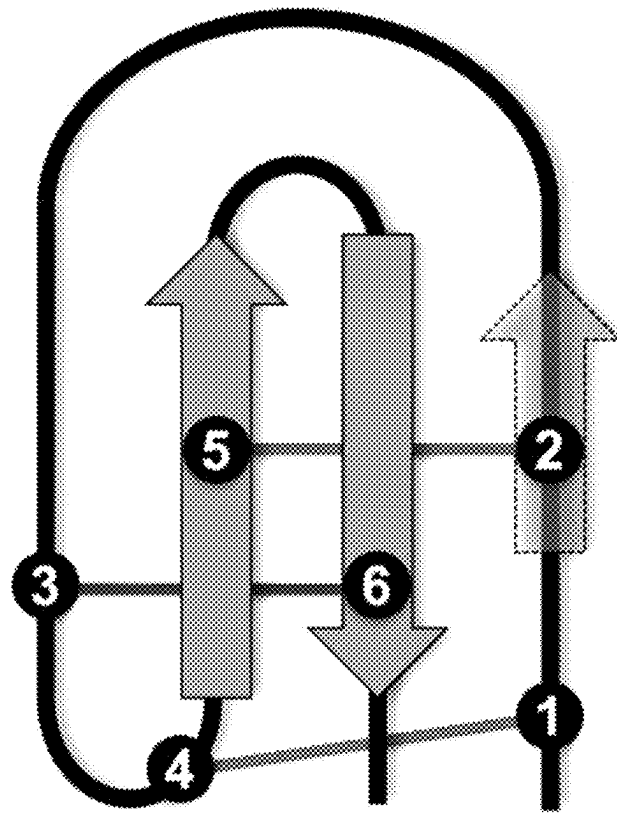


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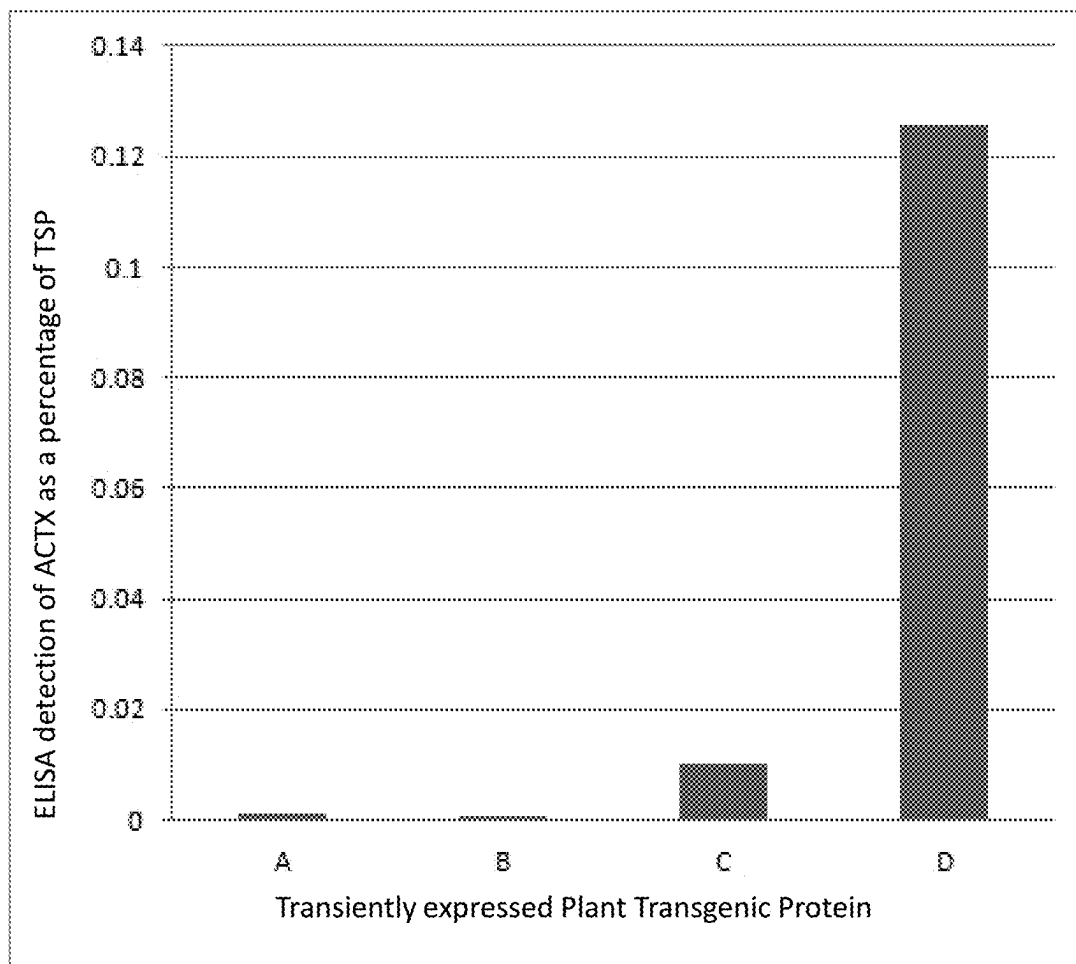


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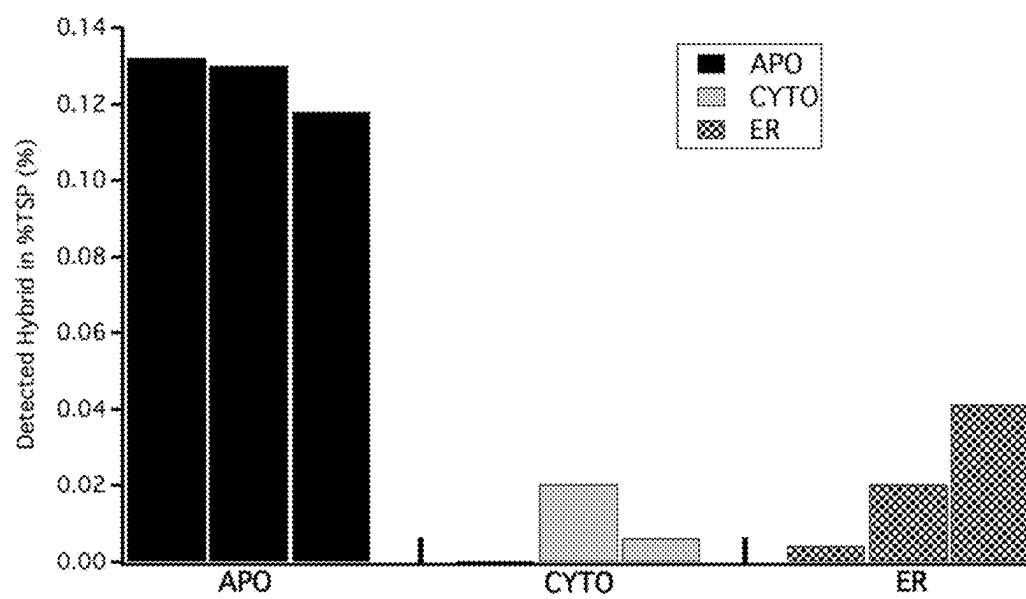


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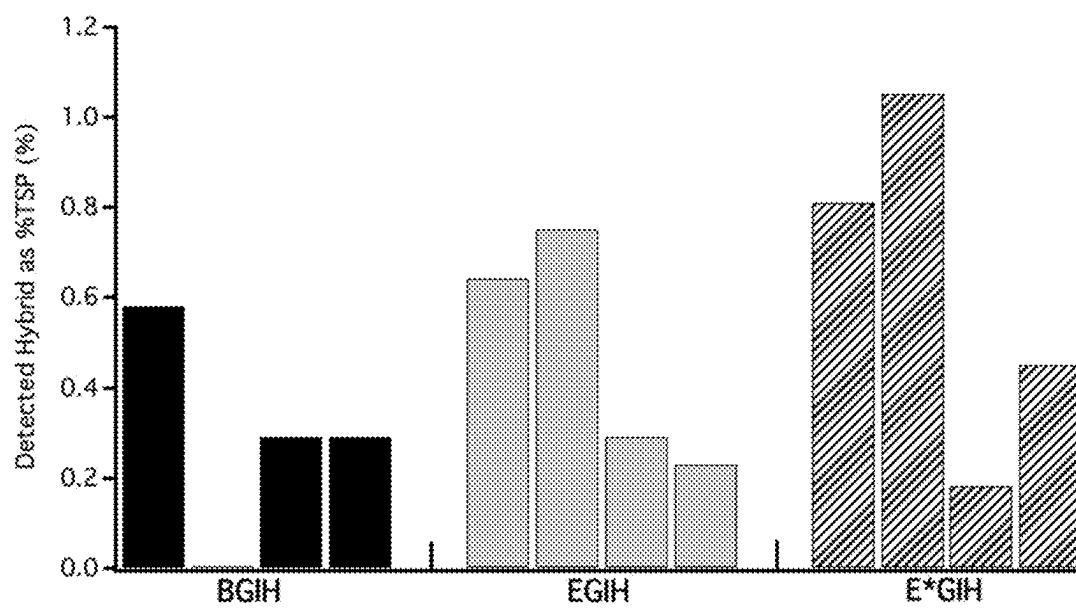


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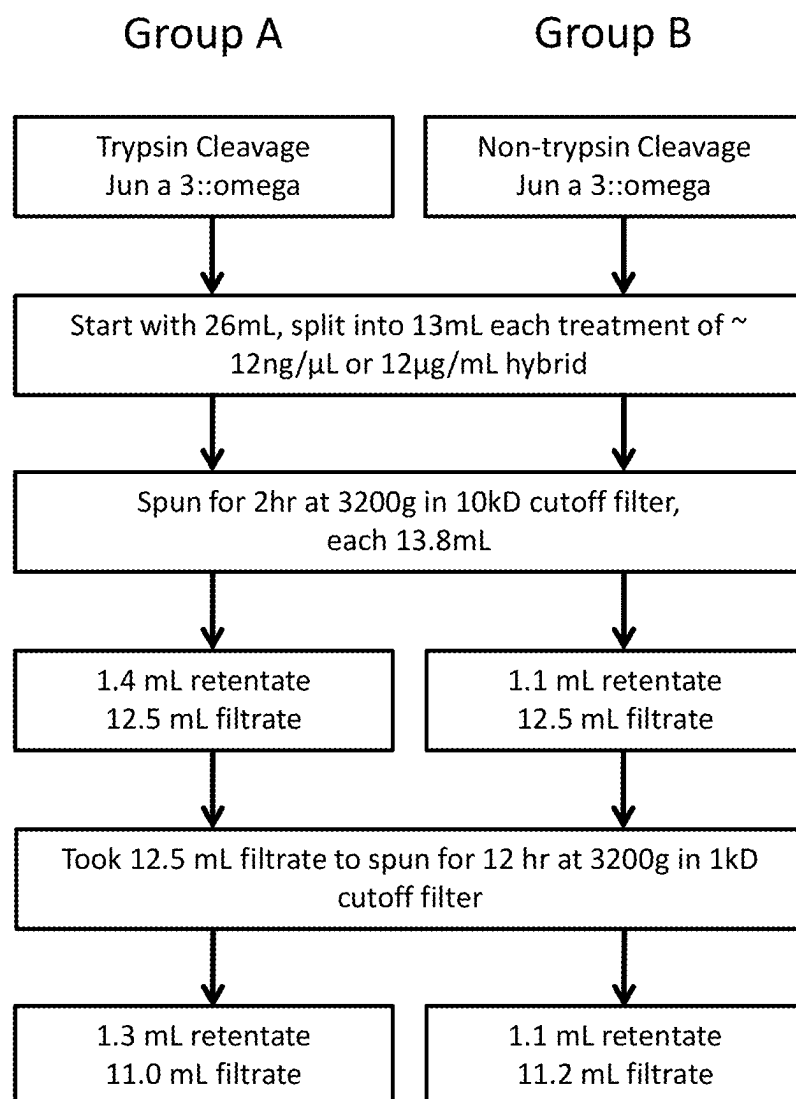


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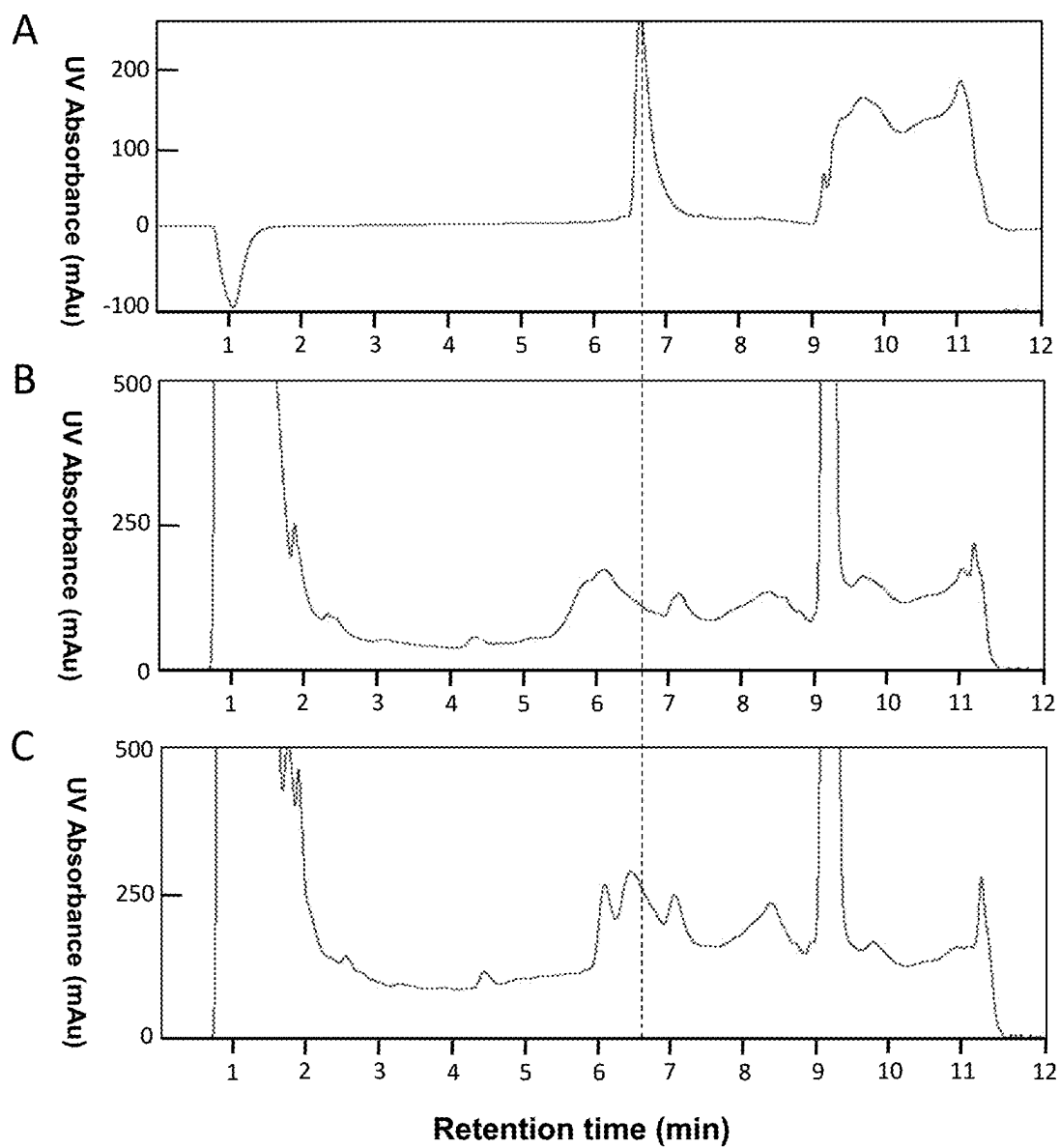


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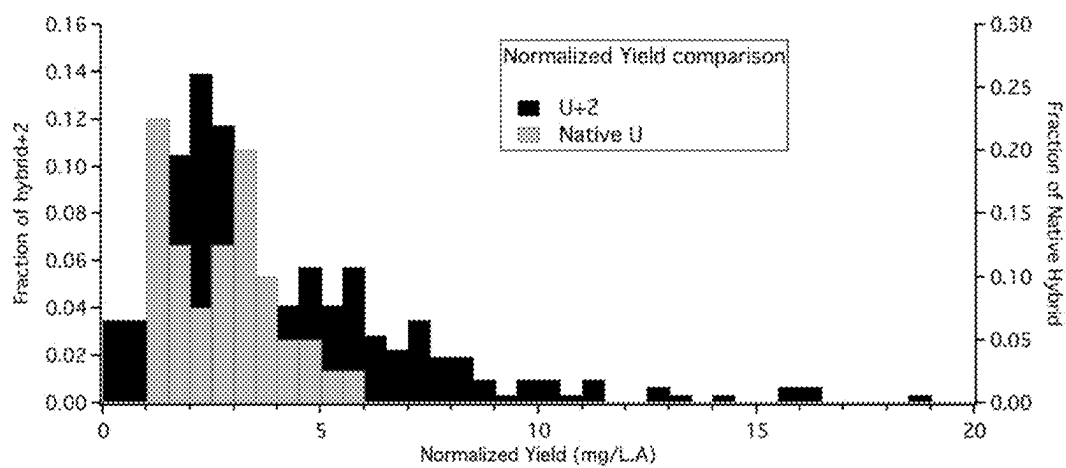


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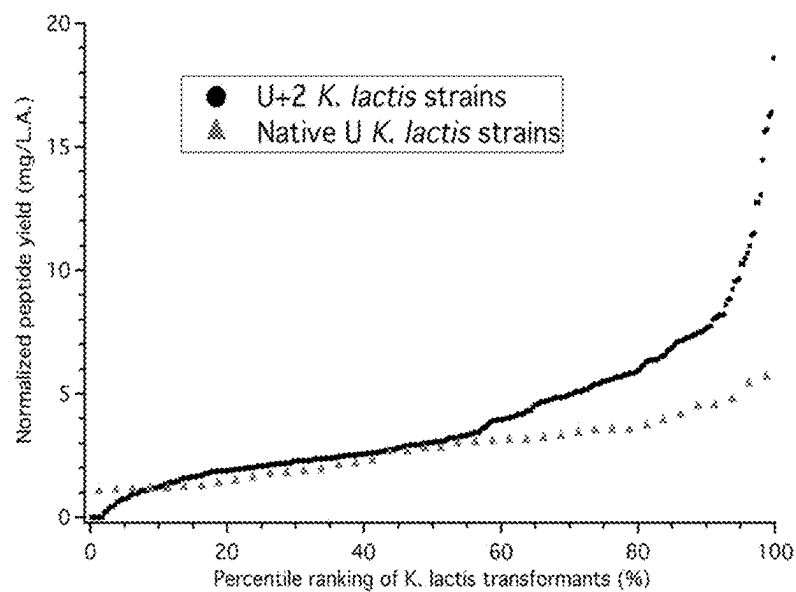


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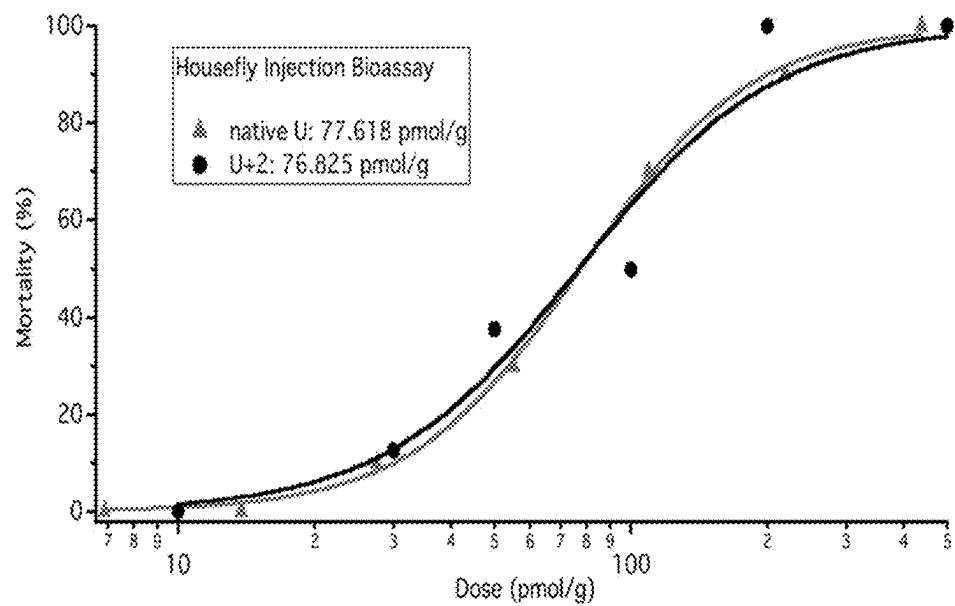


Fig. 14

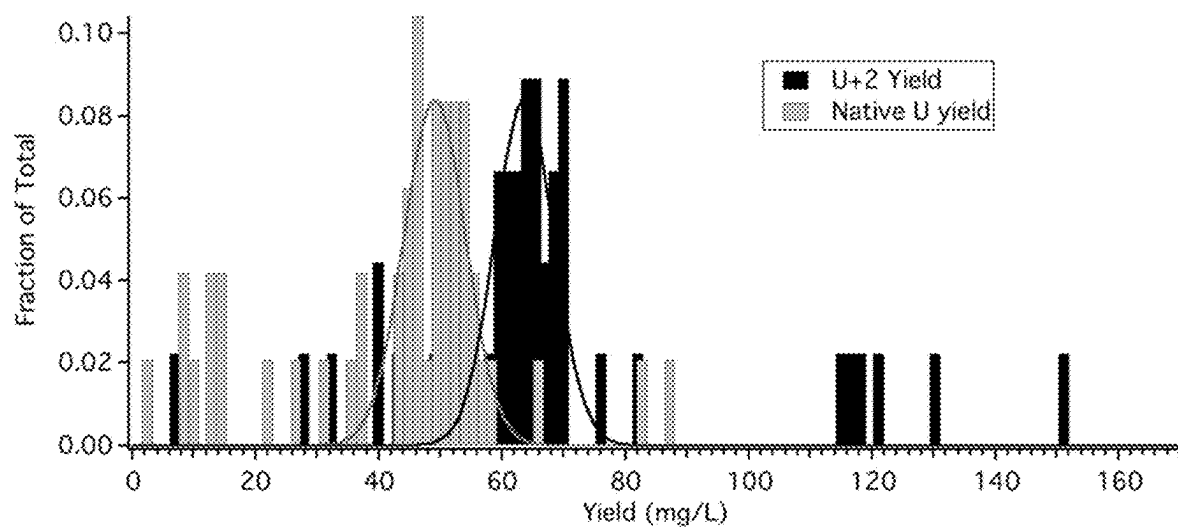


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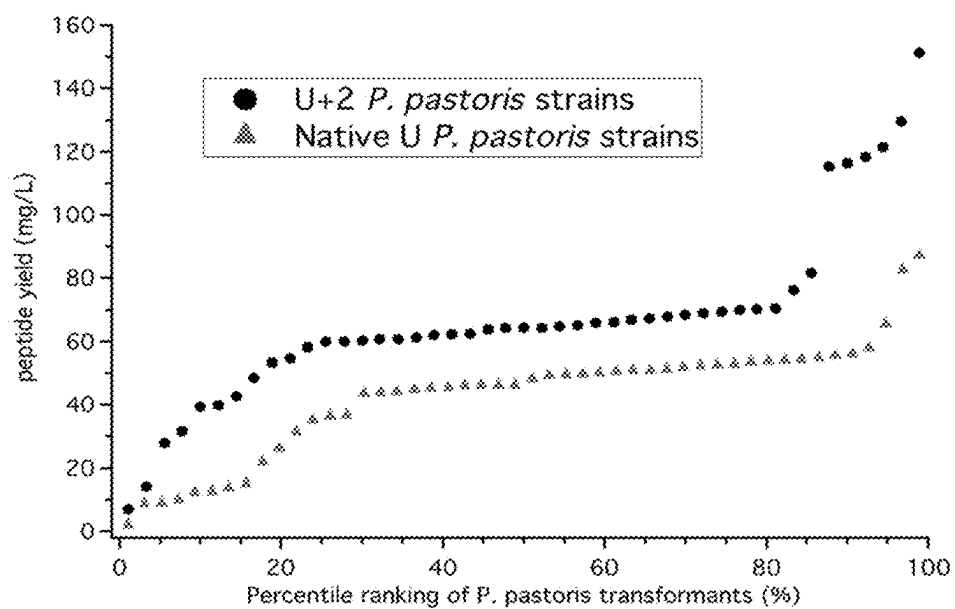


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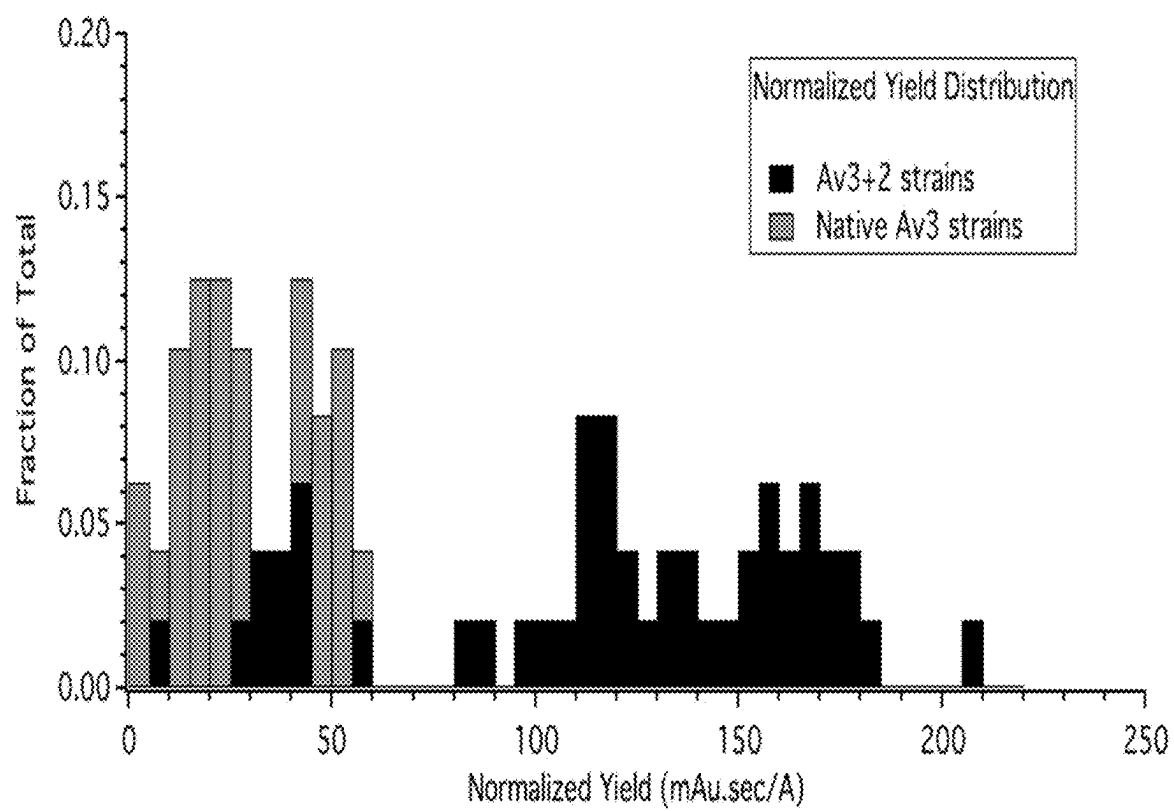


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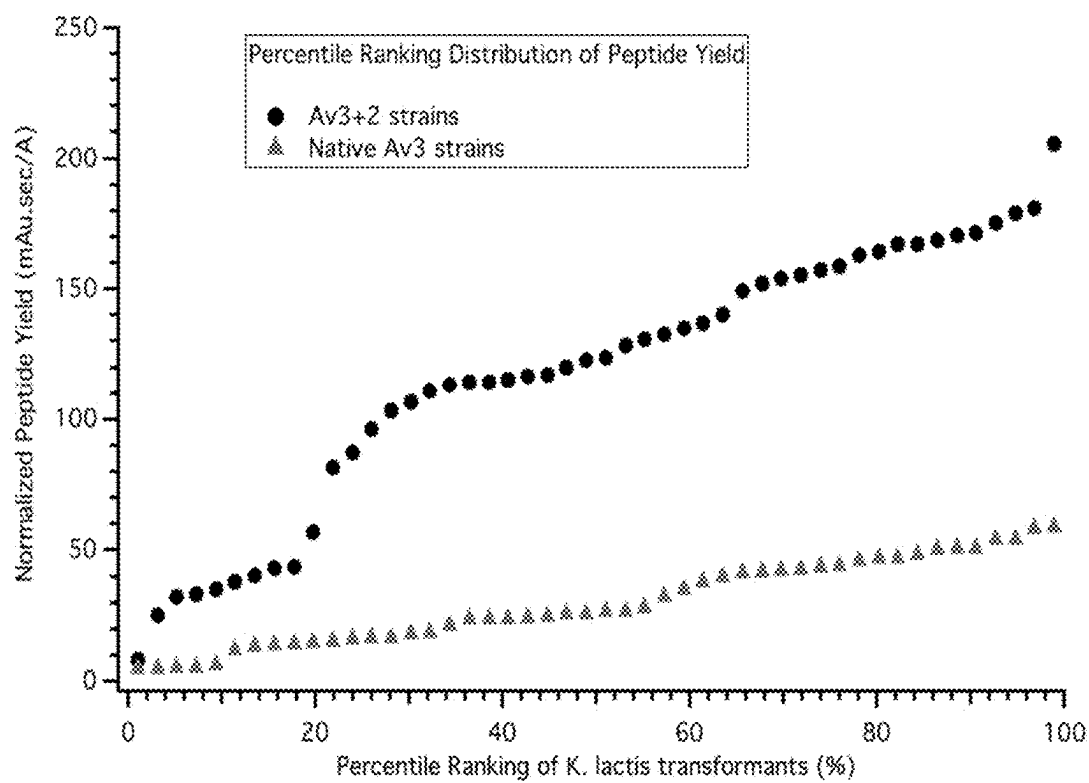


Fig. 18

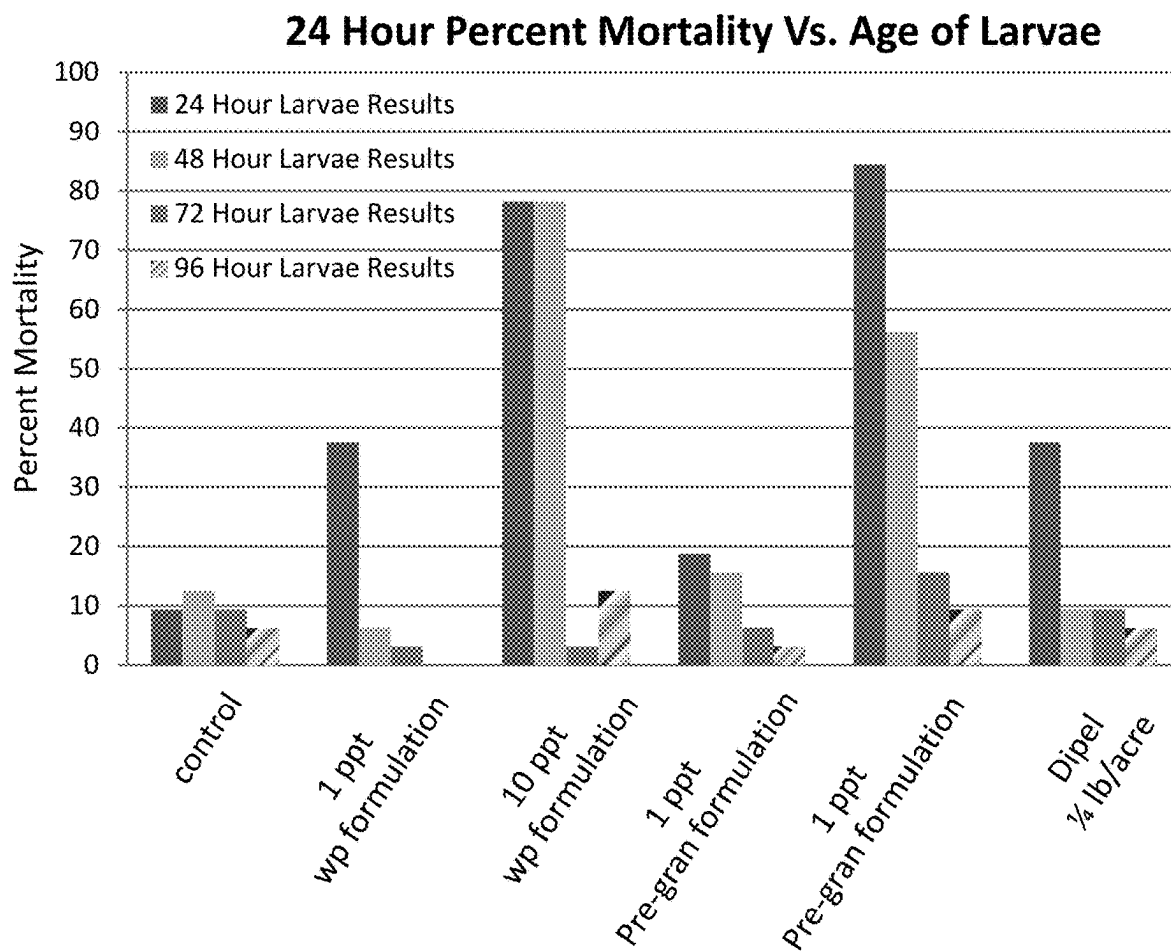


Fig. 19

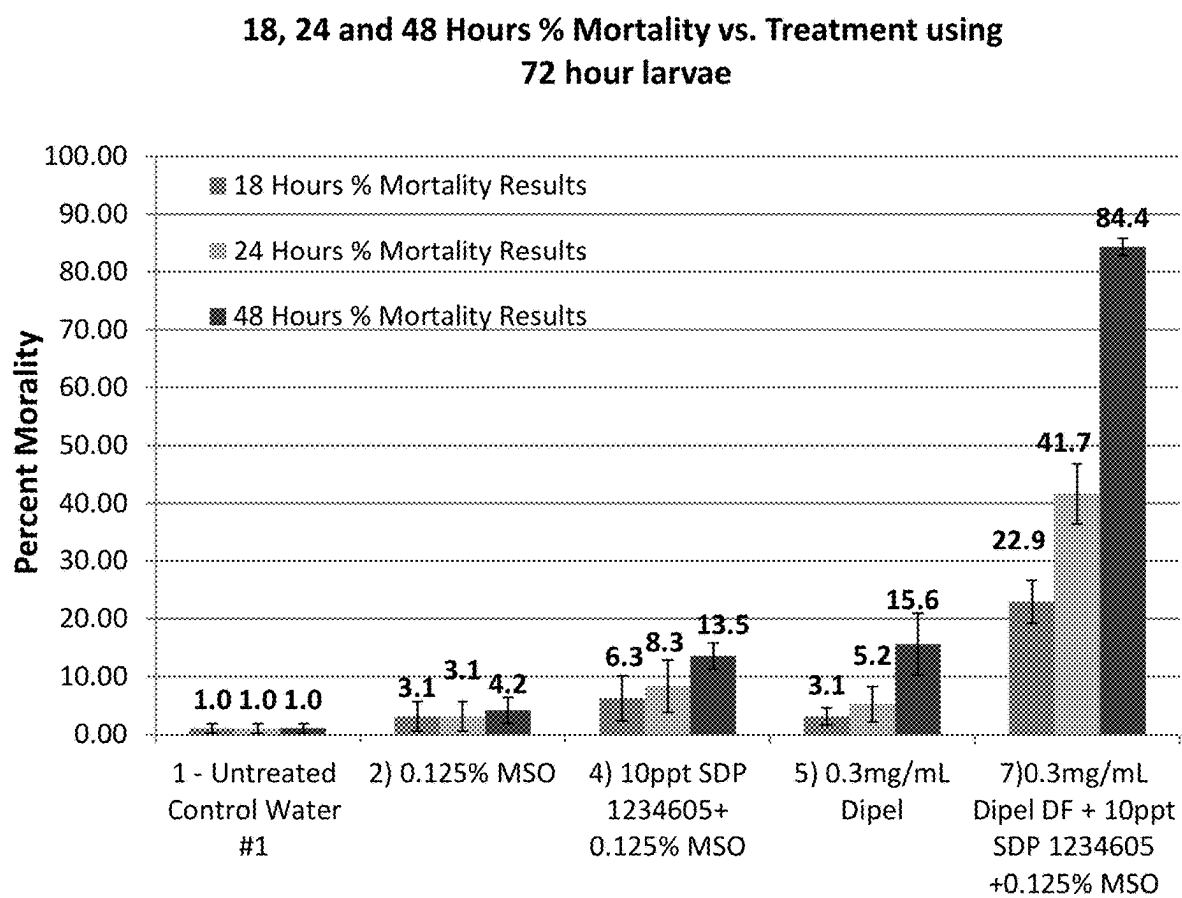
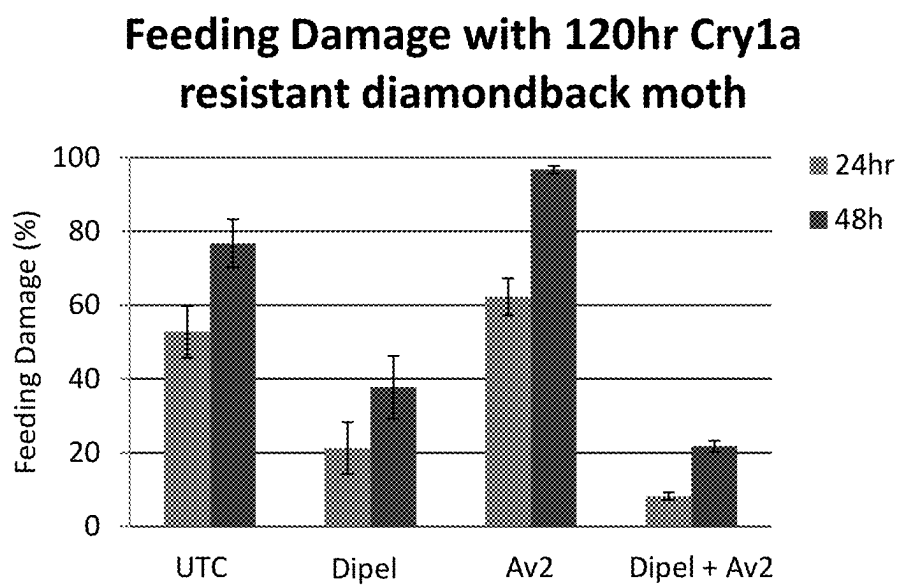


Fig. 20

**Fig. 21**

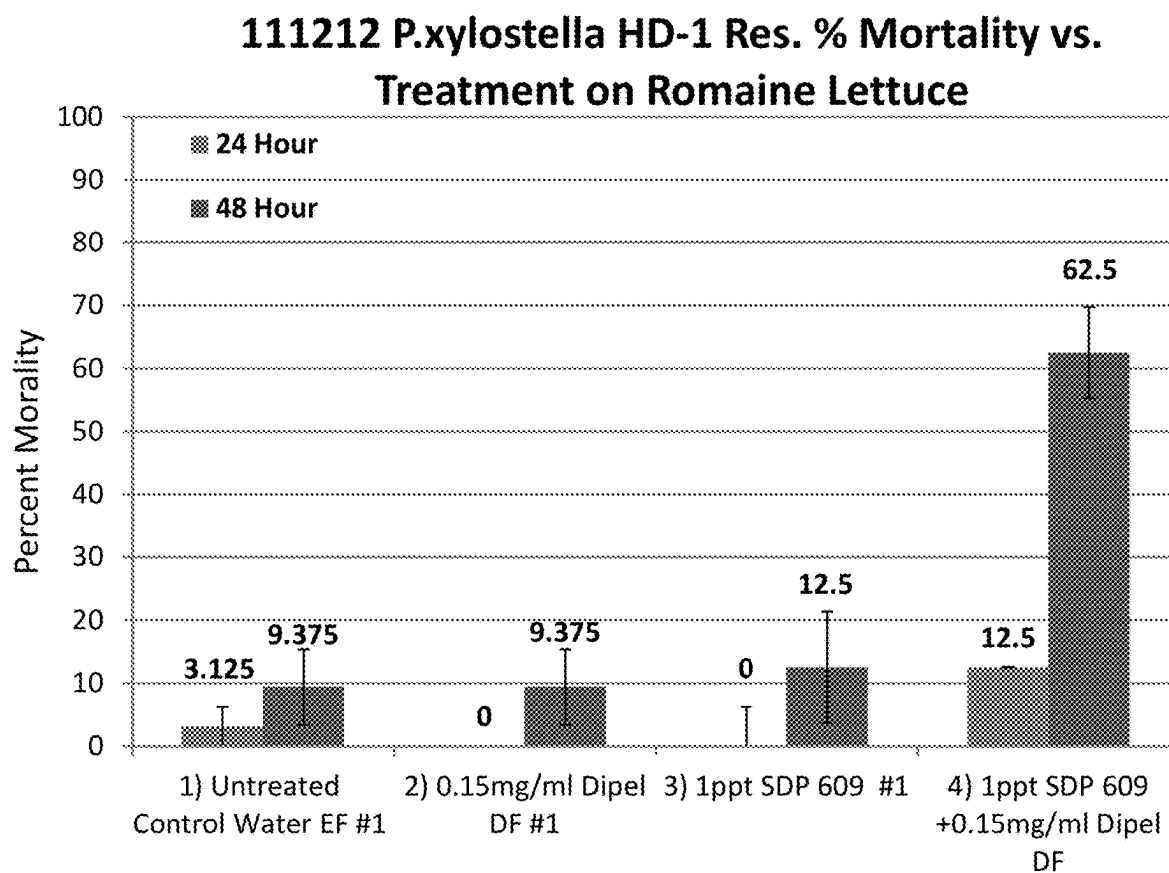


Fig. 22

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INSECTICIDAL PEPTIDE PRODUCTION AND COMBINATION OF CYSTEINE RICH PEPTIDES

REFERENCE TO RELATED APPLICATIONS

This application is a continuation application of U.S. patent application Ser. No. 16/865,193, filed May 1, 2020, which is a divisional application of U.S. patent application Ser. No. 15/390,153, filed on Dec. 23, 2016, which is a divisional application of U.S. patent application Ser. No. 14/383,841, filed on Sep. 8, 2014, which is a 371 of PCT Application No. PCT/US2013/030042, filed on Mar. 8, 2013, which claims the benefit of earlier filed U.S. Provisional Application No. 61/608,921, filed on Mar. 9, 2012, U.S. Provisional Application No. 61/644,212, filed on May 8, 2012, U.S. Provisional Application No. 61/698,261, filed on Sep. 7, 2012, and U.S. Provisional Application No. 61/729,905, filed Nov. 26, 2012, the entire contents of which are incorporated herein by reference.

SEQUENCE LISTING

This application incorporates in its entirety the Sequence Listing XML entitled "225312-515818.xml" (2,061,915 bytes), which was created on Sep. 8, 2022, and filed electronically herewith.

FIELD OF THE INVENTION

New insecticidal proteins, nucleotides, peptides, their expression in plants, methods of producing the peptides, new processes, production techniques, new peptides, new formulations, and combinations of new and known organisms that produce greater yields than would be expected of related peptides for the control of insects are described and claimed.

BACKGROUND

The global security of food produced by modern agriculture and horticulture is challenged by insect pests. Farmers rely on insecticides to suppress insect damage, yet commercial options for safe and functional insecticides available to farmers are diminishing through the removal of dangerous chemicals from the marketplace and the evolution of insect strains that are resistant to all major classes of chemical and biological insecticides. New insecticides are necessary for farmers to maintain crop protection.

Insecticidal peptides are peptides that are toxic to their targets, usually insects or arachnids of some type, and often the peptides can have arthropod origins such as from scorpions or spiders. They may be delivered internally, for example by delivering the toxin directly to the insect's gut or internal organs by injection or by inducing the insect to consume the toxin from its food, for example an insect feeding upon a transgenic plant, and/or they may have the ability to inhibit the growth, impair the movement, or even kill an insect when the toxin is delivered to the insect by spreading the toxin to locus inhabited by the insect or to the insect's environment by spraying, or other means, and then the insect comes into some form of contact with the peptide.

Insecticidal peptides however have enormous problems reaching the commercial market and to date there have been few if any insecticidal peptides approved and marketed for the commercial market, with one notable exception, peptides

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derived from *Bacillus thuringiensis* or Bt. And now there is concern over rising insect resistance to Bt proteins.

Bt proteins, or Bt peptides, are effective insecticides used for crop protection in the form of both plant incorporated protectants and foliar sprays. Commercial formulations of Bt proteins are widely used to control insects at the larval stage. ICK peptides include many molecules that have insecticidal activity. Such ICK peptides are often toxic to naturally occurring biological target species, usually insects or arachnids of some type. Often ICK peptides can have arthropod origins such as the venoms of scorpions or spiders. Bt is the one and only source organism of commercially useful insecticidal peptides. Other classes and types of potential peptides have been identified, such as Trypsin modulating oostatic factor (TMOF) peptides. TMOF peptides have to be delivered to their physiological site of action in various ways, and TMOF peptides have been identified as a potential larvicides, with great potential, see D. Borovsky, Journal of Experimental Biology 206, 3869-3875, but like nearly all other insecticidal peptides, TMOF has not been commercialized or widely used by farmers and there are reasons for this.

The ability to successfully produce insecticidal peptides on a commercial scale, with reproducible peptide formation and folding, at a reasonable and economical price, can be challenging. The wide variety, unique properties and special nature of insecticidal peptides, combined with the huge variety of possible production techniques, can present an overwhelming number of approaches to peptide application and production, but few, if any, are commercially successful.

There are several reasons why so few of the multitude insecticidal peptides that have been identified have ever made it to market. First, most insecticidal peptides are either too delicate or not toxic enough to be used commercially. Second, insecticidal peptides are difficult and costly to produce commercially. Third, many insecticidal peptides quickly degrade and have a short half-life. Fourth, very few insecticidal peptides fold properly when then are expressed by a plant, thus they lose their toxicity in genetically modified organisms (GMOs). Fifth, most of the identified insecticidal peptides are blocked from systemic distribution in the insect and/or lose their toxic nature when consumed by insects. Bt proteins are an exception to this last problem and because they disrupt insect feeding they have been widely used.

Here we present several solutions to these major problems which have prevented commercialization and wide spread use of insecticidal peptides. In the first section, we describe how to create special expression cassettes and systems that allow plants to generate and express properly folded insecticidal peptides that retain their toxicity to insects.

In the second section, we describe how to make a relatively small change to the composition of a peptide and in so doing dramatically increase the rate and amount that can be made through fermentation. This process also simultaneously lowers the cost of commercial industrial peptide production. This section teaches how a protein can be "converted" into a different, more cost effective peptide, that can be produced at higher yields and yet which surprisingly is just as toxic as before it was converted. In the third and final section, we describe how to combine different classes of insecticidal peptides such that they can operate together in a synergistic manner to dramatically change and increase the toxicity and activity of the component peptides when compared to their individual components. This section also provides details and data to support our system, methods and peptide combinations and formulations to deal with a loom-

ing threat of the development and distribution of Bt resistant insects. Bt resistant insects represent the next great threat to the global supply of food and we teach those skilled in the art how to meet and defeat this threat.

SUMMARY OF THE INVENTION

This invention describes how to produce toxic insecticidal peptides in plants so they fold properly when expressed by the plants. It describes how to produce peptides in high yields in laboratory and commercial production environments using various vectors. It describes one class of toxic insecticidal peptide we call CRIPS which stands for Cysteine Rich Insecticidal Peptides (CRIPS). It describes another class of toxic insecticidal peptides we call PFIPS which stands for Pore Forming Insecticidal Proteins (PFIPS). And it describes how novel and synergistic combinations of CRIPS and PFIPS can be fashioned together and used for a variety of purposes, including the protection of crops against of Bt or *Bacillus thuringiensis* peptide resistant insects. We disclose how to make and use combinations of CRIPS and PFIPS to kill and control insects, even Bt resistant insects, at every low doses. Without being bound by theory, our understanding of Bt or *Bacillus thuringiensis* peptides and proteins, allows us to teach one ordinarily skilled in the art, to create novel methods, compositions, compounds (proteins and peptides) and procedures to protect plants and control insects.

We describe and claim a protein comprised of an Endoplasmic Reticulum Signal Peptide (ERSP) operably linked to a Cysteine Rich Insecticidal Protein (CRIP) such as an Inhibitor Cysteine Knot (ICK) motif protein wherein said ERSP is the N-terminal of said protein (ERSP-ICK). A peptide wherein said ERSP is any signal peptide which directs the expressed CRIP to the endoplasmic reticulum of plant cells. A peptide wherein said CRIP is an Inhibitor Cysteine Knot (ICK) protein. A peptide wherein said CRIP is a Non-ICK protein. A peptide wherein said ERSP is a peptide between 5 to 50 amino acids in length, originating from a plant. A peptide operably linked to a Translational Stabilizing Protein (STA), wherein said ERSP is the N-terminal of said protein and a Translational Stabilizing Protein (STA) may be either on the N-terminal side of the CRIP, which is optionally an ICK motif protein (ERSP-STA-ICK); or Non-ICK motif protein (ERSP-STA-Non-ICK) or on the C-terminal side of the ICK or Non-ICK motif protein (ERSP-ICK-STA) or (ERSP-Non-ICK-STA).

We describe and claim a peptide with an N-terminal dipeptide which is added to and operably linked to a known peptide, wherein said N-terminal dipeptide is comprised of one nonpolar amino acid on the N-terminal of the dipeptide and one polar amino acid on the C-terminal of the dipeptide, wherein said peptide is selected from a CRIP (Cysteine Rich Insecticidal Peptide), such as from an ICK peptide, or a Non-ICK peptide. A peptide with an N-terminal dipeptide which is added to and operably linked to a known peptide, where the N-terminal dipeptide is comprised of one nonpolar amino acid on the N-terminal of the dipeptide and one polar amino acid on the C-terminal of the dipeptide. A peptide where the non-polar amino acid from the N-terminal amino acid of the N-terminal dipeptide is selected from glycine, alanine, proline, valine, leucine, isoleucine, phenylalanine and methionine. A peptide where the polar amino acid of the C-terminal amino acid of the N-terminal peptide is selected from serine, threonine, cysteine, asparagine, glutamine, histidine, tryptophan, tyrosine. A peptide where the non-polar amino acid from the N-terminal amino acid of

the N-terminal dipeptide is selected from glycine, alanine, proline, valine, leucine, isoleucine, phenylalanine and methionine and said polar amino acid of the C-terminal amino acid of the N-terminal peptide is selected from serine, threonine, cysteine, asparagine, glutamine, histidine, tryptophan, tyrosine. A peptide where the dipeptide is comprised of glycine-serine.

We describe a composition comprising at least two types of insecticidal protein or peptides wherein one type is a Pore Forming Insecticidal Protein (PFIP) and the other type is a Cysteine Rich Insecticidal Peptide (CRIP). A composition where the CRIP is a ICK and optionally, said ICK is derived from, or originates from, *Hadronyche versuta*, or the Blue Mountain funnel web spider, *Atrax robustus*, *Atrax formidabilis*, *Atrax infensus*, including toxins known as U-ACTX polypeptides, U-ACTX-Hv1a, rU-ACTX-Hv1a, rU-ACTX-Hv1b, or mutants or variants. A composition where the CRIP is a Non-ICK CRIP and optionally said Non-ICK CRIP is derived from, or originates from, animals having Non-ICK CRIPS such as sea anemones, sea urchins and sea slugs, optionally including the sea anemone named *Anemonia viridi*, optionally including the peptides named Av2 and Av3 especially peptides similar to Av2 and Av3 including such peptides listed in the sequence listing or mutants or variants.

We describe a method to control Bt resistant insects comprising, creating composition of at least two types of peptides wherein one type of peptide is a pore forming insecticidal protein (PFIP) and the other type of peptide is a cysteine rich insecticidal peptide (CRIP) and the PFIP and CRIP proteins are selected from any of the compositions described herein and from any of the proteins provided in the sequence listing and then applying said composition to the locus of the insect. A method of controlling Bt resistant insects comprising protecting a plant from Bt resistant insects comprising, creating a plant which expresses a combination of at least two properly folded peptides wherein one type of peptide is a pore forming insecticidal protein (PFIP) and the other type of peptide is a cysteine rich insecticidal peptide (CRIP) and the PFIP and CRIP proteins are selected from any of the compositions described herein and from any of the proteins provided in the sequence listing. A method where the CRIP is administered any time during which the PFIP is affecting the lining of the insect gut. A method where the CRIP is administered following the testing of the insect for Bt resistance and wherein said insect tested positive for Bt resistance. We describe the application of any of the compounds described herein in solid or liquid form to either the insect, the locus of the insect or as a Plant Incorporated Protectant.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1 is a diagram of invention of N-terminal fusion of ERSP (Endoplasmic Reticulum Signal Peptide in diagonal stripes) to a CRIP (Cysteine Rich Insecticidal Protein) such as ICK (Inhibitor Cysteine Knot) motif in vertical stripes).

FIG. 2 is a diagram of invention of N-terminal fusion of ERSP (diagonal stripes) to a CRIP motif insecticidal protein (vertical stripes) that is fused with a STA (Translational Stabilizing Protein in horizontal stripes). There are two possible orientations shown in FIG. 2.

FIG. 3 is a diagram of invention of N-terminal fusion of ERSP (diagonal stripes) fused to a CRIP motif (vertical stripes) that is fused with a translational stabilizing protein (STA) shown in horizontal stripes. The STA is separated from the CRIP motif by an intervening sequence called an

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intervening linker peptide (LINKER) shown in checker-board. Two possible orientations are shown in FIG. 3.

FIG. 4 is a diagram similar to FIG. 3 with the (LINKER-CRIP) motif with the subscript letter "N" to show that the LINKER-CRIP motif can be used once or repeated several time, preferably from 1-10 repeats and even more up to 15, 20 or 25 times are possible.

FIG. 5 is a diagram that shows that the CRIP-LINKER or ICK-LINKER group can also function as a STA-LINKER group. In other words, the combination of CRIP-LINKER or ICK-LINKER can function as a STA-LINKER. In other words one can use two ICK motifs with one LINKER and dispense with the need for a Translational Stabilizing Protein or STA.

FIG. 6 is a diagram of a covalent cross-linking of the cysteines in an inhibitor cysteine knot (ICK) motif protein. The arrows in the diagram represent β sheets: the numbers represent the ICK motif-forming cystine amino acids, numbered in the order of their occurrence in the primary structure from N to C terminus. The thick curved line represents the primary structure of the protein: the thin straight lines represent the covalent cross-linking of the specific cysteines to create an ICK motif. Sometimes the β sheet encompassing cysteine number 2 is not present.

FIG. 7 is a graph of the ELISA detected levels of ACTX (as a percentage of Total Soluble Protein (% TSP) resulting from expression from plant transgenes encoding ACTX as a translational fusion with the various other structural elements.

FIG. 8 is a graph of iELISA detected % TSPs of tobacco transiently expressed GFP fused U-ACTX-Hv1a with different accumulation localization. APO: apoplast localization; CYTO: cytoplasm localization; ER: endoplasmic reticulum localization.

FIG. 9 is a graph of iELISA detected % TSPs of tobacco leaves transiently expressing GFP fused U-ACTX-Hv1a using the FECT expression vectors encoding translational fusions with three different ERSP sequences: BAAS signal peptide (BGIH). Extensin signal peptide (EGIH) and modified Extensin signal peptide (E*GIH).

FIG. 10 is a diagram of the concentration process of trypsin treated and non-trypsin treated Jun a 3 fused Omega-ACTX-Hv1a protein extracted from the transiently transformed tobacco leaves.

FIG. 11 is HPLC chromatographs for the samples containing omega-ACTX-Hv1a. Samples loaded on the HPLC system to produce the chromatographs were as follows: A. 25 μ g synthetic omega-ACTX-Hv1a; B. 500 μ L of Sample B 1 kD filtration retentate; C. 500 μ L of Sample A 1 kD filtration retentate.

FIG. 12 is a graphical representation of the distribution of the normalized peptide yields of both U+2-ACTX-Hv1a (sometimes referred to herein as "U+2") and native U-ACTX-Hv1a (sometimes referred to herein as "native U"), produced in *Kluyveromyces lactis* (*K. lactis*) strains. The U+2 data is shown in black and the native U data is in gray. The x-axis shows the normalized yield in units of milligrams per liter per light absorbance unit at wavelength of 600 nm (mg/L.A.). The left y-scale shows the fraction of U+2 strains. The right y-scale shows the fraction of native U strains.

FIG. 13 is another graphical representation of the distribution of the normalized peptide yields from U+2 and native U-ACTX-Hv1a *K. lactis* strains. Here the y-axis shows the normalized yield (normalized for cell density in the respective cultures as described below) in milligrams per liter per light absorbance unit at wavelength of 600 nm (mg/L.A.) for

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individual strains, and the x-axis corresponds to the percentile rank of the observed yield for each strain, in relation to the yield observed for all other *K. lactis* strains engineered to produce the same peptide isoform.

FIG. 14 is a graphical representation of the dose-response of housefly injection bioassay's with U+2 and native U-ACTX-Hv1a. The U+2 data is marked with black round dots and the native U data is marked with gray triangles. The x-scale shows the dose in units of picomoles per gram of housefly. The y-scale shows the mortality percentage.

FIG. 15 is a graphical representation of the distribution of the peptide yields from U+2 and native U-ACTX-Hv1a produced from *Pichia pastoris* (*P. pastoris*) strains. The U+2 data is shown in black and the native U data is in gray. The x-axis shows the yield in milligrams per liter and the y-scale shows the fraction of total U+2 or native U production from *P. pastoris* strains.

FIG. 16 is another graphical representation of the distribution of the peptide yields of U+2 and native U-ACTX-Hv1a produced from *P. pastoris* strains. Here the y-axis shows the yield in milligrams per liter for individual strains, and the x-axis corresponds to the percentile rank of the observed yield for each strain (in relation to the yields observed for all other *P. pastoris* strains engineered to produce the same peptide isoform).

FIG. 17 is a graphical representation of the distribution of the peptide yields of sea anemone toxin, Av3 and Av3+2, produced from the *K. lactis* expression strains. The native toxin is named Av3 from the sea anemone named *Anemonia viridis*. The modified toxin here is labeled Av3+2. Like the example above we produced the toxic peptides in strains of *Kluyveromyces lactis* or *K. lactis*. The x-axis shows the peptide yield in mAu-sec/A for individual strains, and the y-axis shows the fraction of the strains. In FIG. 17 the native Av3 strain is shown in light grey, the modified high production strain Av3+2 is shown in black.

FIG. 18 shows the difference in the peptide yields of Av3+2 and native Av3 produced from the corresponding *K. lactis* strains by plotting the peptide yields as a function of the percentile rank of the transformants which produce the same peptide. Here the y-axis shows the normalized yield in mAu-sec/A for individual strains, and the x-axis corresponds to the percentile rank of the observed yield for each strain, in relation to the yield observed for all other *K. lactis* strains engineered to produce the same peptide isoform.

FIG. 19 Graph of a foliar bioassay 24 hour percent mortality vs. age of larvae after application and exposure to ICK peptides or Bt proteins.

FIG. 20 Graph of a foliar bioassay measuring percent mortality at 18, 24 and 48 hour post application using Bt proteins or ICK peptides or combination of Bt+ICK peptides on 72 hour larvae.

FIG. 21 Graph of a foliar feeding bioassay measuring foliar damage by insects resistant to Bt, at 24 hr and 48 hr after exposure to Bt proteins or Non-ICK CRIP or their combinations.

FIG. 22 Graph of a foliar feeding bioassay measuring percent mortality at 24 and 48 hour post application using Bt proteins or ICK peptides or their combination on Bt protein resistant *P. xylostella* larvae.

BRIEF DESCRIPTION OF THE SEQUENCE LISTING

This invention includes a sequence listing of 1593 sequences.

SEQ ID NOs: 1-28, 1553-1570, and 1593 are mentioned or referred to in Part 1.

SEQ ID NOs: 29-32, and 1571-1592 are mentioned or referred to in Part 2.

SEQ ID NOs: 33-1042 mentioned or referred to in Part 3.

SEQ ID NOs: 1043-1221 are sequences derived from or having a spider origin.

SEQ ID NOs: 1222-1262 are sequences derived from or having a sea anemone origin.

SEQ ID NOs: 1263-1336 are sequences derived from or having a scorpion origin.

SEQ ID NOs: 1337-1365 are sequences derived from or having a scorpion origin.

SEQ ID NOs: 1366-1446 are sequences derived from or having a Cry or Cyt origin.

SEQ ID NOs: 1447-1552 are sequences derived from or having a VIP origin.

DETAILED DESCRIPTION OF THE INVENTION

Definitions

“ACTX” or “ACTX peptide” means a Family of insecticidal ICK peptides that have been isolated from an Australian funnel-web spiders belonging to the Atracinae subfamily. One such spider is known as the Australian Blue Mountains Funnel-web Spider, which has the scientific name *Hydronyche versuta*. Two examples of ACTX peptides from this species are the Omega and U peptides.

“Agroinfection” means a plant transformation method where DNA is introduced into a plant cell by using Agrobacteria *A. tumefaciens* or *A. rhizogenes*.

“BAAS” means barley alpha-amylase signal peptide. It is an example of an ERSP.

“Binary vector” or “binary expression vector” means an expression vector which can replicate itself in both *E. coli* strains and *Agrobacterium* strains. Also, the vector contains a region of DNA (often referred to as t-DNA) bracketed by left and right border sequences that is recognized by virulence genes to be copied and delivered into a plant cell by *Agrobacterium*.

“Bt,” also known as *Bacillus thuringiensis* or *B. thuringiensis*, means a gram-positive soil bacterium that has been used worldwide for more than sixty years to control agricultural, forestry, and public health insect pests.

“Bt proteins” and “Bt peptides” refer to the same thing here and these are peptides produced by Bt. Such peptides are frequently written as “cry”, “cyt” or “VIP” proteins encoded by the cry, cyt and vip genes. Bt proteins are more usually attributed to insecticidal crystal proteins encoded by the cry genes. Bt proteins are examples of PFIPS (Pore Forming Insecticidal Proteins) see definition below. Examples PFIPS and other Bt proteins are provided in the sequence listing.

“Chimeric gene” means a DNA sequence that encodes a gene derived from portions of one or more coding sequences to produce a new gene.

“Cleavable linker” means a short peptide sequence in the protein that is the target site of proteases that can cleave and separate the protein into two parts or a short DNA sequence that is placed in the reading frame in the ORF and encoding a short peptide sequence in the protein that is the target site of protease that can cleave and separate the protein into two parts.

“Conditioned medium” means the cell culture medium which has been used by cells and is enriched with cell derived materials but does not contain cells.

“Conversion” or “converted” refers to the process of making an HP peptide.

“CRIP” and “CRIPS” is an abbreviation for Cysteine Rich Insecticidal Protein or Proteins. Cysteine rich insecticidal peptides (CRIPS) are peptides rich in cysteine which form disulfide bonds. CRIPS contain at least four (4) sometimes six (6) and sometimes eight (8) cysteine amino acids among proteins or peptides having at least 10 amino acids where the cysteines form two (2), three (3) or four (4) disulfide bonds. The disulfide bonds contribute to the folding, three-dimensional structure, and activity of the insecticidal peptide. The cysteine-cysteine disulfide bonds and the three dimensional structure they form play a significant role in the toxicity of these insecticidal peptides. A CRIP is exemplified by both inhibitory cysteine knot or ICK peptides (usually having 6-8 cysteines) and by examples of toxic peptides having disulfide bonds but that are not considered ICK peptides (Non-ICK CRIPS). Examples of an ICK would be an ACTX peptide from a spider and defined above. Examples of a Non-ICK CRIP would be a peptide like Av2 and Av3 which are peptides first identified from sea anemones. These peptides are examples of a class of compounds that modulate sodium channels in the insect peripheral nervous system (PNS). Non-ICK CRIPS can have 4-8 cysteines which form 2-4 disulfide bonds. These cysteine-cysteine disulfide bonds stabilized toxic peptides (CRIPS) can have remarkable stability when exposed to the environment. Many CRIPS are isolated from venomous animals such as spiders, scorpions, snakes and sea snails and sea anemones and they are toxic to insects. Additional description is provided below.

“Defined medium” means a medium that is composed of known chemical components but does not contain crude proteinaceous extracts or by-products such as yeast extract or peptone.

“Disulfide bond” means a covalent bond between two cysteine amino acids derived by the coupling of two thiol groups on their side chains.

“Double transgene peptide expression vector” or “double transgene expression vector” means a yeast expression vector which contains two copies of the insecticidal peptide expression cassette.

“ELISA” or “iELISA” means a molecular biology protocol in which the samples are fixed to the surface of a plate and then detected as follows: a primary antibody is applied followed by a secondary antibody conjugated to an enzyme which converts a colorless substrate to colored substrate which can be detected and quantified across samples. During the protocol, antibodies are washed away such that only those that bind to their epitopes remain for detection. The samples, in our hands, are proteins isolated from plants, and ELISA allows for the quantification of the amount of expressed transgenic protein recovered.

“Expression ORF” means a nucleotide encoding a protein complex and is defined as the nucleotides in the ORF.

“ER” or “Endoplasmic reticulum” is a subcellular organelle common to all eukaryotes where some post translation modification processes occur.

“ERSP” or “Endoplasmic reticulum signal peptide” is an N-terminus sequence of amino acids that during protein translation of the transgenic mRNA molecule is recognized and bound by a host cell signal-recognition particle, which moves the protein translation ribosome/mRNA complex to the ER in the cytoplasm. The result is the protein translation

is paused until it docks with the ER where it continues and the resulting protein is injected into the ER.

“ersp” means a nucleotide encoding the peptide, ERSP.

“ER trafficking” means transportation of a cell expressed protein into ER for post-translational modification, sorting and transportation.

“FECT” means a transient plant expression system using Foxtail mosaic virus with elimination of coating protein gene and triple gene block.

“GFP” means a green fluorescent protein from the jellyfish *Aequorea victoria*. It is an example of a translational stabilizing protein.

“High Production peptide” or “HP peptide” means a peptide which is capable of being made, or is “converted,” according to the procedures described herein and which, once converted can be produced at increased yields, or higher rates of production, or in greater than normal amounts, in a biological system. The higher rates of production can be from 20 to 400% or greater than can be achieved with a peptide before conversion, using the same or similar production methods that were used to produce the peptide before conversion.

“Hybrid peptide,” aka “hybrid toxin,” aka “hybrid-ACTX-Hv1a,” aka “native hybrid-ACTX-Hv1a,” as well as “U peptide,” aka “U toxin,” aka “native U,” aka “U-ACTX-Hv1a,” aka “native U-ACTX-Hv1a,” all refer to an ACTX peptide, which was discovered from a spider known as the Australian Blue Mountains Funnel-web Spider, *Hydronyche versuta*, and is a dual antagonist to insect voltage-gated Ca^{2+} channels and voltage-gated K^{+} channels.

“IGER” means a name for a short peptide, based on its actual sequence of one letter codes. It is an example of an intervening linker.

“ICK motif,” “ICK motif protein,” “inhibitor cystine knot motif,” “Toxic insect ICK peptides,” “ICK peptides,” “CK” peptides,” “cystine knot motif,” or “cystine knot peptides” means a 16 to 60 amino acid peptide with at least 6 half-cystine core amino acids having three disulfide bridges, wherein the 3 disulfide bridges are covalent bonds and of the six half-cystine residues the covalent disulfide bonds are between the first and fourth, the second and fifth, and the third and sixth half-cystines, of the six core half-cystine amino acids starting from the N-terminal amino acid. In general this type of peptide comprises a beta-hairpin secondary structure, normally composed of residues situated between the fourth and sixth core half-cystines of the motif, the hairpin being stabilized by the structural crosslinking provided by the motif’s three disulfide bonds. Note that additional cysteine/cystine or half-cystine amino acids may be present within the inhibitor cystine knot motif. Examples are provided in the sequence listing.

“ick” means a nucleotide encoding an ICK motif protein.

“ICK motif protein expression ORF” or “expression ORF” means a nucleotide encoding a ICK motif protein complex and is defined as the nucleotides in the ORF.

“ICK motif protein expression vector” or “ICK expression vector,” or “ICK motif expression vector,” means a binary vector which contains an expression ORF. The binary vector also contains the necessary transcription promoter and terminator sequence surrounding the expression ORF to promote expression of the ORF and the protein it encodes.

“Insect” means any arthropod and nematode, including acarids, and insects known to infest all crops, vegetables, and trees and includes insects that are considered pests in the fields of forestry, horticulture and agriculture. Examples of specific crops that might be protected with the methods disclosed herein are soy bean, corn, cotton, alfalfa and the

vegetable crops. A list of specific crops and insects appears towards the end of this document.

“Insect gut environment” or “gut environment” means the specific pH and proteinase conditions found within the fore, mid or hind gut of an insect or insect larva.

“Insect hemolymph environment” means the specific pH and proteinase conditions of found within an insect or insect larva.

“Insecticidal activity” means that on or after exposure of the insect to compounds or peptides, the insect either dies stops or slows its movement or it’s feeding, stops or slows its growth, fails to pupate, cannot reproduce or cannot produce fertile offspring.

“Insecticidal peptide” or “Insecticidal protein” or “toxic peptide” or “toxic protein” means a protein having insecticidal activity when ingested by, in contact with, or injected into an insect.

“Insecticidal peptide production strain screen” means a screening process that identifies the higher-yielding insecticidal peptide production yeast strains from the lower yielding strains. In the described methods herein, it refers to screens that use reverse-phase HPLC or the housefly injection bioassay.

“Integrative expression vector or integrative vector” means a yeast expression vector which can insert itself into a specific locus of the yeast cell genome and stably becomes a part of the yeast genome.

“Intervening linker” means a short peptide sequence in the protein separating different parts of the protein, or a short DNA sequence that is placed in the reading frame in the ORF to separate the upstream and downstream DNA sequences such that during protein translation the proteins encoded in the DNA can achieve their independent secondary and tertiary structure formation. The intervening linker can be either resistant or susceptible to cleavage in plant cellular environments, in the insect and/or lepidopteran gut environment, and in the insect hemolymph and lepidopteran hemolymph environment.

“Known peptide” means a peptide known to have biological activity and may be a mature peptide or any version or fragment thereof including pre and pro peptides and conjugates of active peptides. A preferred known peptide is one with insecticidal activity.

“L” in the proper context means an intervening linker peptide, which links a translational stabilizing protein with an ICK motif protein or a multiple ICK motif protein domain, and links same or different multiple ICK motif protein. When referring to amino acids, “L” can also mean leucine.

“Linker, LINKER” or in some contexts “L” means an intervening linker peptide, which links a translational stabilizing protein with a ICK motif protein or a multiple ICK motif protein domain, and links same or different multiple ICK motif proteins. The linker can have one of (at least) three roles: to cleave in the insect gut environment, to cleave in the plant cell, or to be designed not to intentional cleave.

“l” or linker” means a nucleotide coding for an intervening linker peptide.

“Lepidopteran gut environment” means the specific pH and proteinase conditions of found within the fore, mid or hind gut of a lepidopteran insect or larva.

“Lepidopteran hemolymph environment” means the specific pH and proteinase conditions of found within lepidopteran insect or larva.

“Multiple ICK motif protein domain” means a protein composed of multiple ICK motif proteins which are linked by multiple intervening linker peptides. The ICK motif

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proteins in the multiple ICK motif protein domain can be same or different, and the intervening linker peptides in this domain can also be the same or different.

“Non-ICK CRIPS” can have 4-8 cysteines which form 2-4 disulfide bonds. Non-ICK peptides include cystine knot peptides that are not ICK peptides. Non-ICK peptides may have different connection orders of the cystine bonds than ICKs. Examples of a Non-ICK CRIP are peptides like Av2 and Av3 which are peptides first identified from sea anemones. These anemone peptides are examples of a class of compounds that modulate sodium channels in the insect peripheral nervous system (PNS).

“Non-Polar amino acid” is an amino acid that is weakly hydrophobic and includes glycine, alanine, proline, valine, leucine, isoleucine, phenylalanine and methionine. Glycine or gly is the most preferred non-polar amino acid for the dipeptides of this invention.

“Normalized peptide yield” means the peptide yield in the conditioned medium divided by the corresponding cell density at the point the peptide yield is measured. The peptide yield can be represented by the mass of the produced peptide in a unit of volume, for example, mg per liter or mg/L, or by the UV absorbance peak area of the produced peptide in the HPLC chromatograph, for example, mAu·sec. The cell density can be represented by visible light absorbance of the culture at wavelength of 600 nm (OD600).

“One letter code” means the peptide sequence which is listed in its one letter code to distinguish the various amino acids in the primary structure of a protein. alanine=A, arginine=R, asparagine=N, aspartic acid=D, asparagine or aspartic acid=B, cysteine=C, glutamic acid=E, glutamine=Q, glutamine or glutamic acid=Z, glycine=G, histidine=H, isoleucine=I, leucine=L, lysine=K, methionine=M, phenylalanine=F, proline=P, serine=S, threonine=T, tryptophan=W, tyrosine=Y, valine=V.

“Omega peptide” aka “omega toxin,” aka “omega-ACTX-Hv1a,” aka “native omega-ACTX-Hv1a.” all refer to an ACTX peptide which was first isolated from a spider known as the Australian Blue Mountains Funnel-web Spider, *Hydronyche versuta*, and which is an antagonist to the insect voltage-gated Ca²⁺ channel.

“ORF” or “Open reading frame” or “peptide expression ORF” means that DNA sequence encoding a protein which begins with an ATG start codon and ends with a TGA, TAA or TAG stop codon. ORF can also mean the translated protein that the DNA encodes.

“Operably linked” means that the two adjacent DNA sequences are placed together such that the transcriptional activation of one can act on the other.

“PEP” means Plant Expressed Peptide.

“Peptide expression cassette”, or “expression cassette” means a DNA sequence which is composed of all the DNA elements necessary to complete transcription of an insecticidal peptide in a biological expression system. In the described methods herein, it includes a transcription promoter, a DNA sequence to encode an α -mating factor signal sequence and a Kex 2 cleavage site, an insecticidal peptide transgene, a stop codon and a transcription terminator.

“Peptide expression vector” means a host organism expression vector which contains a heterologous insecticidal peptide transgene.

“Peptide expression yeast strain”, “peptide expression strain” or “peptide production strain” means a yeast strain which can produce a heterologous insecticidal peptide.

“Peptide made special” means a peptide previously having low peptide yield from a biological expression system

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that becomes an HP peptide because of the methods described herein used to increase its yield.

“Peptide transgene” or “insecticidal peptide transgene” means a DNA sequence that encodes an insecticidal peptide and can be translated in a biological expression system.

“Peptide yield” means the insecticidal peptide concentration in the conditioned medium which is produced from the cells of a peptide expression yeast strain. It can be represented by the mass of the produced peptide in a unit of volume, for example, mg per liter or mg/L, or by the UV absorbance peak area of the produced peptide in the HPLC chromatograph, for example, mAu·sec.

“Peritrophic membrane” means a lining inside the insect gut that traps large food particles can aid in their movement through the gut while allowing digestion, but also protecting the gut wall.

“PFIP” means a protein that can form a pore or channel in the cells that line an insect gut, such as gut epithelium cells. Examples of PFIPS are Bt proteins such as cry, crt and VIP other PFIP examples can be found in the sequence listing.

“PIP” or “Plant-incorporated protectant” means an insecticidal protein produced by transgenic plants, and the genetic material necessary for the plant to produce the protein.

“Plant cleavable linker” means a cleavable linker peptide, or a nucleotide encoding a cleavable linker peptide, which contains a plant protease recognition site and can be cleaved during the protein expression process in the plant cell.

“Plant regeneration media” means any media that contains the necessary elements and vitamins for plant growth and plant hormones necessary to promote regeneration of a cell into an embryo which can germinate and generate a plantlet derived from tissue culture. Often the media contains a selectable agent to which the transgenic cells express a selection gene that confers resistance to the agent.

“Plant transgenic protein” means a protein from a heterologous species that is expressed in a plant after the DNA or RNA encoding it was delivered into one or more of the plant cells.

“Polar amino acid” is an amino acid that is polar and includes serine, threonine, cysteine, asparagine, glutamine, histidine, tryptophan and tyrosine: preferred polar amino acids are serine, threonine, cysteine, asparagine and glutamine: with serine being most highly preferred.

“Post-transcriptional gene silencing”, or “PTGS”, means a cellular process within living cells that suppress the expression of a gene.

“Protein” has the same meaning as “Peptide” in this document.

“Recombinant vector” means a DNA plasmid vector into which foreign DNA has been inserted.

“Selection gene” means a gene which confers an advantage for a genomically modified organism to grow under the selective pressure.

“STA”, or “Translational stabilizing protein”, or “stabilizing protein”, or “fusion protein” means a protein with sufficient tertiary structure that it can accumulate in a cell without being targeted by the cellular process of protein degradation. The protein can be between 5 and 50aa (eg another ICK-motif protein), 50 to 250aa (GNA), 250 to 750aa (eg chitinase) and 750 to 1500aa (eg enhancin). The translational stabilizing protein is coded by a DNA sequence for a protein that is fused in frame with a sequence encoding an insecticidal protein in the ORF. The fusion protein can either be upstream or downstream of the Insecticidal protein and can have any intervening sequence between the two sequences as long as the intervening sequence does not

result in a frame shift of either DNA sequence. The translational stabilizing protein can also have an activity which increases delivery of the ICK motif protein across the gut wall and into the hemolymph of the insect. Such a delivery can be achieved by actively trafficking the entire ORF across the gut wall, or by cleavage within the gut environment to separate the ICK motif protein while the translational stabilizing protein damages the peritrophic membrane and/or gut wall to increase diffusion of the ICK motif protein into the hemolymph.

“sta” means a nucleotide encoding a translational stabilizing protein.

“TMOF” “TMOF motif,” or “TMOF proteins” means “trypsin modulating oostatic factor” protein sequences. Examples are provided in the sequence listing. Numerous examples and variants are provided herein. SEQ ID NO: 708 is the wild type TMOF sequence. Other non-limiting variants are provided in SEQ. ID. NO: s 709-721. Other examples would be known or could be created by one skilled in the art.

“TSP” or “total soluble protein” means the total amount of protein that can be extracted from a plant tissue sample and solubilized into the extraction buffer.

“Transgene” means a heterologous DNA sequence encoding a protein which is transformed into a plant.

“Transgenic host cell” means a cell which is transformed with a gene and has been selected for its transgenic status via an additional selection gene.

“Transgenic plant” means a plant that has been derived from a single cell that was transformed with foreign DNA such that every cell in the plant contains that transgene.

“Transient expression system” means an *Agrobacterium tumefaciens*-based system which delivers DNA encoding a disarmed plant virus into a plant cell where it is expressed. The plant virus has been engineered to express a protein of interest at high concentrations, up to 40% of the TSP. In the technical proof, there are two transient expression systems used, a TRBO and a FECT system and the plant cells are leaf tissue of a tobacco plant “*Nicotiana benthamiana*.”

“TRBO” means a transient plant expression system using Tobacco mosaic virus with removal of the viral coating protein gene.

“Trypsin cleavage” means an in vitro assay that uses the protease enzyme trypsin (which recognizes exposed lysine and arginine amino acid residues) to separate a cleavable linker at that cleavage site. It also means the act of the trypsin enzyme cleaving that site.

“U peptide,” U protein” aka “U toxin,” aka “native U,” aka “U-ACTX-Hv1a,” aka “native U-ACTX-Hv1a,” as well as “Hybrid peptide,” aka “hybrid toxin,” aka “hybrid-ACTX-Hv1a,” aka “native hybrid-ACTX-Hv1a,” all refer to a native protein or native toxin, that can be found in nature or is otherwise known, in the case of “U-ACTX-Hv1a,” aka “native U-ACTX-Hv1a,” the protein is a native spider toxin, that was first discovered from a spider with origins in the Australian Blue Mountains and is dual antagonist against insect voltage gated Ca^{2+} channels and K^{+} channels. The spider from which the toxin was discovered is known as the Australian Blue Mountains Funnel-web Spider, which has the scientific name *Hydronyche versuta*.

“U+2 peptide,” “U+2 protein,” “U+2 toxin,” or “U+2,” or “U+2-ACTX-Hv1a” all refer to either a toxin, which has an additional dipeptide operatively linked to the native peptide, and may refer to the spider toxin which is sometimes called the U peptide and other names noted above. The additional dipeptide that is operatively linked to the U peptide and thus indicated as “+2” or “plus 2” can be selected among several

peptides, any of which may result in a “U+2 peptide” with unique properties as discussed herein. These are also sometimes called “high production peptides.” When the term “U+2-ACTX-Hv1a” is used, it refers to a specific high production toxic peptide, comprising a naturally occurring peptide from the Australian Blue Mountains Funnel-web Spider, which has the scientific name *Hydronyche versuta*.

“VIP” proteins were discovered from screening the supernatant of vegetatively grown strains of Bt for possible insecticidal activity. They have little or no similarity to cry proteins and they were named Vegetative Insecticidal Proteins or VIP. Of particular use and preference for use with this document are what have been called VIP3. Vip3 proteins or Vip toxins which have Lepidopteran activity. They are thought to have a similar mode of action as Bt cry peptides. In this document VIP proteins are categorized as a PFIP type of protein.

“Yeast expression vector,” or “expression vector,” or “vector,” means a plasmid which can introduce a heterologous gene and/or expression cassette into yeast cells to be transcribed and translated.

“Yield” refers to the production of a peptide, and increased yields can mean increased amounts of production, increased rates of production, and an increased average or median yield and increased frequency at higher yields.

Section 1. Plant Incorporated Peptides or Plant Expressed Peptides “PIPS” AND PEPS”

Plant-incorporated protectants, or “PIPs”, have presented one solution to the insect pressure faced by farmers. Modern agriculture employs genes from the *Bacillus thuringiensis* expressed as plant transgenic proteins to act as PIPs, but natural resistant insect strains have been detected in the field and threaten this class. Additional PIPs with novel modes of action need to be developed to manage the development of resistance. A novel class of proteins with insecticidal activity having the potential to become PIPs, are called Cysteine Rich Insecticidal Proteins (CRIPs) these proteins have 4, 6 or 8 cysteines and 2, 3 or 4 disulfide bonds. One example of this class of compounds are said to be of the type called inhibitor cysteine knot (ICK) motif protein. ICK motif proteins that have insecticidal activity have potential to be insecticidal proteins and PIPs.

ICK motif proteins are a class of proteins with at least six cysteine residues that form a specific ICK tertiary structure. Covalent cross-linking of the cysteine residues in the ICK motif proteins form disulfide bridges that result in a tertiary structures that makes the protein relatively resistant to proteases and sometimes to extreme physical conditions (pH, temperature, UV light, etc.), and confers activity against ion channels, which might specific to insects. Many ICK motif proteins have evolved in the venom of invertebrates and vertebrates that use the ICK motif proteins as a toxin to immobilize or kill their predators or prey. Such insecticidal peptides often have scorpion, spider and sometimes snake origins. In nature, toxic peptides can be directed to the insect’s gut or to internal organs by injection. In the case of a PIP, the delivery is usually via the insect’s consumption of transgenic protein expressed in plant tissue. Upon this consumption of the toxin from its food, for example an insect feeding upon a transgenic plant, the ICK motif protein may have the ability to inhibit the growth, impair the movement, or even kill an insect.

Toxic peptides however often lose their toxicity when they are expressed in plants. Unless the ICK motif protein is expressed as a properly folded protein it cannot successfully protect a plant or crop from insect damage. In some cases a plant expressed peptide will need to be activated by cleavage

within the insect or during expression process in a plant in order to be active. There is a need for methods and modified peptides and nucleic acids that enable peptides to not only be expressed in a plant but to be expressed, folded properly and in some cases cleaved properly such that the peptide retains its activity against an insect even after expression in a plant. In this section we present several ways to produce active peptides adapted for expression in plants.

We describe various combinations of different peptides operably linked together to make novel protein complexes. The following protein complexes are described. A peptide comprised of an Endoplasmic Reticulum Signal Peptide (ERSP) operably linked to Cysteine Rich Insecticidal Peptide (CRIP) such as an Inhibitor Cystine Knot (ICK) motif protein, which is designated as ERSP-ICK, wherein said ERSP is the N-terminal of said peptide, and where the ERSP peptide is between 3 to 60 amino acids in length, between 5 to 50 amino acids in length, between 20 to 30 amino acids in length and or where the peptide is BAAS, or tobacco extensin signal peptide, or a modified tobacco extensin signal peptide, or Jun a 3 signal peptide of *Juniperus ashei* or *J ashei*.

A peptide comprised of an Endoplasmic Reticulum Signal Peptide (ERSP) operably linked to a Cysteine Rich Insecticidal Peptide (CRIP) such as an Inhibitor Cystine knot (ICK) motif protein, which is designated as ERSP-ICK, wherein the ICK motif protein is between 16 and 60 amino acids in length, between 26 and 48 amino acids in length, between 30 and 44 amino acids in length and or where the ICK motif protein is U-ACTX-Hv1a, or Omega-ACTX-Hv1a, or Kappa-ACTX-Hv1c.

A peptide comprised of an Endoplasmic Reticulum Signal Peptide (ERSP) operably linked to an Inhibitor Cystine Knot (ICK) motif protein, designated as ERSP-ICK, wherein said ERSP and Inhibitor Cystine Knot (ICK) motif protein are combinations of any of the sizes and lengths described herein and/or are comprised of any of the identified sequences taught in this document.

A nucleotide that codes for any of the peptides that are described herein as Endoplasmic Reticulum Signal Peptides (ERSP) and/or Cysteine Rich Insecticidal Peptide (CRIP) such as an Inhibitor Cystine Knot (ICK) motif proteins. An expression ORF comprising any of the nucleotides that code for these peptides. An expression ORF comprising any of the nucleotides that code for these peptides transformed into a transgenic plant genome. A peptide wherein said ICK motif protein is an insecticidal protein. A peptide wherein said insecticidal peptide is any of the ICK motif proteins or peptide described herein. A peptide wherein said insecticidal peptide is any peptide selected from any of the peptides or sources of peptides including *Atrax* or *Hadronyche*. An insecticidal peptide selected from any of the peptides in the Sequence Listing and fragments thereof including mature, pre, and pro peptide versions of said peptides and sequence numbers. A peptide wherein said insecticidal peptide is any peptide selected described or selected from an ACTX protein. A TMOF protein.

The use of any of the peptides or nucleotides described herein to make or transform a plant or plant genome in order to express properly folded toxic peptides in a transformed plant. The use of any of the peptides or nucleotides described herein to make or transform a plant or plant genome in order to express properly folded toxic peptides in the transformed plant and to cause the accumulation of the expressed and properly folded toxic peptides in said plant and to cause an increase the plant's resistance to insect damage.

A method of using the nucleotides of any of the peptides or expression ORFs in a CRIP, an ICK a Non-ICK, motif protein expression vectors to create transgenic plants. An ICK motif protein expression vector comprising any of the nucleotides which express any peptides described herein. An ICK motif protein expression vector incorporated into a transformed plant, comprising nucleotides that code for any of the peptides disclosed herein or that could be made by one skilled in the art given the teaching disclosed herein. A procedure for the generation of transformed plants having or expressing any of the peptides described herein. A plant made by any of the products and processes described herein.

A protein comprised of an Endoplasmic Reticulum Signal Peptide (ERSP) operably linked to an Inhibitor Cystine Knot (ICK) motif protein or cysteine rich peptide, operably linked to an intervening linker peptide (L or Linker), which is designated as ERSP-Linker-ICK. (ERSP-L-ICK), or ERSP-ICK-Linker (ERSP-ICK-L), wherein said ERSP is the N-terminal of said protein and said L or Linker, may be either on the N-terminal side (upstream) of the ICK motif protein or the C-terminal side (downstream) of the ICK motif protein. A protein designated as ERSP-L-ICK, or ERSP-ICK-L, comprising any of the ERSPs or ICK motif proteins described herein and wherein said L can be an uncleavable linker peptide, or a cleavable linker peptide, which may be cleavable in a plant cells during protein expression process or may be cleavable in an insect gut environments and hemolymph environments, and comprised of any of the intervening linker peptide (LINKER) described, or taught by this document including the following sequences: IGER (SEQ ID NO. 1) EEKKN. (SEQ ID NO. 2) and ETMFKHGL (SEQ ID NO. 3).

A nucleotide that codes for any of the peptides described as Endoplasmic Reticulum Signal Peptide (ERSP). Inhibitor Cystine knot (ICK) motif protein and or intervening linker peptide (LINKER) and any and all nucleotides that code for any of these proteins that are used to create transgenic plants.

The use of any of the peptides or nucleotides that code for Endoplasmic Reticulum Signal Peptide (ERSP). Inhibitor Cystine knot (ICK) motif protein and/or intervening linker peptide (LINKER) to make or transform a plant or plant genome in order to express properly folded toxic peptides in a transformed plant. The use of any of the peptides or nucleotides that code for Endoplasmic Reticulum Signal Peptide (ERSP). Inhibitor Cystine knot (ICK) motif protein and/or intervening linker peptide to make or transform a plant or plant genome in order to express properly folded toxic peptides in the transformed plant and to cause the accumulation of the expressed and properly folded toxic peptides in said plant and to cause an increase the plant's resistance to insect damage.

A method of using the nucleotides or expression ORFs that code for Endoplasmic Reticulum Signal Peptide (ERSP). Inhibitor Cystine Knot (ICK) motif protein and/or intervening linker peptide (LINKER) to create transgenic plants. An expression ORF comprising any of the nucleotides which are in an ICK expression vector express any peptides described herein. ERSP. ICK motif protein and/or LINKER. A functional expression ORF in an ICK motif protein expression vector incorporated into a transformed plant, comprising nucleotides that code for any of the peptides disclosed herein that code for ERSP. ICK motif protein and/or LINKER or that could be made by one skilled in the art given the teaching disclosed herein. A procedure for the generation of transformed plants having or expressing any of the peptides described herein. ERSP. ICK motif

protein and/or LINKER. A plant made by any of the products and processes described herein.

A protein comprised of an Endoplasmic Reticulum Signal Peptide (ERSP) operably linked to an Inhibitor Cystine knot (ICK) motif protein operably linked to a Translational Stabilizing Protein (STA), which is designated as ERSP-STA-ICK or ERSP-ICK-STA, wherein said ERSP is the N-terminal of said protein and said STA may be either on the N-terminal side (upstream) of the ICK motif protein of the C-terminal side (downstream) of the ICK motif protein. A protein designated as ERSP-STA-ICK or ERSP-ICK-STA, comprising any of the ERSPs or ICK motif proteins described herein and where STA is comprised of any of the translational stabilizing proteins described, or taught by this document including GFP (Green Fluorescent Protein), GNA (snowdrop lectin), Jun a 3. (*Juniperus ashei*) and many other ICK motif proteins.

A nucleotide that codes for any of the peptides described as Endoplasmic Reticulum Signal Peptide (ERSP). Inhibitor Cystine Knot (ICK) motif protein and/or Translational Stabilizing Protein (STA) and any and all nucleotides having any of these functional groups that code for any of these proteins that are used to create transgenic plants.

The use of any of the peptides or nucleotides that code for Endoplasmic Reticulum Signal Peptide (ERSP). Inhibitor Cystine Knot (ICK) motif protein and/or Translational Stabilizing Protein (STA) to make or transform a plant or plant genome in order to express properly folded toxic peptides in a transformed plant. The use of any of the peptides or nucleotides that code for Endoplasmic Reticulum Signal Peptide (ERSP). Inhibitor Cystine Knot (ICK) motif protein and/or Translational Stabilizing Protein (STA) to make or transform a plant or plant genome in order to express properly folded toxic peptides in the transformed plant and to cause the accumulation of the expressed and properly folded toxic peptides in said plant and to cause an increase the plant's resistance to insect damage.

A method of using the nucleotides or expression ORFs that code for Endoplasmic Reticulum Signal Peptide (ERSP). Inhibitor Cystine knot (ICK) motif protein and/or Translational Stabilizing Protein (STA) in an ICK expression vector to create transgenic plants. An expression ORF comprising any of the nucleotides which express ERSP. ICK motif protein and/or STA. A functional expression ORF in a ICK motif protein expression vector that is incorporated into a transformed plant, comprising nucleotides that code for that code for ERSP. ICK motif protein and/or STA or that could be made by one skilled in the art given the teaching disclosed herein. A procedure for the generation of transformed plants having or expressing ERSP. ICK motif protein and/or STA. A plant made by any of the products and processes described herein.

A protein comprised of an Endoplasmic Reticulum Signal Peptide (ERSP) operably linked to an Inhibitor Cystine Knot (ICK) motif protein operably linked to a Translational Stabilizing Protein (STA) operably linked to an Intervening Linker Peptide (LINKER) which is designated as ERSP-STA-LINKER-ICK. ERSP-ICK-LINKER-STA. ERSP-STA-L-ICK or ERSP-ICK-L-STA, wherein said ERSP is the N-terminal of said protein and said STA may be either on the N-terminal side (upstream) of the ICK motif protein of the C-terminal side (downstream) of the ICK motif protein and said LINKER is between STA and the ICK motif protein. A protein designated as ERSP-STA-LINKER-ICK or ERSP-ICK-LINKER-STA, comprising any of the ERSPs. ICK motif proteins. Intervening Linker Peptides and Translational Stabilizing Proteins described herein.

A nucleotide that codes for any of the peptides described as Endoplasmic Reticulum Signal Peptide (ERSP). Inhibitor Cystine Knot (ICK) motif protein. Intervening Linker Peptide (LINKER) and/or Translational Stabilizing Protein (STA) and any and all nucleotides that code for any of these proteins that are used to create transgenic plants.

The use of any of the peptides or nucleotides that code for Endoplasmic Reticulum Signal Peptide (ERSP). Inhibitor Cystine Knot (ICK) motif protein. Intervening Linker Peptide and/or Translational Stabilizing Protein (STA) to make or transform a plant or plant genome in order to express properly folded toxic peptides in a transformed plant. The use of any of the peptides or nucleotides that code for Endoplasmic Reticulum Signal Peptide (ERSP). Inhibitor Cystine Knot (ICK) motif protein. Intervening Linker Peptide (LINKER) and/or Translational Stabilizing Protein (STA) to make or transform a plant or plant genome in order to express properly folded toxic peptides in the transformed plant and to cause the accumulation of the expressed and properly folded toxic peptides in said plant and to cause an increase the plant's resistance to insect damage.

A method of using the nucleotides or expression ORFs in an ICK expression vector that code for Endoplasmic Reticulum Signal Peptide (ERSP). Inhibitor Cystine Knot (ICK) motif protein. Intervening Linker Peptide and/or Translational Stabilizing Protein (STA) to create transgenic plants. An expression ORF comprising any of the nucleotides in an ICK expression vector which express ERSP. ICK motif protein. LINKER and/or STA. A functional expression ORF in an ICK expression vector incorporated into a transformed plant, comprising nucleotides that code for that code for ERSP. ICK motif protein. LINKER and/or STA or that could be made by one skilled in the art given the teaching disclosed herein. A procedure for the generation of transformed plants having or expressing ERSP. ICK motif protein. LINKER and/or STA. A plant made by any of the products and processes described herein.

A protein comprised of an Endoplasmic Reticulum Signal Peptide (ERSP) operably linked to multiple Inhibitor Cystine Knot (ICK) motif protein domain, which are operably linked by Intervening Linker Peptides (LINKER), operably linked to a Translational Stabilizing Protein (STA) operably linked to an Intervening Linker Peptide, which is designated as ERSP-STA-(LINKER_i-ICK_j)_N or ERSP-(ICK_j-LINKER_i)_N-STA and sometimes as ERSP-STA-(L_i-ICK_j)_N or ERSP-(ICK_j-L_i)_N-STA, wherein said ERSP is the N-terminal of said protein and said STA may be either on the N-terminal side (upstream) of the multiple ICK motif protein domain ((LINKER_i-ICK_j)_N) or the C-terminal side (downstream) of the multiple ICK motif protein domain ((ICK_j-LINKER_i)_N) and said multiple Intervening Peptides (LINKER_i) is between STA and the multiple ICK motif protein domain and between the ICK motif proteins in the multiple ICK motif protein domain. A protein designated as ERSP-STA-(LINKER_i-ICK_j)_N or ERSP-(ICK_j-LINKER_i)_N-STA, comprising any of the ERSPs, ICK motif proteins, Intervening Linker Peptides and Translational Stabilizing Proteins described herein.

A nucleotide that codes for any of the peptides described as Endoplasmic Reticulum Signal Peptide (ERSP), multiple Inhibitor Cystine Knot (ICK) motif protein domain, Intervening Linker Peptide (LINKER) and/or Translational Stabilizing Protein (STA) and any and all nucleotides that code for any of these proteins that are used to create transgenic plants.

The use of any of the peptides or nucleotides that code for Endoplasmic Reticulum Signal Peptide (ERSP), multiple

Inhibitor Cystine Knot (ICK) motif protein domain, Intervening Linker Peptide, (LINKER) and/or Translational Stabilizing Protein (STA) to make or transform a plant or plant genome in order to express properly folded toxic peptides in a transformed plant. The use of any of the peptides or nucleotides that code for Endoplasmic Reticulum Signal Peptide (ERSP), multiple Inhibitor Cystine Knot (ICK) motif protein domain, Intervening Linker Peptide (LINKER) and/or Translational Stabilizing Protein (STA) to make or transform a plant or plant genome in order to express properly folded toxic peptides in the transformed plant and to cause the accumulation of the expressed and properly folded toxic peptides in said plant and to cause an increase the plant's resistance to insect damage.

A method of using the nucleotides or expression ORFs that code for Endoplasmic Reticulum Signal Peptide (ERSP), multiple Inhibitor Cystine Knot (ICK) motif protein domain, Intervening Linker Peptide (LINKER) and/or Translational Stabilizing Protein (STA) to create transgenic plants. An expression ORF comprising any of the nucleotides which express ERSP, multiple ICK motif protein domain, L or LINKER and/or STA. A functional expression ORF incorporated into a transformed plant, comprising nucleotides that code for ERSP, multiple ICK motif protein domain, LINKER and/or STA or that could be made by one skilled in the art given the teaching disclosed herein. A procedure for the generation of transformed plants having or expressing ERSP, multiple ICK motif protein domain, LINKER and/or STA. A plant made by any of the products and processes described herein.

A chimeric gene comprising a promoter active in plants operatively linked to the nucleic acids or expression ORF of the nucleotides described herein. A method of making, producing or using these chimeric genes that are described herein. A recombinant vector comprising the chimeric genes described herein. A method of making, producing or using the recombinant vectors described herein. A transgenic host cell comprising the chimeric genes described herein. A method of making, producing or using the transgenic host cell described herein. A transgenic host cell as described herein which is a transgenic plant cell. A method of making, producing or using the transgenic plant cell described herein. A transgenic plant comprising the transgenic plant cell described herein. A method of making, producing or using the transgenic plants described herein. A transgenic plant as described herein which made from a corn, soybean, cotton, rice, wheat, sorghum, switchgrass, sugarcane, alfalfa, potatoes, tomatoes, tobacco, any of green leafy vegetables, or any of fruit trees. Seed from a transgenic plant as described herein wherein said seed comprises a chimeric gene as described herein. A method of making, producing or using the transgenic plant described herein. A method of making, producing or using the seeds described herein.

Plant expressed inhibitory cysteine knot (ICK) motif proteins from spiders and scorpions have been described (Khan et al, Transgenic Res., 2006, 15: 349-357; Hemindez-Campuzano et al, Toxicon. 2009 January; 53(1):122-8). We describe how to make plant expressed ICK motif proteins that are active and accumulate in plants to insecticidal dose levels. We show that prior descriptions of plant expressed ICK motif proteins were actually descriptions of inactive proteins that had lost their natural toxicity. We describe methods to increase the efficacy of the plant expression, to increase the accumulation of plant expressed proteins and to dramatically increase the insecticidal activity of plant expressed proteins. We describe how to induce the plant expressed ICK motif proteins to enter the Endoplasmic

Reticulum (ER) directed by an Endoplasmic Reticulum Signaling Protein (ERSP) in plant cells, in order for the correct covalent cross-linking of peptide disulfide bridges which generate the essential tertiary ICK motif structure required for insecticidal activity. We further describe the plant expressed, ER-trafficking ICK motif protein complex with a translational stabilizing protein domain (STA) added in order to increase the size of the resulting ICK fusion protein which enhances peptide accumulation in the plant. We further describe the plant expressed, ER-trafficking ICK motif protein, with a translational stabilizing protein added as above, and with an intervening linker peptide (LINKER) added, the latter of which may allow for potential cleavage and the recovery of the active form of the ICK motif protein having insecticidal activity. We further describe the plant expressed polypeptide, which contains ER-trafficking ICK motif protein domain with multiple ICK motif proteins separated by intervening linker peptides (LINKER), with an intervening linker peptide added, with a translation stabilizing protein added, latter of which allows the correctly folded ICK motif protein to accumulate in the plant to the insecticidal dose.

This invention describes the ICK motif protein with insecticidal activity that are plant expressed and which can successfully protect a plant or crop from insect damage. The ICK motif protein expression ORF described herein is a nucleotide which will enable the plant translated peptides to not only be expressed in a plant but also to be expressed and folded properly, and to be accumulated to the insecticidal dose in the plant. An example of a protein expression ORF can be an ICK motif protein expression ORF which is can be described below in equation style and is shown in diagram style in the drawings or figures.

$\text{ersp-sta}-(\text{linker}_i\text{-crip}_j)_N$, or $\text{ersp}-(\text{crip}_j\text{-linker}_i)_N\text{-sta}$
The expression above is merely one example, and similar expressions could be written for other types of CRIP expression ORFs, for example an ICK expression ORF, could be written as:

$\text{ersp-sta}-(\text{linker}_i\text{-ick}_j)_N$, or $\text{ersp}-(\text{ick}_j\text{-linker}_i)_N\text{-sta}$
These expressions, equations or linear diagrams describe a polynucleotide open reading frame (ORF) for one type of CRIP, one which expresses the ICK motif protein complex, which can be described as $\text{ERSP-STA}-(\text{LINKER}_i\text{-ICK})_N$ or $\text{ERSP}-(\text{ICK}_j\text{-LINKER}_i)_N\text{-STA}$, or as $\text{ERSP-STA}-(\text{L}_1\text{-ICK}_j)_N$ or $\text{ERSP}-(\text{ICK}_j\text{-L}_1)_N\text{-STA}$, containing four possible peptide components with dash signs to separate each component. In the diagrams above, the nucleotide component of ersp is a polynucleotide segment encoding a plant endoplasmic reticulum trafficking signal peptide (ERSP). The component of sta is a polynucleotide segment encoding a translation stabilizing protein (STA), which helps the accumulation of the ICK motif protein expressed in plants but may not be necessary in the ICK motif protein expression ORF. The component of linker_i is a polynucleotide segment encoding an intervening linker peptide (L OR LINKER) to separate the ICK motif proteins from each other and from the translation stabilizing protein, and the subscription "i" indicates that different types of linker peptides can be used in the CRIP or ICK motif protein expression ORF. In the case that sta is not used in the ICK motif protein expression ORF, ersp can directly be linked to the polynucleotide encoding an ICK motif protein without a linker. The component of ick_j is a polynucleotide segment encoding an ICK motif protein (ICK), and the subscription "j" indicates different ICK motif proteins; $(\text{linker}_i\text{-ick}_j)_N$ indicates that the structure of the nucleotide encoding an intervening linker peptide and an ICK motif protein can be

repeated "N" times in the same open reading frame in the same ICK motif protein expression ORF, where N can be any integrate number from 1 to 10. N can be from 1 to 10, specifically N can be 1, 2, 3, 4, or 5, and in some embodiments N is 6, 7, 8, 9 or 10. The repeats may contain polynucleotide segments encoding different intervening linkers (LINKER) and different ICK motif proteins. The different polynucleotide segments including the repeats within the same ICK motif protein expression ORF are all within the same translation frame.

Any combination of the four principal components, ersp, sta, linker and crp or ick as in the diagram of the ICK motif protein expression ORF, may be used to create a PEP type ICK motif protein expression ORF as long as a minimum of ersp and at least one copy of crp or ick are used.

I. The ERSR or Ersp Component of the PEPs.

The ICK motif protein expression ORF starts with an ersp at its 5' terminus. For the ICK motif protein to be properly folded and functional when it is expressed from a transgenic plant, it must have an ersp nucleotide fused in frame with the polynucleotide encoding an ICK motif protein. During cellular translation process, translated ERSR can direct the ICK motif protein being translated to insert into the Endoplasmic Reticulum (ER) of the plant cell by binding with a cellular component called a signal-recognition particle. Within the ER the ERSR peptide is cleaved by signal peptidase and the ICK motif protein is released into the ER, where the ICK motif protein is properly folded during the post-translation modification process, for example, the formation of disulfide bonds. Without any additional retention protein signals, the protein is transported through the ER to the Golgi apparatus, where it is finally secreted outside the plasma membrane and into the apoplastic space. ICK motif protein can accumulate at apoplastic space efficiently to reach the insecticidal dose in plants. FIG. 1 shows a representative diagram of a simple two component peptide or nucleotide composed of an ERSR functionally linked to an ICK motif. The ICK could be a suitable CRIP. More complex proteins and polynucleotides utilizing ERSR are diagrammed in FIGS. 2-5 and these figures are further discussed in the discussion of the STA or Translational Stabilizing Protein.

The ERSR peptide is at the N-terminal region of the plant translated ICK motif protein complex and the ERSR portion is composed of about 3 to 60 amino acids. In some embodiments it is 5 to 50 amino acids. In some embodiments it is 10 to 40 amino acids but most often is composed of 15 to 20; 20 to 25; or 25 to 30 amino acids. The ERSR is a signal peptide so called because it directs the transportation of a protein. Signal peptides may also be called targeting signals, signal sequences, transit peptides, or localization signals. The signal peptides for ER trafficking are often 15 to 30 amino acid residues in length and have a tripartite organization, comprised of a core of hydrophobic residues flanked by a positively charged aminoterminal and a polar, but uncharged carboxyterminal region. (Zimmermann, et al, "Protein translocation across the ER membrane", *Biochimica et Biophysica Acta*, 2011, 1808: 912-924).

Many ERSRs are known. Many plant ERSRs are known. It is NOT required that the ERSR be derived from a plant ERSR, non-plant ERSRs will work with the procedures described herein. Many plant ERSRs are however well known and we describe some plant derived ERSRs here. BAAS, for example, is derived from the plant, *Hordeum vulgare*, and has the amino acid sequence as follows:

(SEQ ID NO: 4)

MANKHLSLSLFLVLLGLSASLASG

Plant ERSRs, which are selected from the genomic sequence for proteins that are known to be expressed and released into the apoplastic space of plants, and a few examples are BAAS, carrot extensin, tobacco PR1. The following references provide further descriptions, and are incorporated by reference herein in their entirety. De Loose, M. et al. "The extensin signal peptide allows secretion of a heterologous protein from protoplasts" *Gene*, 99 (1991) 95-100. De Loose, M. et al. described the structural analysis of an extensin—encoding gene from *Nicotiana plumbaginifolia*, the sequence of which contains a typical signal peptide for translocation of the protein to the endoplasmic reticulum. Chen, M. H. et al. "Signal peptide-dependent targeting of a rice alpha-amylase and cargo proteins to plastids and extracellular compartments of plant cells" *Plant Physiology*, 2004 July; 135(3): 1367-77. Epub 2004 Jul. 2. Chen, M. H. et al. studied the subcellular localization of α -amylases in plant cells by analyzing the expression of α -amylase, with and without its signal peptide, in transgenic tobacco. These references and others teach and disclose the signal peptide that can be used in the methods, procedures and peptide, protein and nucleotide complexes and constructs described herein.

II. The CRIP and ICK Motif Protein Component or Crp and Ick of the PEPs.

In our ICK motif protein expression ORF diagram, "ick" means a polynucleotide encoding an "ICK motif protein," or "inhibitor cystine knot motif protein", which is a 16 to 60 amino acid peptide with at least 6 half-cysteine core amino acids having three disulfide bridges, wherein the 3 disulfide bridges are covalent bonds and of the six half-cysteine residues the covalent disulfide bonds are between the first and fourth, the second and fifth, and the third and sixth half-cystines, of the six core half-cysteine amino acids starting from the N-terminal amino acid. The ICK motif protein also comprises a beta-hairpin secondary structure, normally composed of residues situated between the fourth and sixth core half-cysteines of the motif, the hairpin being stabilized by the structural crosslinking provided by the motif's three disulfide bonds. Note that additional cysteine/cysteine or half-cysteine amino acids may be present within the inhibitor cysteine knot motif, as shown in FIG. 6. The CRIP or ICK motif can be repeated in order to increase toxic peptide accumulation in the plant. See FIG. 4 and FIG. 5. This ability to repeat the CRIP or ICK motif, from 1 to 10 times and sometimes up to 15, 20 or 25 times is also shown in the equation like diagram of a CRIP or ICK protein expression ORF described herein as ersp-sta-(linker_i-ick_j)_N, or ersp-(ick_j-linker_i)_N-sta where the number of repeating LINKER-ICK motifs is given by the subscript number N and N is commonly 1-10 but can go even higher in some plants.

A similar expression like ersp-sta-(linker_i-ick_j)_N, or ersp-(ick_j-linker_i)_N-sta could be written and would describe other CRIP peptides. In this section an example of one expression ORF is one used to increase peptide expression in plants and is best exemplified with an ICK protein. In the diagram above, a polynucleotide open reading frame (ORF) which expresses an ICK motif protein complex, which can be described as ERSR-STA-(LINKER_i-ICK_j)_N or ERSR-(ICK_j-LINKER_i)_N-STA, or as ERSR-STA-(L₁-ICK_j)_N or ERSR-(ICK_j-L₁)_N-STA, containing four possible peptide components with dash signs to separate the each component is used. An alternate method of showing this type of construct can be found in the figures. In the diagram and the figures, the nucleotide component of ersp is a polynucleotide segment encoding a plant endoplasmic reticulum trafficking signal peptide (ERSR). The component of sta is a polynucle-

otide segment encoding a translation stabilizing protein (STA), which helps the accumulation of the ICK motif protein expressed in plants but may not be necessary in the ICK motif protein expression ORF. The component of l_i is a polynucleotide segment encoding an intervening linker peptide (L OR LINKER) to separate the ICK motif proteins from each other and from the translation stabilizing protein, and the subscription "i" indicates that different types of linker peptides can be used in the ICK motif protein expression ORF. In the case that sta is not used in the ICK motif protein expression ORF, ersp can directly be linked to the polynucleotide encoding an ICK motif protein without a linker. The component of ick_j is a polynucleotide segment encoding an ICK motif protein (ICK), and the subscription "j" indicates different ICK motif proteins; $(linker_i-ick_j)_N$ indicates that the structure of the nucleotide encoding an intervening linker peptide and an ICK motif protein can be repeated "N" times in the same open reading frame in the same ICK motif protein expression ORF, where N can be any integrate number from 1 to 10, but can go even higher to 15, 20 and 25, these repeats may contain polynucleotide segments encoding different intervening linkers and different ICK or CRIP motif proteins. The different polynucleotide segments including the repeats within the same ICK or CRIP motif protein expression ORF are all within the same translation frame.

This motif is common in peptides isolated from the venom of numerous species. Invertebrate species include spiders, scorpions, cone snail, sea anemone etc., other examples are numerous, even snake venom has been known to have peptides having the ICK motif. An example within spiders that we used is from a class of ACTX peptides from the Australian Blue Mountains Funnel-web Spider, but the procedures described herein are useful and may be applied to any protein with the ICK motif.

Examples of peptide toxins with the ICK motif can be found in the following references. The N-type calcium channel blocker ω -Conotoxin was reviewed by Lew, M. J. et al. "Structure-Function Relationships of ω -Conotoxin GVIA" *Journal of Biological Chemistry*, Vol. 272, No. 18, Issue of May 2, pp. 12014-12023, 1997. A summary of numerous arthropod toxic peptides from different spider and scorpion species was reviewed in, Quintero-Hernandez, V. et al. "Scorpion and Spider Venom Peptides: Gene Cloning and Peptide Expression" *Toxicon*, 58, pp. 644-663, 2011. The three-dimensional structure of Hanatoxin1 using NMR spectroscopy was identified as an inhibitor cysteine knot motif in Takahashi, H. et al. "Solution structure of hanatoxin1, a gating modifier of voltage-dependent K⁺ channels: common surface features of gating modifier toxins" *Journal of Molecular Biology*, Volume 297, Issue 3, 31 Mar. 2000, pp. 771-780. The isolation and identification of cDNA encoding a scorpion venom ICK toxin peptide, Opicalcine1, was published by Zhu, S. et al. "Evolutionary origin of inhibitor cysteine knot peptides" *FASEB J.*, 2003 Sep. 17, (12):1765-7, Epub 2003 Jul. 3. The sequence-specific assignment and the secondary structure identification of BgK, a K⁺ channel-blocking toxin from the sea anemone *Bunodosoma granulifera*, was disclosed by Dauplais, M. et al. "On the convergent evolution of animal toxins" *Journal of Biological Chemistry*, 1997 Feb. 14; 272(7): 4302-9. A review of the composition and pharmacology of spider venoms with emphasis on polypeptide toxin structure, mode of action, and molecular evolution showing cysteine bridges, cysteine knot formations and the "knotting-type" fold was published by Escoubas, P. et al. "Structure and pharmacology of spider venom neurotoxins" *Biochimie*, Vol. 82, Issues 9-10, 10 Sep.

2000, pp. 893-907. The purified peptide, iberiotoxin, an inhibitor of the Ca²⁺-activated K⁺ channel, from scorpion (*Buthus tamulus*) venom was disclosed in Galvez, A. et al. "Purification and characterization of a unique, potent, peptidyl probe for the high conductance calcium-activated potassium channel from venom of the scorpion *Buthus tamulus*" *Journal of Biological Chemistry*, 1990 Jul. 5; 265(19): 11083-90. The purified peptide, charybdotoxin, an inhibitor of the Ca²⁺-activated K⁺ channel, from the venom of the scorpion *Leiurus quinquestriatus* was disclosed in Gimenez-Gallego, G. et al. "Purification, sequence, and model structure of charybdotoxin, a potent selective inhibitor of calcium-activated potassium channels" *Proc Natl Acad Sci*, 1988 May; 85(10): 3329-3333. From these and other publications, one skilled in the art should be able to readily identify proteins and peptides having what we describe as the ICK motif, ICK motif protein or the "inhibitor cysteine knot motif."

The ICK motif protein can be any protein with the ICK motif and is between 16 and 60 amino acids in length, with at least 6 cysteine residues that create covalent cross-linking disulfide bonds in the proper order. See FIG. 6. Some ICK motif peptides have between 26-60 amino acids in length. Some ICK motif proteins are between 16-48 amino acids in length. Some ICK motif proteins are between 26-48 amino acids in length. Some ICK motif proteins are between 30-44 amino acids in length. ICK motif proteins with natural insecticidal activity are preferred but ICK motif proteins with other types of activity such as salt and frost resistance are known to those skilled in the art and are claimed herein. Examples of insecticidal ICK motif proteins include the ACTX peptides and genes, and including all of the peptides and their coding genes known as Magi6.

An example of a protein expression ORF could be an ICK motif protein expression ORF diagrammed below as:

ersp-sta-(linker_i-ick_j)_N, or ersp-(ick_j-linker)_N-sta

A similar expression could be written for other CRIP peptides. In this section this example of an expression ORF is one used to high peptide expression and is best exemplified with an ICK protein. The diagram above a polynucleotide open reading frame (ORF) which expresses an ICK motif protein complex, which can be described as ERSP-STA-(LINKER_i-ICK_j)_N or ERSP-(ICK_j-LINKER_i)_N-STA, or as ERSP-STA-(L_i-ICK_j)_N or ERSP-(ICK_j-L_i)_N-STA, containing four possible peptide components with dash signs to separate the each component. In this diagram, the nucleotide component of ersp is a polynucleotide segment encoding a plant endoplasmic reticulum trafficking signal peptide (ERSP). The component of sta is a polynucleotide segment encoding a translation stabilizing protein (STA), which helps the accumulation of the ICK motif protein expressed in plants but may not be necessary in the ICK motif protein expression ORF. The component of l_i is a polynucleotide segment encoding an intervening linker peptide (L OR LINKER) to separate the ICK motif proteins from each other and from the translation stabilizing protein, and the subscription "i" indicates that different types of linker peptides can be used in the ICK motif protein expression ORF. In the case that sta is not used in the ICK motif protein expression ORF, ersp can directly be linked to the polynucleotide encoding an ICK motif protein without a linker. The component of ick_j is a polynucleotide segment encoding an ICK motif protein (ICK), and the subscription "j" indicates different ICK motif proteins; $(linker_i-ick_j)_N$ indicates that the structure of the nucleotide encoding an intervening linker peptide and an ICK motif protein can be repeated "N" times in the same open reading frame in the

same ICK motif protein expression ORF, where N can be any integrate number from 1 to 10, and the repeats may contain polynucleotide segments encoding different intervening linkers and different ICK or CRIP motif proteins. The different polynucleotide segments including the repeats within the same ICK or CRIP motif protein expression ORF are all within the same translation frame.

Examples of insecticidal ICK motif proteins include the ACTX peptides and genes and include all of the peptides and their coding genes as described in the references provided above and herein. Specific examples of ICK motif proteins and peptides disclosed for purposes of providing examples and not intended to be limiting in any way, are the peptides and their homologues as described above, and in particular peptides and nucleotides which originate from the venoms of Australian Funnel-web spiders. The following documents are incorporated by reference in the United States in their entirety, are known to one skilled in the art, and have all been published. They disclose numerous ICK motif proteins which, their full peptide sequence, their full nucleotide sequence, are specifically disclosed and are incorporated by reference, and in addition the full disclosures are incorporated by reference including all of their sequence listings. See the following: U.S. Pat. No. 7,354,993 B2, issued Apr. 8, 2008, specifically the peptide and nucleotide sequences listed there as sequences 1-39, from 7,354,993 B2, and those named U-ACTX polypeptides, and these and other toxins that can form 2 to 4 intra-chain disulfide bridges, and variants thereof, and the peptides appearing on columns 4 to 9 and in FIG. 2 of 7,354,993 B2. Other specific sequences can be found in EP patent 1 812 464 B1, published and granted 08.10.2008, see Bulletin 2008/41, specifically the peptide and nucleotide sequences listed in the sequence listing, and those the other toxins that can form 2 to 4 intra-chain disulfide bridges, and those sequences listed there as 1-39, and sequences named U-ACTX polypeptides, and variants thereof, and the peptides appearing in paragraphs 0023 to 0055, and appearing in FIG. 1 of EP patent 1 812 464 B1.

Described and incorporated by reference to the peptides identified herein are homologous variants of sequences mentioned, having homology to such sequences or referred to herein, which are also identified and claimed as suitable for making special according to the processes described herein, including all homologous sequences having at least any of the following percent identities to any of the sequences disclosed here or to any sequence incorporated by reference: 50%, 55%, 60%, 65%, 70%, 75%, 80%, 85%, 90%, 95%, 96%, 97%, 98%, or 99% or greater identity or 100% identity to any and all sequences identified in the patents noted above, and to any other sequence identified herein, including each and every sequence in the sequence listing of this application. When the term homologous or homology is used herein with a number such as 50% or greater, then what is meant is percent identity or percent similarity between the two peptides. When homologous or homology is used without a numeric percent then it refers to two peptide sequences that are closely related in the evolutionary or developmental aspect in that they share common physical and functional aspects, like topical toxicity and similar size (i.e., the homolog being within 100% greater length or 50% shorter length of the peptide specifically mentioned herein or identified by reference herein as above).

Described and incorporated by reference to the peptides identified herein are toxic peptides including the following: peptides and its variants found in, isolated from, or derived from spiders of the genus *Atrax* or *Hadronyche*, including the genus species, *Hadronyche versuta*, or the Blue Mountain funnel web spider, *Atrax robustus*, *Atrax formidabilis*, *Atrax infensus*, including toxins known as U-ACTX polypeptides, U-ACTX-Hv1a, rU-ACTX-Hv1a, rU-ACTX-Hv1b,

or mutants or variants, especially peptides of any of these types and especially those less than about 200 amino acids but greater than about 10 amino acids, and especially peptides less than about 150 amino acids but greater than about 20 amino acids, especially peptides less than about 100 amino acids but greater than about 25 amino acids, especially peptides less than about 65 amino acids but greater than about 25 amino acids, especially peptides less than about 55 amino acids but greater than about 25 amino acids, especially peptides of about 37 or 39 or about 36 to 42 amino acids, especially peptides with less than about 55 amino acids but greater than about 25 amino acids, especially peptides with less than about 45 amino acids but greater than about 35 amino acids, especially peptides with less than about 115 amino acids but greater than about 75 amino acids, especially peptides with less than about 105 amino acids but greater than about 85 amino acids, especially peptides with less than about 100 amino acids but greater than about 90 amino acids, including peptide toxins of any of the lengths mentioned here that can form 2, 3 and or 4 or more intrachain disulfide bridges, including toxins that disrupt calcium channel currents, including toxins that disrupt potassium channel currents, especially toxins that disrupt insect calcium channels or Us thereof, especially toxins or variants thereof of any of these types, and any combination of any of the types of toxins described herein that have oral or topical insecticidal activity, can be made special by the processes described herein.

The U peptides from the Australian Funnel Web Spider, genus *Atrax* and *Hadronyche* are particularly suitable and work well when treated by the methods, procedures or processes described by this invention. Examples of such suitable peptides tested and with data are provided herein. The following species are also specifically known to carry toxic peptides suitable for plant expression as PIPs by the process of this invention. The following species are specifically named: *Atrax formidabilis*, *Atrax infensus*, *Atrax robustus*, *Hadronyche infensa*, *Hadronyche versuta*. Any toxic peptides derived from any of the genus listed above and/or genus species and homologous to the U peptide are suitable for plant expression as PIPs according to the process in this invention.

The Examples in this specification are not intended to, and should not be used to limit the invention, they are provided only to illustrate the invention.

As noted above, many peptides are suitable candidates as the subject of the process for the plant expression as PIP. The sequences noted above, below and in the sequence listing are especially suitable peptides that can be expressed in plants as PIP, and some of these have been expressed in plants as PIP according to this invention with the results shown in the examples below.

(SEQ ID NO: 5)

GSQYC VPVDQ PCSLN TQPC DDATC TQERN ENGHT VYCR A

55 Named "U+2-ACTX-Hv1a," it has disulfide bridges at positions: 5-20, 12-25, 19-39. The molecular weight is 4564.85 Daltons. Another example of an ICK motif insecticidal protein:

(SEQ ID NO: 6)

QYCVV VDQPC SLNTQ PCCDD ATCTQ ERNEN GHTVYCR A

60 Named "U-ACTX-Hv1a," it has disulfide bridges at positions: 3-18, 10-23, 17-37.

65 The molecular weight is 4426.84 Daltons.

Additional examples include many sequences in the sequence listing.

II. The Translational Stabilizing Protein Component, STA or Sta.

One of the ICK motif protein expression ORFs, ERSP-ICK, is sufficient to express a properly folded ICK motif peptide in the transformed plant, but in order for effective protection of a plant from pest damage, the plant expressed ICK motif protein needs to be accumulated to the insecticidal level. With transformation of a properly constructed ICK motif protein expression ORF, a transgenic plant can express and accumulate greater amounts of the correctly folded ICK motif protein. When a plant accumulates greater amounts of properly folded toxic peptides it can more easily resist or kill the insects that attack and eat the plants. The translational stabilizing protein can be used to significantly increase the accumulation of the toxic peptide in the plant and thus the potency of the PIP, especially when the PIP has a translational stabilizing protein of its own. See various representations of how the STA may be used in expression ORFs in FIGS. 2-5, and in various linear diagrams or equation like expressions used below. The translational stabilizing protein can be a domain of another protein or it can comprise an entire protein sequence. The translational stabilizing protein is a protein with sufficient tertiary structure that it can accumulate in a cell without being targeted by the cellular process of protein degradation. The protein can be between 5 and 50aa (e.g. another ICK-motif protein), 50 to 250aa (GNA), 250 to 750aa (e.g. chitinase) and 750 to 1500aa (e.g. enhancin).

In addition to FIGS. 2-5 the following linear diagram below describes one of the examples of the ICK motif protein expression ORF that encodes a stabilizing protein fused with ICK motif protein:

ersp-sta-l-ick

The protein, or protein domain can contain proteins that have no useful characteristics other than translation stabilization, or they can have other useful traits in addition to translational stabilization. Useful traits can include: additional insecticidal activity, such as activity that is destructive to the peritrophic membrane, activity that is destructive to the gut wall, and/or activity that actively transports the ICK motif protein across the gut wall. One embodiment of the translational stabilizing protein can be a polymer of fusion proteins involving ICK motif proteins. A specific example of a translational stabilizing protein is provided here to illustrate the use of a translational stabilizing protein. The example is not intended to limit the disclosure or claims in any way. Useful translational stabilizing proteins are well known in the art, and any proteins of this type could be used as disclosed herein. Procedures for evaluating and testing production of peptides are both known in the art and described herein. One example of one translational stabilizing protein is SEQ ID NO:7, one letter code, as follows:

(SEQ ID NO: 7)

ASKGE ELFTG VVPIL VELDG DVNGH KFSVS GEGEG DATYGR
KLTLL FICTT GKLPV PWPTL VTTF S YGVQC FSRYP DHMKR
HDFFK SAMPE GYVQE RTISF KDDGN YKTRA EVKFE GDTLV
NRIEL KGIDF KEDGN ILGHK LEYNY NSHN V YITAD KQKNG
IKANF KIRHN IEDGS VQLAD HYQQN TPIGD GPVLL PDNHY
LSTQS ALSKD PNEKR DHMVL LEFVT AAGIT HGMDE LYK

Named "GFP." The molecular weight is 26736.02 Daltons.

In some embodiments the STA can even be CRIP or ICK as shown in FIG. 5. In these embodiments there is no separate STA protein, the STA protein is the same as the CRIP or ICK used. It could be the identical ICK that is bound with the LINKER, or there could be different ICKs one type bound to the LINKER and the other type acting as the STA. These alternative arrangements are also discussed in the section on LINKERS.

Additional examples of translational stabilizing proteins can be found in the following references, incorporated by reference in their entirety: Kramer, K. J. et al. "Sequence of a cDNA and expression of the gene encoding epidermal and gut chitinases of *Manduca sexta*" *Insect Biochemistry and Molecular Biology*, Vol. 23, Issue 6, September 1993, pp. 691-701. Kramer, K. J. et al. isolated and sequenced a chitinase-encoding cDNA from the tobacco hornworm, *Manduca sexta*. Hashimoto, Y. et al. "Location and nucleotide sequence of the gene encoding the viral enhancing factor of the *Trichoplusia ni* granulosis virus" *Journal of General Virology*, (1991), 72, 2645-2651. Hashimoto, Y. et al. cloned the gene encoding the viral enhancing factor of a *Trichoplusia ni* granulosis virus and determined the complete nucleotide sequence. Van Damme, E. J. M. et al. "Biosynthesis, primary structure and molecular cloning of snowdrop (*Galanthus nivalis* L.) lectin" *European Journal of Biochemistry*, 202, 23-30 (1991). Van Damme, E. J. M. et al. isolated Poly(A)-rich RNA from ripening ovaries of snowdrop lectin (GNA), yielding a single 17-kDa lectin polypeptide upon translation in a wheat-germ cell-free system, called agglutinin. These references and others teach and disclose translational stabilizing proteins that can be used in the methods, procedures and peptide, protein and nucleotide complexes and constructs described herein.

IV. The Intervening Linker Peptide Component, LINKER, Linker, L or if Polynucleotide: Linker or 1 of the PEPs

The ICK motif protein expression ORF described in this invention also incorporates polynucleotide sequences encoding intervening linker peptides between the polynucleotide sequences encoding the ICK motif protein (ick) and the translational stabilizing protein (sta), or between polynucleotide sequences encoding multiple ICK motif proteins domain ((l-ick)_N or (ick-l)_N) if the expression ORF involves multiple ICK motif protein domain expression. The intervening linker peptides (LINKERS) separate the different parts of the expressed ICK motif protein complex and help proper folding of the different parts of the complex during the expression process. In the expressed ICK motif protein complex, different intervening linker peptides can be involved to separate different functional domains. Various representations of proteins with LINKERS are shown in (FIGS. 3-5). The LINKER is attached to a CRIP such as an ICK and this bivalent group can be repeated up to 10 (N=1-10) and possibly even more than 10 times in order to facilitate the accumulation of properly folded insecticidal peptide in the plant that is to be protected.

The intervening linker peptide is usually between 1 and 30 amino acids in length. It is not necessary an essential component in the expressed ICK motif protein complex in plants. A cleavable linker peptide can be designed to the ICK motif protein expression ORF to release the properly folded ICK motif protein from the expressed ICK motif protein complex in the transformed plant to improve the protection the ICK motif protein to the plant from pest damage. One type of the intervening linker peptide is the plant cleavable linker peptide. This type of linker peptides can be completely removed from the expressed ICK motif protein expression complex during the post-translational expression

process in the plant cells. Therefore the properly folded ICK motif protein linked by this type of intervening linker peptides can be released in the plant cells from the expressed ICK motif protein complex during the post-translational expression process. Here we show numerous examples of LINKERS.

Another type of the cleavable intervening linker peptide is not cleavable during the expression process in plants. However, it has a protease cleavage site specific to serine, threonine, cysteine, aspartate proteases or metalloproteases. The type of cleavable linker peptide can be digested by proteases found in the insect and lepidopteran gut environment and/or the insect hemolymph and lepidopteran hemolymph environment to release the ICK motif protein in the insect gut or hemolymph. Here we show numerous examples of LINKERS. These linkers are presented as examples only and should not be considered limiting the invention. Using the information taught by this disclosure it should be a matter of routine for one skilled in the art to make or find other examples of LINKERS that will be useful in this invention.

An example of a cleavable type of intervening linker that illustrates the invention is listed in SEQ ID NO: 1, but cleavable linkers are not limited to this example. SEQ ID NO: 1 (one letter code) is IGER and here we name it "IGER." The molecular weight of this intervening linker or LINKER is 473.53 Daltons.

An intervening linker peptide (LINKER) can also be one without any type of protease cleavage site, i.e. an uncleavable intervening linker peptide. An example of this is the linker ETMFKHGL (SEQ ID NO. 3).

Other examples of intervening linker peptides can be found in the following references, which are incorporated by reference herein in their entirety: A plant expressed serine proteinase inhibitor precursor was found to contain five homogeneous protein inhibitors separated by six same linker peptides in Heath et al. "Characterization of the protease processing sites in a multidomain proteinase inhibitor precursor from *Nicotiana glauca*" *European Journal of Biochemistry*, 1995; 230: 250-257. A comparison of the folding behavior of green fluorescent proteins through six different linkers is explored in Chang, H. C. et al. "De novo folding of GFP fusion proteins: high efficiency in eukaryotes but not in bacteria" *Journal of Molecular Biology*, 2005 Oct. 21; 353(2): 397-409. An isoform of the human GalNAc-Ts family, GalNAc-T2, was shown to retain its localization and functionality upon expression in *N. benthamiana* plants by Daskalova, S. M. et al. "Engineering of *N. benthamiana* L. plants for production of N-acetylgalactosamine-glycosylated proteins" *BMC Biotechnology*, 2010 Aug. 24; 10: 62. The ability of endogenous plastid proteins to travel through stromules was shown in Kwok, E. Y. et al. "GFP-labelled Rubisco and aspartate aminotransferase are present in plastid stromules and traffic between plastids" *Journal of Experimental Botany*, 2004 March; 55(397): 595-604. Epub 2004 Jan. 30. A report on the engineering of the surface of the tobacco mosaic virus (TMV), virion, with a mosquito decapeptide hormone, trypsin-modulating oostatic factor (TMOF) was made by Borovsky, D. et al. "Expression of *Aedes* trypsin-modulating oostatic factor on the virion of TMV: A potential larvicide" *Proc Natl Acad Sci*, 2006 Dec. 12; 103(50): 18963-18968. These references and others teach and disclose the intervening linkers that can be used in the methods, procedures and peptide, protein and nucleotide complexes and constructs described herein.

The ICK motif protein expression ORF described above can be cloned into any plant expression vector for the ICK motif protein expression in plant transiently or stably. Transient Plant Expression Systems

Transient plant expression systems can be used to promptly optimize the structure of the ICK motif protein expression ORF for some specific ICK motif protein expression in plants, including the necessity of some components, codon optimization of some components, optimization of the order of each components, etc. A transient plant expression vector is often derived from a plant virus genome. Plant virus vectors provide advantages in quick and high level of foreign gene expression in plant due to the infection nature of plant viruses. The full length of the plant viral genome can be used as a vector, but often a viral component is deleted, for example the coat protein, and transgenic ORFs are subcloned in that place. The ICK motif protein expression ORF can be subcloned into such a site to create a viral vector. These viral vectors can be introduced into plant mechanically since they are infectious themselves, for example through plant wound, spray-on etc. They can also be transformed into plants by agroinfection by cloning the virus vector into the T-DNA of the crown gall bacterium, *Agrobacterium tumefaciens*, or the hairy root bacterium, *Agrobacterium rhizogenes*. The expression of the ICK motif protein in this vector is controlled by the replication of the RNA virus, and the virus translation to mRNA for replication is controlled by a strong viral promoter, for example, 35S promoter from Cauliflower mosaic virus. Viral vectors with ICK motif protein expression ORF are usually cloned into T-DNA region in a binary vector that can replicate itself in both *E. coli* strains and *Agrobacterium* strains. The transient transformation of a plant can be done by infiltration of the plant leaves with the *Agrobacterium* cells which contain the viral vector for ICK motif protein expression. In the transient transformed plant, it is common for the foreign protein expression to be ceased in a short period of time due to the post-transcriptional gene silencing (PTGS). Sometimes a PTGS suppressing protein gene is necessary to be co-transformed into the plant transiently with the same type of viral vector that drives the expression of with the ICK motif protein expression ORF. This improves and extends the expression of the ICK motif protein in the plant. The most commonly used PTGS suppressing protein is P19 protein discovered from tomato bushy stunt virus (TBSV).

A demonstration of transient plant expression can be found in FIG. 7.

FIG. 7 shows transiently expressed Plant Transgenic Protein. In FIG. 7 reports the relative accumulation of the ICK proteins compared to the % TSP, as detected by ELISA. There are four variations of ICK expression ORFs in FIG. 7 that illustrate the necessity of the ERSP to get proper folding of the ICK and the STA to get accumulation of the protein. Bar A reports a FECT expression system expressing SEQ ID NO: 8 the omega peptide (ICK) without any fusions. Bar B reports a TRBO expression system expressing SEQ ID NO: 9 a BAAS ERSP fused to the omega peptide (ICK). Bar C reports a FECT expression system expressing SEQ ID NO: 10 a GFP (STA) fused to IGER (Linker) fused to Hybrid toxin (ICK). Bar D reports a FECT expression system expressing SEQ ID NO: 11 a BAAS (ERSP) fused to a GFP (STA) fused to IGER (Linker) fused to Hybrid toxin (ICK). The detection levels for Bar A and B show negligible protein detection. In Bar A this is likely due to no proper folding of the ICK which occurs in the ER and in Bar B this is likely due to proper folding but no accumulation due to the lack of a STA. There are detectable levels in Bars C and D. When

the experiment for Bar C [(SEQ ID NO: 10) a GFP (STA) fused to IGER (Linker) fused to Hybrid toxin (ICK)] was performed there was a high level of GFP fluorescence detected (data not shown) indicating much of the TSP was the fusion protein, however, when the ELISA was performed only 0.01% of the TSP was detected, and this is likely due to the lack of proper folding which did not occur as this protein was not targeted to the ER where folding occurs. The antibodies used in ELISA only detect the tertiary structure of a properly folded protein. When the experiment for Bar D [SEQ ID NO: 11 a BAAS (ERSP) fused to a GFP (STA) fused to IGER (Linker) fused to Hybrid toxin (ICK)] was performed there was some GFP fluorescence detected and an accumulation 0.10% of the TSP the ICK peptide fused to GFP. When the data for Bars A, B, C and D is taken together it is apparent that an ERSP in the ICK expression ORF is required to get proper folding and to increase the accumulation of the peptide a STA is required.

We have demonstrated and documented GFP emission of the green fluorescence of GFP-Hybrid fusion protein constructs in tobacco leaves transiently transformed using different FECT vectors designed for targeted expression. We have succeeded in using pFECT-BGIH vector for APO (apoplast localization) accumulation; pFECT-GIH vector for CYTO (cytoplasm localization) accumulation; and pFECT-BGIH-ER vector for ER (endoplasmic reticulum localization) accumulation. Data not shown.

We have demonstrated and documented GFP emission of the green fluorescence of GFP-Hybrid fusion protein constructs in tobacco leaves transiently transformed using different types of ERSP. We have succeeded in demonstrating expression with pFECT-BGIH vector; expression with pFECT-EGIH vector; and expression with pFECT-E*GIH vector. Data not shown.

We have measured levels of peptide accumulation and this is shown in FIGS. 8 and 9. FIG. 8 is a graph of iELISA detected % TSPs of tobacco transiently expressed GFP fused U-ACTX-Hv1a with different accumulation localization. APO: apoplast localization; CYTO: cytoplasm localization; ER: endoplasmic reticulum localization. FIG. 9 is a graph of iELISA detected % TSPs of tobacco leaves transiently expressing GFP fused U-ACTX-Hv1a using the FECT expression vectors encoding translational fusions with three different ERSP sequences: BAAS signal peptide (BGIH), Extensin signal peptide (EGIH) and modified Extensin signal peptide (E*GIH).

Integration of Protein Expression ORF into Plant Genome Using Stable Plant Transformation Technology

The ICK motif protein expression ORF can also be integrated into plant genome using stable plant transformation technology, and therefore ICK motif proteins can be stably expressed in plants and protect the transformed plants from generation to generation. For the stable transformation of plants, the ICK motif protein expression vector can be circular or linear. A few critical components must be included in the vector DNA. The ICK motif protein expression ORF for stable plant transformation should be carefully designed for optimal expression in plants based on the study in the transient plant expression as described above.

The expression of ICK motif protein is usually controlled by a promoter that promotes transcription in some of all cells of the transgenic plant. The promoter can be a strong plant viral promoter, for example, the constitutive 35S promoter from Cauliflower Mosaic Virus (CaMV); it also can be a strong plant promoter, for example, the hydroperoxide lyase promoter (pHPL) from *Arabidopsis thaliana*; the Glycine max polyubiquitin (Gmubi) promoter from soy-

bean; the ubiquitin promoters from different plant species (rice, corn, potato, etc.), etc. A plant transcriptional terminator often occurs after the stop codon of the ORF to halt the RNA polymerase and transcription of the mRNA. To evaluate the ICK motif protein expression, a reporter gene can be included in the ICK motif protein expression vector, for example, beta-glucuronidase gene (GUS) for GUS straining assay, green fluorescent protein (GFP) gene for green fluorescence detection under UV light, etc. For selection of transformed plants, a selection marker gene is usually included in the ICK motif protein expression vector. The marker gene expression product can provide the transformed plant with resistance to specific antibiotics, for example, kanamycin, hygromycin, etc., or specific herbicide, for example, glyphosate etc. If agroinfection technology is adopted for plant transformation, T-DNA left border and right border sequences are also included in the ICK motif protein expression vector to transport the T-DNA portion into the plant. The constructed ICK motif protein expression vector can be transform into plant cells or tissues using many transformation technologies. Agroinfection is a very popular way to transform a plant using an *Agrobacterium tumefaciens* strain or an *Agrobacterium rhizogenes* strain. Particle bombardment (also called Gene Gun, or Biolistics) technology is also very commonly used for plant transformation. Other less commonly used transformation methods include tissue electroporation, silicon carbide whiskers, direct injection of DNA, etc. After transformation, the transformed plant cells or tissues placed on plant regeneration media to regenerate successfully transformed plant cells or tissues into transgenic plants. The evaluation of the integration and expression of the ICK motif protein expression ORF in the transformed plant can be performed as follows.

Evaluation of a Transformed Plant

Evaluation of a transformed plant can be done in DNA level, RNA level and protein level. A stably transformed plant can be evaluated at all of these levels and a transiently transformed plant is usually only evaluated at protein level. To ensure that the ICK expression motif protein expression ORF integrates into the genome of a stably transformed plant, the genomic DNA can be extracted from the stably transformed plant tissues for the PCR evaluation or the Southern blot application. The expression of the ICK motif protein in the stably transformed plant can be evaluated in RNA level, i.e. the total mRNA can be extracted from the transformed plant tissues and the northern blot technique and the RT-PCR technology can applied to evaluate the mRNA level of the ICK motif protein qualitatively or quantitatively. The expression of the ICK motif protein in the transformed plant can also be evaluated in protein level directly. There are many ways to evaluate the ICK motif protein expressed in a transformed plant. If a reporter gene is transformed into the plant along with the ICK motif protein expression ORF, the reporter gene assay can be performed to initially evaluate the expression of the transformed ICK motif protein expression ORF, for example, GUS straining assay for GUS reporter gene expression, green fluorescence detection assay for GFP reporter gene expression, luciferase assay for luciferase reporter gene expression, etc. Moreover, the total expressed protein can be extracted from the transformed plant tissues for the direct evaluation of the expression of the ICK motif protein in the transformed plants. The extracted total expressed protein sample can be used in Bradford assay to evaluate the total protein level in the sample. Analytical HPLC chromatography technology, Western blot technique, or iELISA assay can be adopted to qualitatively or quantitatively evaluate the

ICK motif protein in the extracted total protein sample from the transformed plant tissues. The ICK motif protein expression can also be evaluated by using the extracted total protein sample from the transformed plant tissues in an insect bioassay. Finally, the transformed plant tissue or the whole transformed plant can be tested in insect bioassays to evaluate the ICK motif protein expression and its protection for the plant.

We provide a detailed description and summary of Part I as follows:

We describe a protein comprised of an Endoplasmic Reticulum Signal Peptide (ERSP) operably linked to a Cysteine Rich Insecticidal Protein (CRIP) such as an Inhibitor Cysteine Knot (ICK) motif protein wherein said ERSP is the N-terminal of said protein (ERSP-ICK). The ERSP is any signal peptide which directs the expressed CRIP to the endoplasmic reticulum of plant cells. The CRIP can be an Inhibitor Cysteine Knot (ICK) protein or a Non-ICK protein. The ERSP is a peptide between 5 to 50 amino acids in length, originating from a plant, that is operably linked to a Translational Stabilizing Protein (STA), wherein said ERSP is the N-terminal of said protein and an intervening STA sequence may be either on the N-terminal side of the CRIP, which is optionally an ICK motif protein (ERSP-STA-ICK); or Non-ICK motif protein (ERSP-STA-Non-ICK) or on the C-terminal side of the ICK or Non-ICK motif protein (ERSP-ICK-STA) or (ERSP—Non-ICK—STA). The ERSP is a peptide between 3 to 60 amino acids in length, or a peptide between 5 to 50 amino acids in length, or a peptide between 20 to 30 amino acids in length. It can originate from a plant, Barley Alpha-Amylase Signal peptide (BAAS) with a SEQ ID NO 4. The ERSP can be a peptide that is tobacco extensin signal peptide with a SEQ ID NO 18. The ERSP can be a modified tobacco extensin signal peptide with a SEQ ID NO 19 or a Jun a 3 signal peptide from *Juniperus ashei* with a SEQ ID NO 27.

We describe a CRIP example that is an ICK motif protein is between 16 and 60 amino acids in length, between 26 and 48 amino acids in length, between 30 and 44 amino acids in length, where it is selected from any of the peptides or sources of peptides with inhibitory cysteine knot motif, or a insecticidal peptide and where it is any of the peptides or sources of peptides including *Atrax* or *Hadronyche*, any of the peptides originating from *Hadronyche versuta*, an ACTX peptide. The ICK motif protein is any insecticidal peptide and fragments thereof including mature, pre, and pro peptide versions of said peptides and sequence numbers as well as any mutations, or deletion, or addition of peptide segments but still maintenance of inhibitory cysteine knot structure. The ICK motif protein can be U-ACTX-Hv1a with SEQ ID NO: 6, Omega-ACTX-Hv1a with SEQ ID NO: 24, Kappa-ACTX-Hv1c. An expression ORF comprising any of the nucleotides that code for those peptides. An expression ORF comprising any of the nucleotides that code for the peptides integrated into a transgenic plant genome. The use of any of the peptides or nucleotides described herein to make or transform a plant or plant genome in order to express properly folded insecticidal peptides in a transformed plant and or to make or transform a plant or plant genome in order to express properly folded insecticidal peptides in the transformed plant and to cause the accumulation of the expressed and properly folded insecticidal peptides in said plant and to cause an increase the plant's resistance to insect damage. We describe procedures to use nucleotides to create transgenic plants and transformed plants having or expressing any of the peptides described herein. We describe a transformed plant made by any of these products and processes.

We describe a protein comprised of an Endoplasmic Reticulum Signal Peptide (ERSP) operably linked to a CRIP which is optionally an Inhibitor Cysteine Knot (ICK) motif protein or Non-ICK protein operably linked to a Translational Stabilizing Protein (STA), wherein said ERSP is the N-terminal of said protein and an intervening Translational Stabilizing Protein sequence may be either on the N-terminal side of the ICK motif protein (ERSP-STA-ICK or optionally a (ERSP-Non-ICK-STA) or the C-terminal side of the ICK motif protein (ERSP-ICK-STA) or ERSP-STA-Non-ICK).

We describe such a STA with a molecular weight of 12 kD and above, where said STA can be many proteins, including an ICK motif protein with molecular weight of 12 kD and above, or multiple ICK motif proteins connected with linker peptides (L) with molecular weight of 12 kD and above, for example ERSP-ICK-(L_i-ICK_j)_N or ERSP-(ICK_j-L_i)_N-ICK.

We explain the linker peptides can be the same or different. We say that one STA is a green fluorescence protein (GFP) originating from jellyfish with SEQ ID NO 13 and the STA can be a snowdrop lectin, *Galanthus nivalis* agglutinin (GNA), with SEQ ID NO 28 and that STA can be a *Juniperus ashei* protein, Jun a 3, with SEQ ID NO 26.

We describe a LINKER is any peptide with 4-20 amino acids in length. We describe a LINKER that is any peptide containing a protease recognition site. We describe a LINKER as any peptide containing a plant protease cleavage site. We describe a LINKER is a peptide containing an amino acid sequence of IGER (SEQ ID NO: 1), EEKKN (SEQ ID NO: 2) and (SEQ ID NO: 3). We describe a LINKER as any peptide which can be cleaved in the insect digestive system, or in the insect hemolymph. We describe a LINKERS wherein said LINKER is a peptide containing a trypsin cleavage site.

We describe a nucleotide that codes for any of the proteins described including expression ORFs comprising any of the nucleotides that code for the peptides, as well as expression ORF comprising any of the nucleotides that code for the peptides, integrated into a transgenic plant genome, as well as transformed into a plant or plant genome in order to express properly folded insecticidal peptides in a transformed plant, as well as transformed into a plant or plant genome in order to express properly folded insecticidal peptides in the transformed plant and to cause the accumulation of the expressed and properly folded insecticidal peptides in said plant and to cause an increase the plant's resistance to insect damage. We describe transgenic plants that result from these descriptions and transformed plants having or expressing any of the peptides described herein.

We explain and describe an expression ORF comprising any of the nucleotides that code for the peptides herein as well as an expression ORF integrated into a transgenic plant genome, and one reason this is done is to make or transform a plant or plant genome in order to express properly folded insecticidal peptides in a transformed plant and one reason this is done is to have the transformed plant cause the accumulation of the expressed and properly folded insecticidal peptides in said plant and to cause an increase the plant's resistance to insect damage. We teach how to make the transgenic plants using these procedures and expressing the peptides herein and any other peptides that one skilled in the art would use given the teaching herein and using any of the products and processes described herein.

We teach how to make a protein comprised of an Endoplasmic Reticulum Signal Peptide (ERSP) operably linked to an Inhibitor Cysteine Knot (ICK) motif protein operably linked to translational stabilizing protein (STA), operably

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linked to an intervening linker peptide (L), wherein said ERSP is the N-terminal of said protein, and said LINKER is between STA and the ICK motif protein, and said translational stabilizing protein may be either on the N-terminal side (upstream) of the ICK motif protein or the C-terminal side (downstream) of the ICK motif protein, and described as ERSP-STA-L-ICK, or ERSP-ICK-L-STA. And we explain the aforementioned ERSP, CRIP and ICK, LINKER, STA can be any of the peptides as described herein and any other peptides that one skilled in the art would use given the teaching herein and using any of the products and processes described herein.

We teach how to make a protein comprised of an Endoplasmic Reticulum Signal Peptide (ERSP) operably linked to a multiple Inhibitor Cysteine Knot (ICK) motif protein domain in which ICK motif proteins are linked to each other via intervening linker peptides (L), operably linked to a translational stabilizing protein (STA), operably linked to an intervening linker peptide (L), wherein said ERSP is the N-terminal of said protein, and said LINKER is between STA and the multiple ICK motif proteins domain, and said STA may be either on the N-terminal side (upstream) of the multiple ICK motif protein domain or the C-terminal side (downstream) of the multiple ICK motif protein domain, and described as ERSP-STA-(L_i-ICK_j)_N, or ERSP-(ICK_j-L_i)_N-STA.

We teach how to make the nucleotides that code for these proteins, the expression ORFs, to make a and to integrated into a transgenic plant genome, the chimeric genes, recombinant vectors, transgenic host cells, transgenic plant cells, transgenic plants, transgenic plants of which are corn, soybean, cotton, rice, wheat, sorghum, switchgrass, sugarcane, alfalfa, potatoes, tomatoes, tobacco, any of green leafy vegetables, or any of fruit trees, or any plants and species as mentioned herein, and a seed from a transgenic plant according to these procedures where the seed comprises the chimeric gene.

EXAMPLES

The Examples in this specification are not intended to, and should not be used to, limit the invention; they are provided only to illustrate the invention.

Example 1

Expression Comparison Between Two Transient Plant Expression Systems.

The transient plant transformation technologies were adopted to promptly optimize the ICK motif protein expression ORF for plant expression. Agroinfection technology with a plant viral vector has been used here for the transient plant transformation due to its high efficiency, easiness and inexpensiveness. Two viral transient plant expression systems were evaluated here for the ICK motif protein expression in plants. One was a tobacco mosaic virus overexpression system (TRBO, Lindbo J A, Plant Physiology, 2007, V145: 1232-1240). The TRBO DNA vector has a T-DNA region for agroinfection, which contains a CaMV 35S promoter that drives expression of the tobacco mosaic virus RNA without the gene encoding the viral coating protein. The other viral transient plant expression system was the FECT expression system (Liu Z & Kearney C M, BMC Biotechnology, 2010, 10:88). The FECT vector also contains a T-DNA region for agroinfection, which contains a CaMV 35S promoter that drives the expression of the foxtail mosaic virus RNA without the genes encoding the viral coating

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protein and the triple gene block. Both expression systems use the “disarmed” virus genome, therefore viral plant to plant transmission can be effectively prevented. To efficiently express the introduced heterologous gene, the FECT expression system additionally needs to co-express P19, a RNA silencing suppressor protein from tomato bushy stunt virus, to prevent the post-transcriptional gene silencing (PTGS) of the introduced T-DNA. (The TRBO expression system does not need co-expression of P19). The two transient plant expression systems were tested and compared by transient expression of ICK motif protein in Tobacco (*Nicotiana benthamiana*) as described below.

The ICK motif protein expression ORF was designed to encode a series of translationally fused structural motifs that can be described as follows: N'-ERSP-Sta-L-ICK-C'. Here the ICK motif protein for expression is U-ACTX-Hv1a, which has the following amino acid sequence (N' to C', one letter code):

(SEQ ID NO: 12)
QYCVFVDQPCSLNTQPCDDATCTQERNENGHTVYYCRA

The ERSP motif used here is the Barley Alpha-Amylase Signal peptide (BAAS), which comprises of 24 Amino acids as shown below (N' to C', one letter code):

(SEQ ID NO: 4)
MANKHLSLSLFLVLLGLSASLASG

The stabilizing protein (Sta) in this expression ORF was Green Fluorescent Protein (GFP), which has amino acid sequence as follows (N' to C', one letter code):

(SEQ ID NO: 13)
MASKGEELFTGVVPILVELDGDVNGHKFVSVSGEGEDATYGLTKLFICT
TGKLPVPWPTLVTTFSYGVQCFSRYPDHMKRHDFFKSAPEGYVQERTIS
FKDDGNYKTRAIEVKFEGDTLVNRIELKGIDFKEDGNILGHKLEYNINSHN
VYITADKQKNGIKANFKIRHNIEDGSVQLADHYQNTPIGDGPVLLPDNH
YLSTQSALS KDPNEKRDMVLLLEFVTAAGITHGMDELYK

The linker peptide between GFP and U-ACTX-Hv1a contains the trypsin cleavage site and has an amino acid sequence as shown below (N' to C', one letter code):

(SEQ ID NO: 1)
IGER

According to the ICK motif expression ORF formula, this specific ICK expression ORF can be described as BAAS-GFP-IGER-Hybrid, or BGIH. The BGIH ORF was chemically synthesized by adding Pac I restriction site at its 5' terminus and Avr II restriction site at its 3' terminus. The sequence of the synthetic BGIH is below:

(SEQ ID NO: 14)
TTAATTAATAAGGCTAATAAACACCTGAGTTTGTCACTATTCTCGTGTG
CTCGGTTATCTGCTTCACTTGCAAGCGGAGCTAGCAAAGGAGAAGAACT
TTTCACTGGAGTTGTCCCAATCTTGTGTAATTAGATGGTGATGTTAATG
GGCACAAATTTTCTGTCACTGGAGAGGGTGAAGGTGATGCTACATACGGA

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-continued

AAGCTTACCCCTAAATTTATTGCACTACTGGAAACTACCTGTTCCATG
 GCCAACACTTGTCACTACTTTCTTATGGTGTTCATGCTTTTCCCGTT
 ATCCGGATCATATGAAACGGCATGACTTTTCAAGAGTGCCATGCCCGAA
 GGTATGTACAGGAACGCACTATATCTTTCAAAGATGACGGGAACACAA
 GACGCGTGCTGAAGTCAAGTTTGAAGGTGATACCTTGTTAATCGTATCG
 AGTTAAAGGTATTGATTTTAAAGAAGATGGAACATTCTCGGACACAAA
 CTCGAGTACAACATACTACACAAATGTATACATCACGGCAGACAAAACA
 AAAGAATGGAATCAAAGCTAACTTCAAATTCGCCACAACATTGAAGATG
 GATCCGTTCAACTAGCAGACATTATCAACAAAATACTCCAATTGGCGAT
 GGCCCTGTCCTTTTACCAGACAACCATACCTGTCGACACAATCTGCCCT
 TTCGAAAGATCCCAACGAAAGCGTGACCACATGGTCTTCTTGAGTTTG
 TAACTGCTGCTGGGATTACACATGGCATGGATGAGCTCTACAAAATTGGT
 GAAAGACAATATTGTGTTCCAGTTGATCAACCATGTTCTTAACTACTCA
 ACCATGTTGTGATGATGCTACTTGTACTCAAGAAAGAAATGAAATGGAC
 ATACTGTTTATTATTGTAGAGCTTAACCTAGG

The BGIH ORF was cloned into the Pac I and Avr II restriction sites of the FECT expression vector to create a BGIH expression vector for the FECT transient plant expression system (pFECT-BGIH). To maximize BGIH expression in the FECT expression system, a FECT vector expressing the RNA silencing suppressor protein P19 (pFECT-P19) was generated for co-transformation. To create a BGIH expression vector for TRBO transient plant expression system, a routine PCR procedure was performed to add a Not I restriction site to the 3' terminus of the BGIH ORF described above. The new BGIH ORF was then cloned into Pac I and Not I restriction sites of the TRBO expression vector to create a BGIH expression vector for the TRBO transient plant expression system (pTRBO-BGIH).

An *Agrobacterium tumefaciens* strain, GV3101, was used for the transient expression of BGIH in tobacco leaves by the FECT and TRBO expression systems. To make competent GV3101 cells the following procedure was performed: an overnight culture of GV3101 was used to inoculate 200 mL Luria-Bertani (LB) medium. The cells were then allowed to grow to log phase with OD₆₀₀ between 0.5 and 0.8. Then the cells were pelleted by centrifugation at 5000 rpm for 10 minutes at 4° C. The cells were then washed once with 10 mL prechilled TE buffer (Tris-HCl 10 mM, EDTA 1 mM, pH 8.0), and then resuspended into 20 mL LB medium. The GV3101 cell resuspension was then aliquoted in 250 µL fractions into 1.5 mL microtubes. The aliquots were then snap-frozen in liquid nitrogen and stored at -80° C. freezer for future transformation.

The pFECT-BGIH and pTRBO-BGIH vectors were then transformed into the competent GV3101 cells using a freeze-thaw method as follows: the stored competent GV3101 cells were thawed on ice and then mixed with 1-5 µg pure DNA (pFECT-BGIH or pTRBO-BGIH vector). The cell-DNA mixture was then kept on ice for 5 minutes, then transferred to -80° C. for 5 minutes, and then incubated in a 37° C. water bath for 5 minutes. The freeze-thaw treated cells were then diluted into 1 mL LB medium and shaken on a rocking table for 2-4 hours at room temperature. A 200 µL aliquot of the cell-DNA mixture was then spread onto LB agar plates with the appropriate antibiotics (10 µg/mL rifampicin, 25 µg/mL gentamycin, and 50 µg/mL kanamycin were

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used for both pFECT-BGIH transformation and pTRBO-BGIH transformation) and incubated at 28° C. for two days. Resulting transformant colonies were then picked and culture in 6 mL aliquots of LB medium with the appropriate antibiotics for transformed DNA analysis and making glycerol stocks of the transformed GV3101 cells.

The transient transformation of tobacco leaves was performed using leaf injection with a 3 mL syringe without needle. The transformed GV3101 cells were streaked onto an LB plate with the appropriate antibiotics (as described above) and incubated at 28° C. for two days. A colony of transformed GV3101 cells was inoculated to 5 mL of LB-MESA medium (LB media supplemented with 10 mM MES, 20 µM acetosyringone) and the same antibiotics described above, and grown overnight at 28° C. The cells of the overnight culture were collected by centrifugation at 5000 rpm for 10 minutes and resuspended in the induction medium (10 mM MES, 10 mM MgCl₂, 100 µM acetosyringone) at a final OD₆₀₀ of 1.0. The cells were then incubated in the induction medium for 2 hour to overnight at room temperature and were then ready for transient transformation of tobacco leaves. The treated cells were infiltrated into the underside of attached leaves of *Nicotiana benthamiana* plants by injection, using a 3 mL syringe without a needle attached. For the FECT transient transformation, the pFECT-BGIH transformed GV3101 cells and pFECT-P19 transformed GV3101 cells were mixed together in equal amounts for infiltration of tobacco leaves by injection with a 3 mL syringe. For the TRBO transient transformation, only pTRBO-BGIH transformed GV3101 cells were infiltrated into tobacco leaves. The ICK motif protein expression in tobacco leaves was evaluated at 6-8 days post-infiltration.

The BGIH expression ORF contains a fusion protein of GFP (STA) and U-ACTX-Hv1a (ICK) with an IGER (SEQ ID NO: 1) linker peptide (LINKER) between them. As shown in FIG. 3, the green fluorescence of the expressed GFP portion of the transgenes was detected under U.V. light in tobacco leaves transformed with both the FECT and TRBO vectors. Interestingly, green fluorescence appeared evenly distributed in the FECT vector transformed tobacco leaves (with the exception of the vascular tissues), whereas green fluorescence in the TRBO vector transformed tobacco leaves appeared to accumulate in the vascular tissues which is due to TRBO retaining its viral movement protein and FECT not.

To quantitatively evaluate the ICK motif protein expression, the expressed proteins in the transformed tobacco leaves were extracted by following the procedure described here. 100 mg disks of transformed leaf tissue were collected by punching leaves with the large opening of a 1000 µL pipette tip. The collected leaf tissue was placed into a 2 mL microtube with 5/32" diameter stainless steel grinding balls, and frozen in -80° C. for 1 hour, and then homogenized using a Troemner-Talboys High Throughput Homogenizer. 750 µL ice-cold TSP-SE1 extraction solutions (sodium phosphate solution 50 mM, 1:100 diluted protease inhibitor cocktail, EDTA 1 mM, DIECA 10 mM, PVPP 8%, pH 7.0) was added into the tube and vortexed. The microtube was then left still at room temperature for 15 minutes and then centrifuged at 16,000 g for 15 minutes at 4° C. 100 µL of the resulting supernatant was taken and loaded into pre-Sephadex G-50-packed column in 0.45 µm Millipore Multi-Screen filter microtiter plate with empty receiving Costar microtiter plate on bottom. The microtiter plates were then centrifuged at 800 g for 2 minutes at 4° C. The resulting

filtrate solution, herein called total soluble protein extract (TSP extract) of the tobacco leaves, was ready for the quantitative analysis.

The total soluble protein concentration of the TSP extract was estimated using Pierce Coomassie Plus protein assay. BSA protein standards with known concentrations were used to generate a protein quantification standard curve. 2 μ L of each TSP extract was mixed into 200 μ L of the chromogenic reagent (CPPA reagent) of the Coomassie Plus protein assay kits and let react for 10 minutes. The chromogenic reaction was then evaluated by reading OD595 using a SpectroMax-M2 plate reader using SoftMax Pro as control software. The concentrations of total soluble proteins were 0.788 ± 0.20 μ g/ μ L and 0.533 ± 0.03 μ g/ μ L in the TSP extract from FECT-BGIH expression leaves and TRBO-BGIH expression leaves respectively. These results were used for the calculation of percentage of the expressed U-ACTX-Hv1a in the TSP (% TSP) in the iELISA assay.

Indirect ELISA (iELISA) assay was performed as follows to quantitatively evaluate the ICK motif protein in the tobacco leaves transiently transformed with the FECT and TRBO expression systems. 5 μ L of the leaf TSP extract was diluted into 95 μ L CB2 solution (Immunochemistry Technologies) in the well of an Immulon 2HD 96-well plate, with serial dilutions performed as necessary. Leaf proteins were from the extract samples were then allowed to coat the well walls for 3 hours in the dark at room temperature, and then the CB2 solution was removed, and each well was washed twice with 200 μ L PBS (Gibco). 150 μ L blocking solution (Block BSA in PBS with 5% non-fat dry milk) was then added into each well and incubated for 1 hour, in the dark, at room temperature. After the removal of the blocking solution and a PBS wash of the wells, 100 μ L of rabbit anti-U-ACTX-Hv1a antibody (primary antibody) (1:250 dilution in blocking solution) was added to each well and incubated for 1 hour in the dark at room temperature. The primary antibody was then removed and each well was washed with PBS 4 times. Then 100 μ L of HRP-conjugated goat anti-rabbit antibody (secondary antibody, used at 1:1000 dilution in the blocking solution) was added into each well and incubated for 1 hour in the dark at room temperature. After removal of the secondary antibody and wash of the wells with PBS, 100 μ L substrate solution (a 1:1 mixture of ABTS peroxidase substrate solution A and solution B, KPL) was added to each well, and the chromogenic reaction was allowed to go until sufficient color development was apparent. Then 100 μ L of peroxidase stop solution was added to each well to stop the reaction. The light absorbance of each reaction mixture in the plate was read at 405 nm using a SpectroMax-M2 plate reader, with SoftMax Pro used as control software. Serially diluted known concentrations of pure U-ACTX-Hv1a samples were treated in the same manner as described above in the iELISA assay to generate a mass-absorbance standard curve for quantities analysis. The expressed U-ACTX-Hv1a was detected by iELISA at 3.09 ± 1.83 ng/ μ L in the leaf TSP extracts from the FECT-BGIH transformed tobacco; and 3.56 ± 0.74 ng/ μ L in the leaf TSP extract from the TRBO-BGIH transformed tobacco. Or the expressed U-ACTX-Hv1a is 0.40% total soluble protein (% TSP) for FECT-BGIH transformants and 0.67% TSP in TRBO-BGIH transformants.

In conclusion, both FECT and TRBO transient plant expression systems can be used to express the ICK motif protein in plant. The ICK motif protein expression level in both systems is very close. However, the expression in the FECT system distributes evenly in the agroinfiltrated leaves,

whereas the expression in the TRBO system accumulates in the vascular tissue of the agroinfiltrated leaves.

Example 2

ICK Motif Protein Transient Expression in Tobacco Leaf with Accumulation at Different Subcellular Targets.

Plant expressed ICK motif protein needs to accumulate to a certain level in the plant to effectively protect the plant from insect damage. The accumulation level of the plant expressed ICK motif protein may be affected by its final localization in the plant cells. In this example, we investigated the effects of different subcellular localizations of the plant expressed ICK motif protein on the protein's accumulation level in the plant (using the FECT transient plant expression system). Three subcellular targets were investigated in this example, plant cell wall apoplast (APO), the endoplasmic reticulum (ER) and the cytoplasm (CYTO).

The APO targeted ICK motif protein expression ORF was designed to encode a series of translationally fused structural motifs that can be described as follows: N'-ERSP-Sta-L-ICK-C'. Again the ICK motif protein in this study was U-ACTX-Hv1a, and the BGIH expression ORF in the example 1 was used. The same vector as in the example 1, pFECT-BGIH, was used here.

The CYTO targeted ICK motif protein expression ORF was designed to encode a series of translationally fused structural motifs that can be described as follows: N'-Sta-L-ICK-C'. In this study, the DNA sequence encoding the barley α -amylase signal peptide was removed from the BGIH expression ORF and became the GIH expression ORF, whose open reading frame sequence is below:

(SEQ ID NO: 15)

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ATGGCTAGCAAAGGAGAAGAACTTTCTACTGGAGTTGTCCCAATCTCTGT
TGAATTAGATGGTGATGTTAATGGGCACAAATTTCTGTCTAGTGAGAGG
GTGAAGGTGATGCTACATACGGAAGCTTACCCTTAAATTTATTTGCACT
ACTGGAAAACTACCTGTTCCTATGGCCAACACTTGCTACTACTTTCTCTTA
TGGTGTTCAATGCTTTTCCCGTTATCCGGATCATATGAAACGGCATGACT
TTTTCAAGAGTGCCATGCCCCGAAGTTATGTACAGGAACGCACTATATCT
TTCAAAGATGACGGGAACATAAGACGCGTGCTGAAGTCAAGTTTGAAGG
TGATACCCTTGTTAATCGTATCGAGTTAAAGGTATTGATTTTAAAGAA
ATGGAAACATTCTCGGACACAACTCGAGTACAACATAACTCACACAAT
GTATACATCACGGCAGACAAACAAAGAATGGAATCAAAGCTAACTTCAA
AATTGCCACACAACTGAAGATGGATCCGTTCACTAGCAGACCATTATC
AACAAAATACTCCAATTGGCGATGGCCCTGTCTTTTACCAGACAACCAT
TACCTGTGACACAACTCTGCCTTTTCAAAGATCCCAACGAAAAGCGTGA
CCACATGGTCTCTTGTAGTTTGTAACTGTCTGCTGGGATTACATGGCA
TGGATGAGCTCTACAAATTTGGTGAAGACAATATTGTGTTCCAGTTGAT
CAACCATGTTCTCTTAATACTCAACCATGTTGTGATGATGCTACTTGTAC
TCAAGAAAGAAATGAAATGGACATACTGTTTATTATTGTAGAGCTTAA

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The GIH expression ORF was cloned into the Pac I and Avr II restriction sites of the FECT expression vector to create a GIH expression vector for FECT transient plant expression system (pFECT-GIH) for the CYTO targeting expression of U-ACTX-Hv1a.

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The ER targeted ICK motif protein expression ORF was designed by adding a DNA sequence encoding the ER targeting signal peptide at the C' end of the BGIH expression ORF which was named as BGIH-ER expression ORF. The ER targeting signal peptide used here has the following amino acid sequence (one letter code for amino acid):

KDEL

(SEQ ID NO: 16)

The DNA sequence of the BGIH-ER expression ORF is as follows:

(SEQ ID NO: 17)

ATGGCTAATAAACACCTGAGTTTGTCACTATTCCTCGTGTGCTCGGGTT
 ATCTGCTTCACTTGCAAGCGGAGCTAGCAAAGGAGAAGAACTTTTCACTG
 GAGTTGTCCCAATTCTTGTGAATTAGATGGTGTATTAATGGGCACAAA
 TTTTCTGTGAGGAGAGGGTGAAGGTGATGCTACATACGGAAGCTTAC
 CCTTAAATTTATTTGCACTACTGGAAGAACTACCTGTTCCATGGCCAAACAC
 TTGTCACTACTTTCTCTTATGGTGTCAATGCTTTTCCGTTATCCGGAT
 CATATGAACCGCATGACTTTTCAAGAGTGCCATGCCGCAAGGTATGT
 ACAGGAACGCCTATATCTTTCAAGATGACGGGAACACAGACGCGTG
 CTGAAGTCAAGTTTGAAGGTGATACCTTGTTAATCGTATCGAGTTAAAA
 GGTATTGATTTTAAAGAAGATGAAACATTCTCGGACACAACTCGAGTA
 CAACTATAACTCACACAATGTATACATCACGGCAGACAAACAAAAGATG
 GAATCAAAGCTAACTTCAAAATTCGCCACAACATTGAAGATGGATCCGTT
 CAACTAGCAGACCATTATCAACAAAATACTCCAATTGGCGATGGCCCTGT
 CCTTTTACCAGACAACCATTTACCTGTGACACAATCTGCCCTTTTGAAAG
 ATCCCAACGAAAGCGTGACCACATGGTCTTCTTGAGTTTGTAAGTCT
 GCTGGGATTACACATGGCATGGATGAGCTCTACAAAATTGGTGAAAGACA
 ATATTGTGTTCCAGTTGATCAACCATGTTCTATTAATACTCAACCATGTT
 GTGATGATGCTACTTGTACTCAAGAAAGAAATGAAATGGACATACTGTT
 TATTATTGTAGAGCTAAAGATGAGCTCTAA

The BGIH-ER expression ORF was cloned into the Pac I and Avr II restriction sites of the FECT expression vector to create a BGIH-ER expression vector for FECT transient plant expression system (pFECT-BGIH-ER), for ER-targeted expression of U-ACTX-Hv1a.

All three vectors, pFECT-BGIH, pFECT-GIH and pFECT-BGIH-ER, were transformed into the *Agrobacterium* strain, GV3101, and the resulting transformed GV3101 cells were used for transient transformation into the leaves of *Nicotiana benthamiana* using the methods described in Example 1. All of the three expression ORF's should transiently express a fusion protein, comprising GFP-fused U-ACTX-Hv1a with a trypsin cleavable linker between the two structural domains.

After 6 days of transient tobacco transformation, the expression of GFP-fused U-ACTX-Hv1a was examined initially by detection of green fluorescence under UV light. Green fluorescence was detected at various levels in all of the transformed tobacco leaves. The transformed leaves with CYTO targeted accumulation of GFP fused U-ACTX-Hv1a showed the strongest green fluorescence, and those leaves with APO or ER targeted fusion protein accumulation showed weaker green fluorescence. Thus, the results indicated that CYTO targeted expression may facilitate greater

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accumulation of transgenic GFP fused U-ACTX-Hv1a protein than the APO and ER targeted expression in tobacco leaves. In three replications of this experiment, the transformed tobacco leaves with CYTO targeted expression always showed green fluorescence similar to or stronger than that of the leaves with APO targeted expression, and the weakest green fluorescence was detected in the tobacco leaves transformed with the ER targeted constructs. These initial results indicated that CYTO targeted expression may accumulate as much or more transgenic fusion protein than APO targeted expression, and that ER targeted expression yielded the least accumulation.

Total soluble protein samples were extracted from tobacco leaves transformed with the different FECT vectors (protocol was described in detail in Example 1). Pierce Coomassie Plus protein assay was performed as in the description in Example 1 to determine the concentrations of the total soluble protein in the TSP extracts, yielding the following concentration estimates: 0.31 ± 0.04 $\mu\text{g}/\mu\text{L}$, 0.31 ± 0.03 $\mu\text{g}/\mu\text{L}$ and 0.34 ± 0.05 $\mu\text{g}/\mu\text{L}$ for APO targeted, CYTO targeted and ER targeted expressions respectively (N=3).

The indirect ELISA protocol was then performed using the TSP extracts as described in Example 1 to quantitate the expression level of the U-ACTX-Hv1a protein as a percentage of total soluble protein (% TSP), yielding the following percentage estimates: $0.126 \pm 0.032\%$, $0.049 \pm 0.085\%$ and $0.025 \pm 0.018\%$ for APO targeted, CYTO targeted and ER targeted expressions respectively (N=3). FIG. 8 summarizes this quantification of expressed U-ACTX-Hv1a (as % TSP values) for the various transformed tobacco leaves described above. These results indicated that APO targeted transgene expression resulted in the greatest accumulation of correctly folded ICK motif protein expressed in the leaves.

Overall, although the tobacco leaves transformed to produce CYTO targeted, transgenic GFP fused U-ACTX-Hv1a presented the most potent green fluorescence signal, iELISA results detected the least U-ACTX-Hv1a peptide in these transgenic tobacco leaves, in fact, considerably less than what was detected for leaves transformed for ER targeted expression (which had weakest green fluorescence signal). In iELISA assays, the primary antibody (rabbit anti-U-ACTX-Hv1a antibody) can only bind on the correctly folded U-ACTX-Hv1a peptide.

Example 3

Alternate Signal Peptides for Expression of ICK Motif Proteins in Plants.

Since ER signal peptide may play a role in the protein expression level, two other ERSPs were tested using the FECT expression system described in the prior examples. The two ERSP candidates were tobacco extensin signal peptide, abbreviated as "E" in this study (Memelink et al, the Plant Journal, 1993, V4: 1011-1022), and one of its variants abbreviated as "E*" (Pogue G P et al, Plant Biotechnology Journal, 2010, V8: 638-654). Their amino acid sequences are listed below (N' to C', one letter code, with non-identical residues in bold font):

Extensin signal peptide

(SEQ ID NO: 18)

(EMGKMASL**FASLL**VVLVLSLSLASESSA

Extensin signal peptide variant (E*):

(SEQ ID NO: 19)

MGKMASL**FATFL**VVLVLSLSLASESSA

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A DNA sequence encoding E was designed for tobacco expression as follows:

(SEQ ID NO: 20)
 ATGGGTAAGATGGCTTCTCTGTTGCTTCTCTGCTGGTTGTTCTGGTTTC
 TCTGTCTCTGGCTTCTGAATCTTCTGCT

The E DNA sequence was generated using oligo extension PCR with four synthetic DNA primers. Then, in order to add a Pac I restriction site at its 5' terminus and add part of 5' terminal DNA sequence of GFP at its 3' terminus, a further PCR was performed using the E DNA sequence as a template, yielding a 117 bp DNA fragment. This fragment was then used as the forward PCR primer to amplify the DNA sequence encoding the GFP-IGER linker-U-ACTX-Hv1a ORF from the vector pFECT-BGIH (refer to Example 1 and Example 2), thus producing a U-ACTX-Hv1a expression ORF encoding (from N' to C' terminus) extensin signal peptide-GFP-IGER linker-U-ACTX-Hv1a, following one of our ICK motif protein expression ORF design as ERSP-StaL-ICK. This expression ORF, named "EGIH", has a Pac I restriction site at its 5' terminus and Avr II restriction site at the 3' terminus. EGIH has the following DNA sequence:

(SEQ ID NO: 21)
 TTAATTAATGGGTAAGATGGCTTCTCTGTTGCTTCTCTGCTGGTTGTT
 CTGTTTCTCTGTCTCTGGCTTCTGAATCTTCTGCTGCTAGCAAAGGAGA
 AGAACTTTTCACTGGAGTTGTCCCAATTCTTGTGAATTAGATGGTGATG
 TTAATGGGCACAAATTTTCTGTCTGAGAGGGTGAAGGTGATGCTACA
 TACGGAAGCTTACCCCTAAATTTATTGCACTACTGGAAACTACCTGT
 TCCATGGCCAACTGTGCACTACTTTCTCTTATGGTGTTCATGCTTTT
 CCCGTTATCCGATCATATGAAACGGCATGACTTTTCAAGAGTGCCATG
 CCCGAAGGTTATGTACAGGAACGCACTATATCTTCAAGATGACGGGAA
 CTACAAGACGCGTGCTGAAGTCAAGTTTGAAGGTGATACCCCTGTTAATC
 GTATCGAGTTAAAGGTATTGATTTTAAAGAAGATGGAACATTCTCGGA
 CACAACTCGAGTACAATACTACACAAATGTATACATCACGGCAGA
 CAAACAAAAGAAATGAATCAAAGCTAACTTCAAATTCGCCACAACATTG
 AAGATGGATCCGTTCACTAGCAGACCATTTCAACAAAATACTCCAATT
 GGCGATGGCCCTGTCTTTTACCAGACAACCATTTACCTGTGACACAATC
 TGCCCTTTTCAAAGATCCCAACGAAAAGCGTGACCACATGGTCTCTTTG
 AGTTTGTAACGTCTGCTGGGATTACACATGGCATGGATGAGCTCTACAAA
 ATTGGTGAAAGACAATATTGTGTCCAGTTGATCAACCATGTTCTCTTAA
 TACTCAACCATGTTGTGATGATGCTACTGTACTCAAGAAAGAAATGAAA
 ATGGACATACTGTTTATTATTGTAGAGCTTAACCTAGG

The EGIH DNA sequence was cloned into Pac I and Avr II restriction sites of the FECT vector to generate the pFECT-EGIH vector for transient plant expression of GFP fused U-ACTX-Hv1a protein.

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A DNA sequence encoding the variant extensin signal peptide (E*) was designed for tobacco expression as follows:

(SEQ ID NO: 22)
 ATGGGTAAGATGGCTTCTCTGTTGCTACTTTTCTGGTTGTTCTGGTTTC
 TCTGTCTCTGGCTTCTGAATCTTCTGCT

10 An "E*GIH" DNA sequence, which encoded a translational fusion of (listed from N'to C') variant extensin signal peptide-GFP-IGER linker-U-ACTX-Hv1a protein, was created using the same techniques as described above for the EGIH ORF. The resulting E*GIH ORF has the following
 15 DNA sequence:

(SEQ ID NO: 23)
 TTAATTAATGGGTAAGATGGCTTCTCTGTTGCTACTTTTCTGGTTGTT
 20 CTGTTTCTCTGTCTCTGGCTTCTGAATCTTCTGCTGCTAGCAAAGGAGA
 AGAACTTTTCACTGGAGTTGTCCCAATTCTTGTGAATTAGATGGTGATG
 TTAATGGGCACAAATTTTCTGTCTGAGAGGGTGAAGGTGATGCTACA
 25 TACGGAAGCTTACCCCTAAATTTATTGCACTACTGGAAACTACCTGT
 TCCATGGCCAACTGTGCACTACTTTCTCTTATGGTGTTCATGCTTTT
 CCCGTTATCCGATCATATGAAACGGCATGACTTTTCAAGAGTGCCATG
 30 CCCGAAGGTTATGTACAGGAACGCACTATATCTTCAAGATGACGGGAA
 CTACAAGACGCGTGCTGAAGTCAAGTTTGAAGGTGATACCCCTGTTAATC
 GTATCGAGTTAAAGGTATTGATTTTAAAGAAGATGGAACATTCTCGGA
 35 CACAACTCGAGTACAATACTACACAAATGTATACATCACGGCAGA
 CAAACAAAAGAAATGAATCAAAGCTAACTTCAAATTCGCCACAACATTG
 AAGATGGATCCGTTCACTAGCAGACCATTTCAACAAAATACTCCAATT
 GGCGATGGCCCTGTCTTTTACCAGACAACCATTTACCTGTGACACAATC
 40 TGCCCTTTTCAAAGATCCCAACGAAAAGCGTGACCACATGGTCTCTTTG
 AGTTTGTAACGTCTGCTGGGATTACACATGGCATGGATGAGCTCTACAAA
 ATTGGTGAAAGACAATATTGTGTCCAGTTGATCAACCATGTTCTCTTAA
 45 TACTCAACCATGTTGTGATGATGCTACTGTACTCAAGAAAGAAATGAAA
 ATGGACATACTGTTTATTATTGTAGAGCTTAACCTAGG

The E*GIH DNA sequence was cloned into Pac I and Avr II restriction sites of the FECT vector to generate the pFECT-E*GIH vector for transient plant expression of GFP fused U-ACTX-Hv1a protein.

Three different FECT expression vectors, pFECT-BGIH, pFECT-EGIH and pFECT-E*GIH, were used to transiently express GFP fused U-ACTX-Hv1a protein in tobacco plants to evaluate how the protein expression level is affected by the different ERSPs. The three FECT expression vectors were transformed into Agrobacterium, GV3101, and then the transformed GV3101 was injected into tobacco leaves for transient expression of GFP fused U-ACTX-Hv1a protein in tobacco leaves using the techniques described in Example 1.

The expression levels of GFP fused U-ACTX-Hv1a from three different FECT expression vectors described above are first evaluated visually by detecting green fluorescence under UV light. Green fluorescence from the transiently transformed tobacco leaves from the three different FECT

vectors is visible to the naked eye. All of the leaves showed similar levels of green fluorescence, suggesting that none of the three ERSPs tested contributed to a significant increase in the expression level of GFP fused U-ACTX-Hv1a protein.

Total soluble protein samples were extracted from the tobacco leaves transformed with the three ERSP FECT vectors as described above (protocol is described in detail in Example 1). Pierce Coomassie Plus protein assay was then performed (as described in Example 1) to determine the concentration of the total soluble protein in the resulting TSP samples, yielding values of 0.85 ± 0.68 $\mu\text{g}/\mu\text{L}$, 0.70 ± 0.47 $\mu\text{g}/\mu\text{L}$ and 0.76 ± 0.77 $\mu\text{g}/\mu\text{L}$ for samples corresponding to the BGIH, EGIH and E*GIH expression ORFs respectively (N=4).

Indirect ELISA was then performed using the TSP extracts (as described in Example 1) to quantify the expression level of the U-ACTX-Hv1a protein as a percentage of the total soluble protein (% TSP), yielding values of $0.39 \pm 0.17\%$ (N=3, as one data point was taken out as outlier), $0.48 \pm 0.26\%$ (N=4), and $0.62 \pm 0.38\%$ (N=4) for samples corresponding to the FECT vectors with BGIH, EGIH and E*GIH expression ORFs respectively. FIG. 9 summarizes the estimated U-ACTX-Hv1a levels as percentage in the total soluble protein (% TSP) for all of the samples taken from the tobacco leaves transformed with the three ERSP ORF described above. Although the data of % TSP from three FECT vector transformation looked different, they are not statistically different by Student's t-test. In other words, the three ERSPs did not make difference in the expression level of U-ACTX-Hv1a in the transiently transformed tobacco leaves.

Example 4

Stabilizing Protein Expressed as Fusion Protein to the ICK Motif Protein Helps the Accumulation of ICK Motif Protein in Transformed Plants.

The ICK motif protein for plant expression in this example was omega-ACTX-Hv1a, originating from the Australian Blue Mountains Funnel Web Spider, *Hadronyche versuta*. Omega-ACTX-Hv1a has the following amino acid sequence (one letter code):

(SEQ ID NO: 24)
SPTCIPSGQPCPYNENCCSQSCTFKENENGNTVKRCD

The FECT expression system was used to express omega-ACTX-Hv1a in the tobacco plant, *Nicotiana benthamiana*. Two FECT vectors encoding different omega-ACTX-Hv1a expression ORFs were engineered. One of these expression ORFs encoded omega-ACTX-Hv1a with Barley Alpha-Amylase Signal peptide (BAAS) at its N' terminus without any stabilizing protein. This expression ORF, referred to herein as "BO", was subcloned to yield the FECT expression vector pFECT-BO. The other omega-ACTX-Hv1a expression ORF encodes a translational fusion of omega-ACTX-Hv1a to the protein Jun a 3. The mature Jun a 3 is a ~30 kDa plant defending protein which is also an allergen for some people, is produced by *Juniperus ashei* trees and is used in this ORF as a translational stabilizing protein (STA.) Its amino acid sequence is listed below (one letter code):

(SEQ ID NO: 25)
MARVSELAFLLAATLAI SLHMQEAGVVKFDIKNQC GTYTVWAAGLPGGG
KRLDQGQTWTVNLAAGTASARFWGRTGCTPDASGKGCQTGDCGGQLS
CTVSGAVPATLAEYTSQSDQDYDVSLVDGFNIPLAINPTNAQCTAPAC
KADINAVCPSELKVDGGCNSACNVFKTDQVCCRNNAVVDNCPATNYSKI
FKNQCQAYSAYAKDDTATFACASGTDYSIVFC

The mature Jun a 3 protein is provided below in SEQ ID NO: 26.

(SEQ ID NO: 26)
KFDIKNQC GTYTVWAAGLPGGGKRLDQGQTWTVNLAAGTASARFWGRTGCT
PDASGKGCQTGDCGGQLSCTVSGAVPATLAEYTSQSDQDYDVSLVDGFN
IPLAINPTNAQCTAPACKADINAVCPSELKVDGGCNSACNVFKTDQVCCR
NAYVDNCPATNYSKIFKNQCQAYSAYAKDDTATFACASGTDYSIVFC

The ERSP encoded in the ORF of SEQ. ID. 25 is the Jun a 3 native signal peptide shown below as SEQ. ID 27.

SEQ. ID. 27
MARVSELAFLLAATLAI SLHMQEAGVV

The IGER linker, encoded by the sequence between the omega-ACTX-Hv1a domain and Jun a 3 domains that are encoded in the ORF, is described in detail in Example 1. Taken together, this omega-ACTX-Hv1a expression ORF is referred to as S-Juna3-IGER-Omega, or SJIO. Likewise, the FECT vector into which the SJIO expression ORF was inserted was named pFECT-SJIO.

The two omega-ACTX-Hv1a FECT expression vectors, pFECT-BO and pFECT-SJIO, were used to transiently express omega-ACTX-Hv1a protein in tobacco plants. The two FECT expression vectors were transformed into Agrobacterium strain GV3101, and the resulting GV3101 transformant was injected into tobacco leaves for transient expression of omega-ACTX-Hv1a in tobacco leaves using the techniques described in detail in Example 1.

At day 6 post-tobacco transformation, transformed tobacco leaves were collected and total soluble leaf proteins were extracted from the leaves (refer to Example 1 for detailed methods). Pierce Coomassie Plus protein assay was then performed to determine the concentrations of the total soluble leaf protein, yielding values of 3.047 ± 0.176 $\mu\text{g}/\mu\text{L}$ (N=2) and 2.473 ± 0.209 $\mu\text{g}/\mu\text{L}$ (N=2) for the leaves transformed with constructs encoding pFECT-SJIO and pFECT-BO respectively.

The indirect ELISA protocol was then performed using the TSP extracts above as described in Example 1 to quantitatively evaluate the expression level of the omega-ACTX-Hv1a protein as percentage of the total soluble protein (% TSP), yielding values of $0.133 \pm 0.014\%$ (N=2) and $0.0004 \pm 0.0003\%$ (N=2) for the leaves transformed with the pFECT-SJIO and pFECT-BO vectors respectively. These data indicated that omega-ACTX-Hv1a expressed as a translational fusion to Jun a 3 accumulated to a more than 300-fold higher steady state level than that of omega-ACTX-Hv1a expressed without translational fusion to the Jun a 3 protein.

The example 4 above, the function of the STA could also have been performed with snowdrop lectin (GNA) having the following sequence:

(SEQ ID NO: 28)
 DNILYSGETLSTGEFLNYGSFVFIMQEDCNLVLYDVKPIWATNTGGLSR
 SCFLSMQTDGNLVVYNPSNKPWASNTGGQNGNYVCILQKDRNVVIYGTD
 RWATG

Example 5

A Cleavable Linker Between the Stabilizing Protein Domain and the ICK Motif Protein Domains in an ICK Motif Fusion Protein Expression ORF Enhances the Insecticidal Activity of the Resulting ICK Motif Protein Expressed in a Transgenic Plant.

Since most chewing insects secrete trypsin into their guts to digest food, we designed a fusion protein expression ORF that encoded a trypsin cleavable linker between the stabilizing protein domain and the ICK motif protein domain of the fusion, in order to facilitate release of the ICK motif domain from the intact fusion protein in the insect gut.

The ICK motif protein for plant expression here was omega-ACTX-Hv1a, whose amino acid sequence is as follows (one letter code):

(SEQ ID NO: 24)
 SPTCIPSGQPCPYNNCCSQSCTFKENENGNTVKRCD

The omega-ACTX-Hv1a expression ORF that was used encodes a fusion protein comprising the following domains (N' to C'): Jun a 3 signal peptide:: Jun a 3:: IGER linker:: omega-ACTX-Hv1a, as in the structural formula ERSP-Sta-L-ICK described above. The origin and sequence of Jun a 3 is as described above in Example 4.

The ERSP used here was the Jun a 3 native signal peptide, as described above in Example 4.

The IGER linker, encoded by the sequence between the omega-ACTX-Hv1a domain and Jun a 3 domains that are encoded in the ORF, is described in detail in Example 1. Taken together, this omega-ACTX-Hv1a expression ORF is referred to as S-Juna3-IGER-Omega, or SJIO. Likewise, the FECT vector into which the SJIO expression ORF was inserted was named pFECT-SJIO.

The vector, pFECT-SJIO, was then used to transiently express omega-ACTX-Hv1a protein in tobacco plants. The vector was transformed into *Agrobacterium*, GV3101, and then the transformed GV3101 was injected into tobacco leaves for transient expression of omega-ACTX-Hv1a in the leaves using the techniques described in detail in Example 1.

On day 6 post tobacco leaf transformation, 3.3 g of transformed tobacco leaf was collected and ground in liquid nitrogen. 50 mL of TSP-Se1 buffer was used to extract the total soluble proteins (TSP) from the ground leaves by following the procedure described in Example 1. A total of 26 mL extract was recovered from the TSP extraction procedure, which was then evenly split into two samples, A and B, with 13 mL extract for each group. Sample A was treated with trypsin to release omega-ACTX-Hv1a from the fused Jun a 3 protein by adding 1.3 mL of 1 mg/mL trypsin in 1 mM HCl at 37° C. for 1 hour. Sample B was not treated by trypsin cleavage. To get omega-ACTX-Hv1a in the concentration range of bioactivity, both groups were concentrated in the same way as following. First, the extractions

were loaded into a concentrator with 10 kD cutoff filter membrane and spun at 3200 g for 2 hours. Then 1.4 mL retentate from Sample A and 1.1 mL retentate from Sample B were saved for later tests. The 12.5 mL filtrate from Sample A and 12.5 mL filtrate from Sample B were further concentrated by being spun in concentrators with 1 kD cutoff filter membranes at 3200 g for 16 hours. 1.3 mL retentate was recovered from Sample A and 1.1 mL retentate was recovered from Sample B. Both 1 kD cutoff filtration retentates were saved for later tests. This sample concentration procedure was summarized in FIG. 10. The total TSP extraction from pFECT-SJIO transformed tobacco leaves was split evenly to two samples. One sample (A) was treated by trypsin cleavage and the other (B) was not. Both groups were concentrated by being spun in the concentrators with 10 kD and then 1 kD cutoff filter membranes, and the retentates from the 10 kD and 1 kD cutoff filtration were saved for further tests.

The SJIO expression ORF expressed a fusion protein as following, Jun a 3::IGER::Omega-ACTX-Hv1a, which comprises a total of 266 amino acid residues and has a predicted molecular weight of 28,204.28 Da. The trypsin cleavage of this fusion protein should release an omega-ACTX-Hv1a with molecular weight of 4049.2 Da and Jun a 3::IGER fusion protein with molecular weight of 24,155.1 Da. Therefore, if the trypsin cleavage reaction is complete in the treatment, then the anticipated major components of the filtration samples are as follows:

Sample A 10 kD filtration retentate: Jun a 3::IGER fusion.
 Sample A 1 kD filtration retentate: Omega-ACTX-Hv1a.
 Sample B 10 kD filtration retentate: Jun a 3::IGER::Omega-ACTX-Hv1a fusion.
 Sample B 1 kD filtration retentate: no SJIO expressed protein.

To quantify the omega-ACTX-Hv1a peptide in the retentate samples, iELISA was performed as described in Example 1. The detected omega-ACTX-Hv1a concentrations in the samples were as follows:

Sample A 10 kD filtration retentate: 1.328 ng/μL of omega-ACTX-Hv1a, total 1.86 μg.
 Sample A 1 kD filtration retentate: 2.768 ng/μL of omega-ACTX-Hv1a, total 3.60 μg.
 Sample B 10 kD filtration retentate: 12.656 ng/μL of omega-ACTX-Hv1a, total 13.92 μg.
 Sample B 1 kD filtration retentate: 0.752 ng/μL of omega-ACTX-Hv1a, total 0.83 μg.

As indicated, Omega-ACTX-Hv1a was detected in all filtration samples that were analyzed.

The detected omega-ACTX-Hv1a in the Group A 10 kD filtration retentate is presumably due in large part to physical retention of the uncleaved fusion protein. Likewise the omega-ACTX-Hv1a detected in the Group B 1 kD filtration retentate sample could be due to a low rate of spurious filtration of the uncleaved fusion protein through the 10 kD cutoff filter membrane.

To confirm the trypsin-cleavage reaction was successful, reverse phase High Performance Liquid Chromatography (rpHPLC) was performed to analyze the components in the reserved filtration samples. HPLC was performed using a Varian E218 HPLC system with an Onyx 100 monolithic C18 column (4.6×100 mm), using water with 0.1% trifluoroacetic acid (solvent A) and acetonitrile with 0.1% trifluoroacetic acid (solvent B) as mobile phase components. The omega-ACTX-Hv1a peptide was eluted from the column at a flow rate of 2 mL per minute using a linear gradient of 10-20% solvent B over 10 minutes. Samples of 99% pure

synthetic omega-ACTX-Hv1a were used in rpHPLC to produce a standard curve (relating peak area to mass of peptide injected). FIG. 11 shows three separate elution profiles, 11A, 11B, 11C. As shown in FIG. 11A, the omega-ACTX-Hv1a peptide eluted at 6.5 minutes post-injection. When a 500 μ L sample from Group B 1 kD filtration retentate was loaded into the HPLC system, there was no protein peak between 6 and 7 minutes post-injection in the corresponding HPLC chromatograph (FIG. 11B). When a 500 μ L sample from Group A 1 kD filtration retentate was loaded into the HPLC system, there was a peak at retention time of 6.3 minute (see dotted line in FIG. 11) in the corresponding chromatograph, representing omega-ACTX-Hv1a released from the fusion protein by trypsin cleavage (FIG. 11C). The area of this peak corresponded to a concentration of omega-ACTX-Hv1a of between 16-70 ng/ μ L in the Sample A 1 kD filtration retentate (depending on the approach used to integrate the peak).

The reserved filtration samples were used to perform housefly injection bioassays to test the activity of the omega-ACTX-Hv1a in the fusion protein form and in the released form from the fusion protein. Housefly pupae (*Musca domestica*) were purchased from Benzon Research, Inc. and kept at 25° C. in a plastic box with air holes on the box lid and fly food (1:1 ratio sugar and powder milk) and cotton balls soaked in water in the box. On the day after adult housefly emergence, the flies were immobilized using a CO₂ line and then kept immobile using a CO₂ infusion pad. Flies weighing 12-18 mg were selected for the injection bioassay. To perform housefly injection, a microapplicator loaded with a 1 cc glass syringe with a 30 gauge needle, in which the injection solution was loaded, was used to deliver 0.5 μ L doses into the dorsal thorax of the flies. The injected flies were then put into labeled boxes with air holes, and mortality was scored 24 hours post-injection. The following samples were injected into houseflies (groups of 10 flies were used for each sample):

Water injection as negative control.

Group A 10 kD filtration retentate.

Group A 1 kD filtration retentate.

Group B 10 kD filtration retentate.

Group B 1 kD filtration retentate.

0.13 mg/mL trypsin solution as negative control.

At 24 hrs. post injection, the Sample A 10 kD filtration retentate and Sample A 1 kD filtration retentate caused 100% housefly mortality, while 0% mortality was observed for the flies injected with the other samples. Pure, native sequence omega-ACTX-Hv1a showed an LD₅₀ of 100 pmol/gram of housefly in this housefly injection bioassay; hence, to generate 100% mortality in this paradigm, the concentration of the injected omega-ACTX-Hv1a must at least 25 ng/ μ L. This is consistent with the bioassay results, since HPLC analysis of the Sample A 1 kD filtration retentate indicated a concentration of concentration of omega-ACTX-Hv1a of 16-70 ng/ μ L. Filtration samples that did not comprise material that was treated with trypsin cleavage did not generate mortality in the housefly injection bioassay, indicating that the Jun a 3 fused omega-ACTX-Hv1a was considerably less active than native-sequence omega-ACTX-Hv1a cleaved away from the fusion construct by trypsin. Therefore, the linker region of a plant ICK motif protein expression ORF can show enhanced insecticidal function when designed to be cleavable, such that the ICK motif domain of the ICK fusion protein can be released from the other structural domains of the protein by proteolysis.

Part I. High Production Peptides

The ability to successfully produce insecticidal peptides on a commercial scale, with reproducible peptide formation and folding, and with cost controls can be challenging. The wide variety, unique properties and special nature of peptides, combined with the huge variety of possible production techniques can present an overwhelming number of approaches to peptide production.

There are few if any descriptions, however, that describe how to change a peptide so that it will be produced in a biological system at a much higher rate of production than the peptide is typically produced before it is changed. Here we present a way to change the composition of a peptide and in so doing increase the rate and amount and simultaneously lower the cost of peptide production. We describe novel ways of changing or "converting" one peptide into a different, more cost effective peptide, yet one which surprisingly is just as toxic as before it was converted.

We describe examples of these novel converted peptides, and we show how these methods for altering or converting a peptide can make a significant improvement in the yield of peptides without making significant changes in its activity. The new processes, new peptides, new formulations, and new organisms for producing those peptides are described and claimed herein. A process is described which increases the insecticidal peptide production yield from yeast expression systems by adding a dipeptide at the N terminus of insecticidal peptides. The addition of a dipeptide does not adversely affect the insecticidal activities of insecticidal peptides.

We describe examples of these novel converted peptides, and we show how these methods for altering or converting a peptide can make a significant improvement in the yield of peptides without making significant changes in its activity. The new processes, new peptides, new formulations, and new organisms for producing those peptides are described and claimed herein.

Detailed procedures for making high production peptides.

We describe a process and peptide that can increase peptide production. When followed these techniques will provide a converted peptide by adding a dipeptide at the N-terminus of the native peptide that has better production rate than the native peptide in three different ways. First, the over-all average yield of the dipeptide-native peptide strains is better than that of the native strains; second, the median yield of the dipeptide-native peptide strains is better than that of the native; and third, there are more dipeptide strains at the higher yield range than there are for native peptide strains. The process described here can be used in various in vivo systems, including plants, animals and microbes. The invention requires the addition of a dipeptide to the N-terminus of the native peptide, which is the peptide that was known before the dipeptide is added. The known peptide is then "converted", and it can then be made with greater yields than were previously thought possible. In one embodiment insecticidal peptides are linked to a dipeptide. These dipeptide-native peptide systems can be used in plants that can produce the peptides. Plant produced peptides have a variety of uses from production to simply making a toxic peptide available for consumption by a damaging insect, thus either protecting the plants or possibly providing other benefits.

In one embodiment we describe a process for increasing insecticidal peptide production yield in yeast expression systems by the addition of any dipeptide to the N-terminus of the insecticidal peptide. The dipeptide is composed of a non-polar amino acid and a polar amino acid. The non-polar amino acid may be selected from glycine, alanine, proline, valine, leucine, isoleucine, phenylalanine and methionine.

Glycine is the preferred non-polar amino acid. The polar amino acid may be selected from serine, threonine, cysteine, asparagine, glutamine, histidine, tryptophan and tyrosine. Serine is the preferred polar amino acid. The process and amino acids are described where the non-polar amino acid is at the N-terminus of the dipeptide and in one embodiment the preferred N-terminus of the dipeptide is glycine. The process and amino acids are described where the polar amino acid is at the C-terminus of the dipeptide and in one embodiment the preferred C-terminus of the dipeptide is serine.

In one embodiment of the invention the dipeptide is glycine-serine, gly-ser or GS. These amino acids are typically encoded by the following codons: Gly may be encoded by codons such as GGT, GGC, GGA, GGG and Ser may be encoded by codons such as TCT, TCC, TCA, TCG, AGT, and AGC.

The transgenes of the insecticidal peptides are designed such that their transgene sequences are optimized for the specific expression that may be needed. For example, the transgenes of insecticidal peptides may be optimized for expression in yeast, plants, bacteria, and viruses. Examples of such uses of the invention would include the engineering and optimization of transgenes for crops like maize and soybean, with the purpose of protecting them from insect pests. In one example we design transgenes of insecticidal peptides such that their transgene sequences are optimized for the specific expression in yeast expression systems, using for example, *Kluyveromyces lactis*, *Pichia pastoris*, and *Saccharomyces cerevisiae*. Other suitable yeast expression systems are known in the art. The nucleotide codons for a dipeptide, such as glycine-serine, (gly-ser) are added to the 5' end of the transgene sequences of the mature insecticidal peptides. The transgene sequences are then ligated into appropriate expression vectors, which can provide appropriate selection markers, strong promoter-terminator sets for the specific yeast expression system, signal sequences for secretion, and cleavage sites between the respective signal sequences and mature peptide sequences. The insecticidal peptide expression vectors are then transformed into yeast cells, by means known to one skilled in the art, including either electroporation or chemical transformation methods, in order to generate stable peptide expression yeast strains. When these yeast strains grow in appropriate media, they produce insecticidal peptides modified by the addition of a dipeptide sequence, glycine-serine, to the N-terminus of the mature insecticidal peptides, which are secreted into the growth media. The addition of the dipeptide, glycine-serine, to the N-terminus of the mature insecticidal peptides, significantly improves the yield of the insecticidal peptides without adverse effects on the insecticidal activities of the peptides.

Our data shows that any Cysteine Rich Insecticidal Peptide (CRIP) can be made to grow at significantly higher yields than would otherwise be possible using the procedures we describe here. We have demonstrated the both ICK and non ICK types of CRIPs can have their yields dramatically improved using the High Production techniques we described. Here we provide evidence of dramatic and surprising increases in yields of two very diverse types of CRIPs.

The insecticidal peptides that can be converted may be selected from insecticidal venom, for example the venom of a spider. The spider may be an Australian funnel web spider. The peptides from the genus of *Atrax* or *Hadronyche* are U-ACTX-Hv1a and its analogs and are easily made special using the procedures described herein. Specific peptide

examples from spiders are described in the sequence listing provided herein. These peptides and others can be converted using the procedures described herein.

The insecticidal peptides that can be converted may be selected from sea anemone toxins such as from *Anemone viridis* as described in Example 3. Sea anemones are far removed in their normal habitat from the funnel web spiders of the genus of *Atrax* or *Hadronyche* and the venom from *Anemone viridis* is not considered a ICK type of venom, as is venomous peptides from *Atrax* or *Hadronyche* but in spite of that the venom of the sea urchin, like the U-ACTX-Hv1a toxic peptides and other insecticidal venoms is that they are all a type of venom that we call Cysteine Rich Insecticidal Peptide or CRIP and identified here for the first time as such. The procedures described herein, in all the sections, are expected and believed to work with all of the peptide in the sequence listings and all of the peptides related to those sequences that would be understood by one skilled in the art to be a Cysteine Rich Insecticidal Peptide or CRIP. All such peptides and others can be converted using the procedures described herein.

In addition to the process, we also disclose novel High Production Peptides, herein "HP peptides," comprising a dipeptide bound to one end of a peptide. In our embodiments the peptide is an insecticidal peptide. In one embodiment the dipeptide is added to the N-terminus of the peptide. We have demonstrated success in producing high yield strains with both ICK and non-ICK CRIP peptides. In a further embodiment the dipeptide is composed of a non-polar amino acid and a polar amino acid. In a further embodiment the non-polar amino acid is selected from glycine, alanine, proline, valine, leucine, isoleucine, phenylalanine and methionine, and the polar amino acid is selected from serine, threonine, cysteine, asparagine, glutamine histidine, tryptophan and tyrosine. In one specific embodiment an HP peptide is comprised of a peptide which is modified to have the dipeptide of glycine-serine as the first two amino acids of an otherwise unmodified, mature peptide. HP peptides may be produced by adding glycine-serine to the U peptide and its analogs to create HP peptides.

The modified peptides made by the processes described herein are new and are separately claimed. These peptides are described by all of their properties and not simply their sequence. These peptides are novel and have unique properties. Both HP peptides and the process of making them are disclosed and claimed herein.

Examples of useful peptides are well known and can be found in numerous references. One class of useful peptides is insecticidal peptides. Insecticidal peptides can be identified by their peptide nature and their activity, usually oral or injection insecticidal activity. Here we provide a few examples to better illustrate and describe the invention, but the invention is not limited to these examples. All of these examples and others not shown here are descriptive of new materials, described and claimed here for the first time.

HP (High Production) peptides are defined here as any peptides capable of being produced at greater than normal rates of production using the techniques described herein. Such peptides may have insecticidal activity. Typically, insecticidal peptides show activity when injected into insects but most do not have significant activity when applied to an insect topically. The insecticidal activity of HP peptides is measured in a variety of ways.

Common methods of measurement are widely known to those skilled in the art. Such methods include, but are not limited to determination of median response doses (e.g., LD₅₀, PD₅₀, LC₅₀, ED₅₀) by fitting of dose-response plots

based on scoring various parameters such as: paralysis, mortality, failure to gain weight, etc. Measurements can be made for cohorts of insects exposed to various doses of the insecticidal formulation in question. Analysis of the data can be made by creating curves defined by probit analysis and/or the Hill Equation, etc. In such cases, doses would be administered by hypodermic injection, by hyperbaric infusion, by presentation of the insecticidal formulation as part of a sample of food or bait, etc.

Specific examples of HP peptides disclosed for purposes of providing examples and not intended to be limiting in any way, are the U peptide and its homologues, which origin from the venoms of Australian Funnel-web spiders. The description of these peptides can be found in this document in earlier sections.

The Examples in this specification are not intended to, and should not be used to limit the invention, they are provided only to illustrate the invention.

As noted above, many peptides are suitable candidates as the subject of the process to make special. The sequences noted above, below and in the sequence listing are especially suitable peptides that can be made special, and some of these have been made special according to this invention with the results shown in the examples below.

(one letter code)

SEQ ID NO: 5
GSQYC VPVDQ PCSLN TQPC DDATC TQERN ENGHT VYYCR A.

Named "U+2-ACTX-Hv1a," It has disulfide bridges at positions: 5-20, 12-25, 19-39. The molecular weight is 4564.85 Daltons.

(one letter code)

SEQ ID NO: 29
GSRSC CPCYW GGCPW GQNCY PEGCS GPKV.

Named "Av3+2," It has disulfide bridges at positions: 5-19, 6-13, 8-24. The molecular weight is 3076.47 Daltons.

Preparation of the HP Peptides

The HP peptides described herein can be prepared as below. The open reading frames (ORFs) of the insecticidal peptides are designed such that their nucleotide sequences are optimized for species-specific expression. Shown below is a specific example of a process for increasing insecticidal peptide production yield from yeast expression systems by addition of a dipeptide to the N-terminus of the insecticidal peptide. The dipeptide is composed of a non-polar amino acid and a polar amino acid. The non-polar amino acid may be selected from glycine, alanine, proline, valine, leucine, isoleucine, phenylalanine and methionine and glycine is the preferred non-polar amino acid. The polar amino acid may be selected from serine, threonine, cysteine, histidine, tryptophan, tyrosine, asparagine and glutamine and serine is the preferred polar amino acid. In the example below, the non-polar amino acid is at the N-terminus of the dipeptide and it is glycine. In the example below, the polar amino acid is at the C-terminus of the dipeptide and it is serine.

The insecticidal peptide ORF is designed for secretion from host yeast cells as follows: the ORF starts with a signal peptide sequence, followed by DNA sequence encoding a Kex 2 cleavage site (Lysine-Arginine), followed by the insecticidal peptide transgene with addition of glycine-serine codons at the 5' terminus, and finally ends with a stop codon at the 3' terminus. All these elements will be expressed to a fusion peptide in yeast cells as a single open reading frame. An α -mating factor signal sequence is most

frequently used to facilitate metabolic processing of the recombinant insecticidal peptides through the endogenous secretion pathway of the recombinant yeast, i.e. the expressed fusion peptide will typically enter the Endoplasmic Reticulum, wherein the α -mating factor signal sequence is removed by signal peptidase activity, and then the resulting pro-insecticidal peptide will be trafficked to the Golgi Apparatus, in which the Lysine-Arginine dipeptide mentioned above is completely removed by Kex 2 endoprotease, after which the mature, HP insecticidal peptide, comprising the additional non-native glycine-serine dipeptide at its N-terminus, is secreted out of the cells.

To enhance insecticidal peptide expression level in the recombinant yeast cells, the codons of the insecticidal peptide ORF are usually optimized for expression in the specific host yeast species. Naturally occurring frequencies of codons observed in endogenous open reading frames of a given host organism are not necessarily optimized for high efficiency expression. Furthermore, different yeast species (for example, *Kluyveromyces lactis*, *Pichia pastoris*, *Saccharomyces cerevisiae*, etc.) have different optimal codons for high efficiency expression. Hence, codon optimization should be considered for the peptide ORF, including the sequence elements encoding the signal sequence, the Kex2 cleavage site and the insecticidal peptides, since they are initially translated as one fusion peptide in the recombinant yeast cells.

The codon-optimized peptide expression DNAs are then ligated into appropriate expression vectors for yeast expression. There are many expression vectors available for yeast expression, including episomal vectors and integrative vectors, and they are usually designed for specific yeast strains. One should carefully choose the appropriate expression vector in view of the specific yeast expression system which will be used for the peptide production. Here we used integrative vectors, which will integrate into chromosomes of the transformed yeast cells and be stable through cycles of cell division and proliferation.

The expression vectors usually contain some *E. coli* elements for DNA preparation in *E. coli*, for example, *E. coli* replication origin, antibiotic selection marker, etc. The vectors also contain an array of the sequence elements needed for expression of the transgene of interest, for example, transcriptional promoters, terminators, yeast selection markers, integrative DNA sequences homologous to host yeast DNA, etc. There are many suitable yeast promoters available, including natural and engineered promoters. In our efforts, yeast promoters such as pLAC4, pAOX1, pUPP, pADH1, pTEF, pGal1, etc. have been used. We also used the following commonly used yeast selection markers: acetamide prototrophy selection, zeocin-resistance selection, geneticin-resistance selection, nourseothricin-resistance selection, uracil deficiency selection. Other markers known to one skilled in the art could also be used. The integrative DNA sequences are homologous to targeted genomic DNA loci in the transformed yeast species, and such integrative sequences include pLAC4, 25S rDNA, pAOX1, and TRP2, etc. The locations of insecticidal peptide transgenes can be adjacent to the integrative DNA sequence (Insertion vectors) or within the integrative DNA sequence (replacement vectors).

To get more copies of insecticidal peptide ORF integrated into the host yeast chromosomes, the expression vectors can be designed and generated to contain two or three copies of insecticidal peptide expression cassette. Each copy of the insecticidal peptide expression cassette in the expression vector should contain independent and complete expression

structures including promoter, signal sequence, Kex2 cleavage sequence and, the insecticidal peptide transgene, stop codon transcription terminator.

The peptide expression vectors are then transformed into yeast cells. First, the expression vectors are usually linearized by specific restriction enzyme cleavage to facilitate chromosomal integration via homologous recombination. The linear expression vector is then transformed into yeast cells by a chemical or electroporation method of transformation and integrated into the targeted locus of the yeast genome by homologous recombination. The integration can happen at the same chromosomal locus multiple times; therefore the genome of a transformed yeast cell can contain multiple copies of insecticidal peptide transgenes. The successful transformants can be identified using growth conditions that favor a selective marker engineered into the expression vector and co-integrated into yeast chromosomes with the insecticidal peptide transgenes; examples of such markers include, but aren't limited to, acetamide prototrophy, zeocin resistance, geneticin resistance, nourseothricin resistance, and uracil prototrophy.

Due to the influence of unpredictable and variable factors—such as epigenetic modification of genes and networks of genes, and variation in the number of integration events that occur in individual cells in a population undergoing a transformation procedure—individual yeast transformants of a given transformation process will differ in their capacities to produce a transgenic insecticidal peptide. Therefore, yeast transformants carrying the insecticidal peptide transgenes should be screened for high yield strains. Two effective methods for such screening, each dependent on growth of small-scale cultures of the transformants to provide conditioned media samples for subsequent analysis, use reverse-phase HPLC or housefly injection procedures to analyze conditioned media samples from the transformants.

The transformant cultures are usually performed in 14 mL round bottom polypropylene culture tubes with 5-10 mL defined medium added to each tube, or in 48-well deep well culture plates with 1-2 mL defined medium added to each well. The defined medium, not containing crude proteinaceous extracts or by-products such as yeast extract or peptone, is used for the cultures to reduce the protein background in the conditioned media harvested for the later screening steps. The cultures are performed at the optimal temperature, for example, 23.5° C. for *K. lactis*, for 5-6 days, until the maximum cell density is reached. The insecticidal peptides are now produced from the transformants and secreted out of cells to the growth medium. To prepare samples for the screening, cells are removed from the cultures by centrifugation and the supernatants are collected as the conditioned media, which are then cleaned by filtration through 0.22 µm filter membrane and then made ready for insecticidal peptide production strain screening, a couple of examples of such screening methods are described below.

One of the screening methods is reverse-phase HPLC (rpHPLC) screening of transformants. In this screening method, an HPLC analytic column with bonded phase of C18 is used. Acetonitrile and water are used as mobile phase solvents, and a UV absorbance detector set at 220 nm is used for the peptide detection. Appropriate amounts of the conditioned medium samples are loaded into the rpHPLC system and eluted with a linear gradient of mobile phase solvents. The corresponding peak area of the insecticidal peptide in the HPLC chromatograph is used to quantify the insecticidal peptide concentrations in the conditioned media. Known amounts of pure insecticidal peptide are run through the same rpHPLC column with the same HPLC protocol to

confirm the retention time of the peptide and to produce a standard peptide HPLC curve for the quantification.

A second screening method is the housefly injection assay. Insecticidal peptide can kill houseflies when injected in measured doses through the body wall of the dorsal thorax. The efficacy of the insecticidal peptide can be defined by the median lethal dose of the peptide (LD50), which causes 50% mortality of the injected houseflies. The pure insecticidal peptide is normally used in the housefly injection assay to generate a standard dose-response curve, from which an LD50 value can be determined. Using an LD50 value from the analysis of a standard dose-response curve of the pure insecticidal peptide in question, quantification of the insecticidal peptide produced by a yeast transformant can be achieved using a housefly injection assay performed with serial dilutions of the corresponding conditioned media.

The insecticidal peptide production strain screen can identify the high yield yeast strains from hundreds of transformants. These strains can be fermented in bioreactor to achieve up to 6 g/L yield of the insecticidal peptides when using optimized fermentation media and fermentation conditions. The higher rates of production can be anywhere from 20 to 400, 20 to 100, 20 to 200, 20 to 300, 40 to 100, 40 to 200, 40 to 300, 40 to 400, 60 to 100, 60 to 200, 60 to 300, 60 to 400, 80 to 100, 80 to 200, 80 to 300, 80 to 400, 100 to 150, 100 to 200, 150 to 200, 200 to 250, 250 to 300, 250 to 350, 250 to 400, 300 to 350, 300 to 400% and 350 to 400 or any range of any value provided or even greater yields than can be achieved with a peptide before conversion, using the same or similar production methods that were used to produce the peptide before conversion.

Any of the sequences from the sequence listing, and as far as we know any CRIP could all be used to make high production peptides similar to either the ACTX motifs from the Australian Blue Mountain Funnel-web Spider we call the "U+2" peptide described below, or the Av3+2 peptide of the toxic sea anemone, *Anemone viridis*, that we teach and describe in the examples below by using procedures taught here and the knowledge of one ordinarily skilled in the art. In addition, any other suitable CRIP peptide could be used in a like manner to produce a high production or plus 2, i.e. +2 peptide.

Examples of High Production Peptides

The Examples in this specification are not intended to, and should not be used to limit the invention, they are provided only to illustrate the invention.

Example 1

Expression of native U and U+2-ACTX-Hv1a in *Cluyveromyces lactis* (*K. lactis*).
Insecticidal Peptides to Express:

U + 2-ACTX-Hv1a: (SEQ ID NO: 5)
GSQYCVVDQPCSLNTQPCDDATCTQERNENGHTVYYCRA
and

Native U-ACTX-Hv1a: (SEQ ID NO: 6)
QYCVVDQPCSLNTQPCDDATCTQERNENGHTVYYCRA

To express the two insecticidal peptides above in *K. lactis*, the expression vector, pKLAC1, and the *K. lactis* strain, YCT306, were used, which are available from New England Biolabs, Ipswich, MA, USA. pKLAC1 vector is an integra-

tive expression vector. Once the U+2 and native U-ACTX-Hv1a transgenes were cloned into pKLAC1 and transformed into YCT306, their expression was controlled by the LAC4 promoter. The resulting transformants produced pre-propeptides comprising an α -mating factor signal peptide, a Kex2 cleavage site and mature insecticidal peptides. The α -Mating factor signal peptide guides the pre-propeptides to go through the endogenous secretion pathway and finally the mature insecticidal peptides are released into the growth media.

Codon optimization for U+2-ACTX-Hv1a expression was performed in two rounds. In the first round, based on some common features of high expression DNA sequences, 33 variants of the peptide ORF, expressing an α -Mating factor signal peptide, a Kex2 cleavage site and the U+2-ACTX-Hv1a peptide, were designed and their expression levels were evaluated in the YCT306 strain of *K. lactis*, resulting in an initial *K. lactis* expression algorithm. In the 2nd round of optimization, five more variant U+2-ACTX-Hv1a peptide ORFs were designed based on the initial *K. lactis* expression algorithm to further fine-tune the *K. lactis* expression algorithm, and identified the best ORF for the U+2-ACTX-Hv1a peptide expression in *K. lactis*. This DNA sequence has an open reading frame encoding an α -mating factor signal peptide, a Kex2 cleavage site and a U+2-ACTX-Hv1a peptide. The optimized DNA sequence was cloned into the pKLAC1 vector using Hind III and Not I restriction sites, resulting in the U+2-ACTX-Hv1a expression vector, pLB10V5.

To enable integration of more copies of the optimized U+2-ACTX-Hv1a transgene into the *K. lactis* genome during transformation, generation of a U+2-ACTX-Hv1a expression vector containing two copies of U+2-ACTX-Hv1a expression cassette was processed as follows: A 3,306 bp intact U+2-ACTX-Hv1a expression cassette DNA sequence was synthesized, which comprised an intact LAC4 promoter element, a codon-optimized U+2-ACTX-Hv1a peptide ORF element and a pLAC4 terminator element. This intact expression cassette was then ligated into the pLB10V5 vector between Sal I and Kpn I restriction sites, downstream of the pLAC4 terminator of pLB10V5, resulting in the double transgene U+2-ACTX-Hv1a expression vector, pLB10V5D.

To generate a native U-ACTX-Hv1a expression vector, the pLB10V5 vector was mutagenized by deleting the glycine-serine codons at the 5'-terminus of the U+2-ACTX-Hv1a transgene region, using a Stratagene site-direct mutagenesis kit. This mutagenesis resulted in a new vector, pLB12, containing a single copy of the codon-optimized native U-ACTX-Hv1a expression cassette. To generate a double transgene native U-ACTX-Hv1a expression vector, a Stratagene site-direct mutagenesis kit was used again to remove the glycine-serine codons at the 5'-terminus of the U+2-ACTX-Hv1a transgene region in the 3,306 bp U+2-ACTX-Hv1a expression cassette transgene synthesized previously, followed by ligation to insert the mutagenized cassette into the pLB12 vector between Sal I and Kpn I restriction sites, resulting in the plasmid, pLB12D, an expression vector comprising two intact copies of the codon-optimized native U-ACTX-Hv1a expression cassette.

The double transgene vectors, pLB10V5D and pLB12D, were then linearized using Sac II restriction endonuclease and chemically transformed into YCT306 strain of *K. lactis*, according to instructions provided with a *K. lactis* Protein Expression Kit. The resulting transformants grew on YCB agar plate supplemented with 5 mM acetamide, which only the acetamidase-expressing transformants could use efficiently as a metabolic source of nitrogen.

For insecticidal peptide yield evaluations, 316 colonies were picked from the pLB10V5D transformants plates, and 40 colonies were picked from the pLB12D transformants

plates. Inocula from the colonies were each cultured in 6 mL of the defined *K. lactis* media with 2% pure glycerol added as a carbon source. Cultures were incubated at 23.5° C., with shaking at 280 rpm, for six days, at which point cell densities in the cultures had reached their maximum levels as indicated by light absorbance at 600 nm (OD600). Cells were then removed from the cultures by centrifugation at 4,000 rpm for 10 minutes. The resulting supernatants (conditioned media) were filtered through 0.2 m membranes for HPLC yield analysis.

For the peptide yield evaluation, the filtered conditioned media samples were analyzed on an Agilent 1100 HPLC system equipped with an Onyx monolithic 4.5×100 mm, C18 reverse-phase analytical HPLC column and an auto-injector. HPLC grade water and acetonitrile, both containing 0.1% trifluoroacetic acid, constituted the two mobile phase solvents used for the HPLC analyses. The peak areas of both the native U and U+2-ACTX-Hv1 were measured using HPLC chromatographs and then used to calculate the peptide concentration in the conditioned media, which were then further normalized to the corresponding final cell densities (as determined by OD600 measurements) as normalized peptide yield.

Housefly injection bioassay was used to evaluate the insecticidal activity of the peptides. The conditioned media were serially diluted to generate full dose-response curves from the housefly injection bioassay. Before injection, adult houseflies (*Musca domestica*) were immobilized with CO₂, and 12-18 mg houseflies were selected for injection. A microapplicator, loaded with a 1 cc syringe and 30-gauge needle, was used to inject 0.5 μ L per fly doses of serially diluted conditioned media samples into houseflies through the body wall of the dorsal thorax. The injected houseflies were placed into closed containers with moist filter paper and breathing holes on the lids, and they were examined by mortality scoring at 24 hours post-injection.

Normalized yields were calculated. Peptide yield means the peptide concentration in the conditioned media in units of mg/L. But peptide yields are not always sufficient to accurately compare the strain production rate. Individual strains may have different growth rates, hence when a culture is harvested, different cultures may vary in cell density. A culture with a high cell density may produce a higher concentration of the peptide in the media, even though the peptide production rate of the strain is lower than another strain which has a higher production rate. So the term "normalized yield" is created by dividing the peptide yield with the cell density in the corresponding culture and this allows a better comparison of the peptide production rate between strains. The cell density is represented by the light absorbance at 600 nm with a unit of "A" (Absorbance unit).

Table 1, FIG. 12 and FIG. 13 summarize the U+2- and native U-ACTX-Hv1a normalized peptide yield distributions from the *K. lactis* strains. The overall averaged U+2-ACTX-Hv1a normalized peptide yield from the *K. lactis* strains was 4.06±3.05 mg/L·A, which was statistically significantly higher than the averaged native U-ACTX-Hv1a normalized peptide yield, 2.73±1.25 mg/L·A, by Student's t-test at 99% confidence level. The median normalized peptide yield of the U+2-ACTX-Hv1a *K. lactis* strains was 9.36 mg/L·A, which was almost three times higher than the median yield of native U-ACTX-Hv1a strains (3.35 mg/L·A). The U+2-ACTX-Hv1a peptide expression strains had much higher ratios of the strain counts at high yield level than the native U-ACTX-Hv1a strains. All of these results indicated that the addition of the glycine-serine dipeptide to the N-terminus of the U-ACTX-Hv1a peptide contributes to significant improvement of the predicted yield for yeast transformants expressing this peptide.

Table 1 shows a comparison of peptide yields from *K. lactis* strains.

TABLE 1

U + 2 and native U-ACTX-Hv1a Peptide Yield Comparison								
Normalized Yield Level	U + 2 Yield (total 316 strains)				Native U Yield (total 40 strains)			
	Strain count	Ratio to total	Overall average	Median Yield	Strain count	Ratio to total	Overall average	Median Yield
>2 mg/L · A	242	0.765823	4.06 ± 3.05	9.36	26	0.65	2.73 ± 1.25	3.35
>3 mg/L · A	161	0.509494	(mg/L · A)	(mg/L · A)	18	0.45	(mg/L · A)	(mg/L · A)
>4 mg/L · A	124	0.392405			6	0.15		
>6 mg/L · A	62	0.196203			0	0		
>8 mg/L · A	29	0.0917722			0	0		
>10 mg/L · A	16	0.0506329			0	0		
>12 mg/L · A	9	0.028481			0	0		
>14 mg/L · A	6	0.0189873			0	0		

FIG. 12 shows the histograms of the normalized peptide yield distributions for the U+2 and native U strains. The X scale shows the range of the normalized peptide yield. The Y scale on the left shows the frequency of the U+2 producing strains in the specific range of the normalized yield, and the Y scale on the right shows the frequency of the native U producing strains in the specific range of the normalized yield. The black bars represent the U+2 yield distribution and the grey bars represent the native U yield distribution. For example, the first black bar tells that about 0.03 (3%) of the total U+2 producing strains have normalized yields between 0 and 0.5 mg/L · A. The strain counts are different between native and +2 strains because 316 strains for U+2 were screened and 40 strains for the native peptide were screened.

FIG. 13 shows the distribution of the peptide yields from U+2 and native U-ACTX-Hv1a produced from the *K. lactis* strains. The U+2 data is shown in black and the native U data is in gray. The x-axis shows the yield in milligrams per liter and the y-scale shows the fraction of total U+2 or native U production from *K. lactis* strains. The yield from the U+2 strains, and the number of U+2 strains available that can produce high yields is far higher for the U+2 strains as compared to the native U strains.

Ordinarily one might expect making changes to a peptide sequence that dramatically improves its yield could affect its toxicity. Surprisingly that is not what happens with the dipeptides of this disclosure. Our data indicates the addition of the dipeptide, and especially the glycine-serine dipeptide, to the N-terminus of the U-ACTX-Hv1a peptide, does not lower the effectiveness of the insecticidal activities of the peptide. FIG. 14 shows two dose-response curves for housefly injection bioassays performed with the native and U+2-ACTX-Hv1a conditioned medium samples. The U+2-ACTX-Hv1a has a median lethal dose (LD50) of 76.8 pmol/g, which is consistent with the LD50 of native U-ACTX-Hv1a, 77.6 pmol/g.

Example 2

Peptide yields of transformants of the yeast, *Pichia pastoris* (*P. pastoris*), expressing either U+2-ACTX-Hv1a or U-ACTX-Hv1a were studied.

Two *P. pastoris* vectors, pJUGαKR and pJUZAαKR, were used for the U+2-ACTX-Hv1a or native U-ACTX-Hv1a peptide expression in *P. pastoris*. pJUGαKR and pJUZAαKR are available from Biogrammatix, Carlsbad, California, USA. Both vectors are integrative vectors and use the uracil

phosphoribosyltransferase promoter (pUPP) to enhance the heterologous transgene expression. The only difference between the vectors is that pJUGαKR provides G418 resistance to the host yeast, while pJUZAαKR provides Zeocin resistance.

Pairs of complementary oligonucleotides, encoding the native U-ACTX-Hv1a and U+2-ACTX-Hv1a respectively, were designed and synthesized for sub cloning into the two yeast expression vectors. Hybridization reactions were performed by mixing the corresponding complementary oligonucleotides to a final concentration of 20 μM in 30 mM NaCl, 10 mM Tris-Cl (all final concentrations), pH 8, and then incubating at 95° C. for 20 min, followed by a 9 hour incubation starting at 92° C. and ending at 17° C., with 3° C. drops in temperature every 20 min. The hybridization reactions resulted in two DNA fragments encoding U+2-ACTX-Hv1a and native U-ACTX-Hv1a peptides respectively. The two *P. pastoris* vectors were digested with BsaI-HF restriction enzymes, and the double stranded products of the Uization reactions were then sub cloned into the linearized *P. pastoris* vectors using standard procedures. Following verification of the sequences of the four sub clones, plasmid aliquots were transformed by electroporation into the *P. pastoris* strain, Bg08. The resulting transformed yeast, selected based on resistance to Zeocin or G418 conferred by elements engineered into vectors pJUZAαKR and pJUGαKR, respectively, were cultured and screened as described below. Since no transformant strains had more than one antibiotic resistance marker, and since transformation procedures were performed the same for yeast cells transformed with the U+2-ACTX-Hv1a transgene as for those transformed with the native U-ACTX-Hv1a transgene, it is reasonable to presume that the distributions of transgene copy number were comparable for the two populations of transformants being compared below.

Recipes for media and stocks used for the *P. pastoris* cultures are described as follows:

MSM media recipe
 2 g/L sodium citrate dihydrate
 1 g/L calcium sulfate dihydrate (0.79 g/L anhydrous calcium sulfate)
 42.9 g/L potassium phosphate monobasic
 5.17 g/L ammonium sulfate
 14.33 g/L potassium sulfate
 11.7 g/L magnesium sulfate heptahydrate
 2 mL/L PTM1 trace salt solution
 0.4 ppm biotin (from 500×, 200 ppm stock)
 1-2% pure glycerol or other carbon source

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PTM1 Trace Salts Solution:

Cupric sulfate-5H₂O 6.0 g
Sodium iodide 0.08 g
Manganese sulfate-H₂O 3.0 g
Sodium molybdate-2H₂O 0.2 g
Boric Acid 0.02 g
Cobalt chloride 0.5 g
Zinc chloride 20.0 g
Ferrous sulfate-7H₂O 65.0 g
Biotin 0.2 g
Sulfuric Acid 5.0 ml
Add Water to a final volume of 1 liter

48-well Deep-well plates, sealed after inoculation with sterile, air-permeable tape, were used to culture the insecticidal peptide *P. pastoris* transformants. Colonies on the *P. pastoris* transformant plates were picked and inoculated the deep-well plates with 1 mL media per well, which was composed of MSM+0.2% PTM1+biotin (500× diluted from 200 ppm stock)+1% glycerol (pure). Inoculated plates were grown 5 days at 23.5° C. with 220 rpm shaking in a refrigerated incubator-shaker. 100 µL 5% glycerol were added to each well of the plates at 2, 3, and 4 days post inoculation. On day 5 post-inoculation, conditioned media was harvested by centrifugation at 3700 rpm for 15 minutes, followed by filtration using filter plate with 0.22 µM membrane. Filtered media stored at -20° C. for further analyses.

0.3 mL aliquots of conditioned *P. pastoris* media prepared as described above were analyzed using rphPLC described in EXAMPLE 1 to determine the concentrations of the native U-ACTX-Hv1a or U+2-ACTX-Hv1a peptide present in the media. Results of this analysis are summarized in Table 2, FIG. 15 and FIG. 16. The average peptide yields with a common mean and standard deviation are 67.0±27.9 mg/L for the U+2-ACTX-Hv1a *P. pastoris* strains and 42.9±18.3 mg/L for the native U-ACTX-Hv1a strains. A student's t-test indicated that the probability of such differing distributions of yields is far below 1%. The median yield from the U+2-ACTX-Hv1a strains was 79.0 mg/L, far higher than that from the native U-ACTX-Hv1a strains (44.7 mg/L). It is observed that the U+2-ACTX-Hv1a strains had much higher ratios of the strain counts at high peptide yield level than the native U-ACTX-Hv1a strains. All these results support the conclusion that the extra glycine-serine dipeptide at the N-terminus of the U+2-ACTX-Hv1a significantly improved the capacity of yeast transformants to produce this peptide and secrete it into conditioned media.

Table 2 shows a comparison of peptide yields from *P. pastoris* strains.

TABLE 2

U + 2 and native U-ACTX-Hv1a Peptide Yield Comparison								
U + 2 Yield (total 45 strains)					Native U Yield (total 48 strains)			
Normalized Yield Level	Strain count	Ratio to total	Overall average	Median Yield	Strain count	Ratio to total	Overall average	Median Yield
>30 mg/L	42	93.3%	67.0 ± 27.9 (mg/L)	79.0 (mg/L)	38	79.2%	42.9 ± 18.3 (mg/L)	44.7 (mg/L)
>40 mg/L	39	86.7%			34	70.8%		
>50 mg/L	37	82.2%			19	39.6%		
>60 mg/L	34	75.6%			3	6.3%		
>70 mg/L	11	24.4%			2	4.2%		
>80 mg/L	7	15.6%			2	4.2%		
>90 mg/L	6	13.3%			0	0		
>100 mg/L	6	13.3%			0	0		

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Example 3

Expression of one of the type 3 sea anemone toxins discovered from *Anemone viridis*, native Av3 and Av3+2 in the yeast strain *Kluyveromyces lactis*.

Insecticidal Peptides to Express:

Av3 + 2:

(SEQ ID NO. 29)

GSRSCCPCYWGGCPWQNCYPEGCSGPKV

Native Av3:

(SEQ ID NO. 30)

RSCCPCYWGGCPWQNCYPEGCSGPKV

To express the two non-ICK CRIP peptides above in *Kluyveromyces lactis*, the pKLAC1 vector and the *Kluyveromyces lactis* strain, YCT306, were used as in example 1.

The Av3 and Av3+2 peptide ORF, which encode α-MF::Kex2 cleavage site::Av3 (or Av3+2), were codon-optimized using previously determined *K. lactis* expression algorithm.

The optimized Av3+2 expression ORF sequence is follows:

(SEQ ID NO. 31)

AAGCTTGAAAAAATGAAATTTTCCACTATTTTAGCAGCATCTACAGCTT

TAATCAGTGTGTGCATGGCTGCACCTGTGAGTACCGAAACAGATATAGAC

GACCTTCCAATCTCTGTTCCAGAAGAGGCTTTGATAGGATTCATCGATTT

GACTGGTGATGAAGTTTCATTGTTACCAGTGAATAATGGTACCCATACTG

GATATTTGTTCTTAACACCACAATTGCTGAAGCTGCTTTTGCAGATAAG

GATGATTTGGAGAAAAGAGGTTCTAGATCATGCTGCCCTTGTTACTGGGG

TGGTTGTCCATGGGGACAAAAGTGTATCTGTAAGGATGTTCTGGTCCAA

AGGTATGAGCGGCCCG

This optimized DNA sequence was cloned into pKLAC1 vector using Hind III and Not I restriction sites, resulting in the Av3+2 expression vector, pLB102.

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The optimized native Av3 expression ORF sequence is follows:

(SEQ ID NO. 32)

```

AAGCTTGAAAAAATGAAATTTCCACAATCTTAGCTGCAAGTACTGCTC
TTATTTCTGTTGTGATGGCTGCTCCAGTATCTACCGAAACAGATATCGAT
GATTTGCCAATTTTCAGTCCCTGAAGAGGCACTAATCGGATTCATTGACTT
AACCGGTGATGAAGTGAGTTTGTGCCAGTTAACAACGGTACTCATACAG
GTATATTGTTTTTGAATACCACTATAGCTGAAGCAGCATTTCGCTGATAAA
GATGACTTAGAAAAGAGAAGATCATGCTGCCCTTGTTACTGGGGTGGTTG
TCCATGGGGTCAAATTTGTTATCCAGAGGGTTGTTCTGGACCTAAGGTTT
GAGCGGCCGC
  
```

This optimized DNA sequence was cloned into pKLAC1 vector using Hind III and Not I restriction sites, resulting in the native Av3 expression vector, pLB103.

The expression vectors, pLB102 and pLB103, were then linearized using Sac II restriction endonuclease and transformed into YCT306 strain of *K. lactis*, using the electropo-

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Table 3, FIG. 17 and FIG. 18 summarize the Av3+2 and native Av3 normalized peptide yield distributions from the *K. lactis* strains. The normalized peptide yield is represented by the peptide UV peak area in the HPLC chromatograph divided by the corresponding cell density (represented by the OD600) at the end of the cell culture. The overall averaged normalized peptide yield from the Av3+2 strains was 117.5 ± 50.1 mAu·sec/A, which was statistically significantly higher than that of native Av3 which was 29.8 ± 16.1 mAu·sec/A, by Student's t-test at 99% confidence level. The median normalized peptide yield of the Av3+2 *K. lactis* strains was 106.7 mAu·sec/A, which was more than three times higher than that of native Av3 strains (31.7 mAu·sec/A). The Av3+2 expression strains had much higher ratios of the strain counts at high yield level than the native Av3 strains (table 3). And as shown in FIG. 18, overall at the any percentile of peptide yield, Av3+2 strains had higher yield than native Av3 strains. All of these results indicated that the addition of the glycine-serine dipeptide to the N-terminus of the Av3 peptide contributes to significant improvement of the peptide yield from yeast transformants expressing this peptide.

TABLE 3

Av3 + 2 and native Av3 Peptide Yield Comparison								
Normalized Yield Level	Av3 + 2 Yield (pLB102-YCT, total 48 strains)				Av3 (pLB103-YCT, total 48 strains)			
	Strain count	Ratio to total	Overall average	Median Yield	Strain count	Ratio to total	Overall average	Median Yield
>30 mAu · sec/A	46	0.958	117.5 ± 50.1	106.7	21	0.438	29.8 ± 16.1	31.7
>60 mAu · sec/A	38	0.792	(mAu · sec/A)	(mAu · sec/A)	0	0	(mAu · sec/A)	(mAu · sec/A)
>90 mAu · sec/A	36	0.75			0	0		
>120 mAu · sec/A	25	0.521			0	0		
>150 mAu · sec/A	16	0.333			0	0		
>180 mAu · sec/A	2	0.042			0	0		
>2000 mAu · sec/A	1	0.021			0	0		

ration transformation method. The resulting transformants grew on YCB agar plate supplemented with 5 mM acetamide, which only the acetamidase-expressing transformants could use efficiently as a metabolic source of nitrogen.

For insecticidal peptide yield evaluations, 48 colonies of pLB102 transformants and 48 colonies of pLB103 transformants were picked up and inoculated 2.2 mL of the defined *K. lactis* media with 2% sorbitol added as a carbon source in 48-well deep-well plates with 5 mL volume capacity each well. Cultures were processed at 23.5° C., with shaking at 280 rpm, for six days, when cell densities in the cultures were determined by light absorbance at 600 nm (OD600). Cells were then removed from the cultures by centrifugation at 4000 rpm for 10 minutes. The resulting supernatants (conditioned media) were filtered through 0.2 m membranes for HPLC yield analysis.

For the peptide yield evaluation, the filtered conditioned media samples were analyzed on an Agilent 1100 HPLC system equipped with an Onyx monolithic 4.5×100 mm, C18 reverse-phase analytical HPLC column and an auto-injector. HPLC grade water and acetonitrile, both containing 0.1% trifluoroacetic acid, constituted the two mobile phase solvents used for the HPLC analyses. The native Av3 or Av3+2 peak areas in the resulting HPLC chromatographs were used as indication of the peptide concentration in the conditioned media, which were then further normalized to the corresponding final cell densities (as determined by OD600 measurements) as normalized peptide yield.

Crops and Insects

Specific crops and insects that may be controlled by these methods include the following:

The present invention may be used for transformation of any plant species, including, but not limited to, monocots and dicots. Crops for which a transgenic approach or PEP would be an especially useful approach include, but are not limited to: alfalfa, cotton, tomato, maize, wheat, corn, sweet corn, lucerne, soybean, sorghum, field pea, linseed, safflower, rapeseed, oil seed rape, rice, soybean, barley, sunflower, trees (including coniferous and deciduous), flowers (including those grown commercially and in greenhouses), field lupins, switchgrass, sugarcane, potatoes, tomatoes, tobacco, crucifers, peppers, sugarbeet, barley, and oilseed rape, *Brassica* sp., rye, millet, peanuts, sweet potato, cassava, coffee, coconut, pineapple, citrus trees, cocoa, tea, banana, avocado, fig, guava, mango, olive, papaya, cashew, macadamia, almond, oats, vegetables, ornamentals, and conifers.

"Pest" includes, but is not limited to: insects, fungi, bacteria, nematodes, mites, ticks, and the like.

Insect pests include, but are not limited to, insects selected from the orders Coleoptera, Diptera, Hymenoptera, Lepidoptera, Mallophaga, Homoptera, Hemiptera, Orthoptera, Thysanoptera, Dermaptera, Isoptera, Anoplura, Siphonaptera, Trichoptera, and the like. More particularly, insect pests include Coleoptera, Lepidoptera, and Diptera.

Insects of suitable agricultural, household and/or medical/veterinary importance for treatment with the insecticidal polypeptides include, but are not limited to, members of the following classes and orders:

The order Coleoptera includes the suborders Adephaga and Polyphaga. Suborder Adephaga includes the superfamilies Caraboidea and Gyrinoidea. Suborder Polyphaga includes the superfamilies Hydrophiloidea, Staphylinoidea, Cantharoidea, Cleroidea, Elateroidea, Dascilloidea, Dryopoidea, Byrrhoidea, Cucujoidea, Meloidea, Mordelloidea, Tenebrionoidea, Bostrichoidea, Scarabaeoidea, Cerambycoidea, Chrysomeloidea, and Curculionoidea. Superfamily Caraboidea includes the families Cicindelidae, Carabidae, and Dytiscidae. Superfamily Gyrinoidea includes the family Gyrinidae. Superfamily Hydrophiloidea includes the family Hydrophilidae. Superfamily Staphylinoidea includes the families Silphidae and Staphylinidae. Superfamily Cantharoidea includes the families Cantharidae and Lampyridae. Superfamily Cleroidea includes the families Cleridae and Dermestidae. Superfamily Elateroidea includes the families Elateridae and Buprestidae. Superfamily Cucujoidea includes the family Coccinellidae. Superfamily Meloidea includes the family Meloidae. Superfamily Tenebrionoidea includes the family Tenebrionidae. Superfamily Scarabaeoidea includes the families Passalidae and Scarabaeidae. Superfamily Cerambycoidea includes the family Cerambycidae. Superfamily Chrysomeloidea includes the family Chrysomelidae. Superfamily Curculionoidea includes the families Curculionidae and Scolytidae.

Examples of Coleoptera include, but are not limited to: the American bean weevil *Acanthoscelides obtectus*, the leaf beetle *Agelastica alni*, click beetles (*Agriotes lineatus*, *Agriotes obscurus*, *Agriotes bicolor*), the grain beetle *Ahasverus avena*, the summer schaffer *Amphimallon solstitialis*, the furniture beetle *Anobium punctatum*, *Anthonomus* spp. (weevils), the Pygmy mangold beetle *Atomaria linearis*, carpet beetles (*Anthrenus* spp., *Attagenus* spp.), the cowpea weevil *Callosobruchus maculatus*, the fried fruit beetle *Carpophilus hemipterus*, the cabbage seedpod weevil *Ceutorhynchus assimilis*, the rape winter stem weevil *Ceutorhynchus picipitarsis*, the wireworms *Conoderus vespertinus* and *Conoderus falli*, the banana weevil *Cosmopolites sordidus*, the New Zealand grass grub *Costelytra zealandica*, the June beetle *Cotinis nitida*, the sunflower stem weevil *Cylindrocaptus adspersus*, the larder beetle *Dermestes lardarius*, the corn rootworms *Diabrotica virgifera*, *Diabrotica virgifera virgifera*, and *Diabrotica barberi*, the Mexican bean beetle *Epilachna varivestis*, the old house borer *Hylotropes bajulus*, the lucerne weevil *Hypera postica*, the shiny spider beetle *Gibbium psyllioides*, the cigarette beetle *Lasioderma serricornis*, the Colorado potato beetle *Leptinotarsa decemlineata*, *Lyctus* beetles' (*Lyctus* spp.), the pollen beetle *Meligethes aeneus*, the common cockshaver *Melolontha melolontha*, the American spider beetle *Mezium americanum*, the golden spider beetle *Niptus hololeucus*, the grain beetles *Oryzaephilus surinamensis* and *Oryzaephilus mercator*, the black vine weevil *Otiorynchus sulcatus*, the mustard beetle *Phaedon cochleariae*, the crucifer flea beetle *Phyllotreta cruciferae*, the striped flea beetle *Phyllotreta striolata*, the cabbage steam flea beetle *Psylliodes chrysocephala*, *Ptinus* spp. (spider beetles), the lesser grain borer *Rhizopertha dominica*, the pea and bean weevil *Sitona lineatus*, the rice and granary beetles *Sitophilus oryzae* and *Sitophilus granaries*, the red sunflower seed weevil *Smicronyx fulvus*, the drugstore beetle *Stegobium paniceum*, the yellow mealworm beetle *Tenebrio molitor*, the flour beetles *Tribolium castaneum* and *Tribolium confusum*, warehouse

and cabinet beetles (*Trogoderma* spp.), and the sunflower beetle *Zygogramma exclamations*.

Examples of Dermaptera (earwigs) include, but are not limited to: the European earwig *Forficula auricularia*, and the striped earwig *Labidura riparia*.

Examples of Dictyonera include, but are not limited to: the oriental cockroach *Blatta orientalis*, the German cockroach *Blattella germanica*, the Madeira cockroach *Leucophaea maderae*, the American cockroach *Periplaneta americana*, and the smokybrown cockroach *Periplaneta fuliginosa*.

Examples of Diplonoda include, but are not limited to: the spotted snake millipede *Blaniulus guttulatus*, the flat-back millipede *Brachydesmus superus*, and the greenhouse millipede *Oxidus gracilis*.

The order Diptera includes the Suborders Nematocera, Brachycera, and Cyclorrhapha. Suborder Nematocera includes the families Tipulidae, Psychodidae, Culicidae, Ceratopogonidae, Chironomidae, Simuliidae, Bibionidae, and Cecidomyiidae. Suborder Brachycera includes the families Stratiomyidae, Tabanidae, Therevidae, Asilidae, Mydidae, Bombyliidae, and Dolichopodidae. Suborder Cyclorrhapha includes the Divisions Aschiza and Aschiza. Division Aschiza includes the families Phoridae, Syrphidae, and Conopidae. Division Aschiza includes the Sections Acalyptratae and Calyptratae. Section Acalyptratae includes the families Otitidae, Tephritidae, Agromyzidae, and Drosophilidae. Section Calyptratae includes the families Hippoboscidae, Oestridae, Tachinidae, Anthomyiidae, Muscidae, Calliphoridae, and Sarcophagidae.

Examples of Diptera include, but are not limited to: the house fly (*Musca domestica*), the African tumbu fly (*Cordyllobia anthropophaga*), biting midges (*Culicoides* spp.), bee louse (*Braula* spp.), the beet fly *Pegomyia betae*, blackflies (*Cnephia* spp., *Eusimulium* spp., *Simulium* spp.), bot flies (*Cuterebra* spp., *Gastrophilus* spp., *Oestrus* spp.), crane flies (*Tipula* spp.), eye gnats (*Hippelates* spp.), filth-breeding flies (*Calliphora* spp., *Fannia* spp., *Hermetia* spp., *Lucilia* spp., *Musca* spp., *Muscina* spp., *Phaenicia* spp., *Phormia* spp.), flesh flies (*Sarcophaga* spp., *Wohlfahrtia* spp.); the flit fly *Oscinella frit*, fruitflies (*Dacus* spp., *Drosophila* spp.), head and canon flies (*Hydrotea* spp.), the hessian fly *Mayetiola destructor*, horn and buffalo flies (*Haematobia* spp.), horse and deer flies (*Chrysops* spp., *Haematopota* spp., *Tabanus* spp.), louse flies (*Lipoptena* spp., *Lynchia* spp., and *Pseudolynchia* spp.), medflies (*Ceratitis* spp.), mosquitoes (*Aedes* spp., *Anopheles* spp., *Culex* spp., *Psorophora* spp.), sandflies (*Phlebotomus* spp., *Lutzomyia* spp.), screw-worm flies (*Chrysomya bezziana* and *Cochliomyia hominivorax*), sheep keds (*Melophagus* spp.); stable flies (*Stomoxys* spp.), tsetse flies (*Glossina* spp.), and warble flies (*Hypoderma* spp.).

Examples of Isontera (termites) include, but are not limited to: species from the families Hodotennitidae, Kalotermitidae, Mastotermitidae, Rhinotermitidae, Serritermitidae, Termitidae, Termopsidae;

Examples of Heteroptera include, but are not limited to: the bed bug *Cimex lectularius*, the cotton stainer *Dysdercus intermedius*, the Sunn pest *Eurygaster integriceps*, the tarnished plant bug *Lygus lineolaris*, the green stink bug *Nezara antennata*, the southern green stink bug *Nezara viridula*, and the triatomid bugs *Panstrongylus megistus*, *Rhodnius ecuadoriensis*, *Rhodnius pallidus*, *Rhodnius prolixus*, *Rhodnius robustus*, *Triatoma dimidiata*, *Triatoma infestans*, and *Triatoma sordida*.

Examples of Homoptera include, but are not limited to: the California red scale *Aonidiella aurantii*, the black bean aphid *Aphis fabae*, the cotton or melon aphid *Aphis gossypii*,

the green apple aphid *Aphis pomi*, the citrus spiny whitefly *Aleurocanthus spiniferus*, the oleander scale *Aspidiotus hederae*, the sweet potato whitefly *Bemisia tabaci*, the cabbage aphid *Brevicoryne brassicae*, the pear psylla *Cacopsylla pyricola*, the currant aphid *Cryptomyzus ribis*, the grape phylloxera *Daktulosphaira vitifoliae*, the citrus psylla *Diaphorina citri*, the potato leafhopper *Empoasca fabae*, the bean leafhopper *Empoasca solana*, the vine leafhopper *Empoasca vitis*, the woolly aphid *Eriosoma lanigerum*, the European fruit scale *Eulecanium corni*, the mealy plum aphid *Hyalopterus arundinis*, the small brown planthopper *Laodelphax striatellus*, the potato aphid *Macrosiphum euphorbiae*, the green peach aphid *Myzus persicae*, the green rice leafhopper *Nephotettix cincticeps*, the brown planthopper *Nilaparvata lugens*, gall-forming aphids (*Pemphigus* spp.), the hop aphid *Phorodon humuli*, the bird-cherry aphid *Rhopalosiphum padi*, the black scale *Saissetia oleae*, the greenbug *Schizaphis graminum*, the grain aphid *Sitobion avenae*, and the greenhouse whitefly *Trialeurodes vaporariorum*.

Examples of Isopoda include, but are not limited to: the common pillbug *Armadillidium vulgare* and the common woodlouse *Oniscus asellus*.

The order Lepidoptera includes the families Papilionidae, Pieridae, Lycaenidae, Nymphalidae, Danaidae, Satyridae, Hesperidae, Sphingidae, Saturniidae, Geometridae, Arctiidae, Noctuidae, Lymantriidae, Sesiidae, and Tineidae.

Examples of Lepidoptera include, but are not limited to: *Adoxophyes orana* (summer fruit tortrix moth), *Agrotis ipsolon* (black cutworm), *Archips podana* (fruit tree tortrix moth), *Bucculatrix pyrivorella* (pear leafminer), *Bucculatrix thurberiella* (cotton leaf perforator), *Bupalus piniarius* (pine looper), *Carpocapsa pomonella* (codling moth), *Chilo suppressalis* (striped rice borer), *Choristoneura fumiferana* (eastern spruce budworm), *Cochylis hospes* (banded sunflower moth), *Diatraea grandiosella* (southwestern corn borer), *Earls insulana* (Egyptian bollworm), *Euphestia kuehniella* (Mediterranean flour moth), *Eupoecilia ambiguella* (European grape berry moth), *Euproctis chrysorrhoea* (brown-tail moth), *Euproctis subflava* (oriental tussock moth), *Galleria mellonella* (greater wax moth), *Helicoverpa armigera* (cotton bollworm), *Helicoverpa zea* (cotton bollworm), *Heliothis virescens* (tobacco budworm), *Hofmannophila pseudopretella* (brown house moth), *Homeosoma electellum* (sunflower moth), *Homona magnanima* (oriental tea tree tortrix moth), *Lithocolletis blancardella* (spotted tentiform leafminer), *Lymantria dispar* (gypsy moth), *Malacosoma neustria* (tent caterpillar), *Mamestra brassicae* (cabbage armyworm), *Mamestra configurata* (Bertha armyworm), the hornworms *Manduca sexta* and *Manuduca quinquemaculata*, *Operophtera brumata* (winter moth), *Ostrinia nubilalis* (European corn borer), *Panolis flammea* (pine beauty moth), *Pectinophora gossypiella* (pink bollworm), *Phyllocnistis citrella* (citrus leafminer), *Pieris brassicae* (cabbage white butterfly), *Plutella xylostella* (diamondback moth), *Rachiplusia ni* (soybean looper), *Spilosoma virginica* (yellow bear moth), *Spodoptera exigua* (beet armyworm), *Spodoptera frugiperda* (fall armyworm), *Spodoptera littoralis* (cotton leafworm), *Spodoptera litura* (common cutworm), *Spodoptera praefica* (yellow-striped armyworm), *Sylepta derogata* (cotton leaf roller), *Tineola bisselliella* (webbing clothes moth), *Tineola pellionella* (case-making clothes moth), *Tortrix viridana* (European oak leafroller), *Trichoplusia ni* (cabbage looper), and *Yponomeuta padella* (small ermine moth).

Examples of Orthoptera include, but are not limited to: the common cricket *Acheta domesticus*, tree locusts (*Ana-*

cridium spp.), the migratory locust *Locusta migratoria*, the two-striped grasshopper *Melanoplus bivittatus*, the differential grasshopper *Melanoplus differentialis*, the redlegged grasshopper *Melanoplus femurrubrum*, the migratory grasshopper *Melanoplus sanguinipes*, the northern mole cricket *Neocurtilla hexadactyla*, the red locust *Nomadacris septemfasciata*, the shortwinged mole cricket *Scapteriscus abbreviatus*, the southern mole cricket *Scapteriscus borellii*, the tawny mole cricket *Scapteriscus vicinus*, and the desert locust *Schistocerca gregaria*.

Examples of Phthiraptera include, but are not limited to: the cattle biting louse *Bovicola bovis*, biting lice (*Damalinia* spp.), the cat louse *Felicola subrostrata*, the shortnosed cattle louse *Haematopinus elyosternus*, the tail-switch louse *Haematopinus quadripertusus*, the hog louse *Haematopinus suis*, the face louse *Linognathus ovillus*, the foot louse *Linognathus pedalis*, the dog sucking louse *Linognathus setosus*, the long-nosed cattle louse *Linognathus vituli*, the chicken body louse *Menacanthus stramineus*, the poultry shaft louse *Menopon gallinae*, the human body louse *Pediculus humanus*, the pubic louse *Phthirus pubis*, the little blue cattle louse *Solenopotes capillatus*, and the dog biting louse *Trichodectes canis*.

Examples of Psocoptera include, but are not limited to: the booklice *Liposcelis bostrychophila*, *Liposcelis decolor*, *Liposcelis entomophila*, and *Trogium pulsatorium*.

Examples of Siphonaptera include, but are not limited to: the bird flea *Ceratophyllus gallinae*, the dog flea *Ctenocephalides canis*, the cat flea *Ctenocephalides felis*, the human flea *Pulex irritans*, and the oriental rat flea *Xenopsylla cheopis*.

Examples of Symphyla include, but are not limited to: the garden symphylan *Scutigera immaculata*.

Examples of Thysanura include, but are not limited to: the gray silverfish *Ctenolepisma longicaudata*, the four-lined silverfish *Ctenolepisma quadriseriata*, the common silverfish *Lepisma saccharina*, and the firebrat *Thennobia domestica*.

Examples of Thysanoptera include, but are not limited to: the tobacco thrips *Frankliniella fusca*, the flower thrips *Frankliniella intonsa*, the western flower thrips *Frankliniella occidentalis*, the cotton bud thrips *Frankliniella schultzei*, the banded greenhouse thrips *Hercinothrips femoralis*, the soybean thrips *Neohydatothrips variabilis*, Kelly's citrus thrips *Pezothrips kellyanus*, the avocado thrips *Scirtothrips perseae*, the melon thrips *Thrips palmi*, and the onion thrips *Thrips tabaci*.

Examples of Nematodes include, but are not limited to: parasitic nematodes such as root-knot, cyst, and lesion nematodes, including *Heterodera* spp., *Meloidogyne* spp., and *Globodera* spp.; particularly members of the cyst nematodes, including, but not limited to: *Heterodera glycines* (soybean cyst nematode); *Heterodera schachtii* (beet cyst nematode); *Heterodera avenae* (cereal cyst nematode); and *Globodera rostochiensis* and *Globodera pallida* (potato cyst nematodes). Lesion nematodes include, but are not limited to: *Pratylenchus* spp.

In one embodiment, the insecticidal compositions comprising the polypeptides, polynucleotides, cells, vectors, etc., can be employed to treat ectoparasites. Ectoparasites include, but are not limited to: fleas, ticks, mange, mites, mosquitoes, nuisance and biting flies, lice, and combinations comprising one or more of the foregoing ectoparasites. The term "fleas" includes the usual or accidental species of parasitic flea of the order Siphonaptera, and in particular the

species *Ctenocephalides*, in particular *C. felis* and *C. canis*, rat fleas (*Xenopsylla cheopis*) and human fleas (*Pulex irritans*).

Insect pests of the invention for the major crops include, but are not limited to: Maize: *Ostrinia nubilalis*, European corn borer; *Agrotis ipsilon*, black cutworm; *Helicoverpa zea*, corn earworm; *Spodoptera frugiperda*, fall armyworm; *Diatraea grandiosella*, southwestern corn borer; *Elasmopalpus lignosellus*, lesser cornstalk borer; *Diatraea saccharalis*, sugarcane borer; *Diabrotica virgifera*, western corn rootworm; *Diabrotica longicornis barberi*, northern corn rootworm; *Diabrotica undecimpunctata howardi*, southern corn rootworm; *Melanotus* spp., wireworms; *Cyclocephala borealis*, northern masked chafer (white grub); *Cyclocephala immaculata*, southern masked chafer (white grub); *Popillia japonica*, Japanese beetle; *Chaetocnema pulicaria*, corn flea beetle; *Sphenophorus maidis*, maize billbug; *Rhopalosiphum maidis*, corn leaf aphid; *Anuraphis maidiradicis*, corn root aphid; *Blissus leucopterus leucopterus*, chinch bug; *Melanoplus femurrubrum*, redlegged grasshopper; *Melanoplus sanguinipes*, migratory grasshopper; *Hylemya platura*, seedcorn maggot; *Agromyza parvicornis*, corn blot leafminer; *Anaphothrips obscurus*, grass thrips; *Solenopsis milesta*, thief ant; *Tetranychus urticae*, twospotted spider mite; Sorghum: *Chilo partellus*, sorghum borer; *Spodoptera frugiperda*, fall armyworm; *Helicoverpa zea*, corn earworm; *Elasmopalpus lignosellus*, lesser cornstalk borer; *Feltia subterranea*, granulate cutworm; *Phyllophaga crinita*, white grub; *Eleodes*, *Conoderus*, and *Aeolus* spp., wireworms; *Oulema melanopus*, cereal leaf beetle; *Chaetocnema pulicaria*, corn flea beetle; *Sphenophorus maidis*, maize billbug; *Rhopalosiphum maidis*, corn leaf aphid; *Sipha flava*, yellow sugarcane aphid; *Blissus leucopterus leucopterus*, chinch bug; *Contarinia sorghicola*, sorghum midge; *Tetranychus cinnabarinus*, carmine spider mite; *Tetranychus urticae*, twospotted spider mite; Wheat: *Pseudaletia unipunctata*, army worm; *Spodoptera frugiperda*, fall armyworm; *Elasmopalpus lignosellus*, lesser cornstalk borer; *Agrotis orthogonia*, western cutworm; *Elasmopalpus lignosellus*, lesser cornstalk borer; *Oulema melanopus*, cereal leaf beetle; *Hypera punctata*, clover leaf weevil; *Diabrotica undecimpunctata howardi*, southern corn rootworm; Russian wheat aphid; *Schizaphis graminum*, greenbug; *Macrosiphum avenae*, English grain aphid; *Melanoplus femurrubrum*, redlegged grasshopper; *Melanoplus differentialis*, differential grasshopper; *Melanoplus sanguinipes*, migratory grasshopper; *Mayetiola destructor*, Hessian fly; *Sitodiplosis mosellana*, wheat midge; *Meromyza americana*, wheat stem maggot; *Hylemya coarctata*, wheat bulb fly; *Frankliniella fusca*, tobacco thrips; *Cephus cinctus*, wheat stem sawfly; *Aceria tulipae*, wheat curl mite; Sunflower: *Suleima helianthana*, sunflower bud moth; *Homoeosoma electellum*, sunflower moth; *Zygogramma exclamationis*, sunflower beetle; *Bothyrus gibbosus*, carrot beetle; *Neolasiptera murfeldtiana*, sunflower seed midge; Cotton: *Heliothis virescens*, cotton budworm; *Helicoverpa zea*, cotton bollworm; *Spodoptera exigua*, beet armyworm; *Pectinophora gossypiella*, pink bollworm; *Anthonomus grandis*, boll weevil; *Aphis gossypii*, cotton aphid; *Pseudatomoscelis seriatus*, cotton fleahopper; *Trialeurodes abutilonea*, bandedwinged whitefly; *Lygus lineolaris*, tarnished plant bug; *Melanoplus femurrubrum*, redlegged grasshopper; *Melanoplus differentialis*, differential grasshopper; *Thrips tabaci*, onion thrips; *Frankliniella fusca*, tobacco thrips; *Tetranychus cinnabarinus*, carmine spider mite; *Tetranychus urticae*, twospotted spider mite; Rice: *Diatraea saccharalis*, sugarcane borer; *Spodoptera frugiperda*, fall armyworm; *Helicoverpa zea*,

corn earworm; *Colaspis brunnea*, grape colaspis; *Lissorhoptrus oryzophilus*, rice water weevil; *Sitophilus oryzae*, rice weevil; *Nephotettix nigropictus*, rice leafhopper; *Blissus leucopterus leucopterus*, chinch bug; *Acrosternum hilare*, green stink bug; Soybean: *Pseudoplusia includens*, soybean looper; *Anticarsia gemmatilis*, velvetbean caterpillar; *Plathypena scabra*, green cloverworm; *Ostrinia nubilalis*, European corn borer; *Agrotis ipsilon*, black cutworm; *Spodoptera exigua*, beet armyworm; *Heliothis virescens*, cotton budworm; *Helicoverpa zea*, cotton bollworm; *Epilachna varivestis*, Mexican bean beetle; *Myzus persicae*, green peach aphid; *Empoasca fabae*, potato leafhopper; *Acrosternum hilare*, green stink bug; *Melanoplus femurrubrum*, redlegged grasshopper; *Melanoplus differentialis*, differential grasshopper; *Hylemya platura*, seedcorn maggot; *Sericothrips variabilis*, soybean thrips; *Thrips tabaci*, onion thrips; *Tetranychus turkestanii*, strawberry spider mite; *Tetranychus urticae*, twospotted spider mite; Barley: *Ostrinia nubilalis*, European corn borer; *Agrotis ipsilon*, black cutworm; *Schizaphis graminum*, greenbug; *Blissus leucopterus leucopterus*, chinch bug; *Acrosternum hilare*, green stink bug; *Euschistus servus*, brown stink bug; *Delia platura*, seedcorn maggot; *Mayetiola destructor*, Hessian fly; *Petrobia latens*, brown wheat mite; Oil Seed Rape: *Brevicoryne brassicae*, cabbage aphid; *Phyllotreta cruciferae*, Flea beetle; *Mamestra configurata*, Bertha armyworm; *Plutella xylostella*, Diamond-back moth; *Delia* spp., Root maggots.

In some embodiments, the insecticidal compositions can be employed to treat combinations comprising one or more of the foregoing insects.

The insects that are susceptible to the peptides of this invention include but are not limited to the following: Cyt toxins affect families such as: Blattaria, Coleoptera, Collembola, Diptera, Echinostomida, Hemiptera, Hymenoptera, Isoptera, Lepidoptera, Neuroptera, Orthoptera, Rhabditida, Siphonoptera, Thysanoptera. Genus-Species are indicated as follows: *Actebia-fennica*, *Agrotis-ipsilon*, *A.-segetum*, *Anticarsia-gemmatilis*, *Argyrotaenia-citrana*, *Artogeia-rapae*, *Bombyx-mori*, *Busseola-fusca*, *Cacyreus-marshall*, *Chilo-suppressalis*, *Christoneura-fumiferana*, *C.-occidentalis*, *C.-pinus pinus*, *C.-rosacena*, *Cnaphalocrocis-medinalis*, *Conopomorpha-cramerella*, *Ctenopsuestis-obliquana*, *Cydia-pomonella*, *Danaus-plexippus*, *Diatraea-saccharalis*, *D.-grandiosella*, *Earias-vittella*, *Elasmopalpus-lignosellus*, *Eldana-saccharina*, *Ephestia-kuehniella*, *Epinotia-aporema*, *Epiphyas-postvittana*, *Galleria-mellonella*, *Genus-Species*, *Helicoverpa-zea*, *H.-punctigera*, *H.-armigera*, *Heliothis-virescens*, *Hyphantria-cunea*, *Lambdina-fiscellaria*, *Leguminivora-glycinivorella*, *Lobesia-botrana*, *Lymantria-dispar*, *Malacosoma-disstria*, *Mamestra-brassicacae*, *M. configurata*, *Manduca-sexta*, *Marasmia-patnalis*, *Maruca-vitrata*, *Orgyia-leucostigma*, *Ostrinia-nubilalis*, *O.-furnacalis*, *Pandemis-pyrusana*, *Pectinophora-gossypiella*, *Perileucoptera-coffeella*, *Phthorimaea-operculata*, *Pianotortrix-octo*, *Piatynota-stultana*, *Pieris-brassicacae*, *Plodia-interpunctata*, *Plutella-xylostella*, *Pseudoplusia-includens*, *Rachiplusia-nu*, *Sciropophaga-incertulas*, *Sesamia-calamistis*, *Spilosoma-virginica*, *Spodoptera-exigua*, *S.-frugiperda*, *S.-littoralis*, *S.-exempta*, *S.-litura*, *Tecia-solanivora*, *Thaumetopoea-pityocampa*, *Trichoplusia-ni*, *Wiseana-cervinata*, *Wiseana-copularis*, *Wiseana-jocosa*, *Blattaria-Blattella*, *Collembola-Xenylla*, *C.-Folsomia*, *Echinostomida-Fasciola*, *Hemiptera-Onco-peltrus*, *He.-Bemisia*, *He.-Macrosiphum*, *He.-Rhopalosiphum*, *He.-Myzus*, *Hymenoptera-Diprion*, *Hy.-Apis*, *Hy.-Macrocentrus*, *Hy.-Meteorus*, *Hy.-Nasonia*, *Hy.-Solenopsis*, *Isoptoda-Porcello*, *Isoptera-Reticulitermes*, *Orthoptera-*

Achta, *Prostigmata-Tetranychus*, *Rhabditida-Acroboloides*, *R.-Caenorhabditis*, *R.-Distolabrellus*, *R.-Panagrellus*, *R.-Pristionchus*, *R.-Pratylenchus*, *R.-Ancylostoma*, *R.-Nippostrongylus*, *R.-Panagrellus*, *R.-Haemonchus*, *R.-Meloidogyne*, and *Siphonaptera-Ctenocephalides*.

We describe Part II with the following description and summary:

We describe a peptide with an N-terminal dipeptide which is added to and operably linked to a known peptide, wherein said N-terminal dipeptide is comprised of one nonpolar amino acid on the N-terminal of the dipeptide and one polar amino acid on the C-terminal of the dipeptide, wherein said peptide is selected from a CRIP (Cysteine Rich Insecticidal Peptide), such as from an ICK peptide, or a Non-ICK peptide. The N-terminal dipeptide which is added to and operably linked to a known peptide, wherein said N-terminal dipeptide is comprised of one nonpolar amino acid on the N-terminal of the dipeptide and one polar amino acid on the C-terminal of the dipeptide. The N-terminal dipeptide has a non-polar amino acid as the N-terminal amino acid of the N-terminal dipeptide that can be selected from glycine, alanine, proline, valine, leucine, isoleucine, phenylalanine and methionine and a polar amino acid of the C-terminal amino acid of the N-terminal peptide can be selected from serine, threonine, cysteine, asparagine, glutamine, histidine, tryptophan, tyrosine.

The N-terminal dipeptide can have a non-polar amino acid as the N-terminal amino acid of the N-terminal dipeptide selected from glycine, alanine, proline, valine, leucine, isoleucine, phenylalanine and methionine and said polar amino acid of the C-terminal amino acid of the N-terminal peptide is selected from serine, threonine, cysteine, asparagine, glutamine, histidine, tryptophan, tyrosine. The N-terminal dipeptide can and preferably is comprised of glycine-serine.

We describe a peptide with a N-terminal dipeptide which is added to and operably linked to a known peptide, wherein said N-terminal dipeptide is comprised of one nonpolar amino acid on the N-terminal of the dipeptide and one polar amino acid on the C-terminal of the dipeptide, wherein said peptide is selected from a PFIP (Pore Forming Insecticidal Protein), or it could be selected from a CRIP (Cysteine Rich Insecticidal Peptide), such as from an ICK peptide, or a Non-ICK peptide. The Non-ICK peptide could be a sea anemone, origin peptide like Av2 or Av3 and the preferred dipeptide is comprised of glycine-serine. The ICK peptide could be from a spider like the ACTX peptides and the preferred dipeptide is comprised of glycine-serine. The PFIP could be a Bt protein, like any of those disclosed herein, in the sequence listing and know to one skilled in the art who reads these description and the preferred dipeptide is comprised of glycine-serine.

As noted above we explain that the N-terminal dipeptide is comprised of one nonpolar amino acid on the N-terminal of the dipeptide and one polar amino acid on the C-terminal of the dipeptide and the non-polar amino acid from the N-terminal amino acid of the N-terminal dipeptide can be selected from glycine, alanine, proline, valine, leucine, isoleucine, phenylalanine and methionine and preferably the non-polar amino acid is glycine. And we explain and claim that any of the peptides in the paragraph below and any of the peptides in this paragraph can act independently and should be treated independently and all of the possible combinations are claimed independently.

As noted above we explain that the N-terminal dipeptide is comprised of one nonpolar amino acid on the N-terminal of the dipeptide and one polar amino acid on the C-terminal

of the dipeptide and the polar amino acid of the C-terminal amino acid of the N-terminal peptide is selected from serine, threonine, cysteine, asparagine, glutamine, histidine, tryptophan, tyrosine and preferably the polar amino acid is serine. And we explain and claim that any of the peptides in the paragraph above and any of the peptides in this paragraph can act independently and should be treated independently and all of the possible combinations are claimed independently.

The peptide to which the N-terminal dipeptide is attached can be any peptide, any toxic peptide, any insecticidal peptide, any PFIP, any CRIP, a CRIP that is a ACTX peptide (which is an example of an ICK peptide), CRIP is a sea anemone peptide (which is an example of a Non-ICK peptide), it can be a PFIP, the PFIP can be a Bt protein, the Bt protein can be cry, cyt, VIP and it can be like any of these peptides as disclosed herein, or in the sequence listing, or known by one skilled in the art who reads these descriptions and understands the document.

We specifically note that these procedures are useful and we claim the procedures themselves and the products of the procedures both as independent claims and as process by product claims for making any insecticidal peptide and in particular any peptide selected from any of the peptides or sources of peptides including *Atrax* or *Hadronyche*, as disclosed herein or elsewhere, as well as any insecticidal peptide with fragments thereof including mature, pre, and pro peptide versions of said peptides and sequence numbers and the peptide in SEQ. ID. NO. 5.

These peptides are useful and the procedures can all be made and used where there is one nonpolar amino acid at the N-terminal end and one polar amino acid at the C-terminal end, and the dipeptide of said non-polar amino acid is selected from glycine, alanine, proline, valine, leucine, isoleucine, phenylalanine and methionine, and it is preferably glycine or gly, and the polar amino acid is selected from serine, threonine, cysteine, asparagine, glutamine, histidine, tryptophan and tyrosine and it is preferably serine or ser. The dipeptide gly-ser is most preferred. The dipeptide can be operably linked to any known peptide, any toxic peptide, any insecticidal peptide, any of the peptides including *Atrax* or *Hadronyche*, disclosed herein any insecticidal peptide with fragments thereof including mature, pre, and pro peptide versions of said peptides and sequence numbers, any mature insecticidal peptide, the toxic peptide comprises SEQ. ID. NO; 6, or the toxic peptide comprises GSQYC VPVDQ PCSLN TQPC DDATC TQERN ENGHT VYYCR A (SEQ ID NO: 5).

We also describe and claim the dipeptide Gly-Ser, nucleotides encoding the dipeptide Gly-Ser selected from GGT, GGC, GGA, or GGG, any of which encodes Gly, and TCT, TCC, TCA, TCG, AGT, and AGC, any of which encodes Ser, and those nucleotides linked to any of the proteins and the process and the products of the process. We describe and claim these nucleotides which code for these peptides operably linked to the 5' terminus of the DNA sequence encoding any peptide disclosed herein.

We explain and disclose a process for increasing the yield of insecticidal peptides which are produced from yeast expression systems comprising the addition of any dipeptide to the N-terminus of any insecticidal peptide. The process and product by process for increasing the yield used the dipeptide as discussed in the paragraphs above.

We specifically discuss the procedures, products, process and products by process with any insecticidal peptide that inhibits both voltage-gated Calcium channels and Calcium-activated potassium channels in insects, with peptide origins

from any species of Australian Funnel-web spider, a spider is selected from the Australian Funnel-web spiders of genus *Atrax* or *Hadronyche*, including *Hadronyche versuta*. We also specifically describe and claim insecticidal peptides that are not ICK motif peptide such as peptides with origins from any species of venomous sea anemone, we refer to the proteins as examples of CRIP motif peptide, that are Non-ICK. We disclose and have tested and show that the procedures work with proteins from the sea anemone genus *Anemonia*, and specifically from selected species, *Anemonia viridis*. We believe to a scientific certainty that the methods will work with insecticidal peptides that contain contains 20-100 amino acids and 2-6 disulfide bonds, and with insecticidal peptide is any insecticidal peptide with at least 50%, 55%, 60%, 65%, 70%, 75%, 80%, 85%, 90%, 95%, 96%, 97%, 98%, or 99% or greater sequence identity to SEQ ID NO 5, SEQ ID NO 6, Av2 and Av3.

We specifically describe and claim the procedures when used with any species of yeast, including but not limited to any species of the genera *Saccharomyces*, *Pichia*, *Kluyveromyces*, *Hansenula*, *Yarrowia* or *Schizosaccharomyces* and the species *Saccharomyces* includes any species of *Saccharomyces*, and preferably we disclose the *Saccharomyces* species *Saccharomyces cerevisiae*. We specifically disclose *Saccharomyces cerevisiae* species is selected from following strains: INVSc1, YNN27, S150-2B, W303-1B, CG25, W3124, JRY188, BJ5464, AH22, GRF18, W303-1A and BJ3505. We specifically disclose *Pichia* species including any species of *Pichia* and preferably the *Pichia* species, *Pichia pastoris*, and preferably the *Pichia pastoris* is selected from following strains: Bg08, Y-11430, X-33, GS115, GS190, JC220, JC254, GS200, JC227, JC300, JC301, JC302, JC303, JC304, JC305, JC306, JC307, JC308, YJN165, KM71, MC100-3, SMD1163, SMD1165, SMD1168, GS241, MS105, any pep4 knock-out strain and any prbl knock-out strain, as well as *Pichia pastoris* is selected from following strains: Bg08, X-33, SMD1168 and KM71. We specifically disclose *Kluyveromyces* species includes any species of *Kluyveromyces*, and preferably *Kluyveromyces lactis*, and we teach that the strain of *Kluyveromyces lactis* can be but is not required to be selected from following strains: GG799, YCT306, YCT284, YCT389, YCT390, YCT569, YCT598, MW98-8C, MS1, CBS293.91, Y721, MD2/1, PM6-7A, WM37, K6, K7, 22AR1, 22A295-1, SD11, MG1/2, MSK110, JA6, CMK5, HP101, HP108 and PM6-3C, in addition to *Kluyveromyces lactis* species is selected from GG799 and YCT306.

We specifically describe and claim the procedures when used with any species of yeast, including but not limited to any species of *Hansenula* species including any species of *Hansenula* and preferably *Hansenula polymorpha*. We specifically describe and claim the procedures when used with any species of yeast, including but not limited to any species of *Yarrowia* species including any species of *Yarrowia* and preferably *Yarrowia lipolytica*. We specifically describe and claim the procedures when used with any species of yeast, including but not limited to any species of *Schizosaccharomyces* species including any species of *Schizosaccharomyces* and preferably *Schizosaccharomyces pombe*. Part 3. In this Part we Describe Combinations of "CRIPS" and "PFIPS."

A large number of venom peptides have been characterized as "insecticidal." However, despite numerous reports, few have found any utility in the market as actual or effective insecticides. In fact, only ω -ACTX-Hv1a has been reported to be toxic by oral administration to the American lone star tick *Amblyomma americanum*. No other spider

toxins have been reported to possess oral activity even in the modified gut of ticks. There has been a report that the bioavailability of these peptides may be increased by coupling them to a carrier protein such as snowdrop lectin (*Galanthus nivalis* agglutinin, GNA). Mukherjee, A. K.; Sollod, B. L.; Wikel, S. K.; King, G. F. "Orally active acaricidal peptide tosins from spider venom." *Toxicon* 2006, 47, 182-187. Garlic lectins are reported to increase the absorption of toxins across the insect midgut Fitches, E et al., *Insect Sci.*, 2008, 15, 483-495, Fitches, E., et al., *Insect Biochem. Mol. Biol.* 2008, 38, 905-915. Fitches, E. et al., *J. Insect Physiol.* 2004, 50, 61-71. For example, fusion of the insecticidal spider toxin U2-SGTX-Sf1a (Sf11) to GNA significantly increased its oral toxicity to the tomato moth *Lacnobia oleracea* Down, R. E. et al., *Pest Manag. Sci.* 2006, 62, 77-85, as well as the rice brown planthopper *Nilaparvata lugens* and the peach-potato aphid *Myzus persicae*. Surprisingly, a thioredoxin- ω -HXTX-Hv1a fusion protein was found to be insecticidal in *Helicoverpa armigera* and *Spodoptera littoralis* caterpillars by topical application, Khan, S. A. *Transgenic Res.* 2006, 15, 349-357. (although the fusion protein was applied topically in a solution containing high levels of imidazole, a compound known to have contact insecticidal activity; Pence, R. J. *California Agric.* 1965, 13-15. These efforts and findings clearly indicate the importance of developing means to enhance the oral bioavailability of venom toxins. We think these efforts are also misdirected. In this disclosure we teach that fusion of insecticidal peptides to carrier proteins that bind to the gut of insects is unnecessary. We describe a better way to deliver the "toxin" in insecticidal peptides to insects. Without wishing to be bound by theory, it is our theory that PFIPS, or Pore Forming Insecticidal Proteins, act by selectively binding to receptors in the insect gut. The PFIPS then, in subsequent events, act to disrupt the membrane potential of the epithelial cells lining the gut. When an appropriate CRIP or TMOF is also timely introduced to the gut at the same time the PFIPS are acting on the insect gut, the result is apoptosis and death of the cells lining the gut. Thus, the gut lining is breached and simultaneously the venomous peptides, often large peptides isolated from venom, can pass through the gut and sicken or kill the target insect. Surprisingly, insects that have developed resistance to Bt proteins have no defenses and show no resistance at all to even low levels of Bt, when a PFIP like Bt is administered to an insect in combination with CRIP or TMOF, that is a toxic peptide, but one with properties that do not act like a PFIPS such as Bt. We provide data showing that certain combinations of co-administered CRIPS and PFIPS can provide more than double the killing and stopping power than would be expected from similar concentration applications of either a CRIP OR PFIPS applied individually.

Examples of a PFIP include the cry and VIP proteins from Bt organisms. Bt proteins like the cry proteins disrupt the insect gut membrane allowing for adventitious infection (sepsis) of the insect by gut flora. In the absence of gut microbes, Bt is not insecticidal. Broderick, Nichole PNAS Vol. 103, No. 41 (2006). Hence one would expect that the mechanism shown to cause Bt mortality (infection) would be mitigated in those insects showing Bt resistance, and it is mitigated in those insects. Bt resistant insects show little gut disruption even when fed high levels of Bt proteins, like cry. What we have surprisingly discovered is that somehow even though these insects guts no longer display the dramatic effects of Bt on the gut, that is they are truly resistant, when they are exposed to insecticidal peptides of a certain type, like the CRIPS and TMOF which have a very different mode

of action than PFIPS like Bt, then these very resistant insects have no resistance what so ever. The disappearance of resistance in a "Bt resistant" insect is surprising, and we show this happens, with our data, in the examples provided herein. This result was completely unexpected. Now however we understand, and we can use this knowledge to explain how sublethal amounts of a PFIP protein like Bt, can be "converted" into a lethal cocktail such that if two (2) or more sublethal amounts of insecticidal protein are co-administered, then the combination of proteins becomes lethal to insects which are otherwise thought to be too large, or too resistant to be susceptible to toxic peptides.

It is surprising that insect resistance to PFIPs alone does not confer resistance to the combination of PFIPS with CRIPS and or TMOF. Because of the mechanism of action of the PFIPS one would expect that the PFIP, like a Bt protein, would no longer contribute to the toxic effects of the combination of PFIPS with CRIPS and or TMOF. Instead the opposite happens and the combination has a greater than expected level of activity as shown with our data.

Insects have developed resistance to Bt. Attempts to combat this resistance have resulted in the use of many different subtypes of Bt. We teach here that insect resistance can be overcome by co-application of venom peptides. Since the most common mode of resistance (mode 1, prior ref) Pence, R. J. "The antimetabolite imidazole as a pesticide." *California Agric.* 1965, 13-15. is down regulation of Bt receptors that line the gut, one would expect insect resistance would be maintained in Bt resistant insects because the number of receptors is insufficient to render the insect vulnerable to sepsis by gut flora. What we have discovered and believe, and our data supports our theory in dramatic fashion (see examples below), is that even with Bt resistant insects there remains sufficient membrane abnormalities that exposure to even low levels of Bt, when combined with certain small a "toxic" insecticidal peptides, having a different type of mode of action than Bt, will surprisingly cause Bt resistant insects to stop feeding or die. We believe this is because the gut lining is still disrupted in these resistant insects, just enough, enough to allow the allow passage of the much smaller venom peptides characteristic of either CRIP and TMOF types of insecticidal peptides.

In this document we do not consider TMOF peptides or Trypsin modulating oostatic factor (TMOF) peptides which have been identified as a potential larvicides, see D. Borovsky, *Journal of Experimental Biology* 206, 3869-3875, to be a CRIP type of insecticidal peptide. We define a CRIP peptide as one with various cysteines according to our definitions herein. TMOF peptides does not fit motif that we describe as a CRIP peptide. Please see the definition section toward the beginning of these documents for a definition of CRIP and TMOF. We discuss combining CRIP and or TMOF type of proteins with a different type of protein we describe as PFIPS.

PFIPS are Pore Forming Insecticidal Proteins which are also defined in the definition section. One example of one type of PFIP are various proteins of the widely used group of proteins derived from Bt, such as cry, cyt and VIP. These are effective insecticides used for crop protection in the form of both plant incorporated protectants and foliar sprays. Commercial formulations of such Bt proteins are widely used to control insects at the larval stage.

In contrast to PFIPS, CRIPS such as Inhibitory cysteine knot or ICK peptides are very different group of peptides that also have insecticidal activity, but they act with a very different mode of action. In this document there is no overlap of a PFIP protein with a CRIP protein, the two

groups are separate and distinct. ICK peptides and even Non-ICK peptides are both considered CRIPS in this document. CRIPS are often toxic to naturally occurring biological target species, usually insects or arachnids of some type. Often CRIP peptides can have arthropod origins such as the venoms of scorpions or spiders, this venom origin is very common with ICKs. CRIP may be delivered to their physiological site of action in various ways, for example by delivering the toxin directly to the insect's gut or internal organs by injection, by application to an insect locus and uptake from surface contact, or by inducing the insect to consume the toxin from its food, for example an insect feeding upon a transgenic plant.

The peptides described herein may be formulated as either applied products or through transgenic plants face challenges. It can be difficult to successfully produce such peptides on a commercial scale, with reproducible peptide formation and folding. Cost controls can be challenging. The wide variety, unique properties and special nature of peptides, combined with the huge variety of possible production techniques present an overwhelming number of approaches to peptide production. Commercial products have their own significant challenges. Peptides are often unstable when applied in the environment of a crop. UV irradiation and other factors can cause Bt insecticides to decay rapidly in the environment, often in as little as a few hours. Further, commercial effectiveness can change. Both Bt spray on products and the transgenic Bt proteins used as plant incorporated protectant face emerging insect resistance.

A product is needed that enhances the acute activity, improves resistance performance, or extends the duration of action in order to increase insect control and crop protection.

Here we present combinations of Bt Protein and ICK and TMOF peptides in various combinations. We describe examples of these novel combinations. The new combinations, products, methods, and their formulation and uses thereof are described and claimed herein.

Cysteine Rich Insecticidal Peptides (CRIPS) in Synergistic Combinations

Cysteine rich insecticidal peptides (CRIPS) are peptides rich in cysteine which form disulfide bonds. The cysteine-cysteine disulfide bonds play a significant role in the toxicity of these insecticidal peptides which are exemplified by both inhibitory cysteine knot or ICK peptides and by examples of toxic peptides with disulfide bonds that are not considered ICK peptides (non-ICK CRIPS) such as peptides from the sea anemone, like Av2 and Av3 peptides. These cysteine-cysteine disulfide bonds stabilized toxic peptides (CRIPS) can have remarkable stability when exposed to the environment. Many ICK peptides are isolated from venomous animals such as spiders, scorpions, and snakes and are toxic to insects. TMOF peptides are known to have larvicidal activity. Av2 and Av3 peptides are isolated from sea anemones. We also describe a different group of peptides that act on the lining of the insect gut. We call these PFIPS for Pore Forming Insecticidal Proteins. Most well known examples of a PFIPS are the Bt proteins, well known because of their specific pesticidal activities and commercial applications. Surprisingly, we discovered that, when the combination of these peptides, PFIPS and CRIPS are combined and administered so they act together in the gut (co-administration of the combination not required only the combination of the activity in the gut is needed) they become highly effective at controlling insects. For example, one of the preferred combinations would be to combine a Bt protein with an ICK

peptides, or sea anemone peptides they create a highly effective insecticide with a potency much greater than one would expect.

We describe an insecticidal combination peptide composition comprising both a PFIP (Pore Forming Insecticidal Proteins) in combination with either a CRIP and/or a TMOF type of insecticidal protein. Note that CRIP includes such insecticidal proteins as ICK (Inhibitor Cystine Knot) peptides, and Non-ICK proteins but TMOF peptides are not considered CRIP proteins. CRIP proteins can include Non-ICK proteins like the proteins first identified in sea anemones, for example Av2 or Av3. The composition can be in the ratio of PFIP to CRIP and or TMOF, on a dry weight basis, from about any or all of the following ratios: 99:1, 95:5, 90:10, 85:15, 80:20, 75:25, 70:30, 65:35, 60:40, 55:45, 50:50, 45:55, 40:60, 35:65, 30:70, 25:75, 20:80, 15:85, 10:90, 5:95 and 1:99, or any combination of any two of these values. We also describe a composition where the ratio of PFIP to CRIP or TMOF on a dry weight basis, is selected from about the following ratios: 50:50, 45:55, 40:60, 35:65, 30:70, 25:75, 20:80, 15:85, 10:90, 5:95, 1:99, 0.5:99.5, 0.1:99.9 and 0.01:99.99 or any combination of any two of these values. CRIP, ICK, Non-ICK CRIP and TMOF can be either 100% of the peptide combined with Bt, or either peptide in any combination that totals 100% of both ICK+TMOF peptide can be combined with Bt.

In another embodiment the combination of mixtures of PFIP in combination with CRIP or TMOF peptides includes either or both of the PFIP and CRIP, ICK and non ICK peptides which are derived from more than 1 different types or bacterial strain origins for either one or both of PFIP, ICK and TMOF peptides. By bacterial strain origins we mean the peptides can be described as having been expressed by a bacterial strain that expresses the peptides with the understanding that many peptides are also artificial in the sense that they are no longer all developed from animal or bacterial strains.

We also disclose compositions where either or both of mixtures of PFIP in combination with CRIP or TMOF peptides and or mixtures of PFIP in combination with CRIP plus or with TMOF peptides are derived from between 2 and 5, 2-15, 2-30, 5-10, 5-15, 5-30, 5-50 and various other different types or bacterial strains origins of either one or both of the proteins. We disclose a composition where either or both of the proteins are encoded by from 2 to 15 different types or bacterial strain origins of either one or both of the PFIP in combination with CRIP or TMOF peptides. And any of these combinations of 2-5, 2-15, 2-30, 5-10, 5-15, 5-30, 5-50 and various other different types and mixtures of PFIP in combination with CRIP or TMOF peptides can contribute more than at least 1% of each strain type to the composition.

We disclose compositions of Bt and ICK, Bt and TMOF peptides or BT and ICK+TMOF peptides where the total concentration of Bt and ICK peptide, Bt and TMOF peptides or BT and ICK+TMOF peptides in the composition is selected from the following percent concentrations: 0, 1, 5, 10, 15, 20, 25, 30, 35, 40, 45, 50, 55, 60, 65, 70, 75, 80, 85, 90, 95, 99 or 100%, or any range between any two of these values, and the remaining percentage of the composition is comprised of excipients. We disclose compositions wherein the insecticidal combination peptide is produced using a genetic cassette that further comprises an ERSP (Endoplasmic Reticulum Signal Peptide) operably linked to the insecticidal ICK peptide, wherein said ERSP is linked at the N-terminal of the insecticidal ICK peptide. We disclose compositions wherein the insecticidal combination peptide is produced using a genetic cassette that further comprises

an ERSP (Endoplasmic Reticulum Signal Peptide) operably linked to the insecticidal ICK peptide, wherein said ERSP is linked at the N-terminal of the insecticidal ICK peptide, wherein the ERSP is BAAS.

We disclose compositions wherein said combination of peptides is produced using a genetic cassette that further comprises a dipeptide operably linked to the insecticidal ICK and of TMOF peptide, wherein said dipeptide is linked at the N-terminal of the insecticidal ICK peptide; and wherein the dipeptide is comprised of one nonpolar amino acid on the N-terminal of the dipeptide and one polar amino acid on the C-terminal of the dipeptide, including embodiments where the dipeptide is glycine-serine, including embodiments where the insecticidal ICK peptide is any insecticidal peptide that inhibits both voltage-gated Calcium channels and Calcium-activated potassium channels in insects, including embodiments where the insecticidal ICK peptide originates from any species of Australian Funnel-web spider, including embodiments where the spider is selected from the Australian Funnel-web spiders of genus *Atrax* or *Hadronyche*, including embodiments where the spider is selected from the Australian Funnel-web spiders of genus *Hadronyche*, including embodiments where the spider is selected from the Australian Blue Mountains Funnel-web, *Hadronyche versuta*, including embodiments where the insecticidal ICK peptide is Hybrid-ACTX-Hv1a, including embodiments where the insecticidal ICK peptide contains 20-100 amino acids and 2-4 disulfide bonds, including embodiments where said insecticidal ICK peptide is any insecticidal peptide with at least 50%, 55%, 60%, 65%, 70%, 75%, 80%, 85%, 90%, 95%, 96%, 97%, 98%, or 99% or greater sequence identity to any of the ICK sequences disclosed herein, including embodiments where the insecticidal ICK peptide is selected from publications incorporated by reference, including embodiments where the Bt protein is any insecticidal Bt protein, including embodiments where the Bt protein is a Cry or Cyt protein, including embodiments where the Bt protein is selected from the group consisting of a Cry1, Cry3, TIC851, CryET70, Cry22, TIC901, TIC201, TIC407, TIC417, a binary insecticidal protein CryET80, and CryET76, a binary insecticidal protein TIC100 and TIC101, a combination of an insecticidal protein ET29 or ET37 with an insecticidal protein TIC810 or TIC812 and a binary insecticidal protein PS149B1, including embodiments where the Bt protein is selected from a Cry protein, a Cry1A protein or a Cry1F protein, including embodiments where the Bt protein is a combination Cry1F-Cry1A protein, including embodiments where the Bt protein comprises an amino acid sequence at least 90% identical to SEQ ID NO: 10, 12, 14, 26, 28, or 34 of U.S. Pat. No. 7,304,206, including embodiments where the Bt Protein is DIPEL®, including embodiments where the Bt protein is THURICIDE®.

We disclose a composition comprising the nucleotides of: Bt (*Bacillus thuringiensis*) protein; and an insecticidal ICK (Inhibitor Cystine Knot) peptide, Bt and TMOF peptide or BT and ICK+TMOF peptides in a transformed plant or plant genome; where the ratio of Bt to ICK, Bt and TMOF peptides or BT and ICK+TMOF peptides, on a dry weight basis, is selected from about the following ratios: 99:1, 95:5, 90:10, 85:15, 80:20, 75:25, 70:30, 65:35, 60:40, 55:45, 50:50, 45:55, 40:60, 35:65, 30:70, 25:75, 20:80, 15:85, 10:90, 5:95 and 1:99, or any combination of any two of these values.

We disclose transformed plant or plant genome wherein the ratio of Bt to ICK, Bt and TMOF peptides or BT and ICK+TMOF peptides on a dry weight basis, is selected from

about the following ratios: 50:50, 45:55, 40:60, 35:65, 30:70, 25:75, 20:80, 15:85, 10:90, 5:95, 1:99, 0.5:99.5, 0.1:99.9 and 0.01:99.99 or any combination of any two of these values. The transformed plant or plant genome may have either or both of the Bt and ICK peptides are derived from more than 1 different type or bacterial strain origin of Bt or ICK peptides, or either or both of the Bt and ICK peptides are derived from between 2 and 5 different type or bacterial strain origin of either Bt or ICK peptides or both Bt and ICK peptides are derived from between 2 and 5 different types or strain origins, or either or both of the Bt and ICK peptides are derived from 2 to 15 different type or bacterial strain origins of either or both of Bt and ICK peptides and at least one strain of either Bt or ICK or both Bt and ICK peptides encoded by more than one copy of the Bt or ICK genes, or either or both of the Bt and ICK peptides are derived from more than one different type or bacterial strain origin of Bt and/or ICK peptides where all the strains of Bt and/or ICK peptides contribute more than at least 1% of each strain type to said composition, or either or both of the Bt and ICK peptides are derived from 2 to 5 different type or bacterial strain origins of either or both of Bt and ICK peptides and at least one strain of either Bt or ICK or both Bt and ICK peptides encoded by more than one copy of the Bt or ICK genes, or the total concentration of Bt and ICK peptide in the composition can be selected from the following percent concentrations: 0, 1, 5, 10, 15, 20, 25, 30, 35, 40, 45, 50, 55, 60, 65, 70, 75, 80, 85, 90, 95, 99 or 100%, or any range between any two of these values, and the remaining percentage of the composition is comprised of excipients.

The compositions and plants described herein include an insecticidal combination peptide produced using a genetic cassette that further comprises an ERSP (Endoplasmic Reticulum Signal Peptide) operably linked to the insecticidal ICK peptide, or to a TMOF peptide wherein said ERSP is linked at the N-terminal of the insecticidal ICK or TMOF peptide. In another embodiment the insecticidal combination peptide is produced using a genetic cassette that further comprises an ERSP (Endoplasmic Reticulum Signal Peptide) operably linked to the insecticidal ICK peptide, wherein said ERSP is linked at the N-terminal of the insecticidal ICK peptide, wherein the ERSP is BAAS. In another embodiment the transgenic plant incorporating and expressing the combination peptides from the nucleotides described herein, wherein said combination peptide is produced using a genetic cassette that further comprises nucleotides expressing a dipeptide operably linked to the insecticidal ICK or TMOF peptide, wherein said dipeptide is linked at the N-terminal of the insecticidal ICK peptide; and wherein the dipeptide is comprised of one nonpolar amino acid on the N-terminal of the dipeptide and one polar amino acid on the C-terminal of the dipeptide. In another embodiment the transgenic plant has a dipeptide that glycine-serine. In another embodiment the transgenic plant has insecticidal ICK peptides expressed that are comprised of an insecticidal peptide combination of ICK and Bt proteins. The transgenic plants can have an insecticidal ICK peptide derived from any species of Australian Funnel-web spider, or the Australian Funnel-web spiders of genus *Atrax* or *Hadronyche*, and the Australian Blue Mountains Funnel-web, *Hadronyche versuta*.

We describe and claim a transgenic plant wherein the insecticidal ICK peptide expressed is Hybrid-ACTX-Hv1a, and or the insecticidal ICK peptide expressed may contain 20-100 amino acids and 2-4 disulfide bonds and or the insecticidal ICK peptide is any insecticidal peptide with at least 50%, 55%, 60%, 65%, 70%, 75%, 80%, 85%, 90%,

95%, 96%, 97%, 98%, or 99% or greater sequence identity to any of the ICK peptides described herein. The transgenic plants disclosed can contain any known Bt protein, including peptides where the Bt protein is a Cry or Cyt protein, and/or the Bt protein is selected from the group consisting of a Cry1, Cry3, TIC851, CryET70, Cry22, TIC901, TIC201, TIC407, TIC417, a binary insecticidal protein CryET80, and CryET76, a binary insecticidal protein TIC100 and TIC101, a combination of an insecticidal protein ET29 or ET37 with an insecticidal protein TIC810 or TIC812 and a binary insecticidal protein PS149B1. The Bt protein can be selected from a Cry protein, a Cry1A protein or a Cry1F protein, or a combination Cry1F-Cry1A protein, or it comprises an amino acid sequence at least 90% identical to sequences 10, 12, 14, 26, 28, or 34 of U.S. Pat. No. 7,304,206. We describe a transgenic plant wherein the Bt protein is DIPEL® and we describe a transgenic plant wherein the Bt protein is THURICIDE®.

We specifically describe and claim a transformed plant expressing the peptides described herein where the average concentration of Bt and ICK peptide, Bt and TMOF peptides or Bt and ICK+TMOF peptides, in an average leaf of a transformed plant is about: 1, 5, 10, 15, 20, 25, 30, 35, 40, 45, 50, 55, 60, 65, 70, 75, 80, 85, 90, 95 or 99%, or any range between any two of these values. We specifically describe and claim a transformed plant expressing properly folded toxic peptides in the transformed plant. We specifically describe and claim a transformed plant expressing properly folded combination toxic peptides in the transformed plant and to cause the accumulation of the expressed and properly folded toxic peptides in said plant and to cause an increase in the plant's yield or resistance to insect damage and they control insect pests in crops and forestry. We describe plants made by any of the products and processes described herein.

We describe expression cassettes comprising any of the nucleotides which express any peptides described herein, including embodiments having a functional expression cassette incorporated into a transformed plant, comprising nucleotides that code for any of the peptides disclosed herein or that could be made by one skilled in the art given the teaching disclosed herein. We describe and claim procedures for the generation of transformed plants having or expressing any of the peptides described herein.

We describe the use of any of the peptides or nucleotides described herein, to make a plant or transform these peptides or nucleotides into a plant, and methods and techniques for generating these proteins in plants and/or expression cassettes comprising any of the peptides and methods to transform them into a plant genome and any method of using, making, transforming any of the described peptides or nucleotides into a plant and methods and techniques for generating transformed plants having or expressing any of the peptides and functional expression cassettes in plants comprising any of the disclosed peptides and their corresponding nucleotides and any plants made by the products and processes described herein.

In some embodiments we disclose a chimeric gene comprising a promoter active in plants operatively linked to the nucleic acids or expression cassettes as described herein. We disclose a method of making, producing, or using the combination of genes described herein. We disclose a recombinant vector comprising the combination of genes described herein. We disclose a method of making, producing, or using the recombinant vector. We disclose a transgenic host cell comprising the combination of genes described herein and the method of making, producing or using the transgenic host cell, which can be a transgenic

plant cell and we disclose a method of making, producing or using such a transgenic plant cell as well as the transgenic plant comprising the transgenic plant cell and how to make and use the transgenic plant. We disclose transgenic plant and seed having the properties described herein that is derived from corn, soybean, cotton, rice, sorghum, switchgrass, sugarcane, alfalfa, potatoes or tomatoes. The transgenic seed may have a chimeric gene that we describe herein. We describe methods of making, producing or using the transgenic plant and or seed of this disclosure.

We also describe methods of using the invention and provide novel formulations. The invention is most useful to control insects. We describe a method of controlling an insect comprising: Applying Bt (*Bacillus thuringiensis*) protein to said insect; and Applying an insecticidal ICK (Inhibitor Cystine Knot) peptide to said insect. This method may be used where the Bt protein and the insecticidal ICK peptide, Bt and TMOF peptides or BT and ICK +TMOF peptides are applied together at the same time in the same compositions or separately in different compositions and at different times. The Bt protein and the insecticidal ICK peptide, and or TMOF peptide may be applied sequentially, and it may be applied to (Bt protein)-resistant insects. The ratio of Bt to ICK or TMOF, on a dry weight basis, can be selected from at least about the following ratios: 99:1, 95:5, 90:10, 85:15, 80:20, 75:25, 70:30, 65:35, 60:40, 55:45, 50:50, 45:55, 40:60, 35:65, 30:70, 25:75, 20:80, 15:85, 10:90, 5:95 and 1:99, or any combination of any two of these values. The ratio of Bt to ICK, on a dry weight basis, can be selected from about the following ratios: 50:50, 45:55, 40:60, 35:65, 30:70, 25:75, 20:80, 15:85, 10:90, 5:95, 1:99, 0.5:99.5, 0.1:99.9 and 0.01:99.99 or any combination of any two of these values. Either or both of the Bt and ICK peptides are derived from more than 1 different types or bacterial strain origins of Bt and ICK peptides, Bt and TMOF peptides or BT and ICK+TMOF peptides. Either or both of the Bt and ICK, Bt and TMOF peptides or BT and ICK+TMOF peptides are derived from between 2 and 5 different types or bacterial strain origins of either Bt or ICK peptides or both Bt and ICK peptides. Either or both of the Bt and ICK peptides are derived from 2 to 15 different types or bacterial strain origins of either or both of Bt and ICK peptides and at least one strain of either Bt or ICK or both Bt and ICK peptides are encoded by more than one copy of the Bt or ICK genes. Either one or both of the Bt and ICK peptides are derived from more than 1 different types or bacterial strain origins of Bt and/or ICK peptides with all the strains of Bt and/or ICK peptides contributing more than at least 1% of the peptides from each strain type in said composition. Either or both of the Bt and ICK peptides are derived from 2 to 5 different types or bacterial strain origins of either one or both of Bt and ICK peptides and at least one strain of either Bt or ICK or both Bt and ICK peptides are encoded by more than one copy of the Bt or ICK genes. The total concentration of Bt and ICK, Bt and TMOF peptides or BT and ICK+TMOF peptides peptide in the composition is selected from the following percent concentrations: 0, 1, 5, 10, 15, 20, 25, 30, 35, 40, 45, 50, 55, 60, 65, 70, 75, 80, 85, 90, 95, 99 or 100%, or any range between any two of these values, and the remaining percentage of the composition is comprised of excipients.

The methods can be used where the insecticidal combination peptide is produced using a genetic cassette that further comprises an ERSP (Endoplasmic Reticulum Signal Peptide) operably linked to the insecticidal ICK peptide, or TMOF peptide; wherein said ERSP is linked at the N-terminal of the insecticidal ICK peptide. In some embodiments

the insecticidal combination peptides used are produced using a genetic cassette that further comprises an ERSP (Endoplasmic Reticulum Signal Peptide) operably linked to the insecticidal ICK peptide, wherein said ERSP is linked at the N-terminal of the insecticidal ICK peptide, wherein the ERSP is BAAS.

Any of the peptides and plants described herein can be used to control insects, their growth and damage, especially their damage to plants. The combination Bt protein and insecticidal ICK peptide can be applied by being sprayed on a plant, or the insect's locus, or the locus of a plant in need of protecting.

We also describe formulations comprising: Bt protein; and an insecticidal ICK, and or an insecticidal TMOF peptide which can include any of the compositions described herein or capable of being made by one skilled in the art given this disclosure. Some of the described formulations include the use of a polar aprotic solvent, and or water, and or where the polar aprotic solvent is present in an amount of 1-99 wt %, the polar protic solvent is present in an amount of 1-99 wt %, and the water is present in an amount of 0-98 wt %. The formulations include formulations where the Bt protein is DIPEL® and where the insecticidal ICK peptide is a hybrid-ACTX-Hv1a peptide. The polar aprotic solvent formulations are especially effective when they contain MSO. MSO is a methylated seed oil and surfactant blend that uses methyl esters of soya oil in amounts of between about 80 and 85 percent petroleum oil with 15 to 20 percent surfactant.

This disclosure provides numerous examples of suitable CRIP type peptides, ICK peptides, NON-ICK CRIP peptides, and TMOF peptides in addition to many type of PFIP type peptides such as Bt and VIP proteins and peptides, when combined, provide novel insecticidal products, and these may be referred to herein as "combination peptides." Peptides suitable for use with this invention are described in this document, and specific examples are disclosed in the sequence listing. The peptides in the sequence listing are provided only as examples to illustrate the invention and to provide direction and meaning for one skilled in the art. It should be understood that the sequence listing does not provide a full and complete list of all CRIPS, ICKs, NON-ICK CRIPS, and TMOF nor does it provide a full and complete list of all PFIPS. Insects may be treated with combination peptides applied directly, such as sprayed onto an insect or its locus, or the combination peptides can be applied indirectly, such as delivered in a transgenic plant. First we provide detailed written descriptions and examples of CRIP peptides like ICK (Section I), and these are also provided above. Then we provide detailed written descriptions and examples of TMOF peptide (Section II). Next we provide detailed written descriptions and examples of Bt proteins (Section III). It should be understood that the application provides these examples as a means to illustrate and not limit the bounds of the patent and the claimed invention. Any suitable Bt protein and ICK peptide or TMOF peptide could be combined in the manner described and result in an effective insecticide. After describing the ICK and Bt proteins, applicant describes various pesticide compositions (Section IV). Plant transformations using both ICK and Bt proteins are described (Section V). Descriptions and examples of CRIP and Bt Combinations (Section VI). TMOF and Bt proteins combinations are described (Section VII). We provide non limiting examples and descriptions of how the ICK and Bt proteins have been combined to produce a highly effective insecticide, with results and data provided herein.

Section I. The ICK Motif Peptides or ICK Peptides.

"ICK motif," "ICK motif protein," "inhibitor cystine knot motif," "Toxic insect ICK peptides" or "ICK peptides" means a 16 to 60 amino acid peptide with at least 6 half-cystine core amino acids having three disulfide bridges, wherein the 3 disulfide bridges are covalent bonds and of the six half-cystine residues the covalent disulfide bonds are between the first and fourth, the second and fifth, and the third and sixth half-cystines, of the six core half-cystine amino acids starting from the N-terminal amino acid. The ICK motif also comprises a beta-hairpin secondary structure, normally composed of residues situated between the fourth and sixth core half-cystines of the motif, the hairpin being stabilized by the structural crosslinking provided by the motif's three disulfide bonds. Note that additional cysteine/cystine or half-cystine amino acids may be present within the inhibitor cystine knot motif.

This motif is common in peptides isolated from the venom of numerous species. Invertebrate species include spiders and scorpions, other examples are numerous, even snake venom has been known to have peptides having the ICK motif. Specific examples of insecticidal ICK peptides are the "U peptides" disclosed herein and in published patents and patent applications and its homologues, which have an origin from the venoms of Australian Funnel-web spiders. These proteins are also referred to as ACTX peptides from the Australian Blue Mountains Funnel-web Spider, but the procedures described herein are useful and may be applied to any protein with the ICK motif. The following documents are incorporated by reference in the United States in their entirety, are known to one skilled in the art, and have all been published.

Examples of peptide toxins with the ICK motif protein can be found in the following references. The N-type calcium channel blocker ω -Conotoxin was reviewed by Lew, M. J. et al. "Structure-Function Relationships of ω -Conotoxin GVIA" Journal of Biological Chemistry, Vol. 272, No. 18, Issue of May 2, pp. 12014-12023, 1997. A summary of numerous arthropod toxic ICK peptides different spider and scorpion species was reviewed in, Quintero-Hernandez, V. et al. "Scorpion and Spider Venom Peptides: Gene Cloning and Peptide Expression" Toxicon, 58, pp. 644-663, 2011. The three-dimensional structure of Hanatoxin1 using NMR spectroscopy was identified as an inhibitor cystine knot motif in Takahashi, H. et al. "Solution structure of hanatoxin1, a gating modifier of voltage-dependent K⁺ channels: common surface features of gating modifier toxins" Journal of Molecular Biology, Volume 297, Issue 3, 31 Mar. 2000, pp. 771-780. The isolation and identification of cDNA encoding a scorpion venom ICK toxin peptide, Opicalcine1, was published by Zhu, S. et al. "Evolutionary origin of inhibitor cystine knot peptides" FASEB J., 2003 Sep. 17, (12):1765-7, Epub 2003 Jul. 3. The sequence-specific assignment and the secondary structure identification of BgK, a K⁺ channel-blocking toxin from the sea anemone *Bunodosoma granulifera*, was disclosed by Dauplais, M. et al. "On the convergent evolution of animal toxins" Journal of Biological Chemistry, 1997 Feb. 14; 272(7): 4302-9. A review of the composition and pharmacology of spider venoms with emphasis on polypeptide toxin structure, mode of action, and molecular evolution showing cystine bridges, cystine knot formations and the "knotting-type" fold was published by Escoubas, P. et al. "Structure and pharmacology of spider venom neurotoxins" Biochimie, Vol. 82, Issues 9-10, 10 Sep. 2000, pp. 893-907. The purified peptide, iberiotoxin, an inhibitor of the Ca²⁺-activated K⁺ channel, from scorpion (*Buthus tamulus*) venom was dis-

closed in Galvez, A. et al. "Purification and characterization of a unique, potent, peptidyl probe for the high conductance calcium-activated potassium channel from venom of the scorpion *Buthus tamulus*" Journal of Biological Chemistry, 1990 Jul. 5; 265(19): 11083-90. The purified peptide, charybdotoxin, an inhibitor of the Ca²⁺-activated K⁺ channel, from the venom of the scorpion *Leiurus quinquestriatus* was disclosed in Gimenez-Gallego, G. et al. "Purification, sequence, and model structure of charybdotoxin, a potent selective inhibitor of calcium-activated potassium channels" Proc Natl Acad Sci, 1988 May; 85(10): 3329-3333. From these and other publications, one skilled in the art should be able to readily identify proteins and peptides having what we describe as the ICK motif, ICK motif protein or the "inhibitor cystine knot motif."

The ICK motif protein can be any protein with the ICK motif and is between 16 and 60 amino acids in length, with at least 6 cysteine residues that create covalent cross-linking disulfide bonds in the proper order. Some ICK motif peptides have between 26-60 amino acids in length. Some ICK motif proteins are between 16-48 amino acids in length. Some ICK motif proteins are between 26-48 amino acids in length. Some ICK motif proteins are between 30-44 amino acids in length. ICK motif proteins with natural insecticidal activity are preferred but ICK motif proteins with other types of activity such as salt and frost resistance are known to those skilled in the art and are claimed herein. Examples of insecticidal ICK motif proteins include the ACTX peptides and genes, and including all of the peptides and their coding genes known as Magi6.

Examples of insecticidal ICK motif proteins include the ACTX peptides and genes and include all of the peptides and their coding genes as described in the references provided above and herein. Specific examples of ICK motif proteins and peptides disclosed for purposes of providing examples and not intended to be limiting in any way, are the peptides and their homologues as described above, and in particular peptides and nucleotides which originate from the venoms of Australian Funnel-web spiders. The following documents are incorporated by reference in the United States in their entirety, are known to one skilled in the art, and have all been published. They disclose numerous ICK motif proteins which, their full peptide sequence, their full nucleotide sequence, are specifically disclosed and are incorporated by reference, and in addition the full disclosures are incorporated by reference including all of their sequence listings. Their sequence listings are known and published. See the following: U.S. Pat. No. 7,354,993 B2, issued Apr. 8, 2008, specifically the peptide and nucleotide sequences listed in the sequence listing, and numbered SEQ ID NOs: 33-71, from 7,354,993 B2, and those named U-ACTX polypeptides, and these and other toxins that can form 2 to 4 intra-chain disulfide bridges, and variants thereof, and the peptides appearing on columns 4 to 9 and in FIG. 2 of 7,354,993 B2. Other specific sequences can be found in EP patent 1 812 464 B1, published and granted 08.10.2008, see Bulletin 2008/41, specifically the peptide and nucleotide sequences listed in the sequence listing, and other toxins that can form 2 to 4 intra-chain disulfide bridges, and those sequences numbered SEQ ID NOs: 33-71, and sequences named U-ACTX polypeptides, and variants thereof, and the peptides appearing in paragraphs 0023 to 0055, and appearing in EP patent 1 812 464 B1, see FIG. 1 of EP 1, 1 812 464 B1.

Described and incorporated by reference in order to disclose the peptides identified herein are homologous variants of sequences mentioned, having homology to such

sequences or referred to herein, which are also identified and claimed as suitable for making special according to the processes described herein, including all homologous sequences having at least any of the following percent identities to any of the sequences disclosed here or to any sequence incorporated by reference: 50%, 55%, 60%, 65%, 70%, 75%, 80%, 85%, 90%, 95%, 96%, 97%, 98%, or 99% or greater identity or 100% identity to any and all sequences identified in the patents noted above, and to any other sequence identified herein, including each and every sequence in the sequence listing of this application. When the term homologous or homology is used herein with a number such as 50% or greater, then what is meant is percent identity or percent similarity between the two peptides. When homologous or homology is used without a numeric percent then it refers to two peptide sequences that are closely related in the evolutionary or developmental aspect in that they share common physical and functional aspects, like topical toxicity and similar size (i.e., the homolog being within 100% greater length or 50% shorter length of the peptide specifically mentioned herein or identified by reference herein as above).

Described and incorporated by reference to describe the peptides identified herein are toxic ICK peptides including the following: the U peptide and its variants; found in, isolated from, or derived from, spiders of the genus *Atrax* or *Hadronyche*, including the genus species, *Hadronyche versuta*, or the Blue Mountain funnel web spider, *Atrax robustus*, *Atrax formidabilis*, *Atrax infensus*, including toxins known as U-ACTX polypeptides, U-ACTX-Hv1a, rU-ACTX-Hv1a, rU-ACTX-Hv1b, or mutants or variants, especially peptides of any of these types and especially those less than about 200 amino acids but greater than about 10 amino acids, and especially peptides less than about 150 amino acids but greater than about 20 amino acids, especially peptides less than about 100 amino acids but greater than about 25 amino acids, especially peptides less than about 65 amino acids but greater than about 25 amino acids, especially peptides less than about 55 amino acids but greater than about 25 amino acids, especially peptides of about 37 or 39 or about 36 to 42 amino acids, especially peptides with less than about 55 amino acids but greater than about 25 amino acids, especially peptides with less than about 45 amino acids but greater than about 35 amino acids, especially peptides with less than about 115 amino acids but greater than about 75 amino acids, especially peptides with less than about 105 amino acids but greater than about 85 amino acids, especially peptides with less than about 100 amino acids but greater than about 90 amino acids, including peptide toxins of any of the lengths mentioned here that can form 2, 3 and or 4 or more intrachain disulfide bridges, including toxins that disrupt calcium channel currents, including toxins that disrupt potassium channel currents, especially toxins that disrupt insect calcium channels or Us thereof, especially toxins or variants thereof of any of these types, and any combination of any of the types of toxins described herein that have oral or topical insecticidal activity, can be made special by the processes described herein.

The U peptides from the Australian Funnel Web Spider, genus *Atrax* and *Hadronyche* are particularly suitable and work well when placed in combination according to the methods, procedures or processes described by this invention. Examples of such suitable peptides tested and with data are provided herein. The following species are also specifically known to carry toxic ICK peptides suitable for being made special by the process of this invention. The following species are specifically named: *Atrax formidabilis*, *Atrax*

infensus, *Atrax robustus*, *Hadronyche infensa*, *Hadronyche versuta*. Any toxic ICK peptides derived from any of the genus listed above and/or genus species and homologous to the U peptide are suitable for being made special according to the process in this invention.

The Examples in this specification are not intended to, and should not be used to limit the invention, they are provided only to illustrate the invention.

As noted above, many peptides are suitable candidates for combinations with Bt protein. The sequences noted above, below and in the sequence listing are especially suitable peptides that can be made special, and some of these have been made special according to this invention with the results shown in the examples below.

Examples of toxic ICK insect peptides are well known and can be found in numerous references. They can be identified by their peptidic nature and their activity, usually oral or injection insecticidal activity. Here we provide a few examples to better illustrate and describe the invention, but the invention is not limited to these examples. All of these examples and others not shown here are descriptive of new materials, described and claimed here for the first time.

Toxic ICK insect peptides are peptides of greater than 5 amino acid residues and less than 3,000 amino acid residues. They range in molecular weight from about 550 Da to about 350,000 Da. Toxic ICK insect peptides have some type of insecticidal activity. Typically they show activity when injected into insects but most do not have significant activity when applied to an insect topically. The insecticidal activity of toxic ICK insect peptides is measured in a variety of ways. Common methods of measurement are widely known to those skilled in the art. Such methods include, but are not limited to determination of median response doses (e.g., LD₅₀, PD₅₀, LC₅₀, ED₅₀) by fitting of dose-response plots based on scoring various parameters such as: paralysis, mortality, failure to gain weight, etc. Measurements can be made for cohorts of insects exposed to various doses of the insecticidal formulation in question. Analysis of the data can be made by creating curves defined by probit analysis and/or the Hill Equation, etc. In such cases, doses would be administered by hypodermic injection, by hyperbaric infusion, by presentation of the insecticidal formulation as part of a sample of food or bait, etc.

Toxic ICK insect peptides or ICK peptides are defined here as all peptides shown to be insecticidal upon delivery to insects either by hypodermic injection, hyperbaric infusion, or upon per os delivery to an insect (i.e., by ingestion as part of a sample of food presented to the insect). This class of peptides thus comprises, but is not limited to, many peptides produced naturally as components of the venoms of spiders, mites, scorpions, snakes, snails, etc. This class also comprises, but is not limited to, various peptides produced by plants (e.g., various lectins, ribosome inactivating proteins, and cysteine proteases), and various peptides produced by entomopathogenic microbes (e.g. the Cry1/Bt protein family of proteins produced by various *Bacillus* species.)

The insecticidal peptides may be selected from insecticidal venom, for example the venom of a spider. The spider may be an Australian funnel web spider. The peptides from may be from the genus of *Atrax* or *Hadronyche*, including U-ACTX-Hv1a and its analogs. Specific peptide examples from spiders are described in the sequence listing provided herein. These peptides can be combined with Bt protein using the procedures described herein.

ICK Peptide Sequence Examples

The following documents are incorporated by reference in the US in their entirety, in other jurisdictions where allowed

and they are of common knowledge given their publication. In addition they are incorporated by reference and known specifically for their sequence listings to the extent they describe peptide sequences. See the following: US Patents:

U.S. Pat. No. 5,763,568, issued Jun. 9, 1998, incorporated herein in its entirety, specifically the sequences in the sequence listing, and those numbered 33-58, and those known as "kappa" or "omega" toxins, including those that can form 2-4 intrachain disulphide bridges, and the peptides appearing on columns 2 and 4, and Table 5, and in FIG. 5, FIG. 15, FIG. 16, FIG. 17, FIG. 18.

U.S. Pat. No. 5,959,182, issued Sep. 28, 1999, incorporated herein in its entirety, specifically the sequences in the sequence listing, and those numbered 33-58 and those known as "kappa" or "omega" toxins, including toxins that can form 2-4 intrachain disulphide bridges, and the peptides appearing on columns 2 and 4, and Table 5, and in FIG. 5, FIG. 15, FIG. 16, FIG. 17, FIG. 18.

U.S. Pat. No. 6,583,264 B2, issued Jun. 24, 2003, and U.S. Pat. No. 7,173,106 B2, issued Feb. 6, 2007, incorporated herein in its entirety, specifically sequence number 1, named "omega-atracotoxin-Hv2a or ω -atracotoxin-Hv2a, including toxins that can form 2-4 intrachain disulphide bridges.

U.S. Pat. No. 7,279,547 B2, issued Oct. 9, 2007, incorporated herein in its entirety, specifically the sequences in the sequence listing, and those numbered 33-67, and variants of ω -atracotoxin-Hv2a, toxins that can form 2-4 intrachain disulphide bridges, and the peptides appearing on columns 4-8 of the specification, and in FIG. 3 and FIG. 4.

U.S. Pat. No. 7,354,993 B2, issued Apr. 8, 2008, incorporated herein in its entirety, specifically the peptide sequences listed in the sequence listing, and those numbered 33-71, and those named U-ACTX polypeptides, toxins that can form 2-4 intrachain disulphide bridges, and variants thereof, and the peptides appearing on columns 4-9 of the specification and in FIG. 1.

EP patent 1 812 464 B1, published and granted 08.10.2008 Bulletin 2008/41, incorporated herein in its entirety, specifically the peptide sequences listed in the sequence listing, toxins that can form 2-4 intrachain disulphide bridges, and those as numbered 33-71, and those named U-ACTX polypeptides, and variants thereof, and the peptides appearing in paragraphs 0023 to 0055, and appearing in FIG. 1.

Described and incorporated by reference to the peptides identified herein are homologous variants of sequences mentioned, have homology to such sequences or referred to herein which are also identified and claimed as suitable for making special according to the processes described herein including but not limited to all homologous sequences including homologous sequences having at least any of the following percent identities to any of the sequences disclosed her or to any sequence incorporated by reference: 30%, 35%, 40%, 45%, 50%, 55%, 60%, 65%, 70%, 75%, 80%, 85%, 90% or 95% or greater identity to any and all sequences identified in the patents noted above, and to any other sequence identified herein, including each and every sequence in the sequence listing of this application. When the term homologous or homology is used herein with a number such as 30% or greater then what is meant is percent identity or percent similarity between the two peptides. When homologous or homology is used without a numeric percent then it refers to two peptide sequences that are closely related in the evolutionary or developmental aspect in that they share common physical and functional aspects like topical toxicity and similar size within 100% greater length or 50% shorter length or peptide.

Described and incorporated by reference to the peptides identified herein that are derived from any source mentioned in the US and EP patent documents referred to above,

including but not limited to the following: toxins isolated from plants and insects, especially toxins from spiders, scorpions and plants that prey on or defend themselves from insects, such as, funnel web spiders and especially Australian funnel web spiders, including toxins found in, isolated from or derived from the genus *Atrax* or *Hadronyche*, including the genus species, *Hadronyche versuta*, or the Blue Mountain funnel web spider, *Atrax robustus*, *Atrax formidabilis*, *Atrax infensus* including toxins known as "atracotoxins," " ω -atracotoxins," "kappa" atracotoxins, "omega" atracotoxins also known as ω -atracotoxin, U-ACTX polypeptides, U-ACTX-Hv1a, rU-ACTX-Hv1a, rU-ACTX-Hv1b, or mutants or variants, especially peptides of any of these types and especially those less than about 200 amino acids but greater than about 10 amino acids, and especially peptides less than about 150 amino acids but greater than about 20 amino acids, especially peptides less than about 100 amino acids but greater than about 25 amino acids, especially peptides less than about 65 amino acids but greater than about 25 amino acids, especially peptides less than about 55 amino acids but greater than about 25 amino acids, especially peptides of about 37 or 39 or about 36 to 42 amino acids, especially peptides with less than about 55 amino acids but greater than about 25 amino acids, especially peptides with less than about 45 amino acids but greater than about 35 amino acids, especially peptides with less than about 115 amino acids but greater than about 75 amino acids, especially peptides with less than about 105 amino acids but greater than about 85 amino acids, especially peptides with less than about 100 amino acids but greater than about 90 amino acids, including peptide toxins of any of the lengths mentioned here that can form 2, 3 and or 4 or more intrachain disulphide bridges, including toxins that disrupt calcium channel currents, including toxins that disrupt potassium channel currents, especially insect calcium channels or hybrids thereof, especially toxins or variants thereof of any of these types, and any combination of any of the types of toxins described herein that have topical insecticidal activity, can be made special by the processes described herein.

Venomous peptides from the Australian Funnel Web Spider, genus *Atrax* and *Hadronyche* are particularly suitable and work well when treated by the methods, procedures or processes described by this invention. These spider peptides, like many other toxic ICK peptides, including especially are toxic scorpion and toxic plant peptides, become topically active or toxic when treated by the processes described by this invention. Examples of suitable peptides tested and resulting data are provided herein. In addition to the organisms mentioned above, the following species are also specifically known to carry toxins suitable for being made special by the process of this invention. The following species are specifically named: *Agelenopsis aperta*, *Androctonus australis Hector*, *Antraxformidabilis*, *Antrax infensus*, *Atrax robustus*, *Bacillus thuringiensis*, *Bothus martensii* Karsch, *Bothus occitanus tunetanus*, *Buthacus arenicola*, *Buthotus judaicus*, *Buthus occitanus mardochei*, *Centruroides noxius*, *Centruroides suffusus suffusus*, *Hadronyche infensa*, *Hadronyche versuta*, *Hadronyche versutus*, *Hololena curta*, *Hotentotta judaica*, *Leiurus quinquestriatus*, *Leiurus quinquestriatus hebraeus*, *Leiurus quinquestriatus quinquestriatus*, *Oldenlandia affinis*, *Scorpio maurus palmatus*, *Tityus serrulatus*, *Tityus zuluani*. Any peptidic toxins from any of the genus listed above and or genus species are suitable for being made special according to the process in this invention.

The Examples in this specification are not intended to, and should not be used to limit the invention, they are provided only to illustrate the invention.

As noted above, many peptides are suitable candidates as the subject of the process to make special. The sequences

noted above, below and in the sequence listing are especially suitable peptides that can be made special, and many of these have been made special according to this invention with the results shown in the examples below.

The Examples in this specification are not intended to, and should not be used to limit the invention, they are provided only to illustrate the invention.

As noted above, many peptides are suitable candidates as the subject of the process for the plant expression as PIP. The sequences noted above, below and in the sequence listing are especially suitable peptides that can be expressed in plants as PEP, and some of these have been expressed in plants as PEP according to this invention with the results shown in the examples below.

(one letter code)

SEQ ID NO: 1042

GSQYC VPDQ PCSLN TQPC DDATC TQERN ENGHT VYYCR A.

Named "U+2-ACTX-Hv1a," It has disulfide bridges at positions: 5-20, 12-25, 19-39. The molecular weight is 4564.85 Daltons.

Another example of an ICK motif insecticidal protein is SEQ ID NO: 1010.

(one letter code)

SEQ ID NO: 661

QYCVV VDQPC SLNTQ PCCDD ATCTQ ERNEN GHTVY YCRA

SEQ ID NO: 661, named "Hybrid-ACTX-Hv1a," has disulfide bridges at positions: 3-18, 10-23, 17-37. The molecular weight is 4426.84 Daltons.

(one letter code)

SEQ ID NO: 593

SPTCI PSGQP CPYNE NCCSQ SCTFK ENENGNTVVKR CD

(three letter code)

SEQ ID NO: 593

Ser Pro Thr Cys Ile Pro Ser Gly Gln Pro Cys Pro

Tyr Asn Glu Asn Cys Cys Ser Gln Ser Cys Thr Phe

Lys Glu Asn Glu Asn Gly Asn Thr Val Lys Arg Cys

Asp

Named "ω-ACTX-Hv1a" it has disulfide bridges at positions: 4-18, 11-22 and 17-36. The molecular weight is 4096.

(one letter code)

SEQ ID NO: 650

GSSPT CIPSG QPCPY NENCC SQSCT FKENE NGNTV KRCD

(three letter code)

SEQ ID NO: 650

Gly Ser Ser Pro Thr Cys Ile Pro Ser Gly Gln Pro

Cys Pro Tyr Asn Glu Asn Cys Cys Ser Gln Ser Cys

Thr Phe Lys Glu Asn Glu Asn Gly Asn Thr Val Lys

Arg Cys Asp

Named "ω-ACTX-Hv1a+2" it has disulfide bridges at positions: 6-20, 13-24 and 19-38. The molecular weight is 4199.

(one letter code)

SEQ ID NO: 651

GSAIC TGADR PCAAC CCCCC GTSCCK AESNG VSYCR KDEP

5 (three letter code)

SEQ ID NO: 651

Gly Ser Ala Ile Cys Thr Gly Ala Asp Arg Pro Cys

Ala Ala Cys Cys Pro Cys Cys Pro Gly Thr Ser Cys

10 Lys Ala Glu Ser Asn Gly Val Ser Tyr Cys Arg Lys

Asp Glu Pro

15 Named "rK-ACTX-Hv1c" it has disulfide bridges at positions: 5-19, 12-24, 15-16, 18-34. The molecular weight is 3912.15

(three letter code)

SEQ ID NO: 652

20 Gly Ser Gln Tyr Cys Val Pro Val Asp Gln Pro Cys

Ser Leu Asn Thr Gln Pro Cys Cys Asp Asp Ala Thr

Cys Thr Gln Glu Arg Asn Glu Asn Gly His Thr Val

25 Tyr Tyr Cys Arg Ala

Named "rU-ACTX-Hv1a ("Hybrid")+2" it has disulfide bridges at positions: 5-20, 12-25, 19-39. The molecular weight is 4570.51.

Other ICK peptides are provided in the sequence listing. SEQ ID NOs: 534-707 are ICK peptide sequences and include the "kappa"/"omega" toxins and the "hybrid" toxins. SEQ ID NO: 593 is omega-ACTX-Hv1a. SEQ ID NO: 661 is hybrid-ACTX-Hv1a or U-ACTX-Hv1a.

30 Section II. The TMOF Motif Peptides or TMOF Peptides.

"TMOF motif," or "TMOF proteins" means trypsin modulating oostatic factor peptide. Numerous examples and variants are provided. SEQ ID NO: 708 is the wild type TMOF sequence. Other non-limiting variants are provided in SEQ. ID. NO:s 709-721. Other examples would be known or could be created by one skilled in the art.

Section III. Bt Proteins

Bt are the initials for a bacteria called *Bacillus thuringiensis*. The Bt bacteria produces a family of peptides that are toxic to many insects. The Bt toxic peptides are well known for their ability to produce parasporal crystalline protein inclusions (usually referred to as crystals) that fall under two major classes of toxins; cytolytins (Cyt) and crystal Bt proteins (Cry). Since the cloning and sequencing of the first crystal proteins genes in the early-1980s, many others have been characterized and are now classified according to the nomenclature of Crickmore et al. (1998). Generally Cyt proteins are toxic towards the insect orders Coleoptera (beetles) and Diptera (flies), and Cry proteins target Lepidoptera (moths and butterflies). Cry proteins bind to specific receptors on the membranes of mid-gut (epithelial) cells resulting in rupture of those cells. If a Cry protein cannot find a specific receptor on the epithelial cell to which it can bind, then it is not toxic. Bt strains can have different complements of Cyt and Cry proteins, thus defining their host ranges. The genes encoding many Cry proteins have been identified.

50 Currently there are four main pathotypes of insecticidal Bt parasporal peptides based on order specificity: Lepidoptera-specific (CryI, now Cry1), Coleoptera-specific (CryIII, now Cry3), Diptera-specific (CryIV, now Cry4, Cry 10, Cry11; and CytA, now Cyt1A), and CryII (Now Cry2), the only
65

family known at that time to have dual (Lepidoptera and Diptera) specificity. Cross-order activity is now apparent in many cases.

The nomenclature assigns holotype sequences a unique name which incorporates ranks based on the degree of divergence, with the boundaries between the primary (Arabic numeral), secondary (uppercase letter), and tertiary (lower case letter) rank representing approximately 95%, 78% and 45% identities. A fourth rank (another Arabic number) is used to indicate independent isolations of holotype toxin genes with sequences that are identical or differ only slightly. Currently, the nomenclature distinguishes 174 holotype sequences that are grouping in 55 cry and 2 cyt families (Crickmore, N., Zeigler, D. R., Schnepf, E., Van Rie, J., Lereclus, D., Daum, J., Bravo, A., Dean, D. H., *B. thuringiensis* toxin nomenclature). Any of these crystal proteins and the genes that produce them may be used to produce a suitable Bt related toxin for this invention.

Also included in the descriptions of this invention are families of highly related crystal proteins produced by other bacteria: Cry16 and Cry17 from *Clostridium bifermentans* (Barloy et al., 1996, 1998), Cry 18 from *Bacillus popilliae* (Zhang et al., 1997), Cry43 from *Paenibacillus lentimorbis* (Yokoyama et al., 2004) and the binary Cry48/Cry49 produced by *Bacillus sphaericus* (Jones et al., 2008). Other crystalline or secreted pesticidal proteins, such as the S-layer proteins (Peña et al., 2006) that are included here are, genetically altered crystal proteins, except those that were modified through single amino acid substitutions (e.g., Lambert et al., 1996). Any of these genes may be used to produce a suitable Bt related toxin for this invention.

Examples of Bt

In particular, isolated nucleic acid molecules corresponding to Bt protein nucleic acid sequences are provided. Additionally, amino acid sequences corresponding to the polynucleotides are encompassed. In particular, the present invention provides for an isolated nucleic acid molecule comprising a nucleotide sequence encoding the amino acid sequence shown in US 2009/0099081, published on Apr. 18, 2009, all of which is herein incorporated by reference in its entirety, and all sequences identified by number specifically incorporated by reference. SEQ ID NO: 9, 11, 13, 15, or 18, or a nucleotide sequence set forth in SEQ ID NO: 1, 2, 4, 6, 7, 8, 10, 12, 14, 16, or 17, as well as variants and fragments thereof. Nucleotide sequences that are complementary to a nucleotide sequence of the invention, or that hybridize to a sequence of the invention are also encompassed.

Nucleotide sequences encoding the proteins of the present invention include the sequence set forth in US 2009/0099081, published on Apr. 18, 2009, SEQ ID NO: 1, 2, 4, 6, 7, 8, 10, 12, 14, 16, or 17, and variants, fragments, and complements thereof. By "complement" is intended a nucleotide sequence that is sufficiently complementary to a given nucleotide sequence such that it can hybridize to the given nucleotide sequence to thereby form a stable duplex. The corresponding amino acid sequence for the Bt protein encoded by this nucleotide sequence are set forth in SEQ ID NO: 33-533.

Nucleic acid molecules that are fragments of these Bt protein encoding nucleotide sequences are also encompassed by the present invention (for example, US 2009/0099081, published on Apr. 18, 2009, all of which is herein incorporated by reference in its entirety, and all sequences identified by number specifically incorporated by reference. SEQ ID NO: 8 is a fragment of SEQ ID NO: 4 and 12; SEQ ID NO: 4 is a fragment of SEQ ID NO: 2). By "fragment" is intended a portion of the nucleotide sequence encoding a

Bt protein. A fragment of a nucleotide sequence may encode a biologically active portion of a Bt protein, or it may be a fragment that can be used as a hybridization probe or PCR primer using methods disclosed below. Nucleic acid molecules that are fragments of a Bt protein nucleotide sequence comprise at least about 50, 100, 200, 300, 400, 500, 600, 700, 800, 900, 1000, 1050, 1100, 1150, 1200, 1250, 1300, 1350, 1400, 1450, 1500, 1550, 1600, 1650, 1700, 1750, 1800, 1850, 1860, 1870, 1880, 1885 contiguous nucleotides, or up to the number of nucleotides present in a full-length Bt-protein encoding nucleotide sequence disclosed herein (for example, 1890 nucleotides for US 2009/0099081, published on Apr. 18, 2009, Here these are provided as SEQ ID NO: 1 and 2, 1806 nucleotides for SEQ ID NO: 4, 1743 nucleotides for SEQ ID NO: 6, 7, 8, and 16, 1809 nucleotides for SEQ ID NO: 10, and 1752 nucleotides for SEQ ID NO: 12 and 14, in the sequence listing) depending upon the intended use. By "contiguous" nucleotides is intended nucleotide residues that are immediately adjacent to one another. Fragments of the nucleotide sequences of the present invention will encode protein fragments that retain the biological activity of the Bt protein protein and, hence, retain pesticidal activity. By "retains activity" is intended that the fragment will have at least about 30%, at least about 50%, at least about 70%, 80%, 90%, 95% or higher of the pesticidal activity of the Bt protein. Methods for measuring pesticidal activity are well known in the art. See, for example, Czapla and Lang (1990) J. Econ. Entomol. 83:2480-2485; Andrews et al. (1988) Biochem. J. 252:199-206; Marrone et al. (1985) J. of Economic Entomology 78:290-293; and U.S. Pat. No. 5,743,477, all of which are herein incorporated by reference in its entirety, and all sequences identified by number specifically incorporated by reference.

A fragment of a Bt protein encoding nucleotide sequence that encodes a biologically active portion of a protein of the invention will encode at least about 15, 25, 30, 50, 75, 100, 125, 150, 175, 200, 250, 300, 350, 400, 450, 500, 550, 560, 570, 575, 580, 585, 590, 595, 600 contiguous amino acids, or up to the total number of amino acids present in a full-length Bt protein protein of the invention (for example, 580 amino acids for SEQ ID NO: 41, 602 amino acids for SEQ ID NO: 43, and 583 amino acids for SEQ ID NO: 45 and 47).

Preferred Bt protein proteins of the present invention are encoded by a nucleotide sequence sufficiently identical to the nucleotide sequence of US 2009/0099081, published on Apr. 18, 2009, all of which is herein incorporated by reference in its entirety, and all sequences identified by number specifically incorporated by reference, sequences 1, 2, 4, 6, 7, 8, 10, 12, 14, 16, or 17. By "sufficiently identical" is intended an amino acid or nucleotide sequence that has at least about 60% or 65% sequence identity, about 70% or 75% sequence identity, about 80% or 85% sequence identity, about 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99% or greater sequence identity compared to a reference sequence using one of the alignment programs described herein using standard parameters. One of skill in the art will recognize that these values can be appropriately adjusted to determine corresponding identity of proteins encoded by two nucleotide sequences by taking into account codon degeneracy, amino acid similarity, reading frame positioning, and the like.

The invention also encompasses variant nucleic acid molecules (for example, US 2009/0099081, published on Apr. 18, 2009, all of which is herein incorporated by reference in its entirety, and all sequences identified by

number specifically incorporated by reference, sequence 2 is a variant of sequences 1; sequence 7 and 8 are variants of sequences 6; sequence 10 is a variant of sequence 4 and 12; and sequence 14 is a variant of sequence 12). "Variants" of the Bt protein encoding nucleotide sequences include those sequences that encode the Bt protein disclosed herein but that differ conservatively because of the degeneracy of the genetic code as well as those that are sufficiently identical as discussed above.

Naturally occurring allelic variants can be identified with the use of well-known molecular biology techniques, such as polymerase chain reaction (PCR) and hybridization techniques as outlined below. Variant nucleotide sequences also include synthetically derived nucleotide sequences that have been generated, for example, by using site-directed mutagenesis but which still encode the Bt protein proteins disclosed in the present invention as discussed below. Variant proteins encompassed by the present invention are biologically active, that is they continue to possess the desired biological activity of the native protein, that is, retaining pesticidal activity. By "retains activity" is intended that the variant will have at least about 30%, at least about 50%, at least about 70%, or at least about 80% of the pesticidal activity of the native protein. Methods for measuring pesticidal activity are well known in the art. See, for example, Czaplak and Lang (1990) J. Econ. Entomol. 83: 2480-2485; Andrews et al. (1988) Biochem. J. 252:199-206; Marrone et al. (1985) J. of Economic Entomology 78:290-293; and U.S. Pat. No. 5,743,477, all of which are herein incorporated by reference in their entirety, and all sequences identified by number specifically incorporated by reference. Examples of the Generation of Synthetic and Variant Bt Genes

In one aspect of the invention, synthetic axmi-004 sequences were generated, for example synaxmi-004 US 2009/0099081, published on Apr. 18, 2009, all of which is herein incorporated by reference in its entirety, and all sequences identified by number specifically incorporated by reference, (sequence 1) and synaxmi-004B (sequence 2). These synthetic sequences have an altered DNA sequence relative to the axmi-004 sequence (sequence 3) recited in U.S. Pat. No. 7,355,099, all of which is herein incorporated by reference in its entirety, and all sequences identified by number specifically incorporated by reference), and encode the original AXMI-004 protein. Likewise, synaxmi-004B-2M (sequence 4) was designated and encodes the axmi-004 alternate start site (herein referred to as axmi-004B-2M and set forth in sequence 5) originally identified in U.S. patent application Ser. No. 10/782,020.

In another aspect of the invention, a third start site was identified in the axmi-004 coding sequence. This coding region is designated axmi-004B-3M (US 2009/0099081, published on Apr. 18, 2009, all of which is herein incorporated by reference in its entirety, and all sequences identified by number specifically incorporated by reference, sequence 16) and encodes the AXMI-004B-3M amino acid sequence set forth in sequence 9. Synthetic sequences encoding the AXMI-004B-3M protein were also designated. These synthetic nucleotide sequences were designated synaxmi-004B-3M, synaxmi-004C-3M, and synaxmi-004D-3M and are set forth in sequences 6, 7, and 8, respectively. In another aspect of the invention, modified versions of the nucleotide sequence encoding AXMI-004B-3M protein were designed such that additional N-terminal residues are added to the encoded protein. These sequence are designated synaxmi-004B-3M-alt1 (US 2009/0099081, published on Apr. 18, 2009, sequence 10), synaxmi-004B-3M-alt2 (sequence 12),

synaxmi-004B-3M-alt3 (sequence 14), and synaxmi-004B-3M-alt4 (sequence 17). The encoded proteins are designated AXMI-004B-3M-ALT1 (sequence 11), AXMI-004B-3M-ALT2 (sequence 13), AXMI-004B-3M-ALT3 (sequence 15), and AXMI-004B-3M-ALT4 (sequence 18).

Other Bt proteins and gene descriptions can be found in the following. Each and every patent publication referred to below with a note as to the Bt toxin to which the publication refers to, is hereby incorporated by reference in its entirety. These documents have also published and they and their sequences are in the public domain.

More Examples of Bt genes, proteins, and the patent documents that describe them are found in Tables 4, 5, and 6 below. The patent documents in Tables 4, 5, 6, in particular the US Patents and US applications, are hereby incorporated by reference in their entirety.

TABLE 4

Bt Toxins	
Toxin	Patents or patent publication No.
Cry1	U.S. 2003046726, U.S. Pat. No. 6,833,449, CN1260397, U.S. 201026939, U.S. 2006174372, U.S. 2006174372, U.S. 642241, U.S. Pat. No. 6,229,004, U.S. 2004194165, U.S. Pat. No. 6,573,240, U.S. Pat. No. 5,424,409, U.S. Pat. No. 5,407,825, U.S. Pat. No. 5,135,867, U.S. Pat. No. 5,055,294, WO2007107302, U.S. Pat. No. 6,855,873, WO2004020636, U.S. 2007061919, U.S. Pat. No. 6,048,839, U.S. 2007061919, AU784649B, U.S. 2007061919, U.S. Pat. No. 6,150,589, U.S. Pat. No. 5,679,343, U.S. Pat. No. 5,616,319, U.S. Pat. No. 5,322,687, WO2007107302, U.S. 2006174372, U.S. 2005091714, U.S. 2004058860, U.S. 2008020968, U.S. Pat. No. 6,043,415, U.S. Pat. No. 5,942,664, WO2007107302, U.S. 2007061919, U.S. Pat. No. 6,172,281, Cry1
Cry1	WO03082910, MX9606262, U.S. Pat. No. 5,530,195, U.S. Pat. No. 5,407,825, U.S. Pat. No. 5,045,669, U.S. 2006174372, U.S. 2007061919, U.S. 2007061919, U.S. 2007061919, Cry1
Cry1	U.S. 2007061919, CN1401772, U.S. Pat. No. 6,063,605, U.S. 2007061919, AU784649B, U.S. Pat. No. 5,723,758, U.S. Pat. No. 5,616,319, U.S. Pat. No. 5,356,623, U.S. Pat. No. 5,322,687, Cry1
Cry1	U.S. Pat. No. 5,723,758, CN1942582, WO9840490, U.S. 2007061919, UA75570, MXPA03006130, U.S. 2003167517, U.S. Pat. No. 6,107,278, U.S. Pat. No. 6,096,708, U.S. Pat. No. 5,073,632, U.S. Pat. No. 7,208,474, U.S. Pat. No. 7,244,880, Cry3
Cry3	U.S. 2002152496, RU2278161, U.S. 2003054391, Cry3
Cry3	U.S. Pat. No. 5,837,237, U.S. Pat. No. 5,723,756, U.S. Pat. No. 5,683,691, U.S. Pat. No. 5,104,974, U.S. Pat. No. 4,996,155, Cry3
Cry3	U.S. Pat. No. 5,837,237, U.S. Pat. No. 5,723,756, WO9840491, U.S. 2004018982, U.S. Pat. No. 6,166,195, U.S. 2001010932, U.S. Pat. No. 5,985,831, Cry5
Cry5	U.S. Pat. No. 5,824,792, U.S. 528153, WO2007062064, U.S. 2001010932, U.S. Pat. No. 5,824,792, Cry6
Cry6	WO2007062064, U.S. 2004018982, U.S. Pat. No. 5,973,231, U.S. Pat. No. 5,874,288, U.S. Pat. No. 5,236,843, U.S. 683106, Cry6
Cry6	U.S. 2004018982, U.S. Pat. No. 6,166,195, U.S. Pat. No. 6,048,839, U.S. Pat. No. 5,683,691, U.S. Pat. No. 5,378,625, U.S. 518709, Cry7
Cry7	CN195215, Cry8
Cry8	

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TABLE 4-continued

Bt Toxins	
Toxin	Patents or patent publication No.
Cry8	U.S. 200301796
Cry8	WO2006053473, U.S. 2007245430,
Cry8	WO200605347
Cry9	U.S. 2007061919,
Cry9	WO200506620
Cry9	U.S. 2007061919, U.S. Pat. No. 6,448,226,
	U.S. 2005097635, WO2005066202,
	U.S. Pat. No. 6,143,550, U.S. Pat. No. 6,028,246,
	U.S. Pat. No. 6,727,409,
Cry9	U.S. 2005097635, WO2005066202,
Cry9	U.S. Pat. No. 6,570,005,
Cry9	AU784649B, U.S. 2007074308,
	U.S. 736180
Cry11	MXPA0200870
Cry12	U.S. 2004018982, U.S. Pat. No. 6,166,195,
	U.S. Pat. No. 6,077,937, U.S. Pat. No. 5,824,792,
	U.S. Pat. No. 5,753,492,
Cry13	U.S. 2004018982, U.S. 6,166,195,
	U.S. Pat. No. 6,077,937, U.S. Pat. No. 5,824,792,
	U.S. Pat. No. 5,753,492,
Cry14	JP2007006895, U.S. Pat. No. 5,831,011,
Cry21	U.S. Pat. No. 5,831,011, U.S. Pat. No. 5,670,365,
Cry22	U.S. 2006218666, U.S. 2001010932,
	MXPA01004361, U.S. Pat. No. 5,824,792,
Cry22	U.S. 2003229919,
Cry23	U.S. 2006051822, U.S. 2003144192,
	UA75317, U.S. Pat. No. 6,399,330,
	U.S. Pat. No. 6,326,351, U.S. Pat. No. 6,949,626,
Cry26	U.S. 200315001
Cry28	U.S. 200315001
Cry31	CA2410153,
Cry34	U.S. 200316752
Cry35	U.S. 2003167522,
Cry37	U.S. 2006051822, U.S. 2003144192,
	UA75317, U.S. 6,399,330, U.S. 6,326,351,
	U.S. Pat. No. 6,949,626,
Cry43	U.S. 200527164
Cyt1	WO2007027776,
Cyt1	U.S. 6150165,
Cyt2	U.S. 2007163000, EP1681351,
	U.S. Pat. No. 6,686,452, U.S. Pat. No. 6,537,756,

TABLE 5

Hybrid Insecticidal Crystal Proteins and Patents.	
Patents ^a	Holotype Toxin ^b
U.S. 2008020967	Cry29Aa
U.S. 2008040827	Cry1Ca
U.S. 2007245430	Cry8Aa
U.S. 2008016596	Cry8Aa
U.S. 2008020968	Cry1Cb

TABLE 6

Patents Relating to Other Hybrid Insecticidal Crystal Proteins	
Source toxins ^a	Patents ^b
Cry1A, Cry1C	U.S. Pat. No. 5,593,881, U.S. Pat. No. 5,932,209
Cry1C, Cry1A, Cry1F	U.S. Pat. No. 6,962,705, U.S. Pat. No. 7,250,501,
	U.S. 2004093637, WO0114562, WO0214517,
	U.S. Pat. No. 6,156,573
Cry23A, Cry37A	U.S. Pat. No. 7,214,788
Cry1A	U.S. Pat. No. 7,019,197
Cry1A, Cry1B	U.S. Pat. No. 6,320,100
Cry1A, Cry1C	AU2001285900B
Cry23A, Cry37A	U.S. 2007208168
Cry3A, Cry1I, Cry1B	WO0134811

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TABLE 6-continued

Patents Relating to Other Hybrid Insecticidal Crystal Proteins	
Source toxins ^a	Patents ^b
Cry3A, Cry3B, Cry3C	U.S. 2004033523
Cry1A, Cry1C, Cry1E,	U.S. Pat. No. 6,780,408
Cry1G	
Cry1A, Cry1F	U.S. 2008047034

10 The sequence listing includes Bt sequences SEQ. ID. NO:s 33-533. These sequences include examples of Bt protein Cry and Cyt protein sequences. Examples are numerous and one skilled in the art would know of many other

15 examples of various Bt sequences that are suitable substitutes for those in this disclosure.

Section IV. Pesticide Compositions and Increasing Plant Yields

20 The active ingredients of the present invention are normally applied in the form of compositions and can be applied to the crop area or plant to be treated, simultaneously or in succession, with other compounds. These compounds can be fertilizers, weed killers, cryoprotectants, surfactants, detergents, pesticidal soaps, dormant oils, polymers, and/or

25 time-release or biodegradable carrier formulations that permit long-term dosing of a target area following a single application of the formulation. They can also be selective herbicides, chemical insecticides, virucides, microbicides, amoebicides, pesticides, fungicides, bacteriocides, nemato-

30 cides, molluscicides or mixtures of several of these preparations, if desired, together with further agriculturally acceptable carriers, surfactants or application-promoting adjuvants customarily employed in the art of formulation. Suitable carriers and adjuvants can be solid or liquid and

35 correspond to the substances ordinarily employed in formulation technology, e.g. natural or regenerated mineral substances, solvents, dispersants, wetting agents, tackifiers, binders or fertilizers. Likewise the formulations may be prepared into edible "baits" or fashioned into pest "traps" to permit feeding or ingestion by a target pest of the pesticidal formulation.

Methods of applying an active ingredient of the present invention or an agrochemical composition of the present invention that contains at least one of the pesticidal proteins

45 produced by the bacterial strains of the present invention include leaf application, seed coating and soil application. The number of applications and the rate of application depend on the intensity of infestation by the corresponding pest.

50 The composition may be formulated as a powder, dust, pellet, granule, spray, emulsion, colloid, solution, or such like, and may be prepared by such conventional means as desiccation, lyophilization, homogenation, extraction, filtration, centrifugation, sedimentation, or concentration of a culture of cells comprising the polypeptide. In all such compositions that contain at least one such pesticidal polypeptide, the polypeptide may be present in a concentration of from about 1% to about 99% by weight.

Lepidopteran or coleopteran pests may be killed or reduced in numbers in a given area by the methods of the invention, or may be prophylactically applied to an environmental area to prevent infestation by a susceptible pest. Preferably the pest ingests, or is contacted with, a pesticidally-effective amount of the polypeptide. By "pesticidally-effective amount" is intended an amount of the pesticide that is able to bring about death to at least one pest,

65 or to noticeably reduce pest growth, feeding, or normal

physiological development. This amount will vary depending on such factors as, for example, the specific target pests to be controlled, the specific environment, location, plant, crop, or agricultural site to be treated, the environmental conditions, and the method, rate, concentration, stability, and quantity of application of the pesticidally-effective polypeptide composition. The formulations may also vary with respect to climatic conditions, environmental considerations, and/or frequency of application and/or severity of pest infestation.

The pesticide compositions described may be made by formulating either the bacterial cell, crystal and/or spore suspension, or isolated protein component with the desired agriculturally-acceptable carrier. The compositions may be formulated prior to administration in an appropriate means such as lyophilized, freeze-dried, desiccated, or in an aqueous carrier, medium or suitable diluent, such as saline or other buffer. The formulated compositions may be in the form of a dust or granular material, or a suspension in oil (vegetable or mineral), or water or oil/water emulsions, or as a wettable powder, or in combination with any other carrier material suitable for agricultural application. Suitable agricultural carriers can be solid or liquid and are well known in the art. The term "agriculturally-acceptable carrier" covers all adjuvants, inert components, dispersants, surfactants, tackifiers, binders, etc. that are ordinarily used in pesticide formulation technology; these are well known to those skilled in pesticide formulation. The formulations may be mixed with one or more solid or liquid adjuvants and prepared by various means, e.g., by homogeneously mixing, blending and/or grinding the pesticidal composition with suitable adjuvants using conventional formulation techniques. Suitable formulations and application methods are described in U.S. Pat. No. 6,468,523, herein incorporated by reference.

Methods for Increasing Plant Yield

Methods for increasing plant yield are provided. The methods comprise introducing into a plant or plant cell a polynucleotide comprising a pesticidal sequence disclosed herein. As defined herein, the "yield" of the plant refers to the quality and/or quantity of biomass produced by the plant. By "biomass" is intended any measured plant product. An increase in biomass production is any improvement in the yield of the measured plant product. Increasing plant yield has several commercial applications. For example, increasing plant leaf biomass may increase the yield of leafy vegetables for human or animal consumption. Additionally, increasing leaf biomass can be used to increase production of plant-derived pharmaceutical or industrial products. An increase in yield can comprise any statistically significant increase including, but not limited to, at least a 1% increase, at least a 3% increase, at least a 5% increase, at least a 10% increase, at least a 20% increase, at least a 30%, at least a 50%, at least a 70%, at least a 100% or a greater increase in yield compared to a plant not expressing the pesticidal sequence.

In specific methods, plant yield is increased as a result of improved pest resistance of a plant expressing a pesticidal protein disclosed herein. Expression of the pesticidal protein results in a reduced ability of a pest to infest or feed on the plant, thus improving plant yield.

Section V. Plant Transformations

Any combination of the principal components ICK motif protein and or TMOF motif protein and Bt protein, can be combined in a PIP. We also disclose the addition of ERSP (Endoplasmic Reticulum Signal Peptide) and a translational stabilizing protein and intervening linker in order to create

a superior PIP (Plant-incorporated protectant) and expressed as a PEP (Plant Expressed Peptide) as long as a minimum of both Bt and ICK motif protein are used, it is preferred to use these two peptides in combination with ERSP. TMOF motif can also be used with or replacing the ICK motif. These compositions can be created, used as a PEP and expressed as a PIP.

We describe methods to increase the efficacy of the plant expression, to increase the accumulation of plant expressed proteins and to dramatically increase the insecticidal activity of plant expressed proteins. We describe targeting of the ICK motif protein to the Endoplasmic Reticulum (ER) by an Endoplasmic Reticulum Signaling Protein (ERSP) in plants, in order to provide for the correct covalent cross-linking of peptide disulfide bridges which generate the essential tertiary ICK motif structure required for insecticidal activity. We further describe targeting of the ICK motif protein to the ER by an ERSP in plants, with a translational stabilizing protein domain added in order to increase the size of the resulting ICK fusion protein which enhances peptide accumulation in the plant. We further describe targeting of the ICK motif protein to the ER by an ERSP in plants, with a translation stabilizing protein added as above, and with an intervening peptide sequence added, the latter of which allows for potential cleavage and the recovery of the active form of the ICK motif protein having insecticidal activity.

This invention describes the ICK motif proteins with insecticidal activity that are plant expressed and which can successfully protect a plant or crop from insect damage. The methods taught herein will enable peptides to not only be expressed in a plant but to be expressed and folded properly, so that they retain their insecticidal activity even after expression in the plant.

We describe how the open reading frame (ORF) of a target peptide, such as an ICK motif peptide, must be modified in order for the desired biological activity to remain after plant expression of the ICK motif peptide. In one embodiment we describe a Plant Incorporated Protectant, or PIP, that expresses an active insecticidal protein. The PIP insecticidal protein is comprised of an Endoplasmic Reticulum Signal Peptide (ERSP) operably linked to a Cysteine Rich Insecticidal Peptide (CRIP) or Inhibitor Cystine Knot (ICK) motif protein, wherein the ERSP is the N-terminal of the linked ERSP+ICK motif protein. The PIP insecticidal protein is then incorporated into a plant of choice to give insect resistance to the plant. The plant cells will express and accumulate the properly folded ICK motif insecticidal protein. When an insect consumes the plant cells, the properly folded ICK motif insecticidal protein will be delivered inside the insect where it will have insecticidal activity and cause the insect either to slow or to stop its feeding, slow its movements, and slow or stop reproduction, all of which provides protection for the plant from insect damage.

We describe transient expression systems to express various plant expression cassettes. One expressed transgene we use is Green Fluorescent Protein or GFP, which is detectable visually when excited by UV light. The GFP transient expression system we used for the evaluation of plant transgenic proteins is for all practical purposes-equivalent to use of a stable transgenic plant system for these types of evaluations.

The CRIP, ICK, TMOF, Sea Anemone Motif can be Linked to the ERSP.

For the ICK motif insecticidal protein to be properly folded when it is expressed from a transgenic plant, it must have an ERSP fused in frame with the ICK motif insecticidal protein. This can also be done with a TMOF motif. This can

be accomplished in several ways. See FIGS. 1, 2 and 3. The protein should be routed through the ER where the correct covalent bond connections for proper disulfide bond formation are formed. Without wishing to be bound by theory, we believe the ER routing results in the correct tertiary structure of the ICK motif protein. It is commonly postulated that such routing is achieved by a cellular component called a signal-recognition particle: the signal-recognition particle binds to the ribosome translating the protein, it pauses translation, and it transports the ribosome/mRNA complex to a translocator pore in the ER, where the ribosome then continues the translation and threads the resulting protein into the ER. Within the ER the ERSP is cleaved and the protein is acted upon by posttranslational modification processes in the ER. Once such process involves protein disulfide isomerases, a class of proteins that catalyze the formation of disulfide bonds. Without any additional retention protein signals, the protein is transported through the ER to the golgi apparatus, where it is finally secreted outside the plasma membrane and into the apoplastic space. Without wishing to be bound by theory, we think proteins, such as insecticidal proteins, that have an ICK motif, need to be routed through the ER, in order for the proteins to have correct disulfide bond formation, if they are expressed in plants.

The ERSP (Endoplasmic Reticulum Signaling Protein).

In addition to the text below, see Part I-I (The EERSP or ersp component of the PEPs).

The ERSP is the N-terminal region of the ERSP+ICK motif protein complex and the ERSP portion is composed of about 3 to 60 amino acids. In some embodiments it is 5 to 50 amino acids. In some embodiments it is 10 to 40 amino acids but most often is composed of 15 to 20; 20 to 25; or 25 to 30 amino acids. The ERSP is a signal peptide so called because it directs the transport of a protein. Signal peptides may also be called targeting signals, signal sequences, transit peptides, or localization signals. The signal peptides for ER targeting are often 15 to 30 amino acid residues in length and have a tripartite organization, comprised of a core of hydrophobic residues flanked by a positively charged aminoterminal and a polar, but uncharged carboxyterminal region. See: Zimmermann, Richard; Eyrisch, Susanne; Ahmad, Mazen and Helms, Volkhard: "Protein translocation across the ER membrane" *Biochimica et Biophysica Acta* 1808 (2011) 912-924, Elsevier.

About half and often more of the ERSP is usually comprised of hydrophobic amino acids, but the percentage of amino acids in an ERSP that are hydrophobic can vary. Without wishing to be bound by any theory of how the invention works, we think the hydrophobic amino acids stick in the membrane of the ER after translation and this allows the signal peptide peptidase to cleave the ERSP off of the translated protein, releasing the ICK motif protein into the ER. Many ERSPs are known. Many plant ERSPs are known. It is NOT required that the ERSP be derived from a plant ERSP, non-plant ERSPs will work with the procedures described herein. Many plant ERSPs are however well known and we describe some plant derived ERSPs here. BAAS, for example, is derived from the plant, *Hordeum vulgare*.

One example of a ERSP used here is BAAS, the sequence of BAAS is MANKH LSLSL FLVLL GLSAS LASG (SEQ ID NO: 1035—one letter code).

This peptide, named "BAAS" is cleaved from the ICK motif upon the protein's translation into the ER. The molecular weight is 2442.94 Daltons. FIGS. 1-3 shows a

representation of an ICK motif protein linked to an ERSP. These figures could equally represent a TMOF motif protein linked to an ERSP.

Plant ERSPs, which are selected from the genomic sequence for proteins that are known to be expressed and released into the apoplastic space of plants, and a few examples are BAAS, carrot extensin, tobacco PR1. The following references provide further descriptions, and are incorporated by reference herein in their entirety. De Loose, M. et al. "The extension signal peptide allows secretion of a heterologous protein from protoplasts" *Gene*, 99 (1991) 95-100. De Loose, M. et al. described the structural analysis of an extensin-encoding gene from *Nicotiana plumbaginifolia*, the sequence of which contains a typical signal peptide for translocation of the protein to the endoplasmic reticulum. Chen, M. H. et al. "Signal peptide-dependent targeting of a rice alpha-amylase and cargo proteins to plastids and extracellular compartments of plant cells" *Plant Physiology*, 2004 July; 135(3): 1367-77. Epub 2004 Jul. 2. Chen, M. H. et al. studied the subcellular localization of α -amylases in plant cells by analyzing the expression of α -amylase, with and without its signal peptide, in transgenic tobacco. These references and others teach and disclose translational stabilizing proteins that can be used in the methods, procedures and peptide, protein and nucleotide complexes and constructs described herein.

The Translational Stabilizing Protein.

In addition to the text below, see Part I-III (The translational stabilizing protein component, STA or sta).

The procedures described above refer to providing a ERSP+CRIP where ERSP+CRIP could be ERSP+ICK, ERSP+Non-ICK, ERSP+Av (SEA ANOMONE) or the procedures could refer to ERSP+TMOF, or they could refer to ERSP+CRIP and a TMOF sufficient to make a plant produce properly folded peptides. We also suggest that in order to more fully protect a plant from some insects, more than just proper folding is sometimes needed. With a properly constructed expression cassette, a plant can be induced to make and accumulate even greater amounts of toxic peptide. When a plant accumulates greater amounts of properly folded toxic CRIP or TMOF peptides it can more easily resist or kill the insects that attack and eat the plants. One way to increase the insecticidal activity of the PIP is with translational stabilizing proteins. The translational stabilizing protein can be used to significantly increase the accumulation of the toxic peptide in the plant and thus the potency of the PIP, especially when the PIP has a translational stabilizing protein of its own. The procedures described herein can provide for the accumulation in the plant of large amounts of the now properly folded transgenic plant proteins. Transgenic plants expressing both an ICK motif insecticidal protein and a translational stabilizing protein, demonstrate dramatically improved accumulation of toxic ICK peptides over systems without a translational stabilizing protein. Representative PIPs with a translational stabilizing protein are described herein.

Experiments comparing plant expressed peptides both with and without a translational stabilizing protein show dramatic differences. The protein expression of an ICK-motif protein without a translational stabilizing protein can be very low. When a translational stabilizing protein is fused to the ICK-motif protein, there are higher levels of detectable accumulation. The translational stabilizing protein can be a domain of another protein or it can comprise an entire protein sequence. The translational stabilizing protein is a protein with sufficient tertiary structure that it can accumulate in a cell without being targeted by the cellular process

of protein degradation. The protein can be between 5 and 50aa (eg another ICK-motif protein), 50 to 250aa (GNA), 250 to 750aa (eg chitinase) and 750 to 1500aa (eg enhancin).

The translational stabilizing protein, (or protein domain) can contain proteins that have no useful characteristics other than translation stabilization, or they can have other useful traits in addition to translational stabilization. One embodiment of the translation stabilization protein can be multiple ICK-motif proteins in tandem. Useful traits can include: additional insecticidal activity, such as activity that is destructive to the peritrophic membrane, activity that is destructive to the gut wall, and/or activity that actively transports the ICK motif protein across the gut wall. One embodiment of the translational stabilizing protein can be a polymer of fusions proteins involving ICK motif proteins. One embodiment of the translational stabilizing protein can be a polymer of fusions proteins involving TMOF motif proteins. A specific example of a translational stabilizing protein is provided here to illustrate the use of a translational stabilizing protein. The example is not intended to limit the disclosure or claims in any way. Useful translational stabilizing proteins are well known in the art, and any proteins of this type could be used as disclosed herein. Procedures for evaluating and testing production of peptides are both known in the art and described herein. One example of one translational stabilizing protein is SEQ ID NO:1036, one letter code, as follows:

(one letter code)

SEQ ID NO: 1036
 ASKGE ELFTG VVPIL VELDQ DVNGH KFSVS GEGEG DATYG
 KLTLK FICTT GKLPV PWPTL VTTFQ YGVQC FSRYP DHMKR
 HDFFK SAMPE GYVQE RTISF KDDGN YKTRA EVKFE GDTLV
 NRIEL KGIDF KEDGN ILGHK LEYNY NSHNV YITAD KQKNG
 IKANF KIRHN IEDGS VQLAD HYQQN TPIGD GPVLL PDNHY
 LSTQS ALSKD PNEKR DHMVL LEFVT AAGIT HGMDE LYK.

Seq. ID No. 1036 is Named "GFP". The molecular weight is 26736.02 Daltons.

Additional examples of translational stabilizing proteins can be found in the following references, incorporated by reference in their entirety: Kramer, K. J. et al. "Sequence of a cDNA and expression of the gene encoding epidermal and gut chitinases of *Manduca sexta*" *Insect Biochemistry and Molecular Biology*, Vol. 23, Issue 6, September 1993, pp. 691-701. Kramer, K. J. et al. isolated and sequenced a chitinase-encoding cDNA from the tobacco hornworm, *Manduca sexta*. Hashimoto, Y. et al. "Location and nucleotide sequence of the gene encoding the viral enhancing factor of the *Trichoplusia ni* granulosis virus" *Journal of General Virology*, (1991), 72, 2645-2651. Hashimoto, Y. et al. cloned the gene encoding the viral enhancing factor of a *Trichoplusia ni* granulosis virus and determined the complete nucleotide sequence. Van Damme, E. J. M. et al. "Biosynthesis, primary structure and molecular cloning of snowdrop (*Galanthus nivalis* L.) lectin" *European Journal of Biochemistry*, 202, 23-30 (1991). Van Damme, E. J. M. et al. isolated Poly(A)-rich RNA from ripening ovaries of snowdrop lectin, yielding a single 17-kDa lectin polypeptide upon translation in a wheat-germ cell-free system. These references and others teach and disclose translational stabilizing proteins that can be used in the methods, procedures and peptide, protein and nucleotide complexes and constructs described herein.

The Intervening Linker

In addition to the text below, see Part I-IV (The Intervening Linker Peptide component, LINKER, linker, L or if polynucleotide; linker or 1 of the PEPs).

This invention also incorporates an intervening linker between ICK motif protein and the translational stabilizing protein. The intervening linker is between 1 and 30 amino acids. It can have either no cleavage sites or a protease cleavage site specific to serine-, threonine-, cysteine-, and aspartate proteases or metalloproteases. The cleavable linker can be the point of digestion by proteases found in the lepidopteran gut environment and/or the lepidopteran hemolymph environment. An example of the additional component to illustrate this invention is listed below, but it is not limited to this example.

The example for an intervening linker is IGER (SEQ ID NO: 1037) Named "IGER" The molecular weight of this intervening linker is 473.53 Daltons.

Other examples of intervening linkers can be found in the following references, which are incorporated by reference herein in their entirety: A comparison of the folding behavior of green fluorescent proteins through six different linkers is explored in Chang, H. C. et al. "De novo folding of GFP fusion proteins: high efficiency in eukaryotes but not in bacteria" *Journal of Molecular Biology*, 2005 Oct. 21; 353(2): 397-409. An isoform of the human GalNAc-Ts family, GalNAc-T2, was shown to retain its localization and functionality upon expression in *N. benthamiana* plants by Daskalova, S. M. et al. "Engineering of *N. benthamiana* L. plants for production of N-acetylgalactosamine-glycosylated proteins" *BMC Biotechnology*, 2010 Aug. 24; 10: 62. The ability of endogenous plastid proteins to travel through stromules was shown in Kwok, E. Y. et al. "GFP-labelled Rubisco and aspartate aminotransferase are present in plastid stromules and traffic between plastids" *Journal of Experimental Botany*, 2004 March; 55(397): 595-604. Epub 2004 Jan. 30. A report on the engineering of the surface of the tobacco mosaic virus (TMV), virion, with a mosquito decapeptide hormone, trypsin-modulating oostatic factor (TMOF) was made by Borovsky, D. et al. "Expression of Aedes trypsin-modulating oostatic factor on the virion of TMV: A potential larvicide" *Proc Natl Acad Sci*, 2006 Dec. 12; 103(50): 18963-18968. These references and others teach and disclose translational stabilizing proteins that can be used in the methods, procedures and peptide, protein and nucleotide complexes and constructs described herein. Other Plant Transformations are More Well Known.

Methods of the invention involve introducing a nucleotide construct into a plant. By "introducing" is intended to present to the plant the nucleotide construct in such a manner that the construct gains access to the interior of a cell of the plant. The methods of the invention do not require that a particular method for introducing a nucleotide construct to a plant is used, only that the nucleotide construct gains access to the interior of at least one cell of the plant. Methods for introducing nucleotide constructs into plants are known in the art including, but not limited to, stable transformation methods, transient transformation methods, and virus-mediated methods.

By "plant" is intended whole plants, plant organs (e.g., leaves, stems, roots, etc.), seeds, plant cells, propagules, embryos and progeny of the same. Plant cells can be differentiated or undifferentiated (e.g. callus, suspension culture cells, protoplasts, leaf cells, root cells, phloem cells, pollen).

"Transgenic plants" or "transformed plants" or "stably transformed" plants or cells or tissues refers to plants that

have incorporated or integrated exogenous nucleic acid sequences or DNA fragments into the plant cell. These nucleic acid sequences include those that are exogenous, or not present in the untransformed plant cell, as well as those that may be endogenous, or present in the untransformed plant cell. "Heterologous" generally refers to the nucleic acid sequences that are not endogenous to the cell or part of the native genome in which they are present, and have been added to the cell by infection, transfection, microinjection, electroporation, microprojection, or the like.

Transformation of plant cells can be accomplished by one of several techniques known in the art. The Bt-protein gene of the invention may be modified to obtain or enhance expression in plant cells. Typically a construct that expresses such a protein would contain a promoter to drive transcription of the gene, as well as a 3' untranslated region to allow transcription termination and polyadenylation. The organization of such constructs is well known in the art. In some instances, it may be useful to engineer the gene such that the resulting peptide is secreted, or otherwise targeted within the plant cell. For example, the gene can be engineered to contain a signal peptide to facilitate transfer of the peptide to the endoplasmic reticulum. It may also be preferable to engineer the plant expression cassette to contain an intron, such that mRNA processing of the intron is required for expression.

Typically this "plant expression cassette" will be inserted into a "plant transformation vector". This plant transformation vector may be comprised of one or more DNA vectors needed for achieving plant transformation. For example, it is a common practice in the art to utilize plant transformation vectors that are comprised of more than one contiguous DNA segment. These vectors are often referred to in the art as "binary vectors". Binary vectors as well as vectors with helper plasmids are most often used for *Agrobacterium*-mediated transformation, where the size and complexity of DNA segments needed to achieve efficient transformation is quite large, and it is advantageous to separate functions onto separate DNA molecules. Binary vectors typically contain a plasmid vector that contains the cis-acting sequences required for T-DNA transfer (such as left border and right border), a selectable marker that is engineered to be capable of expression in a plant cell, and a "gene of interest" (a gene engineered to be capable of expression in a plant cell for which generation of transgenic plants is desired). Also present on this plasmid vector are sequences required for bacterial replication. The cis-acting sequences are arranged in a fashion to allow efficient transfer into plant cells and expression therein. For example, the selectable marker gene and the Bt-protein are located between the left and right borders. Often a second plasmid vector contains the trans-acting factors that mediate T-DNA transfer from *Agrobacterium* to plant cells. This plasmid often contains the virulence functions (Vir genes) that allow infection of plant cells by *Agrobacterium*, and transfer of DNA by cleavage at border sequences and vir-mediated DNA transfer, as is understood in the art (Hellens and Mullineaux (2000) *Trends in Plant Science* 5:446-451). Several types of *Agrobacterium* strains (e.g. LBA4404, GV3101, EHA101, EHA105, etc.) can be used for plant transformation. The second plasmid vector is not necessary for transforming the plants by other methods such as microprojection, microinjection, electroporation, polyethylene glycol, etc.

In general, plant transformation methods involve transferring heterologous DNA into target plant cells (e.g. immature or mature embryos, suspension cultures, undifferentiated callus, protoplasts, etc.), followed by applying a

maximum threshold level of appropriate selection (depending on the selectable marker gene) to recover the transformed plant cells from a group of untransformed cell mass. Explants are typically transferred to a fresh supply of the same medium and cultured routinely. Subsequently, the transformed cells are differentiated into shoots after placing on regeneration medium supplemented with a maximum threshold level of selecting agent. The shoots are then transferred to a selective rooting medium for recovering rooted shoot or plantlet. The transgenic plantlet then grows into a mature plant and produces fertile seeds (e.g. Hiei et al. (1994) *The Plant Journal* 6:271-282; Ishida et al. (1996) *Nature Biotechnology* 14:745-750). Explants are typically transferred to a fresh supply of the same medium and cultured routinely. A general description of the techniques and methods for generating transgenic plants are found in Ayres and Park (1994) *Critical Reviews in Plant Science* 13:219-239 and Bommineni and Jauhar (1997) *Maydica* 42:107-120. Since the transformed material contains many cells; both transformed and non-transformed cells are present in any piece of subjected target callus or tissue or group of cells. The ability to kill non-transformed cells and allow transformed cells to proliferate results in transformed plant cultures. Often, the ability to remove non-transformed cells is a limitation to rapid recovery of transformed plant cells and successful generation of transgenic plants.

Transformation protocols as well as protocols for introducing nucleotide sequences into plants may vary depending on the type of plant or plant cell, i.e., monocot or dicot, targeted for transformation. Generation of transgenic plants may be performed by one of several methods, including, but not limited to, microinjection, electroporation, direct gene transfer, introduction of heterologous DNA by *Agrobacterium* into plant cells (*Agrobacterium*-mediated transformation), bombardment of plant cells with heterologous foreign DNA adhered to particles, ballistic particle acceleration, aerosol beam transformation (U.S. Published Application No. 20010026941; U.S. Pat. No. 4,945,050; International Publication No. WO 91/00915; U.S. Published Application No. 2002015066), *Lec1* transformation, and various other non-particle direct-mediated methods to transfer DNA.

Methods for transformation of chloroplasts are known in the art. See, for example, Svab et al. (1990) *Proc. Natl. Acad. Sci. USA* 87:8526-8530; Svab and Maliga (1993) *Proc. Natl. Acad. Sci. USA* 90:913-917; Svab and Maliga (1993) *EMBO J.* 12:601-606. The method relies on particle gun delivery of DNA containing a selectable marker and targeting of the DNA to the plastid genome through homologous recombination. Additionally, plastid transformation can be accomplished by transactivation of a silent plastid-borne transgene by tissue-preferred expression of a nuclear-encoded and plastid-directed RNA polymerase. Such a system has been reported in McBride et al. (1994) *Proc. Natl. Acad. Sci. USA* 91:7301-7305.

Following integration of heterologous foreign DNA into plant cells, one then applies a maximum threshold level of appropriate selection in the medium to kill the untransformed cells and separate and proliferate the putatively transformed cells that survive from this selection treatment by transferring regularly to a fresh medium. By continuous passage and challenge with appropriate selection, one identifies and proliferates the cells that are transformed with the plasmid vector. Molecular and biochemical methods can then be used to confirm the presence of the integrated heterologous gene of interest into the genome of the transgenic plant.

The cells that have been transformed may be grown into plants in accordance with conventional ways. See, for example, McCormick et al. (1986) Plant Cell Reports 5:81-84. These plants may then be grown, and either pollinated with the same transformed strain or different strains, and the resulting hybrid having constitutive expression of the desired phenotypic characteristic identified. Two or more generations may be grown to ensure that expression of the desired phenotypic characteristic is stably maintained and inherited and then seeds harvested to ensure expression of the desired phenotypic characteristic has been achieved. In this manner, the present invention provides transformed seed (also referred to as "transgenic seed") having a nucleotide construct of the invention, for example, an expression cassette of the invention, stably incorporated into their genome. ICK and TMOF Expression in Plants.

As noted above, there are many alternatives that could be used for the components of ERSP, ICK motif protein, TMOF motif, translational stabilizing protein and intervening linker.

Evaluation of Plant Transformations

Following introduction of heterologous foreign DNA into plant cells, the transformation or integration of heterologous gene in the plant genome is confirmed by various methods such as analysis of nucleic acids, proteins and metabolites associated with the integrated gene.

PCR analysis is a rapid method to screen transformed cells, tissue or shoots for the presence of incorporated gene at the earlier stage before transplanting into the soil (Sambrook and Russell (2001) Molecular Cloning: A Laboratory Manual. Cold Spring Harbor Laboratory Press, Cold Spring Harbor, N.Y.). PCR is carried out using oligonucleotide primers specific to the gene of interest or Agrobacterium vector background, etc.

Plant transformation may be confirmed by Southern blot analysis of genomic DNA (Sambrook and Russell, 2001, supra). In general, total DNA is extracted from the transformant, digested with appropriate restriction enzymes, fractionated in an agarose gel and transferred to a nitrocellulose or nylon membrane. The membrane or "blot" is then probed with, for example, radiolabeled .sup.32P target DNA fragment to confirm the integration of introduced gene into the plant genome according to standard techniques (Sambrook and Russell, 2001, supra).

In Northern blot analysis, RNA is isolated from specific tissues of transformant, fractionated in a formaldehyde agarose gel, and blotted onto a nylon filter according to standard procedures that are routinely used in the art (Sambrook and Russell, 2001, supra). Expression of RNA encoded by the Bt-protein is then tested by hybridizing the filter to a radioactive probe derived from a Bt-protein, by methods known in the art (Sambrook and Russell, 2001, supra).

Western blot, biochemical assays and the like may be carried out on the transgenic plants to confirm the presence of protein encoded by the Bt-protein gene by standard procedures (Sambrook and Russell, 2001, supra) using antibodies that bind to one or more epitopes present on the Bt-protein.

Pesticidal Activity in Plants

In another aspect of the invention, one may generate transgenic plants expressing a Bt-protein that has pesticidal activity. Methods described above by way of example may be utilized to generate transgenic plants, but the manner in which the transgenic plant cells are generated is not critical to this invention. Methods known or described in the art such as Agrobacterium-mediated transformation, biolistic transformation, and non-particle-mediated methods may be

used. Plants expressing a Bt-protein may be isolated by common methods described in the art, for example by transformation of callus, selection of transformed callus, and regeneration of fertile plants from such transgenic callus. In such process, one may use any gene as a selectable marker so long as its expression in plant cells confers ability to identify or select for transformed cells.

A number of markers have been developed for use with plant cells, such as resistance to chloramphenicol, the aminoglycoside G418, hygromycin, or the like. Other genes that encode a product involved in chloroplast metabolism may also be used as selectable markers. For example, genes that provide resistance to plant herbicides such as glyphosate, bromoxynil, or imidazolinone may find particular use. Such genes have been reported (Stalker et al. (1985) J. Biol. Chem. 263:6310-6314 (bromoxynil resistance nitrilase gene); and Sathasivan et al. (1990) Nucl. Acids Res. 18:2188 (AHAS imidazolinone resistance gene). Additionally, the genes disclosed herein are useful as markers to assess transformation of bacterial or plant cells. Methods for detecting the presence of a transgene in a plant, plant organ (e.g., leaves, stems, roots, etc.), seed, plant cell, propagule, embryo or progeny of the same are well known in the art. In one embodiment, the presence of the transgene is detected by testing for pesticidal activity.

Fertile plants expressing a Bt-protein may be tested for pesticidal activity, and the plants showing optimal activity selected for further breeding. Methods are available in the art to assay for pest activity. Generally, the protein is mixed and used in feeding assays. See, for example Marrone et al. (1985) J. of Economic Entomology 78:290-293. Section VI. Descriptions and Examples of CRIP and Bt Protein Combinations

The Bt and ICK peptides may inhibit the growth, impair the movement, or even kill an insect when the combination of toxin is appropriately delivered to the locus inhabited by the insect. SDP 1234604, 1234605 and 609 are spray-dried powder preparations of hybrid+2-ACTX-Hv1a peptide, here "Hv1a peptide." The spray-dried Hv1a peptide powders are made from the peptide, various excipients and fermentation by-products. The '604 and '605 formulations use the same peptide, only the excipients are different. The concentration of the active hybrid peptide was quantified at about 26% weight/weight in both the '604 and '605 powders. The concentration of the active hybrid peptide was quantified at about 35% weight/weight in the 609 powders. The Hv1a peptide in each powder was quantified using a C18 rpHPLC methods known by those skilled in the art.

Inhibitory cysteine knot or ICK peptides can have remarkable stability when exposed to the environment. Many ICK peptides are isolated from venomous animals such as spiders, scorpions, and snakes. Bt proteins are well known because of their specific pesticidal activities. Surprisingly, we have found that, when Bt proteins are selectively mixed with ICK peptides, the combination of Bt and ICK peptides produces a highly effective insecticide with a potency much greater than expected.

We describe an insecticidal combination peptide composition comprising both a Bt (*Bacillus thuringiensis*) protein; and an insecticidal ICK (Inhibitor Cystine Knot) peptide. The composition can be in the ratio of Bt to ICK, on a dry weight basis, from about any or all of the following ratios: 99:1, 95:5, 90:10, 85:15, 80:20, 75:25, 70:30, 65:35, 60:40, 55:45, 50:50, 45:55, 40:60, 35:65, 30:70, 25:75, 20:80, 15:85, 10:90, 5:95 and 1:99, or any combination of any two of these values. We also describe a composition where the ratio of Bt to ICK, on a dry weight basis, is selected

from about the following ratios: 0:50, 45:55, 40:60, 35:65, 30:70, 25:75, 20:80, 15:85, 10:90, 5:95, 1:99, 0.5:99.5, 0.1:99.9 and 0.01:99.99 or any combination of any two of these values.

The procedures described herein can be applied to any PFIP or CRIP peptide. The combination of PFIP and CRIP peptides includes either or both of the PFIP and CRIP peptides being derived from more than 1 different types or bacterial strain origins for either one or both of PFIP and CRIP peptides. By bacterial strain origins we mean the peptides can be described as having been expressed by a bacterial strain that expresses the peptides with the understanding that many PFIP peptides including many Bt proteins are also artificial in the sense that they are no longer all developed from bacterial strains.

In another embodiment the combination of PFIP and CRIP peptides includes either or both of the PFIP such as Bt in combination with ICK, Non-ICK and TMOF peptides being derived from more than 1 different types or bacterial strain origins for either one or both of Bt and ICK peptides. By bacterial strain origins we mean the peptides can be described as having been expressed by a bacterial strain that expresses the peptides with the understanding that many Bt proteins are also artificial in the sense that they are no longer all developed from bacterial strains.

We also disclose compositions where either or both of the PFIP such as Bt in combination with ICK, Non-ICK and TMOF peptides are derived from between 2 and 5, 2-15, 2-30, 5-10, 5-15, 5-30, 5-50 and various other different types or bacterial strains origins of either one or both of Bt or ICK peptides. We disclose a composition where either or both of the Bt and ICK peptides are encoded by from 2 to 15 different types or bacterial strain origins of either one or both of Bt and ICK peptides. And any of these combinations of 2-5, 2-15, 2-30, 5-10, 5-15, 5-30, 5-50 and various other different types and mixtures of Bt and ICK peptides can contribute more than at least 1% of each strain type to the composition.

We disclose composition of Bt and ICK peptides where the total concentration of PFIP such as Bt in combination with ICK, Non-ICK and TMOF peptides in the composition is selected from the following percent concentrations: 1, 5, 10, 15, 20, 25, 30, 35, 40, 45, 50, 55, 60, 65, 70, 75, 80, 85, 90, 95 or 99%, or any range between any two of these values, and the remaining percentage of the composition is comprised of excipients. We disclose compositions wherein the insecticidal combination peptide is produced using a genetic cassette that further comprises an ERSP (Endoplasmic Reticulum Signal Peptide) operably linked to the ICK, Non-ICK and/or TMOF peptides insecticidal ICK peptide, wherein said ERSP is linked at the N-terminal of the insecticidal ICK peptide. We disclose compositions wherein the insecticidal combination peptide is produced using a genetic cassette that further comprises an ERSP (Endoplasmic Reticulum Signal Peptide) operably linked to the insecticidal ICK peptide, wherein said ERSP is linked at the N-terminal of the insecticidal ICK peptide, wherein the ERSP is BAAS.

We disclose compositions wherein said combination peptide is produced using a genetic cassette that further comprises a dipeptide operably linked to the insecticidal ICK peptide, wherein said dipeptide is linked at the N-terminal of the insecticidal ICK peptide; and wherein the dipeptide is comprised of one nonpolar amino acid on the N-terminal of the dipeptide and one polar amino acid on the C-terminal of the dipeptide, including embodiments where the dipeptide is glycine-serine, including embodiments where the insecti-

cidal ICK peptide is any insecticidal peptide that inhibits both voltage-gated Calcium channels and Calcium-activated potassium channels in insects, including embodiments where the insecticidal ICK peptide originates from any species of Australian Funnel-web spider, including embodiments where the spider is selected from the Australian Funnel-web spiders of genus *Atrax* or *Hadronyche*, including embodiments where the spider is selected from the Australian Funnel-web spiders of genus *Hadronyche*, including embodiments where the spider is selected from the Australian Blue Mountains Funnel-web, *Hadronyche versuta*, including embodiments where the insecticidal ICK peptide is Hybrid-ACTX-Hv1a, including embodiments where the insecticidal ICK peptide contains 20-100 amino acids and 2-4 disulfide bonds, including embodiments where said insecticidal ICK peptide is any insecticidal peptide with at least 50%, 55%, 60%, 65%, 70%, 75%, 80%, 85%, 90%, 95%, 96%, 97%, 98%, or 99% or greater sequence identity to any of the ICK sequences disclosed herein, including embodiments where the insecticidal ICK peptide is selected from publications incorporated by reference, including embodiments where the Bt protein is any insecticidal Bt protein, including embodiments where the Bt protein is a Cry or Cyt protein, including embodiments where the Bt protein is selected from the group consisting of a Cry1, Cry3, TIC851, CryET70, Cry22, TIC901, TIC201, TIC407, TIC417, a binary insecticidal protein CryET80, and CryET76, a binary insecticidal protein TIC100 and TIC101, a combination of an insecticidal protein ET29 or ET37 with an insecticidal protein TIC810 or TIC812 and a binary insecticidal protein PS149B1, including embodiments where the Bt Protein is selected from a Cry protein, a Cry1A protein or a Cry1F protein, including embodiments where the Bt protein is a combination Cry1F-Cry1A protein, including embodiments where the Bt protein comprises an amino acid sequence at least 90% identical to SEQ ID NO: 10, 12, 14, 26, 28, or 34 of U.S. Pat. No. 7,304,206, including embodiments where the Bt Protein is DIPEL®, including embodiments where the Bt protein is THURICIDE®.

We disclose a composition comprising the nucleotides of: Bt (*Bacillus thuringiensis*) Protein; and an insecticidal ICK (Inhibitor Cystine Knot) protein, in a transformed plant or plant genome; where the ratio of Bt to ICK, on a dry weight basis, is selected from about the following ratios: 99:1, 95:5, 90:10, 85:15, 80:20, 75:25, 70:30, 65:35, 60:40, 55:45, 50:50, 45:55, 40:60, 35:65, 30:70, 25:75, 20:80, 15:85, 10:90, 5:95 and 1:99, or any combination of any two of these values.

We disclose transformed plant or plant genome wherein the ratio of PFIP such as Bt to ICK, Non-ICK and TMOF peptides; and preferably Bt to ICK, or Bt to an Anomone toxin, on a dry weight basis, is selected from about the following ratios: 50:50, 45:55, 40:60, 35:65, 30:70, 25:75, 20:80, 15:85, 10:90, 5:95, 1:99, 0.5:99.5, 0.1:99.9 and 0.01:99.99 or any combination of any two of these values. The transformed plant or plant genome may have either or both of the Bt and ICK or Bt and Anomone proteins are derived from more than 1 different type or bacterial strain origin of Bt or ICK proteins, or either or both of the Bt and ICK proteins are derived from between 2 and 5 different types or strain origins, or either or both of the Bt and ICK proteins are derived from 2 to 15 different type or bacterial strain origins of either or both of Bt and ICK proteins and at least one strain of either Bt or ICK or both

Bt and ICK proteins encoded by more than one copy of the Bt or ICK genes, or either or both of the Bt and ICK proteins are derived from more than one different type or bacterial strain origin of Bt and/or ICK proteins where all the strains of Bt and/or ICK proteins contribute more than at least 1% of each strain type to said composition, or either or both of the Bt and ICK proteins are derived from 2 to 5 different type or bacterial strain origins of either or both of Bt and ICK proteins and at least one strain of either Bt or ICK or both Bt and ICK proteins encoded by more than one copy of the Bt or ICK genes, or the total concentration of Bt and ICK protein in the composition can be selected from the following percent concentrations: 1, 5, 10, 15, 20, 25, 30, 35, 40, 45, 50, 55, 60, 65, 70, 75, 80, 85, 90, 95 or 99%, or any range between any two of these values, and the remaining percentage of the composition is comprised of excipients.

The compositions and plants described herein include an insecticidal combination protein produced using a genetic cassette that further comprises an ERSP (Endoplasmic Reticulum Signal Peptide) operably linked to the insecticidal ICK peptide, wherein said ERSP is linked at the N-terminal of the insecticidal ICK peptide. In another embodiment the insecticidal combination peptide is produced using a genetic cassette that further comprises an ERSP (Endoplasmic Reticulum Signal Peptide) operably linked to the insecticidal ICK peptide, wherein said ERSP is linked at the N-terminal of the insecticidal ICK peptide, wherein the ERSP is BAAS. In another embodiment the transgenic plant incorporating and expressing the combination peptides from the nucleotides described herein, wherein said combination peptide is produced using a genetic cassette that further comprises nucleotides expressing a dipeptide operably linked to the insecticidal ICK peptide, wherein said dipeptide is linked at the N-terminal of the insecticidal ICK peptide; and wherein the dipeptide is comprised of one nonpolar amino acid on the N-terminal of the dipeptide and one polar amino acid on the C-terminal of the dipeptide. In another embodiment the transgenic plant has a dipeptide that glycine-serine. In another embodiment the transgenic plant has insecticidal ICK peptides expressed that are comprised of an insecticidal peptide combination of ICK and Bt proteins. The transgenic plants can have an insecticidal ICK peptide derived from any species of Australian Funnel-web spider, or the Australian Funnel-web spiders of genus *Atrax* or *Hadronyche*, and the Australian Blue Mountains Funnel-web, *Hadronyche versuta*.

We describe and claim a transgenic plant wherein the insecticidal ICK peptide expressed is Hybrid-ACTX-Hv1a, and or the insecticidal ICK peptide expressed may contain 20-100 amino acids and 2-4 disulfide bonds and or the insecticidal ICK peptide is any insecticidal peptide with at least 50%, 55%, 60%, 65%, 70%, 75%, 80%, 85%, 90%, 95%, 96%, 97%, 98%, or 99% or greater sequence identity to any of the ICK peptides described herein. The transgenic plants disclosed can contain any known Bt protein, including peptides where the Bt protein is a Cry or Cyt protein, and/or the Bt protein is selected from the group consisting of a Cry1, Cry3, TIC851, CryET70, Cry22, TIC901, TIC201, TIC407, TIC417, a binary insecticidal protein CryET80, and CryET76, a binary insecticidal protein TIC100 and TIC101, a combination of an insecticidal protein ET29 or ET37 with an insecticidal protein TIC810 or TIC812 and a binary insecticidal protein PS149B1. The Bt protein can be selected from a Cry protein, a Cry1A protein or a Cry1F protein, or a combination Cry1F-Cry1A protein, or it comprises an amino acid sequence at least 90% identical to SEQ ID NO: 10, 12, 14, 26, 28, or 34 of U.S. Pat. No. 7,304,206. We

describe a transgenic plant wherein the Bt protein is DIPEL® and we describe a transgenic plant wherein the Bt protein is THURICIDE®.

We specifically describe and claim a transformed plant expressing the peptides described herein where the average concentration of Bt and ICK peptide, in an average leaf of a transformed plant is about: 1, 5, 10, 15, 20, 25, 30, 35, 40, 45, 50, 55, 60, 65, 70, 75, 80, 85, 90, 95 or 99%, or any range between any two of these values. We specifically describe and claim a transformed plant expressing properly folded toxic peptides in the transformed plant. We specifically describe and claim a transformed plant expressing properly folded combination toxic peptides in the transformed plant and to cause the accumulation of the expressed and properly folded toxic peptides in said plant and to cause an increase in the plant's yield or resistance to insect damage and they control insect pests in crops and forestry. We describe plants made by any of the products and processes described herein.

We describe expression cassettes comprising any of the nucleotides which express any peptides described herein, including embodiments having a functional expression cassette incorporated into a transformed plant, comprising nucleotides that code for any of the peptides disclosed herein or that could be made by one skilled in the art given the teaching disclosed herein. We describe and claim procedures for the generation of transformed plants having or expressing any of the peptides described herein.

We describe the use of any of the peptides or nucleotides described herein, to make a plant or transform these peptides or nucleotides into a plant, and methods and techniques for generating these proteins in plants and/or expression cassettes comprising any of the peptides and methods to transform them into a plant genome and any method of using, making, transforming any of the described peptides or nucleotides into a plant and methods and techniques for generating transformed plants having or expressing any of the peptides and functional expression cassettes in plants comprising any of the disclosed peptides and their corresponding nucleotides and any plants made by the products and processes described herein.

In some embodiments we disclose a chimeric gene comprising a promoter active in plants operatively linked to the nucleic acids or expression cassettes as described herein. We disclose a method of making, producing, or using the combination of genes described herein. We disclose a recombinant vector comprising the combination of genes described herein. We disclose a method of making, producing, or using the recombinant vector. We disclose a transgenic host cell comprising the combination of genes described herein and the method of making, producing or using the transgenic host cell, which can be a transgenic plant cell and we disclose a method of making, producing or using such a transgenic plant cell as well as the transgenic plant comprising the transgenic plant cell and how to make and use the transgenic plant. We disclose transgenic plant and seed having the properties described herein that is derived from corn, soybean, cotton, rice, sorghum, switchgrass, sugarcane, alfalfa, potatoes or tomatoes. The transgenic seed may have a chimeric gene that we describe herein. We describe methods of making, producing or using the transgenic plant and or seed of this disclosure.

We also describe methods of using the invention and provide novel formulations. The invention is most useful to control insects. We describe a method of controlling an insect comprising: Applying Bt (*Bacillus thuringiensis*) protein to said insect; and Applying an insecticidal ICK (Inhibitor Cystine Knot) peptide to said insect. This method may be

used where the Bt protein and the insecticidal ICK peptide are applied together at the same time in the same compositions or separately in different compositions and at different times. The Bt protein and the insecticidal ICK peptide may be applied sequentially, and it may be applied to (Bt protein)-resistant insects. The ratio of Bt to ICK, on a dry weight basis, can be selected from at least about the following ratios: 99:1, 95:5, 90:10, 85:15, 80:20, 75:25, 70:30, 65:35, 60:40, 55:45, 50:50, 45:55, 40:60, 35:65, 30:70, 25:75, 20:80, 15:85, 10:90, 5:95 and 1:99, or any combination of any two of these values. The ratio of Bt to ICK, on a dry weight basis, can be selected from about the following ratios: 50:50, 45:55, 40:60, 35:65, 30:70, 25:75, 20:80, 15:85, 10:90, 5:95, 1:99, 0.5:99.5, 0.1:99.9 and 0.01:99.99 or any combination of any two of these values. Either or both of the Bt and ICK peptides are derived from more than 1 different types or bacterial strain origins of Bt and ICK peptides. Either or both of the Bt and ICK peptides are derived from between 2 and 5 different types or bacterial strain origins of either Bt or ICK peptides or both Bt and ICK peptides. Either or both of the Bt and ICK peptides are derived from 2 to 15 different types or bacterial strain origins of either or both of Bt and ICK peptides and at least one strain of either Bt or ICK or both Bt and ICK peptides are encoded by more than one copy of the Bt or ICK genes. Either one or both of the Bt and ICK peptides are derived from more than 1 different types or bacterial strain origins of Bt and/or ICK peptides with all the strains of Bt and/or ICK peptides contributing more than at least 1% of the peptides from each strain type in said composition. Either or both of the Bt and ICK peptides are derived from 2 to 5 different types or bacterial strain origins of either one or both of Bt and ICK peptides and at least one strain of either Bt or ICK or both Bt and ICK peptides are encoded by more than one copy of the Bt or ICK genes. The total concentration of Bt and ICK peptide in the composition is selected from the following percent concentrations: 1, 5, 10, 15, 20, 25, 30, 35, 40, 45, 50, 55, 60, 65, 70, 75, 80, 85, 90, 95 or 99%, or any range between any two of these values, and the remaining percentage of the composition is comprised of excipients.

The methods can be used where the insecticidal combination peptide is produced using a genetic cassette that further comprises an ERSP (Endoplasmic Reticulum Signal Peptide) operably linked to the insecticidal ICK peptide, wherein said ERSP is linked at the N-terminal of the insecticidal ICK peptide. In some embodiments the insecticidal combination peptides used are produced using a genetic cassette that further comprises an ERSP (Endoplasmic Reticulum Signal Peptide) operably linked to the insecticidal ICK peptide, wherein said ERSP is linked at the N-terminal of the insecticidal ICK peptide, wherein the ERSP is BAAS.

Any of the peptides and plants described herein can be used to control insects, their growth and damage, especially their damage to plants. The combination Bt Protein and insecticidal ICK peptide can be applied by being sprayed on a plant, or the insect's locus, or the locus of a plant in need of protecting.

We also describe formulations comprising: Bt Protein; and an insecticidal ICK peptide which can include any of the compositions described herein or capable of being made by one skilled in the art given this disclosure. Some of the described formulations include the use of a polar aprotic solvent, and or water, and or where the polar aprotic solvent is present in an amount of 1-99 wt %, the polar protic solvent is present in an amount of 1-99 wt %, and the water is present in an amount of 0-98 wt %. The formulations include

formulations where the Bt protein is DIPEL® and where the insecticidal ICK peptide is a hybrid-ACTX-Hv1a peptide. The polar aprotic solvent formulations are especially effective when they contain MSO. The examples below are intended to illustrate and not limit the invention in any manner.

Section VI. Descriptions and Examples of TMOF and Bt Combinations

The Bt and TMOF peptides may inhibit the growth, impair the movement, or even kill an insect when the combination of toxin is appropriately delivered to the locus inhabited by the insect. The spray-dried powders are made from the peptide, various excipients and fermentation by-products.

We describe an insecticidal combination peptide composition comprising both a Bt (*Bacillus thuringiensis*) protein; and an insecticidal TMOF peptide. The composition can be in the ratio of Bt to TMOF, on a dry weight basis, from about any or all of the following ratios: 99:1, 95:5, 90:10, 85:15, 80:20, 75:25, 70:30, 65:35, 60:40, 55:45, 50:50, 45:55, 40:60, 35:65, 30:70, 25:75, 20:80, 15:85, 10:90, 5:95 and 1:99, or any combination of any two of these values. We also describe a composition where the ratio of Bt to TMOF, on a dry weight basis, is selected from about the following ratios: 0:50, 45:55, 40:60, 35:65, 30:70, 25:75, 20:80, 15:85, 10:90, 5:95, 1:99, 0.5:99.5, 0.1:99.9 and 0.01:99.99 or any combination of any two of these values.

In another embodiment the combination of Bt and TMOF peptides includes either or both of the Bt and TMOF peptides being derived from more than 1 different types or bacterial strain origins for either one or both of Bt and TMOF peptides. By bacterial strain origins we mean the peptides can be described as having been expressed by a bacterial strain that expresses the peptides with the understanding that many Bt proteins are also artificial in the sense that they are no longer all developed from bacterial strains.

We also disclose compositions where either or both of the Bt and TMOF peptides are derived from between 2 and 5, 2-15, 2-30, 5-10, 5-15, 5-30, 5-50 and various other different types or bacterial strains origins of either one or both of Bt or TMOF peptides. We disclose a composition where either or both of the Bt and TMOF peptides are encoded by from 2 to 15 different types or bacterial strain origins of either one or both of Bt and TMOF peptides. And any of these combinations of 2-5, 2-15, 2-30, 5-10, 5-15, 5-30, 5-50 and various other different types and mixtures of Bt and TMOF peptides can contribute more than at least 1% of each strain type to the composition.

We disclose composition of Bt and TMOF where the total concentration of Bt and TMOF peptide in the composition is selected from the following percent concentrations: 1, 5, 10, 15, 20, 25, 30, 35, 40, 45, 50, 55, 60, 65, 70, 75, 80, 85, 90, 95 or 99%, or any range between any two of these values, and the remaining percentage of the composition is comprised of excipients. We disclose compositions wherein the insecticidal combination peptide is produced using a genetic cassette that further comprises an ERSP (Endoplasmic Reticulum Signal Peptide) operably linked to the insecticidal TMOF peptide, wherein said ERSP is linked at the N-terminal of the insecticidal TMOF peptide. We disclose compositions wherein the insecticidal combination peptide is produced using a genetic cassette that further comprises an ERSP (Endoplasmic Reticulum Signal Peptide) operably linked to the insecticidal TMOF peptide, wherein said ERSP is linked at the N-terminal of the insecticidal TMOF peptide, wherein the ERSP is BAAS.

We disclose compositions wherein said combination peptide is produced using a genetic cassette that further comprises a dipeptide operably linked to the insecticidal TMOF peptide, wherein said dipeptide is linked at the N-terminal of the insecticidal TMOF peptide; and wherein the dipeptide is comprised of one nonpolar amino acid on the N-terminal of the dipeptide and one polar amino acid on the C-terminal of the dipeptide, including embodiments where the dipeptide is glycine-serine, including embodiments where the insecticidal TMOF peptide is any includes embodiments where the insecticidal TMOF peptide is at least 50%, 55%, 60%, 65%, 70%, 75%, 80%, 85%, 90%, 95%, 96%, 97%, 98%, or 99% or greater sequence identity to any of the TMOF sequences disclosed herein, including embodiments where the Bt Protein is any insecticidal Bt Protein, including embodiments where the Bt Protein is a Cry or Cyt protein, including embodiments where the Bt Protein is selected from the group consisting of a Cry1, Cry3, TIC851, CryET70, Cry22, TIC901, TIC201, TIC407, TIC417, a binary insecticidal protein CryET80, and CryET76, a binary insecticidal protein TIC100 and TIC101, a combination of an insecticidal protein ET29 or ET37 with an insecticidal protein TIC810 or TIC812 and a binary insecticidal protein PS149B1, including embodiments where the Bt protein is selected from a Cry protein, a Cry1A protein or a Cry1F protein, including embodiments where the Bt protein is a combination Cry1F-Cry1A protein, including embodiments where the Bt protein comprises an amino acid sequence at least 90% identical to SEQ ID NO: 10, 12, 14, 26, 28, or 34 of U.S. Pat. No. 7,304,206, including embodiments where the Bt Endotoxin is DIPEL®, including embodiments where the Bt Protein is THURICIDE®.

We disclose a composition comprising the nucleotides of: Bt (*Bacillus thuringiensis*) protein; and an insecticidal TMOF peptide, in a transformed plant or plant genome; where the ratio of Bt to TMOF, on a dry weight basis, is selected from about the following ratios: 99:1, 95:5, 90:10, 85:15, 80:20, 75:25, 70:30, 65:35, 60:40, 55:45, 50:50, 45:55, 40:60, 35:65, 30:70, 25:75, 20:80, 15:85, 10:90, 5:95 and 1:99, or any combination of any two of these values.

We disclose transformed plant or plant genome wherein the ratio of Bt to TMOF, on a dry weight basis, is selected from about the following ratios: 50:50, 45:55, 40:60, 35:65, 30:70, 25:75, 20:80, 15:85, 10:90, 5:95, 1:99, 0.5:99.5, 0.1:99.9 and 0.01:99.99 or any combination of any two of these values. The transformed plant or plant genome may have either or both of the Bt and TMOF peptides are derived from more than 1 different type or bacterial strain origin of Bt or TMOF peptides, or either or both of the Bt and TMOF peptides are derived from between 2 and 5 different type or bacterial strain origin of either Bt or TMOF peptides or both Bt and TMOF peptides are derived from between 2 and 5 different types or strain origins, or either or both of the Bt and TMOF peptides are derived from 2 to 15 different type or bacterial strain origins of either or both of Bt and TMOF peptides and at least one strain of either Bt or TMOF or both Bt and TMOF peptides encoded by more than one copy of the Bt or TMOF genes, or either or both of the Bt and TMOF peptides are derived from more than one different type or bacterial strain origin of Bt and/or TMOF peptides where all the strains of Bt and/or TMOF peptides contribute more than at least 1% of each strain type to said composition, or either or both of the Bt and TMOF peptides are derived from 2 to 5 different type or bacterial strain origins of either or both of Bt and TMOF peptides and at least one strain of either Bt or TMOF or both Bt and TMOF peptides encoded by more than one copy of the Bt or TMOF genes, or the total concentration

of Bt and TMOF peptide in the composition can be selected from the following percent concentrations: 1, 5, 10, 15, 20, 25, 30, 35, 40, 45, 50, 55, 60, 65, 70, 75, 80, 85, 90, 95 or 99%, or any range between any two of these values, and the remaining percentage of the composition is comprised of excipients.

The compositions and plants described herein include an insecticidal combination peptide produced using a genetic cassette that further comprises an ERSP (Endoplasmic Reticulum Signal Peptide) operably linked to the insecticidal TMOF peptide, wherein said ERSP is linked at the N-terminal of the insecticidal TMOF peptide. In another embodiment the insecticidal combination peptide is produced using a genetic cassette that further comprises an ERSP (Endoplasmic Reticulum Signal Peptide) operably linked to the insecticidal TMOF peptide, wherein said ERSP is linked at the N-terminal of the insecticidal TMOF peptide, wherein the ERSP is BAAS. In another embodiment the transgenic plant incorporating and expressing the combination peptides from the nucleotides described herein, wherein said combination peptide is produced using a genetic cassette that further comprises nucleotides expressing a dipeptide operably linked to the insecticidal TMOF peptide, wherein said dipeptide is linked at the N-terminal of the insecticidal TMOF peptide; and wherein the dipeptide is comprised of one nonpolar amino acid on the N-terminal of the dipeptide and one polar amino acid on the C-terminal of the dipeptide. In another embodiment the transgenic plant has a dipeptide that is glycine-serine. In another embodiment the transgenic plant has insecticidal TMOF peptides expressed that are comprised of an insecticidal peptide combination of TMOF and Bt proteins. The transgenic plants can have an insecticidal TMOF peptide derived from any TMOF species.

We describe and claim a transgenic plant wherein the insecticidal TMOF peptide expressed is may contain 20-100 amino acids and or the insecticidal TMOF peptide is any insecticidal peptide with at least 50%, 55%, 60%, 65%, 70%, 75%, 80%, 85%, 90%, 95%, 96%, 97%, 98%, or 99% or greater sequence identity to any of the TMOF peptides described herein. The transgenic plants disclosed can contain any known Bt Protein, including peptides where the Bt Protein is a Cry or Cyt protein, and/or the Bt Protein is selected from the group consisting of a Cry1, Cry3, TIC851, CryET70, Cry22, TIC901, TIC201, TIC407, TIC417, a binary insecticidal protein CryET80, and CryET76, a binary insecticidal protein TIC100 and TIC101, a combination of an insecticidal protein ET29 or ET37 with an insecticidal protein TIC810 or TIC812 and a binary insecticidal protein PS149B1. The Bt Protein can be selected from a Cry protein, a Cry1A protein or a Cry1F protein, or a combination Cry1F-Cry1A protein, or it comprises an amino acid sequence at least 90% identical to SEQ ID NO: 10, 12, 14, 26, 28, or 34 of U.S. Pat. No. 7,304,206. We describe a transgenic plant wherein the Bt Protein is DIPEL® and we describe a transgenic plant wherein the Bt Protein is THURICIDE®.

We specifically describe and claim a transformed plant expressing the peptides described herein where the average concentration of Bt and TMOF peptide, in an average leaf of a transformed plant is about: 1, 5, 10, 15, 20, 25, 30, 35, 40, 45, 50, 55, 60, 65, 70, 75, 80, 85, 90, 95 or 99%, or any range between any two of these values. We specifically describe and claim a transformed plant expressing properly folded toxic peptides in the transformed plant. We specifically describe and claim a transformed plant expressing properly folded combination toxic peptides in the transformed plant and to cause the accumulation of the expressed and properly

folded toxic peptides in said plant and to cause an increase in the plant's yield or resistance to insect damage and they control insect pests in crops and forestry. We describe plants made by any of the products and processes described herein.

We describe expression cassettes comprising any of the nucleotides which express any peptides described herein, including embodiments having a functional expression cassette incorporated into a transformed plant, comprising nucleotides that code for any of the peptides disclosed herein or that could be made by one skilled in the art given the teaching disclosed herein. We describe and claim procedures for the generation of transformed plants having or expressing any of the peptides described herein.

We describe the use of any of the peptides or nucleotides described herein, to make a plant or transform these peptides or nucleotides into a plant, and methods and techniques for generating these proteins in plants and/or expression cassettes comprising any of the peptides and methods to transform them into a plant genome and any method of using, making, transforming any of the described peptides or nucleotides into a plant and methods and techniques for generating transformed plants having or expressing any of the peptides and functional expression cassettes in plants comprising any of the disclosed peptides and their corresponding nucleotides and any plants made by the products and processes described herein.

In some embodiments we disclose a chimeric gene comprising a promoter active in plants operatively linked to the nucleic acids or expression cassettes as described herein. We disclose a method of making, producing, or using the combination of genes described herein. We disclose a recombinant vector comprising the combination of genes described herein. We disclose a method of making, producing, or using the recombinant vector. We disclose a transgenic host cell comprising the combination of genes described herein and the method of making, producing or using the transgenic host cell, which can be a transgenic plant cell and we disclose a method of making, producing or using such a transgenic plant cell as well as the transgenic plant comprising the transgenic plant cell and how to make and use the transgenic plant. We disclose transgenic plant and seed having the properties described herein that is derived from corn, soybean, cotton, rice, sorghum, switchgrass, sugarcane, alfalfa, potatoes or tomatoes. The transgenic seed may have a chimeric gene that we describe herein. We describe methods of making, producing or using the transgenic plant and or seed of this disclosure.

We also describe methods of using the invention and provide novel formulations. The invention is most useful to control insects. We describe a method of controlling an insect comprising: Applying Bt (*Bacillus thuringiensis*) Protein to said insect; and Applying an insecticidal TMOF peptide to said insect. This method may be used where the Bt protein and the insecticidal ICK peptide are applied together at the same time in the same compositions or separately in different compositions and at different times. The Bt Protein and the insecticidal TMOF peptide may be applied sequentially, and it may be applied to (Bt Protein)-resistant insects. The ratio of Bt to TMOF, on a dry weight basis, can be selected from at least about the following ratios: 99:1, 95:5, 90:10, 85:15, 80:20, 75:25, 70:30, 65:35, 60:40, 55:45, 50:50, 45:55, 40:60, 35:65, 30:70, 25:75, 20:80, 15:85, 10:90, 5:95 and 1:99, or any combination of any two of these values. The ratio of Bt to TMOF, on a dry weight basis, can be selected from about the following ratios: 50:50, 45:55, 40:60, 35:65, 30:70, 25:75, 20:80, 15:85, 10:90, 5:95, 1:99, 0.5:99.5, 0.1:99.9 and 0.01:99.99

or any combination of any two of these values. Either or both of the Bt and TMOF peptides are derived from more than 1 different types or bacterial strain origins of Bt and TMOF peptides. Either or both of the Bt and TMOF peptides are derived from between 2 and 5 different types or bacterial strain origins of either Bt or TMOF peptides or both Bt and TMOF peptides. Either or both of the Bt and TMOF peptides are derived from 2 to 15 different types or bacterial strain origins of either or both of Bt and TMOF peptides and at least one strain of either Bt or TMOF or both Bt and TMOF peptides are encoded by more than one copy of the Bt or TMOF genes. Either one or both of the Bt and TMOF peptides are derived from more than 1 different types or bacterial strain origins of Bt and/or TMOF peptides with all the strains of Bt and/or TMOF peptides contributing more than at least 1% of the peptides from each strain type in said composition. Either or both of the Bt and TMOF peptides are derived from 2 to 5 different types or bacterial strain origins of either one or both of Bt and TMOF peptides and at least one strain of either Bt or TMOF or both Bt and TMOF peptides are encoded by more than one copy of the Bt or TMOF genes. The total concentration of Bt and TMOF peptide in the composition is selected from the following percent concentrations: 1, 5, 10, 15, 20, 25, 30, 35, 40, 45, 50, 55, 60, 65, 70, 75, 80, 85, 90, 95 or 99%, or any range between any two of these values, and the remaining percentage of the composition is comprised of excipients.

The methods can be used where the insecticidal combination peptide is produced using a genetic cassette that further comprises an ERSP (Endoplasmic Reticulum Signal Peptide) operably linked to the insecticidal TMOF peptide, wherein said ERSP is linked at the N-terminal of the insecticidal TMOF peptide. In some embodiments the insecticidal combination peptides used are produced using a genetic cassette that further comprises an ERSP (Endoplasmic Reticulum Signal Peptide) operably linked to the insecticidal TMOF peptide, wherein said ERSP is linked at the N-terminal of the insecticidal TMOF peptide, wherein the ERSP is BAAS.

Any of the peptides and plants described herein can be used to control insects, their growth and damage, especially their damage to plants. The combination Bt protein and insecticidal TMOF peptide can be applied by being sprayed on a plant, or the insect's locus, or the locus of a plant in need of protecting.

We also describe formulations comprising: Bt proteins; and an insecticidal TMOF peptide which can include any of the compositions described herein or capable of being made by one skilled in the art given this disclosure. Some of the described formulations include the use of a polar aprotic solvent, and or water, and or where the polar aprotic solvent is present in an amount of 1-99 wt %, the polar protic solvent is present in an amount of 1-99 wt %, and the water is present in an amount of 0-98 wt %. The formulations include formulations where the Bt protein is DIPEL® and where the insecticidal TMOF peptide is a peptide like any of the TMOF peptides provided in the sequence listing. The polar aprotic solvent formulations are especially effective when they contain MSO. The examples below are intended to illustrate and not limit the invention in any manner.

To summarize, we describe in Part 3, the following:

A composition comprising at least two types of insecticidal protein or peptides wherein one type is a Pore Forming Insecticidal Protein (PFIP) and the other type is a Cysteine Rich Insecticidal Peptide (CRIP). Where the composition can comprise at least two types of insecticidal peptides wherein one type is Pore Forming Insecticidal Protein

(PFIP), wherein said PFIP is a Bt protein and the other type is Cysteine Rich Insecticidal Peptide (CRIP), wherein said CRIP is an ICK protein, wherein said ICK protein is derived from the funnel web spider. We describe a process of: a) evaluation and optional testing of an insect or a sample of insects to determine whether or not the insects show resistance to a PFIP and b) when the result of said evaluation leads to the conclusion that said sample of insects are resistant to a PFIP then c) the application of one or more CRIPS and optionally the CRIPS can be an ICK from *Hadronyche versuta*, or the Blue Mountain funnel web spider, *Atrax robustus*, *Atrax formidabilis*, *Atrax infensus*, including toxins known as U-ACTX polypeptides, U-ACTX-Hv1a, rU-ACTX-Hv1a, rU-ACTX-Hv1b, or mutants or variants, or the CRIP can be a Non-ICK from sea anemones, from the sea anemone named *Anemonia viridi*, the peptides named Av2 and Av3 especially peptides of similar to these in the sequence listing. We describe a method of controlling Insects including Bt resistant insects comprising, creating composition of at least two types of peptides wherein one type of peptide is a pore forming insecticidal peptide (PFIP) and the other type of peptide is a cysteine rich insecticidal peptide (CRIP) and the PFIP and CRIP proteins are selected from any of the compositions described herein and from any of the proteins provided in the sequence listing and then applying said composition to the locus of the insect. We describe a method of controlling Insects including Bt resistant insects comprising protecting a plant from Bt resistant insects comprising, creating a plant which expresses a combination of at least two properly folded peptides wherein one type of peptide is a pore forming insecticidal peptide (PFIP) and the other type of peptide is a cysteine rich insecticidal peptide (CRIP) and the PFIP and CRIP proteins are selected from any of the compositions described herein and from any of the proteins provided in the sequence listing. We describe a process of: a) evaluation and optional testing of an insect or a sample of insects to determine whether or not the insects show resistance to a PFIP and b) when the result of said evaluation leads to the conclusion that said sample of insects are resistant to a PFIP then c) the application of one or more CRIPS and optionally d) the application of a combination of PFIP and CRIP, in either concurrent or sequential applications.

We describe a composition comprising at least two types of insecticidal protein or peptides wherein one type is a Pore Forming Insecticidal Protein (PFIP) and the other type is a Cysteine Rich Insecticidal Peptide (CRIP). A composition where the CRIP is a ICK and optionally, said ICK is derived from, or originates from, *Hadronyche versuta*, or the Blue Mountain funnel web spider, *Atrax robustus*, *Atrax formidabilis*, *Atrax infensus*, including toxins known as U-ACTX polypeptides, U-ACTX-Hv1a, rU-ACTX-Hv1a, rU-ACTX-Hv1b, or mutants or variants. A composition where the CRIP is a Non-ICK CRIP and optionally said Non-ICK CRIP is derived from, or originates from, animals having Non-ICK CRIPS such as sea anemones, sea urchins and sea slugs, optionally including the sea anemone named *Anemonia viridi*, optionally including the peptides named Av2 and Av3 especially peptides similar to Av2 and Av3 including such peptides listed in the sequence listing or mutants or variants. A method of using the composition control Insects including Bt resistant insects comprising, creating composition of at least two types of peptides wherein one type of peptide is a pore forming insecticidal protein (PFIP) and the other type of peptide is a cysteine rich insecticidal peptide (CRIP) and the PFIP and CRIP proteins are selected from any of the compositions described herein and from any of the proteins

provided in the sequence listing and then applying said composition to the locus of the insect. A method controlling Insects including Bt resistant insects comprising protecting a plant from Bt resistant insects comprising, creating a plant which expresses a combination of at least two properly folded peptides wherein one type of peptide is a pore forming insecticidal protein (PFIP) and the other type of peptide is a cysteine rich insecticidal peptide (CRIP) and the PFIP and CRIP proteins are selected from any of the compositions described herein and from any of the proteins provided in the sequence listing. A method of controlling insects including Bt resistant insects where the CRIP is administered any time during which the PFIP is affecting the lining of the insect gut. A method of controlling insects including Bt resistant insects where the CRIP is administered following the testing of the insect for Bt resistance and wherein said insect tested positive for Bt resistance. The application or delivery of any of the compounds described herein in solid or liquid form to either the insect, the locus of the insect or as a Plant Incorporated Protectant.

We describe a composition comprising at least two types of insecticidal peptides wherein one type is a pore forming insecticidal protein (PFIP), wherein said PFIP is a cry protein and the other type is a cysteine rich insecticidal peptide (CRIP), wherein said CRIP is an ICK protein, wherein said ICK protein is derived from the funnel web spider. We describe a composition comprising at least two types of insecticidal peptides wherein one type is a pore forming insecticidal peptide (PFIP), wherein said PFIP has as its origin the Bt organism and the other type is a cysteine rich insecticidal peptide (CRIP), wherein said CRIP is a Non-ICK protein. We describe a composition comprising at least two types of insecticidal peptides wherein one type is a pore forming insecticidal peptide (PFIP) and the other type is a TMOF. We describe a method of protecting a plant from Insects including Bt resistant insects comprising creating a Plant Incorporating a combination of at least two different types of peptides wherein one type of peptide is a pore forming insecticidal peptide (PFIP) and the other type is a cysteine rich insecticidal peptide (CRIP). We describe a method of protecting a plant from Insects including Bt resistant insects comprising, creating a plant which expresses a combination of at least two properly folded peptides wherein one type of peptide is a pore forming insecticidal peptide (PFIP) and the other type of peptide is a cysteine rich insecticidal peptide (CRIP) and the PFIP and CRIP proteins are selected from any of the compositions described herein and from any of the proteins provided in the sequence listing.

We describe an insecticidal combination peptide composition comprising Cysteine Rich Insecticidal protein (CRIP); such as an insecticidal ICK (Inhibitor Cystine Knot) peptide like a spider peptide or Non-ICK like a sea anemone toxin combined with a with pore forming insecticidal protein (PFIP) like a Bt peptide, such as cry, cyp or VIP; or a or a Cysteine Rich Insecticidal protein (CRIP); such as an insecticidal ICK (Inhibitor Cystine Knot) peptide combined with a with a TMOF (trypsin modulating oostatic factor) peptide. Note the CRIP can be a Non-ICK protein like a sea anemone peptide, such as Av2 and Av3 and other similar sequences in the Sequence Listing. We describe such compositions where the ratio of Bt to CRIP, Bt to ICK, Bt to non-ICK CRIP, Bt to TMOF, or Bt to ICK and TMOF on a dry weight basis, is selected from about the following ratios: 99:1, 95:5, 90:10, 85:15, 80:20, 75:25, 70:30, 65:35, 60:40, 55:45, 50:50, 45:55, 40:60, 35:65, 30:70, 25:75, 20:80, 15:85, 10:90, 5:95 and 1:99, or any combination of any two of these values.

Alternatively where the ratio of Bt to CRIP, Bt to ICK, Bt to non-ICK CRIP, Bt to TMOF, and TMOF, and sea anemone on a dry weight basis, is selected from about the following ratios: 50:50, 45:55, 40:60, 35:65, 30:70, 25:75, 20:80, 15:85, 10:90, 5:95, 1:99, 0.5:99.5, 0.1:99.9 and 0.01:99.99 or any combination of any two of these values. Alternatively where ratio of Bt to CRIP, Bt to ICK, Bt to non-ICK CRIP, Bt to TMOF, or Bt to ICK and TMOF, and sea anemone peptides are derived from more than 1 different types or bacterial strain origins of either one or both of Bt and ICK peptides. Alternatively where the Bt, ICK, non-ICK CRIP, sea anemone peptides and TMOF peptides are derived from between 2 and 5 different types or bacterial strains origins of either one or both of Bt, ICK, non-ICK CRIP, sea anemone peptides and TMOF peptides are derived from between 2 and 5 different strains. Alternatively where either or both of the Bt, ICK, non-ICK CRIP, sea anemone peptides and TMOF peptides are derived from 2 to 5 different types or bacterial strain origins of either one or all of Bt, ICK, non-ICK CRIP, sea anemone peptides and TMOF peptides. Alternatively where either or both of the Bt, ICK, non-ICK CRIP, sea anemone peptides and TMOF peptides are encoded by from 2 to 15 different types or bacterial strain origins of either one or all of Bt, ICK, non-ICK CRIP, sea anemone peptides and TMOF peptides. Alternatively where one or all of the Bt, ICK, non-ICK CRIP, sea anemone peptides and TMOF peptides are derived from 2 to 15 different types or bacterial strain origins of either one or all of Bt, ICK, and TMOF peptides and at least one strain of either Bt, ICK, non-ICK CRIP, sea anemone peptides and TMOF peptides or both Bt, ICK, non-ICK CRIP, sea anemone peptides and TMOF peptides and Bt and ICK, Bt and TMOF, or Bt and ICK+TMOF peptides are encoded by more than one copy of the Bt or ICK genes. Alternatively where either or both of the Bt, CRIP, ICK, non-ICK CRIP, sea anemone peptides and TMOF peptides are derived from 2 to 15 strains or bacterial types of Bt and/or ICK, non-ICK CRIP, sea anemone peptides and TMOF peptides with all the strains of Bt and/or ICK peptides contributing more than at least 1% of each strain type to said composition.

We describe a composition of Bt and ICK, non-ICK CRIP, sea anemone peptides and TMOF peptides of numbers 1-9 where the total concentration of Bt and CRIP peptide in the composition is selected from the following percent concentrations: 1, 5, 10, 15, 20, 25, 30, 35, 40, 45, 50, 55, 60, 65, 70, 75, 80, 85, 90, 95 or 99%, or any range between any two of these values, and the remaining percentage of the composition is comprised of excipients. We describe a composition wherein the insecticidal combination peptide is produced using a genetic cassette that further comprises an ERSP (Endoplasmic Reticulum Signal Peptide) operably linked to the insecticidal CRIP peptide, wherein said ERSP is linked at the N-terminal of the insecticidal CRIP peptide. We describe a composition wherein the insecticidal combination peptide is produced using a genetic cassette that further comprises an ERSP (Endoplasmic Reticulum Signal Peptide) operably linked to the insecticidal ICK peptide, wherein said ERSP is linked at the N-terminal of the insecticidal CRIP peptide, wherein the ERSP is BAAS. We describe a composition wherein said combination peptide is produced using a genetic cassette that further comprises a dipeptide operably linked to the insecticidal CRIP peptide, wherein said dipeptide is linked at the N-terminal of the insecticidal CRIP peptide; and wherein the dipeptide is comprised of one nonpolar amino acid on the N-terminal of

the dipeptide and one polar amino acid on the C-terminal of the dipeptide. We describe a composition wherein said dipeptide is glycine-serine.

We describe a composition wherein the insecticidal CRIP peptide is any insecticidal peptide that inhibits both voltage-gated Calcium channels and Calcium-activated potassium channels in insects, and wherein the insecticidal CRIP peptide originates from any species of Australian Funnel-web spider, and wherein said spider is selected from the Australian Funnel-web spiders of genus *Atrax* or *Hadronyche*, and wherein said spider is selected from the Australian Funnel-web spiders of genus *Hadronyche*, and wherein said spider is selected from the Australian Blue Mountains Funnel-web, *Hadronyche versuta*, and wherein the insecticidal CRIP peptide is Hybrid-ACTX-Hv1a, and wherein said insecticidal CRIP peptide contains 20-100 amino acids and 2-4 disulfide bonds, wherein said insecticidal CRIP peptide is any insecticidal peptide with at least 50%, 55%, 60%, 65%, 70%, 75%, 80%, 85%, 90%, 95%, 96%, 97%, 98%, or 99% or greater sequence identity to any of the peptides in the sequence listing.

We describe insecticidal CRIP peptide is from Bt protein and where the Bt protein is a Cry or Cyt protein, or selected from the group consisting of a Cry1, Cry3, TIC851, CryET70, Cry22, TIC901, TIC201, TIC407, TIC417, a binary insecticidal protein CryET80, and CryET76, a binary insecticidal protein TIC100 and TIC101, a combination of an insecticidal protein ET29 or ET37 with an insecticidal protein TIC810 or TIC812 and a binary insecticidal protein PS149B1. We describe Bt protein selected from a Cry protein, a Cry1A protein or a Cry1F protein. We describe wherein said Bt protein is a combination Cry1F-Cry1A protein, DIPEL® or THURICIDE® and where the Bt protein is derived from *Bacillus thuringiensis* kurstaki.

We describe compositions comprising the nucleotides of a PFIP such as Bt (*Bacillus thuringiensis*) protein; and a CRIP such as an insecticidal ICK (Inhibitor Cystine Knot) peptide, or a Non-ICK peptide; in a transformed plant or plant genome; and where the ratio of Bt to ICK, on a dry weight basis, is selected from about the following ratios: 99:1, 95:5, 90:10, 85:15, 80:20, 75:25, 70:30, 65:35, 60:40, 55:45, 50:50, 45:55, 40:60, 35:65, 30:70, 25:75, 20:80, 15:85, 10:90, 5:95 and 1:99, or any combination of any two of these values, or where the composition of number 33, in a transformed plant or plant genome and wherein the ratio of Bt to ICK, on a dry weight basis, is selected from about the following ratios: 0:50, 45:55, 40:60, 35:65, 30:70, 25:75, 20:80, 15:85, 10:90, 5:95, 1:99, 0.5:99.5, 0.1:99.9 and 0.01:99.99 or any combination of any two of these values.

We describe a composition where either or both of the encoded Bt and ICK peptides are derived from more than 1 different type or bacterial strain origin of Bt or ICK peptides, where either or both of the encoded Bt and ICK peptides are derived from between 2 and 5 different type or bacterial strain origin of either Bt or ICK peptides or both Bt and ICK peptides are derived from between 2 and 5 different types or strain origins, where either or both of the encoded Bt and ICK peptides are derived from 2 to 15 different type or bacterial strain origins of either or both of Bt and ICK peptides and at least one strain of either Bt or ICK or both Bt and ICK peptides encoded by more than one copy of the Bt or ICK genes, where either or both of the encoded Bt and ICK peptides are derived from more than one different type or bacterial strain origin of Bt and/or ICK peptides where all the strains of Bt and/or ICK peptides contribute more than at least 1% of each strain type to said composition, where either or both of the encoded Bt and ICK peptides are

derived from 2 to 5 different type or bacterial strain origins of either or both of Bt and ICK peptides and at least one strain of either Bt or ICK or both Bt and ICK peptides encoded by more than one copy of the Bt or ICK genes.

We describe a composition where the total concentration of transgenically expressed Bt and ICK peptide resulting from the composition is selected from the following percent concentrations: 1, 5, 10, 15, 20, 25, 30, 35, 40, 45, 50, 55, 60, 65, 70, 75, 80, 85, 90, 95 or 99%, or any range between any two of these values, and the remaining percentage of the composition is comprised of excipients. We describe a composition where the insecticidal combination peptide is produced using a genetic cassette that further comprises an ERSP (Endoplasmic Reticulum Signal Peptide) operably linked to the insecticidal ICK peptide, wherein said ERSP is linked at the N-terminal of the insecticidal ICK peptide, and where the insecticidal combination peptide is produced using a genetic cassette that further comprises an ERSP (Endoplasmic Reticulum Signal Peptide) operably linked to the insecticidal CRIP peptide, wherein said ERSP is linked at the N-terminal of the insecticidal CRIP peptide, wherein the ERSP is BAAS.

We describe a transgenic plant incorporating and expressing the combination peptides disclosed herein where said combination peptide is produced using a genetic cassette that further comprises nucleotides expressing a dipeptide operably linked to the insecticidal CRIP (peptide), wherein said dipeptide is encoded so that it is covalently linked at the N-terminal of the insecticidal CRIP; and wherein the dipeptide is comprised of one nonpolar amino acid on the N-terminal of the dipeptide and one polar amino acid on the C-terminal of the dipeptide. We describe a transgenic plant wherein the transformed peptide includes a dipeptide with an N terminal glycine-serine. We describe transgenic plant wherein the insecticidal peptides expressed are any insecticidal peptide combination of CRIP and PFIP (or Bt peptides) that allows the peptide to both enter the gut and then inhibits both voltage-gated Calcium channels and Calcium-activated potassium channels in insects.

We describe a transgenic plant wherein the recombinantly produced insecticidal CRIP peptide is derived from an Australian Funnel-web spider or sea anemone and we describe and provide either real or notional examples of transformed plants, transformed with a CRIP from a spider is selected from the Australian Funnel-web spiders of genus *Atrax* or *Hadronyche* or a sea anemone is selected from *Anemonia viridis*. The transgenic plant can have insecticidal ICK peptide expressed that is Hybrid-ACTX-Hv1a. The CRIP can be an ICK or Non-ICK that when expressed contains 20-100 amino acids and 2-4 disulfide bonds. The PIP peptides can have at least 50%, 55%, 60%, 65%, 70%, 75%, 80%, 85%, 90%, 95%, 96%, 97%, 98%, or 99% or greater sequence identity to SEQ ID NO: 33 and or peptide selected from SEQ ID NO: 33-1032.

We describe a transgenic plant wherein the Bt protein is any insecticidal Bt protein and where the Bt protein is a Cry or Cyt protein, and where the Bt protein is selected from the group consisting of a Cry1, Cry3, TIC851, CryET70, Cry22, TIC901, TIC201, TIC407, TIC417, a binary insecticidal protein CryET80, and CryET76, a binary insecticidal protein TIC100 and TIC101, a combination of an insecticidal protein ET29 or ET37 with an insecticidal protein TIC810 or TIC812 and a binary insecticidal protein PS149B1 and where the Bt protein is selected from a Cry protein, a CryIA protein or a Cry1F protein, and where the Bt protein is a combination Cry1F-Cry1A protein, and/or DIPEL® and or THURICIDE®.

We describe a transgenic plant wherein the average concentration of Bt and ICK/Non-ICK peptide, in an average leaf of a transformed plant is about: 1, 5, 10, 15, 20, 25, 30, 35, 40, 45, 50, 55, 60, 65, 70, 75, 80, 85, 90, 95 or 99% of total recoverable soluble protein, or any range between any two of these values, and where the transformed plant expressing the peptides properly folded toxic peptides in the transformed plant, and where it causes the accumulation of the expressed and properly folded toxic peptides in said plant and to cause an increase in the plant's yield or resistance to insect damage. We describe these compositions and procedures to control insects.

We describe expression cassettes comprising any of the nucleotides which express any peptides mentioned here. We describe a functional expression cassette incorporated into a transformed plant, comprising nucleotides that code for any of the peptides disclosed herein or that could be made by one skilled in the art given the teaching disclosed herein. We describe procedures for the generation of transformed plants having or expressing any of the combination peptides described herein. We describe a plant made by any of the products and processes described herein.

We describe the use of any of the peptides or nucleotides described herein, to make a plant or transform these peptides or nucleotides into a plant, and methods and techniques for generating these proteins in plants and/or expression cassettes comprising any of the peptides and methods to transform them into a plant genome and any method of using, making, transforming any of the described peptides or nucleotides into a plant and methods and techniques for generating transformed plants having or expressing any of the peptides and functional expression cassettes in plants comprising any of the disclosed peptides and their corresponding nucleotides and any plants made by the products and processes described herein.

We describe a chimeric gene comprising a promoter active in plants operatively linked to the nucleic acids or expression cassettes as described herein and the methods of making, producing, or using the combination of genes described herein. We describe a recombinant vector comprising the combination of genes described herein. We describe a method of making, producing, or using the recombinant vectors, a transgenic host cell comprising the combination of genes, the transgenic host cell which is a transgenic plant cell, the transgenic plant and transgenic plants which are corn, soybean, cotton, rice, sorghum, switchgrass, sugarcane, alfalfa, potatoes or tomatoes, and the seeds for these and other plants, and where the seed comprises a chimeric gene.

We describe methods of controlling an insect or the locus of an insect comprising: applying a PFIP, like Bt (*Bacillus thuringiensis*) protein to said insect; followed with an application of any or any combination of the following: a cysteine rich insecticidal peptide (CRIP) to said insect and in combination or in the alternative, applying an insecticidal ICK (Inhibitor Cystine Knot) peptide to said insect and in combination or in the alternative, applying a Non-ICK CRIP peptide to said insect and in combination or in the alternative, applying a TMOF peptide to said insect, applying a sea anemone peptide to said insect.

We explain that Bt protein and the insecticidal CRIP, ICK and or TMOF peptide are applied such that they work together, but they do not have to be applied at the same time. The PFIP like a Bt protein and the insecticidal CRIP, ICK and or TMOF peptide can be are applied concurrently or sequentially.

We explain the amounts as follows: the ratio of Bt to CRIP, Bt to ICK, Bt to non-ICK CRIP, Bt to TMOF, or Bt to ICK and TMOF; on a dry weight basis, is selected from about the following ratios: 99:1, 95:5, 90:10, 85:15, 80:20, 75:25, 70:30, 65:35, 60:40, 55:45, 50:50, 45:55, 40:60, 35:65, 30:70, 25:75, 20:80, 15:85, 10:90, 5:95 and 1:99, or any combination of any two of these values; alternatively, the ratio of Bt to CRIP, Bt to ICK, Bt to non-ICK CRIP, Bt to TMOF, or Bt to ICK and TMOF; on a dry weight basis, is selected from about the following ratios: 50:50, 45:55, 40:60, 35:65, 30:70, 25:75, 20:80, 15:85, 10:90, 5:95, 1:99, 0.5:99.5, 0.1:99.9 and 0.01:99.99 or any combination of any two of these values.

We explain both or all of the Bt+CRIP; Bt+ICK, Bt+Non-ICK CRIP, Bt+TMOF or Bt+ICK+TMOF; are derived from more than 1 different types or bacterial strain origins of Bt, o ICK, and TMOF peptides and or both of the Bt and CRIP, ICK, non-ICK CRIP, Bt and TMOF or Bt and ICK+TMOF; Bt+sea anemone peptides are derived from between 2 and 5 different types or bacterial strain origins of either one, two or more of Bt, CRIP, ICK, non-ICK CRIP, sea anemone peptides or TMOF peptides, and or either one, two or all Bt, ICK and TMOF peptides are derived from 2 to 15 different types or bacterial strain origins of either or both of Bt and ICK peptides and at least one strain of either one, two or all of Bt, CRIP, ICK, non-ICK CRIP, sea anemone peptides or TMOF peptides are encoded by more than one copy one, two or all of Bt, CRIP, ICK, non-ICK CRIP, sea anemone peptides or TMOF genes.

We explain that one, two or all Bt, ICK and TMOF peptides are derived from more than 1 different types or bacterial strain origins of one, two or all Bt, ICK and TMOF peptides with all the strains of one, two or all Bt, ICK and TMOF peptides contributing more than at least 1% of the peptides from each strain type in said composition. The total concentration of Bt and CRIP peptide in the composition is selected from the following percent concentrations: 1, 5, 10, 15, 20, 25, 30, 35, 40, 45, 50, 55, 60, 65, 70, 75, 80, 85, 90, 95 or 99%, or any range between any two of these values, and the remaining percentage of the composition is comprised of excipients.

We either provide or provide enough information that one skilled in the art could make a formulation comprising: a PFIP such as a Bt protein; and a CRIP such as an insecticidal ICK or Non-ICK peptide; and/or a TMOF peptide. We explain such formulations could be made using a polar aprotic solvent and a polar protic solvent and further comprising water. In some formulations the polar aprotic solvent is present in an amount of 1-99 wt %, the polar protic solvent is present in an amount of 1-99 wt %, and the water is present in an amount of 0-98 wt %, and it can further comprise MSO.

Example 1

Foliar Bioassay Using SDP 1234604 and 1234605 Against *Spodoptera exigua* on Mud Lakes Farms Romaine Lettuce

Purpose: The purpose of this experiment is designed to determine the percent mortality which occurs against *S. exigua* when SDP 1234604 (wp formulation) and 605 (pre-gran formulation) are sprayed against 1st, 2nd, 3rd and 4th Instar larvae in a foliar leaf disk bioassay.

Assay Preparation and Treatment Formulation: *S. exigua* eggs were received from Benzon Research. Eggs were placed at 10° C. in the wine cooler for two days then moved to the VWR Low Temperature Incubator set at 28° C. and 2-30% Relative Humidity on a rack under LED lights, until

freshly hatched neonate were ~24 hr old for the first experiment. Mud Lake Farms Lettuce was received on Jul. 9, 2012 and stored at 4° C. in a refrigerator until used. For each instar, larvae were placed on mud lakes farms lettuce after 24 hours in the incubator. Lettuce leaves were cut and placed into a medium square polyethylene container and larvae were tapped into the container. After 24 hours, larvae were removed from the old lettuce and fresh lettuce was replaced so that larvae were not reared on less than superior tissue. This occurred once a day, for three days, until larvae were 96 hours old. Lettuce leaves were cut into disks using a 2¼ inch arch which has been sanitized with 70% ethanol and cleaned to remove any leaf tissue from previous assays. Leaf disks were punched on a true bamboo cutting board. A very dilute 12 ppm bleach solution (1/500th dilution of 6ppt hypochlorite {Clorox Bleach}Stock) was used to sanitize the leaf tissue without damaging leaf disks before the quadruple rinse. Leaf disks were subjected to the 12 ppm bleach treatment by placing the cut leaf disk in a 12 ppm solution of bleach in a large rectangular polyethylene container (covered with a lid) and shaking at 3500 rpm on an orbital shaker for 1.5 minutes. Bleach solution was then drained from the bin and leaves were rinsed in bins with dH₂O four times to remove residual bleach with slight agitation in diH₂O on the orbital shaker. Leaf disks were placed onto the paper towels and covered with additional paper towels so that they do not dry out. Only the flattest, circular and uniform disks were then hand dried with Kimwipes to remove any remaining water and placed into labeled Tupperware containers abaxial side up for spraying. During this time, formulations were made (as described in the table that follows) for the spray solutions of spray dried powders on the leaf disks in 50 mL Falcon tube being sure to fill tubes with deionized H₂O before adding the precisely massed amount of spray dried powders. Spraying was performed in the Labconco fume hood in E207 starting with the ventral side of the leaf disk. For spraying, a double action, internal mix airbrush (Paasch Airbrush Company, Chicago IL) with the airline set at a rate of 200 µL/second (20 psi). Leaf disks were sprayed in a circular fashion with the airbrush perpendicular to the leaf surface so that a fine mist covered the entire leaf surface evenly (~3-4 seconds). Between each treatment spray, the cup containing spray solution was rinsed with dH₂O to remove any residues from previous treatments. After spraying, drying was allowed for one hour then disks were flipped so their adaxial side was now orientated facing up in the Tupperware Container and sprayed in the same manner. After spraying the adaxial side, an hour was allowed for drying and leaf disks were placed in labeled petri dishes with 2 90 mm Whatman 3 Qualitative Filter Papers (GE Healthcare UK Limited, Amersham Place Little Chalfont, Buckinghamshire, HP7 9NA, UK) at the bottom that have been wetted with 4 mL of diH₂O using a Eppendorf Repeater Plus and a 25 mL tip. Petri dishes were covered and randomized before ~7-9 freshly hatched neonates *S. exigua* were applied to each leaf disk using a #0 fine haired brush by obtaining a white board and emptying a container of 24, 48, 72 or 96 hr neonates onto it. Plates were sealed with parafilm and placed randomly on the rack for statistical purposes at 27° C. The assay was scored over the following day at 18, 24, 40 and 48 hours by observing mortality and noting any differences between untreated and treated leaves.

FIG. 19 shows the percent mortality results of four experiments recorded for each experiment at 18, 24, 40 and 48 hours. The non-spray dried control treatment showed the lowest average mortality of any treatments. The majority of

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insect mortality is observed at the 18 hour scoring and additional mortality is observed at 40 and 48 hours shown by the 40 and 48 hour scoring. Healthy insects have noticeable green, chlorophyll like color, fast evasion response when prodded with paint brush and average growth for 48 hours. Percent mortality results of 72 and 96 hour larvae are significantly reduced compared to the 24 and 48 hour old larvae. Clearly, both Bt protein and Hybrid peptide treatments alone are ineffective in controlling older insects.

Example 2

Foliar Bioassay using SDP 1234605 against *Spodoptera exigua* on Mud Lakes Farms Romaine Lettuce.

Purpose: The purpose of this experiment is designed to determine the percent mortality which occurs against *S. exigua* when SDP 1234605 is sprayed against 72 hour old larvae in a foliar leaf disk bioassay and when DIPEL® DF is co-sprayed with SDP 1234605.

Assay Preparation and Treatment Formulation: See preparation in Example 1. *S. exigua* eggs were received from Benzon Research. Petri dishes were covered and randomized before ~7-9 freshly hatched neonates *S. exigua* were applied to each leaf disk using a #0 fine haired brush by obtaining a white board and emptying a container of 72 hr old larvae onto it. Plates were sealed with parafilm and placed randomly on the rack for statistical purposes at 27° C. The assay was scored over the following day at 18, 24 and 48 hours by observing mortality and noting any differences between untreated and treated leaves.

FIG. 20 shows a column graph Example 2 data at 18, 24 and 48 hours. Individually 10 parts per thousand (ppt) of Hybrid peptide in formulation '605 and DIPEL® at 300 parts per million (ppm) show little improvement over either the untreated control or surfactant mortalities. However, when combined the resultant mortality at 48 hours of 84.4% surprisingly exceeds that which would be expected from the additive effects of the individual treatments (29.1%). The synergy of the individual components is at least 2.9 fold (84.4/29.1). It is unexpected that an insecticidal protein that kills through sepsis would be synergistic with an insecticidal peptide that modulates ion channels in the CNS.

Example 3

We investigated the potential additive and/or synergistic affects of combinations of *Bacillus thuringiensis* (Bt) proteins and the Av2 peptide from sea anemones. We used the Bt product: DIPEL® DF which is commercially available and commercially available Av2 a toxic sea anemone peptide.

Methods: Small leaf disks (~2 cm) were cut into the inner leaves of cabbage purchased from a local grocery store. Disks were dipped into 400 µL of treatment and placed on 4.25 cm #4 filter disks (Whatman) in the bottom of ~4.5 cm condiment cups. Four disks were prepared per treatment. 75 µL of water was applied to a second smaller 3.2 cm #1 filter disk (Whatman) atop the larger filter disk. Leaf disks were allowed to dry approximately ten minutes before adding four 120 hr old CryTa resistant *Plutella xylostella* per leaf disk. Condiment cups were sealed with non-perforated lids. Treatments were placed in the incubator and scored for mortality and feeding damage at 24 and 48 hrs. Due to large consumption of leaf disks in many treatments, an additional 3.2 cm untreated leaf disk was added at 24 hr to ensure larval starvation did not occur.

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At 24 and 48 hrs, pictures of leaf disks were taken using an Iphone 4S (Apple Inc.), and saved. Individual leaf disk photos were cropped from the group treatment photo and assigned random numbers. Using the program Image J, leaf area eaten was calculated. The image was opened in image J and the scale in the photo was set. To set the scale, a known distance in the photo in centimeters (cm) was drawn using the segment line tool and measured in units of pixels. For this experiment, the known diameter of filter paper disk is 1.5 cm for #1 filter disk and 4.5 cm for the #4 Whatman Filter disk. Using this known length in cm, pixel units are converted in the image to centimeters. Once the scale is set, a freehand selection tool is used to draw around the area where leaf tissue remains. This process was repeated for all photos being sure to log area calculated by image J in the lab notebook. For this experiment the control area of uneaten leaf disk is 2.54 cm² and calculations were made to determine % area eaten.

Treatments:

150 PPM DIPEL® DF: 200 µL 300 PPM DIPEL® DF+200 µL water

1 PPT Av2: 0.1 mg Av2 in 100 µL water (combined four vials 1 PPT Av2 for necessary 400 µL treatments)

150 PPM DIPEL® DF+1PPT Av2: 100 µL 150 PPM DIPEL® DF was added to 0.1 mg Av2 (four vials were combined for necessary 400 µL treatment)

FIG. 21 shows the percent feeding damage resulting from Bt protein resistant diamondback moth larvae (120 hrs old) on cabbage leaf disks. Scoring at both 24 hours and 48 hours shows significant improvement over treatment with DIPEL® alone. While these insects are resistant to Bt, they do still feed to a limited extent without mortality. The combination treatment results in significantly improved protection of the foliar material. Further, treatment with Av2 alone has no effect on feeding damage and it is only in combination with the Bt protein that its effect is made apparent. This is consistent with increased bioavailability of Av2 made possible by the Bt protein.

Example 4

Foliar Bioassay using SDP 1234609 and DIPEL® DF against Earthbound Farms Romaine Lettuce.

Purpose: The purpose of this experiment is to determine the percent mortality which occurs against Bt resistant (HD-1) *P. xylostella* when SDP 1234609 is sprayed against 120 hour old larvae in a foliar leaf disk bioassay and when DIPEL® DF is co-sprayed with SDP 1234609.

Assay Preparation and Treatment Formulation: See preparation in Example 1.

FIG. 22 shows a column graph Example 4 data at 24 and 48 hours. Individually 1 parts per thousand (ppt) of Hybrid peptide in formulation '609 and DIPEL® at 150 parts per million (ppm) show little improvement over either the untreated control or surfactant mortalities. However, when combined the resultant mortality at 48 hours of 62.5% surprisingly exceeds that which would be expected from the additive effects of the individual treatments (21.8%). The synergy of the individual components is at least 2.86 fold (62.5/21.8). Again, it is unexpected that an insecticidal protein that kills through sepsis would be synergistic with an insecticidal peptide that modulates ion channels in the CNS.

SEQUENCE LISTING

The patent contains a lengthy sequence listing. A copy of the sequence listing is available in electronic form from the USPTO web site (<https://seqdata.uspto.gov/?pageRequest=docDetail&DocID=US12410218B2>). An electronic copy of the sequence listing will also be available from the USPTO upon request and payment of the fee set forth in 37 CFR 1.19(b)(3).

The invention claimed is:

1. A combination comprising a Bt toxin selected from the group consisting of: Cry1a, Cry1b, Cry1c, Cry2a, and combinations thereof, and a U+2-ACTX-Hv1a peptide having the amino acid sequence set forth in SEQ ID NO: 5, wherein the Bt toxin and the U+2-ACTX-Hv1a peptide are not fused together as a single fusion protein, wherein the combination is formulated in separate compositions, and wherein the separate compositions of the combination come together and have insecticidal activity.

2. The combination of claim 1, wherein the separate compositions are formulated as powders, dusts, pellets, granules, sprays, emulsions, colloids, solutions, or combinations thereof.

3. The combination of claim 1, wherein the separate compositions are formulated using the same excipients or different excipients.

4. The combination of claim 1, wherein the separate compositions are formulated using the same excipients or different excipients.

5. The combination of claim 1, wherein the combination comprises the Bt toxin integrally expressed in a plant, and the U+2-ACTX-Hv1a peptide is formulated as a composition comprising at least one excipient.

6. The combination of claim 1, wherein the combination comprises the U+2-ACTX-Hv1a peptide integrally expressed in a plant, and the Bt toxin is formulated as a composition comprising at least one excipient.

7. The combination of claim 1, wherein the combination comprises the U+2-ACTX-Hv1a peptide, and the Bt toxin both integrally expressed in a plant.

8. The combination of claim 1, wherein Btk toxin is a *Bacillus thuringiensis* ssp. *kurstaki* strain EVB-113-19 Btk toxin.

9. The combination of claim 1, wherein the ratio of the Bt toxin to the U+2-ACTX-Hv1a peptide is about 10,000:1, 5,000:1, 1,000:1, 500:1, 250:1, 200:1, 100:1, 99:1, 95:5, 90:10, 85:15, 80:20, 75:25, 70:30, 65:35, 60:40, 55:45, 1:1, 45:55, 40:60, 35:65, 30:70, 25:75, 20:80, 15:85, 10:90, 5:95, 1:99, 1:100, 1:200, 1:250, 1:500, 1:1,000, 1:5,000, or 1:10,000.

10. The combination of claim 9, wherein the ratio of Bt toxin to U+2-ACTX-Hv1a peptide is from about 1:1 to about 1:5000.

11. The combination of claim 10, wherein the ratio of Bt toxin to U+2-ACTX-Hv1a peptide is about 1:4000.

12. The combination of claim 9, wherein the ratio of Bt toxin to U+2-ACTX-Hv1a peptide is from about 1:1 to about 1:10.

13. The combination of claim 12, wherein the ratio of Bt toxin to U+2-ACTX-Hv1a peptide is about 1:9.2.

14. The combination of claim 12, wherein the ratio of Bt toxin to U+2-ACTX-Hv1a peptide is about 1:8.75.

15. A method of using the combination of claim 1 to control insects comprising, providing a combination of a Bt toxin selected from the group consisting of: Cry1a, Cry1b, Cry1c, Cry2a, and combinations thereof, and a U+2-ACTX-Hv1a peptide having the amino acid sequence set forth in SEQ ID NO:5; wherein the Bt toxin is not fused to the U+2-ACTX-Hv1a peptide CRIP; wherein the combination is formulated in separate compositions and then applying said combination to the locus of an insect, and wherein the separate compositions of the combination come together and have insecticidal activity.

16. The method of claim 15, wherein the separate compositions are formulated as powders, dusts, pellets, granules, sprays, emulsions, colloids, solutions, or combinations thereof.

17. The method of claim 15, wherein the separate compositions are formulated using the same excipients or different excipients.

18. The method of claim 15, wherein the combination comprises the Bt toxin integrally expressed in a plant, and the U+2-ACTX-Hv1a peptide is formulated as a composition comprising at least one excipient.

19. The method of claim 15, wherein the combination comprises the U+2-ACTX-Hv1a peptide integrally expressed in a plant, and the Bt toxin is formulated as a composition comprising at least one excipient.

20. The method of claim 15, wherein the combination comprises the U+2-ACTX-Hv1a peptide, and the Bt toxin both integrally expressed in a plant.

21. The method of claim 15, wherein the insects are selected from the group consisting of: Achema Sphinx Moth (Hornworm) (*Eumorpha achemon*); Alfalfa Caterpillar (*Colias eurytheme*); Almond Moth (*Caudra cautella*); Amorbia Moth (*Amorbia humerosana*); Armyworm (*Spodoptera* spp., e.g. *exigua*, *frugiperda*, *littoralis*, *Pseudaletia unipuncta*); Artichoke Plume Moth (*Platyptilia carduidactyla*); Azalea Caterpillar (*Datana major*); Bagworm (*Thyridopteryx*); *ephemeraeformis*; Banana Moth (*Hypercompe scribonia*); Banana Skipper (*Erionota thrax*); Blackheaded Budworm (*Acleris gloverana*); California Oakworm (*Phryganidia californica*); Spring Cankerworm (*Paleacrita merriicata*); Cherry Fruitworm (*Grapholita packardii*); China Mark Moth (*Nymphula stagnata*); Citrus Cutworm (*Xylomyges curialis*); Codling Moth (*Cydia pomonella*); Cranberry Fruitworm (*Acrobasis vaccinii*); Cross-striped Cabbageworm (*Evergestis rimosalis*); Cutworm (*Noctuid* species, *Agrotis ipsilon*); Douglas Fir Tussock Moth (*Orgyia pseudotsugata*); Ello Moth (Hornworm) (*Erinnyis ello*); Elm Spanworm (*Ennomos subsignaria*); European Grapevine Moth (*Lobesia botrana*); European Skipper (*Thymelicus lineola* (Essex Skipper); Fall Webworm (*Melissopus latiferreanus*); Filbert Leafroller (*Archips rosanus*); Fruittree Leafroller (*Archips argyrospilia*); Grape Berry Moth (*Paralobe-*

sia viteana; Grape Leafroller (*Platynota stultana*); Grapeleaf Skeletonizer (*Harrisina americana* (ground only)); Green Cloverworm (*Plathypena scabra*); Greenstriped Mapleworm (*Dryocampa rubicunda*); Gummosos-Batrachedra; Comosae (Hodges); Gypsy Moth (*Lymantria dispar*); Hemlock Looper (*Lambdina fiscellaria*); Hornworm (*Manduca* spp.); Imported Cabbageworm (*Pieris rapae*); Io Moth (*Automeris io*); Jack Pine Budworm (*Choristoneura pinus*); Light Brown Apple Moth (*Epiphyas postvittana*); Melonworm (*Diaphania hyalinata*); Mimosa Webworm (*Homadaula anisocentra*); Obliquebanded Leafroller (*Choristoneura rosaceana*); Oleander Moth (*Syntomeida epilais*); Omnivorous Leafroller (*Platynota stultana*); Omnivorous Looper (*Sabulodes aegrotata*); Orangedog (*Papilio cressphontes*); Orange Tortrix (*Argyrotaenia citrana*); Oriental Fruit Moth (*Grapholita molesta*); Peach Twig Borer (*Anarsia lineatella*); Pine Butterfly (*Neophasia menapia*); Podworm (*Heliocoverpa zea*); Redbanded Leafroller (*Argyrotaenia velutinana*); Redhumped Caterpillar (*Schizura concinna*); Rindworm Complex (Various Leps.); Saddleback Caterpillar (*Sibine stimulea*); Saddle Prominent Caterpillar (*Heterocampa guttivitta*); Saltmarsh Caterpillar (*Estigmene acrea*); Sod Webworm (*Crambus* spp.); Spanworm (*Ennomos subsignaria*); Fall Cankerworm (*Alsophila pometaria*); Spruce Budworm (*Choristoneura fumiferana*); Tent Caterpillar (Various Lasiocampidae); Thecla-Thecla Basilides (Geyr) (*Thecla basilides*); Tobacco Hornworm (*Manduca sexta*); Tobacco Moth (*Ephestia elutella*); Tufted Apple Budmoth (*Platynota idaeusalis*); Twig Borer (*Anarsia lineatella*); Variegated Cutworm (*Peridroma saucia*); Variegated Leafroller (*Platynota flavedana*); Velvetbean Caterpillar (*Anticarsia gemmatilis*); Walnut Caterpillar (*Datana integerrima*); Webworm (*Hyphantria cunea*); Western Tussock Moth (*Orgyia vetusta*); Southern Cornstalk Borer (*Diatraea crambidoides*); Corn Earworm; Sweet potato weevil; Pepper weevil; Citrus root weevil; Strawberry root weevil; Pecan weevil; Filbert weevil; Ricewater weevil; Alfalfa weevil; Clover weevil; Tea shot-hole borer; Root weevil; Sugarcane beetle; Coffee berry borer; Annual blue grass weevil (*Lissonotus maculicollis*); Asiatic garden beetle (*Maladera castanea*); European chafer (*Rhizotrogus majalis*); Green June beetle (*Cotinis nitida*); Japanese beetle (*Popillia japonica*); May or June beetle (*Phyllophaga* sp.); Northern masked chafer (*Cyclocephala borealis*); Oriental beetle (*Anomala orientalis*); Southern masked chafer (*Cyclocephala lurida*); Billbug (*Curculionoidea*); *Aedes aegypti*; *Busseola fusca*; *Chilo suppressalis*; *Culex pipiens*; *Culex quinquefasciatus*; *Diabrotica virgifera*; *Diatraea saccharalis*; *Helicoverpa armigera*; *Helicoverpa zea*; *Heliothis virescens*; *Leptinotarsa decemlineata*; *Ostrinia furnacalis*; *Ostrinia nubilalis*; *Pectinophora gossypiella*; *Plodia interpunctella*; *Plutella xylostella*; *Pseudoplusia includens*; *Spodoptera exigua*; *Spodoptera frugiperda*; *Spodoptera littoralis*; *Trichoplusia ni*; and *Xanthogaleruca luteola*.

22. A method to control *Bacillus thuringiensis*-toxin-resistant insects using the combination of claim 1, the method comprising, providing the combination and then applying said combination to the locus of an insect.

23. The method of claim 22, wherein the *Bacillus thuringiensis*-toxin-resistant insects are selected from the group consisting of: *Aedes aegypti*; *Busseola fusca*; *Chilo suppressalis*; *Culex pipiens*; *Culex quinquefasciatus*; *Diabrotica virgifera*; *Diatraea saccharalis*; *Helicoverpa armigera*; *Helicoverpa zea*; *Heliothis virescens*; *Leptinotarsa decemlineata*; *Ostrinia furnacalis*; *Ostrinia nubilalis*; *Pectinophora gossypiella*; *Plodia interpunctella*; *Plutella xylo-*

stella; *Pseudoplusia includens*; *Spodoptera exigua*; *Spodoptera frugiperda*; *Spodoptera littoralis*; *Trichoplusia ni*; and *Xanthogaleruca luteola*.

24. A method of combating, controlling, or inhibiting a pest comprising, applying a pesticidally effective amount of the combination of claim 1 to the locus of the pest, or to a plant or animal susceptible to an attack by the pest.

25. The method of claim 24, wherein the pest is selected from the group consisting of: Achema Sphinx Moth (Hornworm) (*Eumorpha achemon*); Alfalfa Caterpillar (*Colias eurytheme*); Almond Moth (*Caudra cautella*); Amorbia Moth (*Amorbia humerosana*); Armyworm (*Spodoptera* spp., e.g. *exigua*, *frugiperda*, *littoralis*, *Pseudaletia unipuncta*); Artichoke Plume Moth (*Platyptilia carduidactyla*); Azalea Caterpillar (*Datana major*); Bagworm (*Thyridopteryx*); ephemeriformis); Banana Moth (*Hypercompe scribonia*); Banana Skipper (*Erionota thrax*); Blackheaded Budworm (*Acleris gloverana*); California Oakworm (*Phryganidia californica*); Spring Cankerworm (*Paleacrita merriacata*); Cherry Fruitworm (*Grapholita packardii*); China Mark Moth (*Nymphula stagnata*); Citrus Cutworm (*Xylomyges curialis*); Codling Moth (*Cydia pomonella*); Cranberry Fruitworm (*Acrobasis vaccinii*); Cross-striped Cabbageworm (*Evergestis rimosalis*); Cutworm (*Noctuid* species, *Agrotis ipsilon*); Douglas Fir Tussock Moth (*Orgyia pseudotsugata*); Ello Moth (Hornworm) (*Erinnyis ello*); Elm Spanworm (*Ennomos subsignaria*); European Grapevine Moth (*Lobesia botrana*); European Skipper (*Thymelicus lineola* (Essex Skipper)); Fall Webworm (*Melissopus latiferreanus*); Filbert Leafroller (*Archips rosanus*); Fruittree Leafroller (*Archips argyrospilia*); Grape Berry Moth (*Paralobesia viteana*); Grape Leafroller (*Platynota stultana*); Grapeleaf Skeletonizer (*Harrisina americana* (ground only)); Green Cloverworm (*Plathypena scabra*); Greenstriped Mapleworm (*Dryocampa rubicunda*); Gummosos-Batrachedra; Comosae (Hodges); Gypsy Moth (*Lymantria dispar*); Hemlock Looper (*Lambdina fiscellaria*); Hornworm (*Manduca* spp.); Imported Cabbageworm (*Pieris rapae*); Io Moth (*Automeris io*); Jack Pine Budworm (*Choristoneura pinus*); Light Brown Apple Moth (*Epiphyas postvittana*); Melonworm (*Diaphania hyalinata*); Mimosa Webworm (*Homadaula anisocentra*); Obliquebanded Leafroller (*Choristoneura rosaceana*); Oleander Moth (*Syntomeida epilais*); Omnivorous Leafroller (*Platynota stultana*); Omnivorous Looper (*Sabulodes aegrotata*); Orangedog (*Papilio cressphontes*); Orange Tortrix (*Argyrotaenia citrana*); Oriental Fruit Moth (*Grapholita molesta*); Peach Twig Borer (*Anarsia lineatella*); Pine Butterfly (*Neophasia menapia*); Podworm (*Heliocoverpa zea*); Redbanded Leafroller (*Argyrotaenia velutinana*); Redhumped Caterpillar (*Schizura concinna*); Rindworm Complex (Various Leps.); Saddleback Caterpillar (*Sibine stimulea*); Saddle Prominent Caterpillar (*Heterocampa guttivitta*); Saltmarsh Caterpillar (*Estigmene acrea*); Sod Webworm (*Crambus* spp.); Spanworm (*Ennomos subsignaria*); Fall Cankerworm (*Alsophila pometaria*); Spruce Budworm (*Choristoneura fumiferana*); Tent Caterpillar (Various Lasiocampidae); Thecla-Thecla Basilides (Geyr) (*Thecla basilides*); Tobacco Hornworm (*Manduca sexta*); Tobacco Moth (*Ephestia elutella*); Tufted Apple Budmoth (*Platynota idaeusalis*); Twig Borer (*Anarsia lineatella*); Variegated Cutworm (*Peridroma saucia*); Variegated Leafroller (*Platynota flavedana*); Velvetbean Caterpillar (*Anticarsia gemmatilis*); Walnut Caterpillar (*Datana integerrima*); Webworm (*Hyphantria cunea*); Western Tussock Moth (*Orgyia vetusta*); Southern Cornstalk Borer (*Diatraea crambidoides*); Corn Earworm; Sweet potato weevil; Pepper weevil; Citrus root weevil; Strawberry root weevil; Pecan

weevil; Filbert weevil; Ricewater weevil; Alfalfa weevil;
 Clover weevil; Tea shot-hole borer; Root weevil; Sugarcane
 beetle; Coffee berry borer; Annual blue grass weevil (*Lis-*
tronotus maculicollis); Asiatic garden beetle (*Maladera cas-*
tanea); European chafer (*Rhizotroqus majalis*); Green June 5
 beetle (*Cotinis nitida*); Japanese beetle (*Popillia japonica*);
 May or June beetle (*Phyllophaga* sp.); Northern masked
 chafer (*Cyclocephala borealis*); Oriental beetle (*Anomala*
orientalis); Southern masked chafer (*Cyclocephala lurida*);
 Billbug (*Curculionoidea*); *Aedes aegypti*; *Busseola fusca*; 10
Chilo suppressalis; *Culex pipiens*; *Culex quinquefasciatus*;
Diabrotica virgifera; *Diatraea saccharalis*; *Helicoverpa*
armigera; *Helicoverpa zea*; *Heliothis virescens*; *Leptino-*
tarsa decemlineata; *Ostrinia furnacalis*; *Ostrinia nubilalis*;
Pectinophora gossypiella; *Plodia interpunctella*; *Plutella* 15
xylostella; *Pseudoplusia includens*; *Spodoptera exigua*;
Spodoptera frugiperda; *Spodoptera littoralis*; *Trichoplusia*
ni; and *Xanthogaleruca luteola*.

26. The method of claim 25, wherein the pest is selected
 from the group consisting of: *Aedes aegypti*; *Busseola fusca*; 20
Chilo suppressalis; *Culex pipiens*; *Culex quinquefasciatus*;
Diabrotica virgifera; *Diatraea saccharalis*; *Helicoverpa*
armigera; *Helicoverpa zea*; *Heliothis virescens*; *Leptino-*
tarsa decemlineata; *Ostrinia furnacalis*; *Ostrinia nubilalis*;
Pectinophora gossypiella; *Plodia interpunctella*; *Plutella* 25
xylostella; *Pseudoplusia includens*; *Spodoptera exigua*;
Spodoptera frugiperda; *Spodoptera littoralis*; *Trichoplusia*
ni; and *Xanthogaleruca luteola*.

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